

GUV_Analysis_summary

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Details

DESCRIPTION

Loads the reorganised data from `1Data/colour_dance_reorg.csv`, saved from `GUV_organisedata.R`, and generates plots to show the how dances changed as a function of light stimulus.

INPUTS

`colour_dance_reorg.csv`, `Edrich1979_brightness-accuracy.csv`

OUTPUTS

`2Results/phototaxis_modlist.RData`, `2Results/MLE_paired_diffs.csv`, `2Results/MLEdetails_paired_diffs.RData`, `2Results/LRT_paired_diffs.csv`, `2Results/BF_paired_diffs.csv`,

REFERENCES

Edrich, W., Neumeyer, C. and von Helversen, O. (1979). “Anti-sun orientation” of bees with regard to a field of ultraviolet light. *J. Comp. Physiol.* 134, 151–157.

Rossel, S. and Wehner, R. (1984). Celestial orientation in bees: the use of spectral cues. *J. Comp. Physiol. A* 155, 605–613.

Setting up

First load the functions and packages required for this project. These can all be found in `GUV_functions.R`, within the same Github repository.

```
source('GUV_functions.R')
```

Then load the data that will be used. This includes our own dance data, reorganised by `GUV_organisedata.R` as `colour_dance_reorg.csv`, as well as data from Edrich *et al.* (1979) on the relationship between accuracy (mean vector length), wavelength and intensity. This was manually extracted from Fig. 2 and saved in `"Edrich1979_brightness-accuracy.csv"`.

```
#Dance angle data
cd = read.table(file = '1Data/colour_dance_reorg.csv',
                header = T,
                sep = ',')

#Accuracy data
ed = read.table(file = '1Data/Edrich1979_brightness-accuracy.csv',
                header = T,
```

```

        sep = ',')
#Inspect each dataset
head(cd)

```

```

##   measured_angle run      date bearing      angle      ID brightn
## 1           270   1 2016.05.31      90  1.57079633 2016.05.31.14.24      1
## 2              5   2 2016.05.31     355 -0.08726646 2016.05.31.14.24      1
## 3             20   3 2016.05.31     340 -0.34906585 2016.05.31.14.24      1
## 4            320   4 2016.05.31      40  0.69813170 2016.05.31.14.24      1
## 5             20   5 2016.05.31     340 -0.34906585 2016.05.31.14.24      1
## 6             35   6 2016.05.31     325 -0.61086524 2016.05.31.14.24      1
##   colour      cet_time  sun_az  sun_el  sun_el_rad  sun_az_rad
## 1      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556  -2.647205
## 2      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556  -2.647205
## 3      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556  -2.647205
## 4      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556  -2.647205
## 5      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556  -2.647205
## 6      u 2016-05-31 14:24:00 208.3263 58.81733  1.026556  -2.647205

```

```
head(ed)
```

```

##      xx intensity      nn accuracy wavelength
## 1 -31.208  2.15e+09 13.010      0.14        354
## 2 -20.263  3.69e+09 14.249      0.15        535
## 3  -8.217  6.67e+09 19.135      0.20        354
## 4  -1.342  9.36e+09 15.350      0.16        535
## 5  15.118  2.10e+10 30.769      0.33        354
## 6  24.548  3.35e+10 14.731      0.16        535

```

Format data

Ensure that each variable has the correct data format for subsequent analysis.

```

cd = within(cd,
{
  ID = as.factor(ID) # beedance identifier as a factor
  date = as.factor(date) # date as a factor
  signed_angle = Mod360.180(bearing) # bearing between -180 and 180
  angle = circular(rad(signed_angle), # bearing between -pi and pi
                    rotation = 'clock') # circular format suppresses later warnings
})
u_id = with(cd, unique(ID)) # unique beedances
length(u_id) #169 beedances

```

```
## [1] 169
```

Identify dances in which all four stimulus conditions were used.

```
luc = sapply(u_id,
             FUN = IndCond,
             dt = cd)

#Most individuals that made it to bright green
full_ids = u_id[luc == 4]
length(full_ids)#19 individuals
```

```
## [1] 19
```

```
#extract just those individuals

#find data for full condition individuals
full_cd = subset(cd, ID %in% full_ids)
```

Check dance length

Ensure that each variable has the correct data format for subsequent analysis.

```
mruns = aggregate(run ~ ID,
                  data = cd,
                  FUN = max)
summary(mruns$run)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  14.00   26.00   33.00   36.83   43.00   100.00
```

Plot the raw dance data

Plot an example individual to show effect

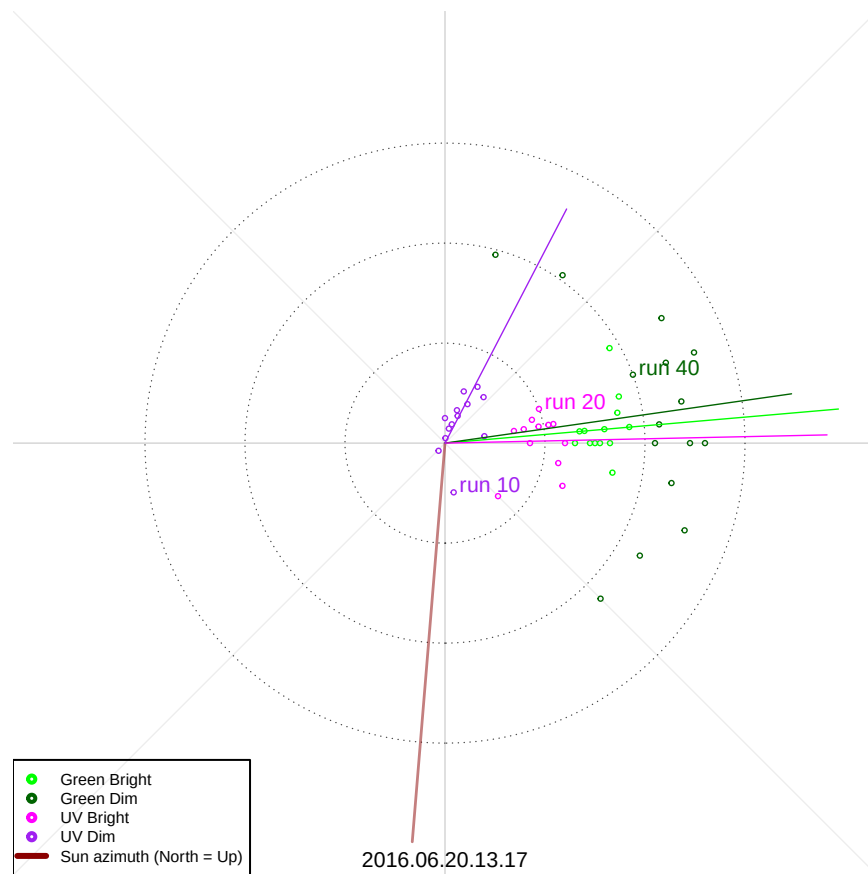
For example we can look at the dance recorded on 2016.06.20 at 13:17.

```
par(pty = 's')
par(mfrow = c(1,1), mar = c(0,0,0,0))
PltDance(ids = '2016.06.20.13.17',
         dtf = cd,
         cx = 0.7)
with(subset(cd, ID %in% '2016.06.20.13.17' & run %in% c(10, 20, 40)),
     {
text(x = run*sin(as.numeric(angle)),
     y = run*cos(as.numeric(angle)),
     labels = paste(' run', run),
     col = c('green', 'darkgreen', 'magenta', 'purple')
         [ifelse(colour %in% 'g',
                 yes = ifelse(brightn %in% 'h', yes = 1, no = 2),
                 no = ifelse(brightn %in% 'h', yes = 3, no = 4))],
     cex = 0.7,
     adj = c(0,0))
})
```

```

)
legend(x = 'bottomleft',
      legend = c('Green Bright',
                  'Green Dim',
                  'UV Bright',
                  'UV Dim',
                  'Sun azimuth (North = Up)'),
      col = c('green',
              'darkgreen',
              'magenta',
              'purple',
              'darkred'),
      pch = c(21, 21, 21, 21, NA),
      lty = c(NA, NA, NA, NA, 1),
      lwd = c(2, 2, 2, 2, 3),
      cex = 0.5
)

```



A plot including all 169 dances can be found at the end of this markdown.

Accuracy effects

Calculate the mean vectors for each set of waggle-runs under each lighting condition.

```

#calculate mean vectors
mean_vectors = aggregate(angle~ID*brightn*colour+sun_az+sun_el_rad, # N.B. Including sun azimuth drops
                          data = cd,
                          FUN = rho.circular
)
#correct names
mean_vectors = within(mean_vectors,
                      {mean_vector = angle; rm(angle)} # anlge now indicates a mean vector, not an angl
)
#add kappa
mle_estimates = aggregate(angle~ID*brightn*colour+sun_az+sun_el,
                          data = cd,
                          FUN = MLE_est
)
#Rayleigh tests
rayleigh_tests = aggregate(angle~ID*brightn*colour+sun_az+sun_el,
                          data = cd,
                          FUN = Rayleigh_p
)
#add to the summary table and
#calculate inverse softplus kappa
mean_vectors = within(mean_vectors,
                      {
                        mu = deg(mle_estimates$angle[, 'mu'])
                        kappa = mle_estimates$angle[, 'kappa']
                        iskappa = inv_softplus(kappa)
                      }
)
#find the full condition individuals
mean_vectors_full = subset(mean_vectors,
                          subset = ID %in% full_ids)
rayleigh_tests_full = subset(rayleigh_tests,
                          subset = ID %in% full_ids)
#check how many dances would count as "oriented"
#all dances
with(rayleigh_tests,
     round(mean(angle <0.05) *100,digits = 2)
)

```

```
## [1] 88.43
```

```

#all dances with balanced data
with(rayleigh_tests_full,
     round(mean(angle <0.05) *100,digits = 2)
)

```

```
## [1] 88.24
```

```

#bright green (sun spot) dances
with(subset(rayleigh_tests,
            subset = brightn %in% 'h' & colour %in% 'g'),
     round(mean(angle <0.05) *100,digits = 2)
)

```

```
## [1] 94.44
```

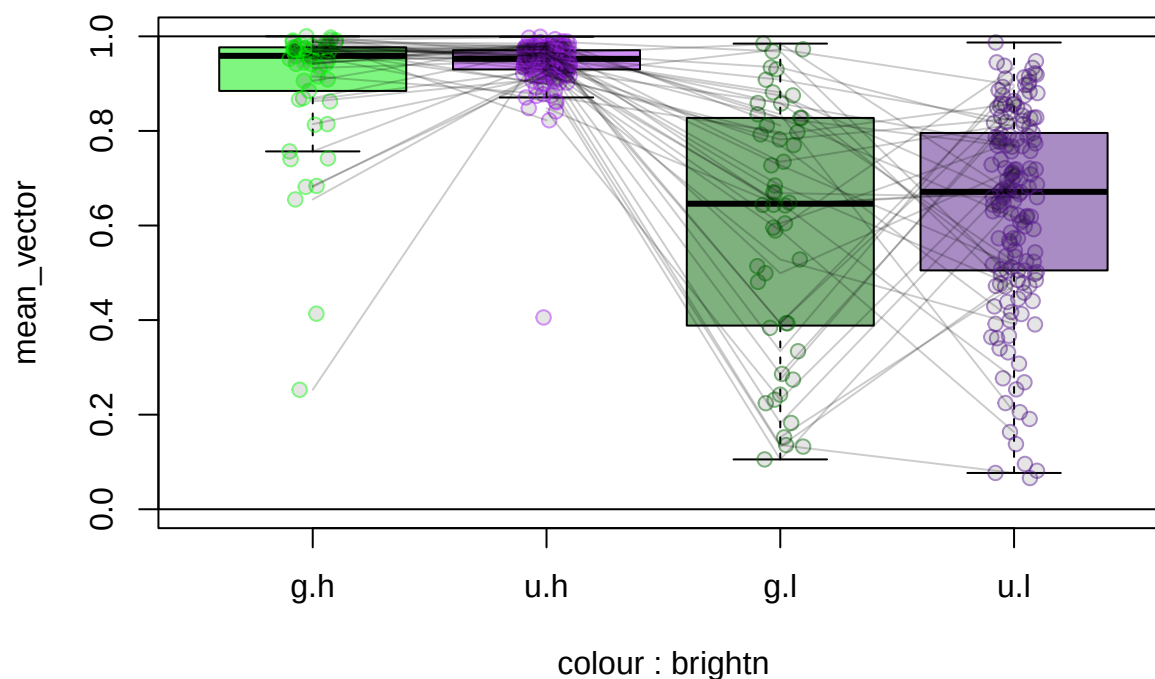
```
#bright green (sun spot) dances with balanced data
with(subset(rayleigh_tests_full,
            subset = brightn %in% 'h' & colour %in% 'g'),
     round(mean(angle <0.05) *100,digits = 2)
    )
```

```
## [1] 88.24
```

Effect of brightness on mean vector length

Entire dataset The first thing we notice is that in the low intensity conditions the accuracy is generally lower than for the high intensity conditions.

```
boxplot(mean_vector~colour*brightn,
        data = mean_vectors,
        ylim = c(0,1),
        col = adjustcolor(c('green2',
                             'purple',
                             'darkgreen',
                             'purple4'), alpha.f = 0.5),
        outline = FALSE)
stripchart(mean_vector~colour*brightn,
           data = mean_vectors,
           add = TRUE,
           vertical = TRUE,
           method = 'jitter',
           col = adjustcolor(c('green2',
                               'purple',
                               'darkgreen',
                               'purple4'), alpha.f = 0.5),
           bg = gray(level = 0,
                     alpha = 0.1),
           pch = 21)
#Identify stimuli
stim = apply(X = expand.grid(c = c('g','u'),b = c('h','l')),
            FUN = paste,
            collapse = '',
            MARGIN = 1)
#Plot lines linking dances by the same individual
invisible(
  {
    lapply(X = u_id,
          FUN = Plt_mvec
        )
  })
abline(h = c(0,1))
```



Print summary statistics Use the quantile function to summarise the distribution of mean vectors within each combination of stimulus colour and brightness.

```
mv_agg = aggregate(mean_vector~colour*brightn,
  data = mean_vectors,
  FUN = quantile,
  probs = c(0.5, 0.25, 0.75))
rownames(mv_agg) = stim
mv_agg = within(mv_agg, rm(colour, brightn))
mv_agg = t(apply(mv_agg, MARGIN = 1, unlist))
print(mv_agg,
  digits = 3)
```

##	mean_vector.50%	mean_vector.25%	mean_vector.75%
## gh	0.959	0.890	0.977
## uh	0.952	0.930	0.971
## gl	0.646	0.391	0.827
## ul	0.671	0.505	0.795

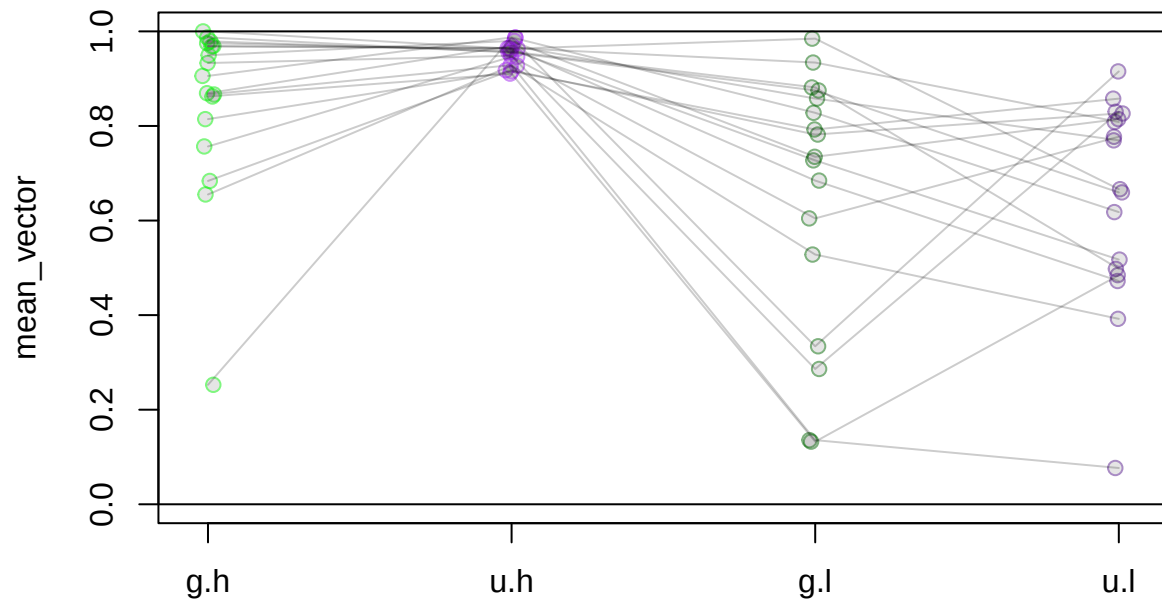
Within individuals (balanced cases only) This effect is even clearer if we look at just the full condition dances.

```

stripchart(mean_vector~colour*brightn,
  data = mean_vectors_full,
  # add = TRUE,
  vertical = TRUE,
  method = 'jitter',
  jitter = 0.02,
  ylim = c(0,1),
  col = adjustcolor(c('green2',
                      'purple',
                      'darkgreen',
                      'purple4'), alpha.f = 0.5),

  bg = gray(level = 0,
            alpha = 0.1),
  pch = 21)
#Plot lines linking dances by the same individual
invisible(
  {
    lapply(X = full_ids,
      FUN = Plt_mvec
    )
  })
abline(h = c(0,1))

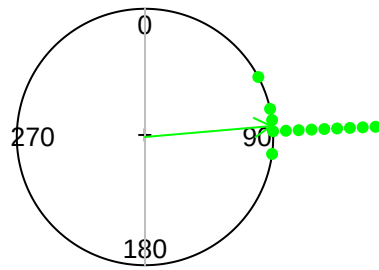
```



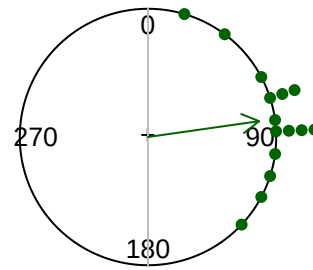
Plot an example individual to show effect

For example we can look at the dance recorded on 2016.06.20 at 13:17.

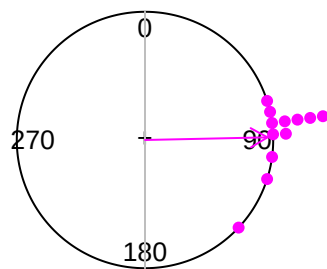
```
example_subs = subset(x = cd,
                      ID %in% '2016.06.20.13.17')
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'g' & brightn %in% 'h')$bearing,
      col = 'green',
      shrink = 1.5,
      title = 'Green Bright')
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'g' & brightn %in% 'l')$bearing,
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim')
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'u' & brightn %in% 'h')$bearing,
      col = 'magenta',
      shrink = 1.5,
      title = 'UV Bright')
PCfun(angles = subset(example_subs,
                      subset = colour %in% 'u' & brightn %in% 'l')$bearing,
      col = 'purple',
      shrink = 1.5,
      title = 'UV Dim')
```



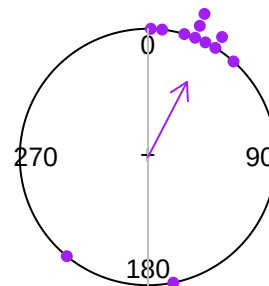
Green Bright



Green Dim



UV Bright



UV Dim

For this individual orientation is at a fixed angle to the light source (0°), but changes in consistency (green, dim), mean direction and bimodality (UV, dim) in response to changes in light source.

Non-parametric tests on accuracy

Here non-parametric or parametric statistics could be revealing.

```
#Identify all effects as factors for aligned ranks transform
mean_vectors_full = within(mean_vectors_full,
  {
    colour = as.factor(colour)
    brightn = as.factor(brightn)
    ID = as.factor(ID)
  }
)
mean_vectors = within(mean_vectors,
  {
    colour = as.factor(colour)
    brightn = as.factor(brightn)
    ID = as.factor(ID)
  }
)
mv_full_art = art(formula = mean_vector~colour*brightn +
  (1+colour+brightn|ID),#insufficient DF to estimate interaction
  data = mean_vectors_full)
sm_mv_full_art = summary(mv_full_art)
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

#check if rank alignment was successful
message('\nSuccessful rank alignment:\n',
        all(sapply(sm_mv_full_art[11], round) == 0),
        '\n')
```

```
##
## Successful rank alignment:
## TRUE
```

```
print(anova(mv_full_art), digits = 3)
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')

## Analysis of Variance of Aligned Rank Transformed Data
##
## Table Type: Analysis of Deviance Table (Type III Wald F tests with Kenward-Roger df)
## Model: Mixed Effects (lmer)
## Response: art(mean_vector)
##
##              F Df Df.res      Pr(>F)
## 1 colour      0.00452  1    16    0.94724
## 2 brightn     56.97631  1    16 1.1689e-06 ***
## 3 colour:brightn 2.99074  1    16    0.10298
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can successfully fit an aligned ranks model to compare across individuals. Note that the “boundary” level random effects indicate that individual effects were too small to estimate. In the individual linear models (e.g. `summary(artlm(mv_full_art, "brightn"))$varcor`) there may still be substantial individual differences. The primary effect appears to be that of brightness, but there may also be effects of interest in the post-hoc contrasts.

```
cn_mv_full_art = sapply(X = c('brightn', 'colour', 'colour:brightn'),
                        FUN = ART_extract,
                        mod = mv_full_art,
                        adjust = 'sidak')
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## NOTE: Results may be misleading due to involvement in interactions
```

```
print(cn_mv_full_art)
```

```
## $brightn
## contrast estimate SE df t.ratio p.value
## h - l          25.5 3.37 16   7.548 <.0001
##
## Results are averaged over the levels of: colour
## Degrees-of-freedom method: kenward-roger
##
## $colour
## contrast estimate SE df t.ratio p.value
## g - u          -0.353 5.25 16  -0.067  0.9472
##
## Results are averaged over the levels of: brightn
## Degrees-of-freedom method: kenward-roger
##
## $`colour:brightn`
## contrast estimate SE df t.ratio p.value
## g,h - g,l      17.71 3.51 28.3   5.038  0.0001
## g,h - u,h     -11.35 4.84 22.2  -2.347  0.1579
## g,h - u,l      22.00 6.29 16.0   3.497  0.0178
## g,l - u,h     -29.06 4.02 16.0  -7.231 <.0001
## g,l - u,l       4.29 4.84 22.2   0.888  0.9455
## u,h - u,l      33.35 3.51 28.3   9.490 <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: sidak method for 6 tests
```

The post-hoc tests indicate a large difference between bright and dim conditions, as expected, but very little evidence for differences between colours. The interactions (for which there was low support in the overall model) reflect the pattern we expect, that all bright conditions have higher accuracy than all dim conditions, and that while the two dim conditions have similar accuracy, the bright UV bright condition may be slightly more accurate than the bright green condition ($p = 0.158$).

Parametric beta-distributed model of accuracy

To investigate this with greater statistical power we would need an appropriate parametric model for bounded variables. One approach is to use the beta distribution, which is defined between (but not at) 0 and 1.

```
### Maximal formula with all random effects -----
#set up model formula with indiv effects on light type
frm_bmod_maxRE = bf(mean_vector~brightn*colour + # light types affect accuracy
                    (1 + brightn*colour|ID), # each individual is more or less oriented
                    phi ~ brightn*colour + # light types affect variance in accuracy
                    (1 + brightn*colour|ID), # some individuals vary more in accuracy than others
                    family = brms::Beta())

#suggest prior values
pr_mu_Intercept = list(dist = 'normal',
                       mu = 0,
                       sd = 1)#deconstrain to beat divergences?
pr_phi_Intercept = list(dist = 'normal',
```

```

        mu = 1,
        sd = 1)
pr_mu_b = list(dist = 'normal',
              mu = 0,
              sd = 1)#deconstrain to beat divergences?
pr_phi_b = list(dist = 'normal',
              mu = 0,
              sd = 1)
pr_mu_sd_Intercept = list(dist = 'student_t',
                        df = 3,
                        mu = 0,
                        sd = 0.1)#constrain to beat divergences?
pr_phi_sd_Intercept = list(dist = 'student_t',
                        df = 3,
                        mu = 0,
                        sd = 0.1)#constrain to beat divergences?
pr_mu_sd = list(dist = 'student_t',
              df = 3,
              mu = 0,
              sd = 0.1)#constrain to beat divergences?
pr_phi_sd = list(dist = 'student_t',
              df = 3,
              mu = 0,
              sd = 0.1)#constrain to beat divergences?

### Assign priors -----

#set priors on:
# - all coefficients to a normal(0,1) = 95% probability of values 0.12-0.87
# - all grouping paramters to a student_t(df = 3, mu = 0, sigma = 2.5) = values < 5.88
# - all other parameters (?) to a normal(0,10)= 95% probability of values 0-1 +- 1e-7

pr_expression = expression({
  prior = sapply(class,
                switch,
                Intercept = with(pr_mu_Intercept,
                                {paste0(dist, '(', mu, ',', sd, ')')}),
                                #the general prior for coefficients (should centre on 0)
                b = with(pr_mu_b,
                        {paste0(dist, '(', mu, ',', sd, ')')}),
                        #the general prior for standard deviations (should be long-tailed)
                sd = with(pr_mu_sd,
                        {paste0(dist, '(', df, ',', mu, ',', sd, ')')}),
                cor = 'lkj(1)', # The only supported prior for correlation matrices is the 'lkj' prior
                'normal(0,1)' # with this many params need more informative priors
  )
#special priors
#mu sd intercept
prior[dpar %in% 'sd' & class %in% 'sd' & coef %in% ''] =
  with(pr_mu_sd_Intercept,
        {paste0(dist, '(', df, ',', mu, ',', sd, ')')})
#phi b
prior[dpar %in% 'phi' & class %in% 'b'] =

```

```

with(pr_phi_b,
      {paste0(dist, '(', mu, ',', sd, ')')})
#phi intercept
prior[dpar %in% 'phi' & class %in% 'b' & coef %in% ''] =
  with(pr_phi_Intercept,
        {paste0(dist, '(', mu, ',', sd, ')')})
#phi sd
prior[dpar %in% 'phi' & class %in% 'sd'] =
  with(pr_phi_sd,
        {paste0(dist, '(', df, ',', mu, ',', sd, ')')})
#phi sd intercept
prior[dpar %in% 'phi' & class %in% 'sd' & coef %in% ''] =
  with(pr_phi_sd_Intercept,
        {paste0(dist, '(', df, ',', mu, ',', sd, ')')})
}
)

#model with all random effects
priors_bmod_mxRE = within(get_prior(formula = frm_bmod_maxRE,
                                   data = mean_vectors_full),
                          expr = eval(pr_expression))
#remove excess priors generated by get_prior but not used by brms
priors_bmod_mxRE = subset(priors_bmod_mxRE, !(coef %in% ''))
print(priors_bmod_mxRE) #all parameters are accounted for

```

Beta model for accuracy

```

##           prior class           coef group resp dpar nlpar lb ub tag
##      normal(0,1)      b      brightnl
##      normal(0,1)      b brightnl:colouru
##      normal(0,1)      b      colouru
## student_t(3,0,0.1)    sd      brightnl      ID
## student_t(3,0,0.1)    sd brightnl:colouru      ID
## student_t(3,0,0.1)    sd      colouru      ID
## student_t(3,0,0.1)    sd      Intercept      ID
##      normal(0,1)      b      brightnl          phi
##      normal(0,1)      b brightnl:colouru          phi
##      normal(0,1)      b      colouru          phi
## student_t(3,0,0.1)    sd      brightnl      ID      phi
## student_t(3,0,0.1)    sd brightnl:colouru      ID      phi
## student_t(3,0,0.1)    sd      colouru      ID      phi
## student_t(3,0,0.1)    sd      Intercept      ID      phi
## source
## default
## default
## default
## default
## default
## default
## default
## default
## default
## default

```

```
## default
## default
## default
## default
```

```
## Run the models -----
system.time( #takes about 2 minutes
{
md_beta = brm(formula = frm_bmod_maxRE,
               prior = priors_bmod_mxRE,
               data = within(mean_vectors_full,
                             {
                               mean_vector = (mean_vector * (length(mean_vector) - 1) + 0.5) / length(m
                             }
                             ),
               family = Beta(link = 'logit',
                             link_phi = 'log'),
               iter = 2e3, # only small number necessary
               control = list( adapt_delta = 0.99 ), # divergences are hard to beat, so set very high
               cores = 4,
               backend = 'cmdstanr',
               silent = 2, #don't print lots of iteration information
               refresh = 0
             )
}
)
```

```
## Warning: 66 of 4000 (2.0%) transitions ended with a divergence.
## See https://mc-stan.org/misc/warnings for details.
```

```
## Loading required namespace: rstan
```

```
## user system elapsed
## 66.873 2.956 50.400
```

```
#some divergent transitions, but rhat and ESS look good
rh_beta = rhat(md_beta)
sm_beta = summary(md_beta)
```

```
## Warning: There were 66 divergent transitions after warmup. Increasing
## adapt_delta above 0.99 may help. See
## http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

```
print(summary(rh_beta), digits= 4)#all 1.01
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.9998 1.0021 1.0036 1.0059 1.0077 1.0272
```

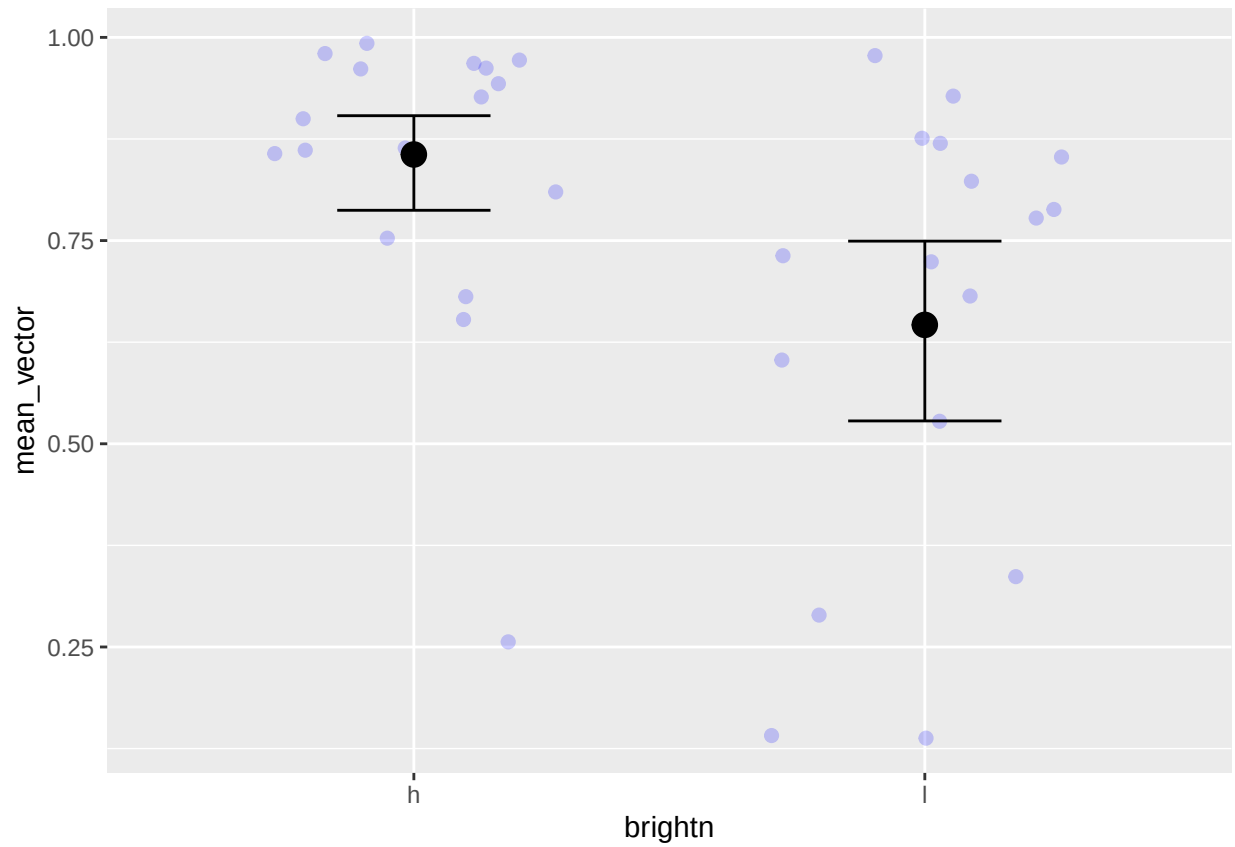
```
print(sm_beta$fixed[, c(1, 3:4)], digits = 3)
```

##	Estimate	l-95% CI	u-95% CI
## Intercept	1.780	1.309	2.2383
## phi_Intercept	1.821	1.239	2.3895
## brightnl	-1.171	-1.767	-0.5518
## colouru	0.973	0.437	1.4985
## brightnl:colouru	-0.936	-1.669	-0.2081
## phi_brightnl	-0.744	-1.503	0.0542
## phi_colouru	2.071	1.065	3.0289
## phi_brightnl:colouru	-0.999	-2.300	1.8911

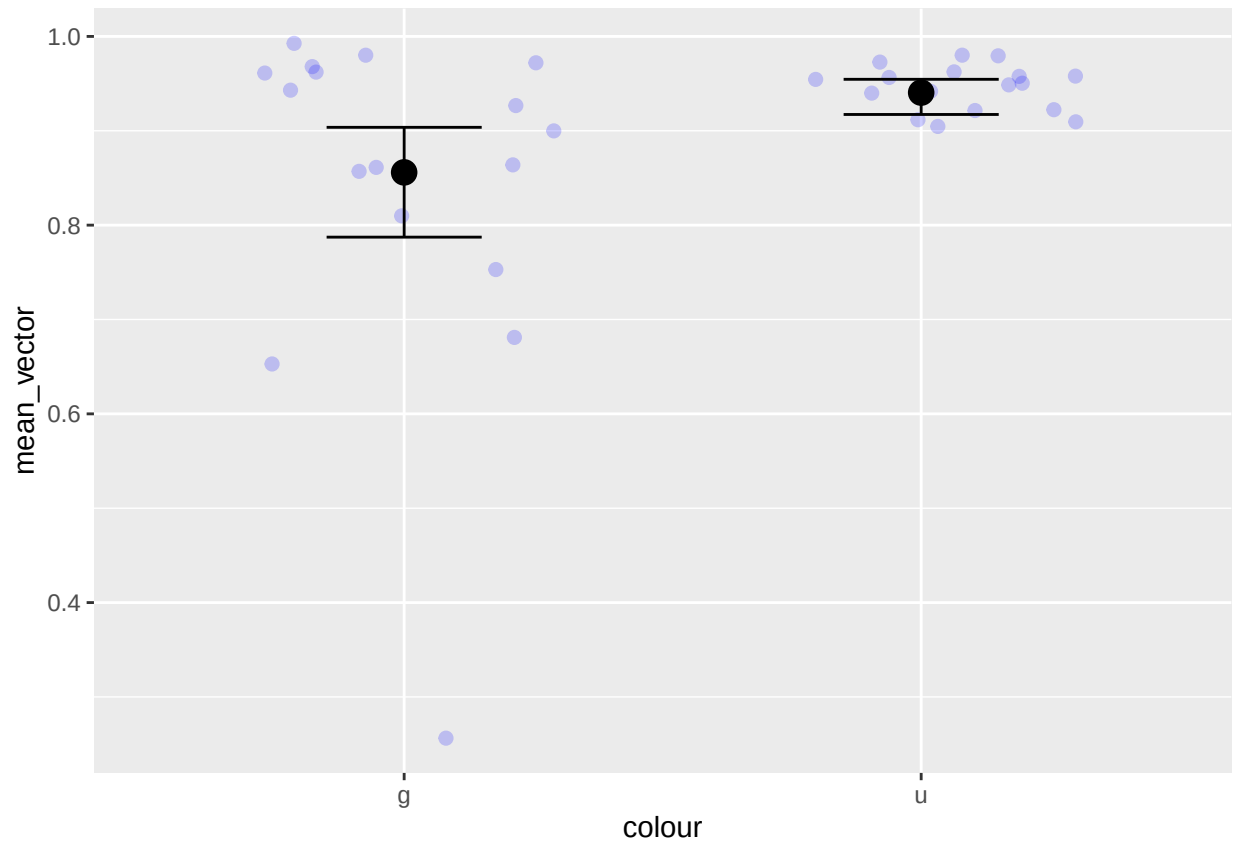
Extract the model's predictions and compare with the original data.

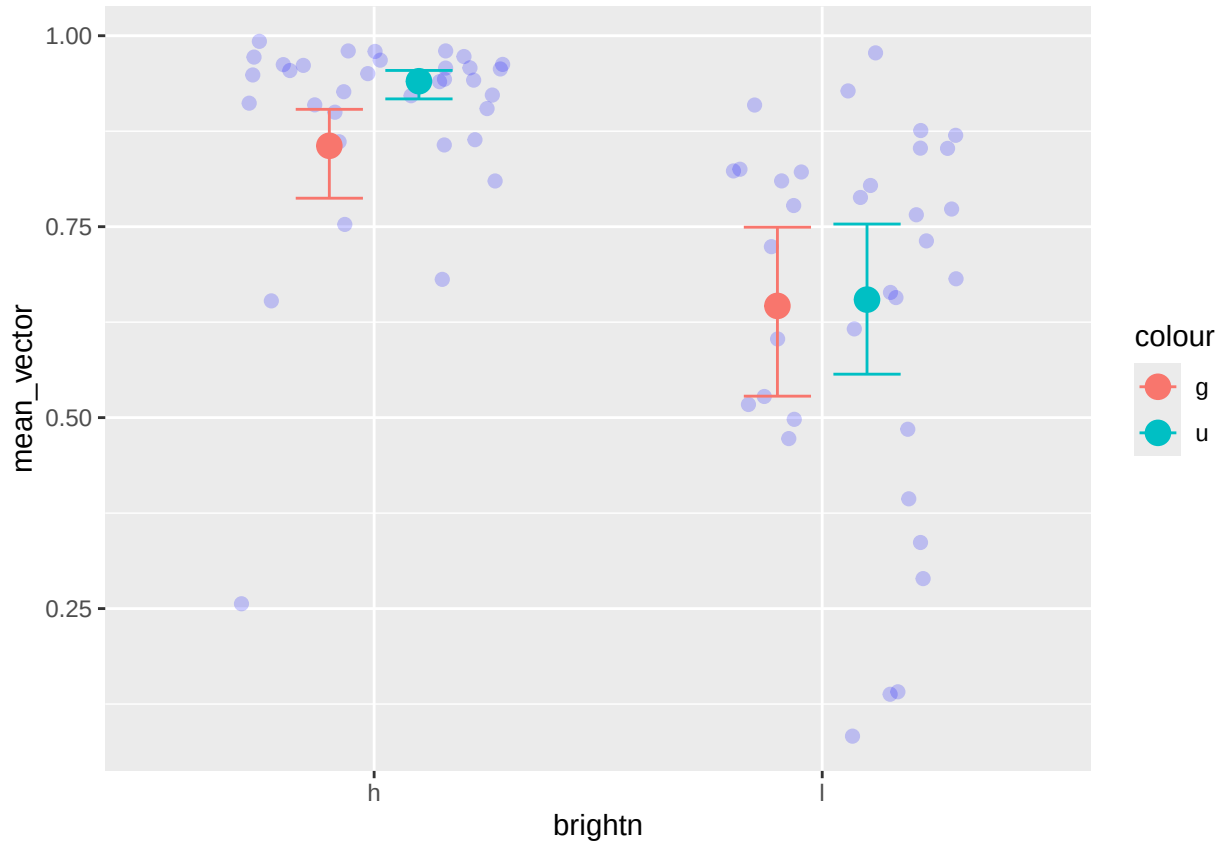
```
# Extract model predictions -----
beta_cond_eff = conditional_effects(md_beta)
#Plot including individual data points
plot(x = beta_cond_eff,
     points = TRUE,
     point_args = list(width = 0.3,
                       col = adjustcolor('blue',
                                         alpha.f = 0.2),
                       width = 0.3
                     ),
     ylim = c(0,1),
     ask = FALSE
  )
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```

```
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
## Ignoring unknown labels:
## * fill : "NA"
## * colour : "NA"
```





This seems to suggest that dances were more accurate in the bright UV condition, which the aligned-ranks test was sceptical of. We can quantify this difference by extracting and comparing the parameter estimates from each (post-warmup) iteration of the model.

```
draws_beta = prepare_predictions(md_beta,
                                re_formula = NA) #just population level effects
#for the average vector length
draws_mu = draws_beta$dparams$mu$fe$b
#for variance in vector length
draws_phi = draws_beta$dparams$phi$fe$b
#N.B. Could also just calculate predictions, then subtract
contr_mu = with(data.frame(draws_mu),
  {
    data.frame(
      #bright green -> dim green = low brightness effect
      gh_gl = b_brightnl,
      #bright UV -> dim UV = dim + UVdim - UV effect
      uvh_uvl = b_brightnl + `b_brightnl.colouru` - b_colouru,
      #bright green -> bright UV = UV effect
      gh_uvh = b_colouru,
      #bright green -> dim UV = UV + dim + UVdim
      gh_uvl = b_brightnl + b_colouru + `b_brightnl.colouru`,
      #dim green -> dim UV = UV effect + UVdim - dim effect
      gl_uvl = b_colouru + `b_brightnl.colouru` - b_brightnl
    )
  }
)
```

```

    )
#On the original scale
contr_rho = with(data.frame(draws_mu),
{
  data.frame(
    #bright green -> dim green = low brightness effect
    gh_gl = plogis(b_Intercept + b_brightnl) -
      plogis(b_Intercept),
    #bright UV -> dim UV = dim + UVdim - UV effect
    uvh_uvl = plogis(b_Intercept + b_brightnl + `b_brightnl.colouru`) -
      plogis(b_Intercept + b_colouru),
    #bright green -> bright UV = UV effect
    gh_uvh = plogis(b_Intercept + b_colouru) -
      plogis(b_Intercept),
    #bright green -> dim UV = UV + dim + UVdim
    gh_uvl = plogis(b_Intercept + b_brightnl +
      b_colouru + `b_brightnl.colouru`)-
      plogis(b_Intercept),
    #dim green -> dim UV = UV effect + UVdim - dim effect
    gl_uvl = plogis(b_Intercept + b_colouru +
      `b_brightnl.colouru`) -
      plogis(b_Intercept + b_brightnl)
  )
}
)

contr_phi = with(data.frame(draws_phi),
{
  data.frame(
    #bright green -> dim green = low brightness effect
    gh_gl = b_phi_brightnl,
    #bright UV -> dim UV = dim + UVdim - UV effect
    uvh_uvl = b_phi_brightnl + `b_phi_brightnl.colouru` - b_phi_colouru,
    #bright green -> bright UV = UV effect
    gh_uvh = b_phi_colouru,
    #bright green -> dim UV = UV + dim + UVdim
    gh_uvl = b_phi_brightnl + b_phi_colouru + b_phi_brightnl.colouru,
    #dim green -> dim UV = UV effect + UVdim - dim effect
    gl_uvl = b_phi_colouru + `b_phi_brightnl.colouru` - b_phi_brightnl
  )
}
)

contr_all = cbind(contr_mu, contr_phi)
quant_all = apply(X = contr_all,
  MARGIN = 2,
  FUN = quantile,
  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
)
quant_mu = apply(X = contr_mu,
  MARGIN = 2,
  FUN = quantile,
  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
)

```

```

quant_rho = apply(X = contr_rho,
                  MARGIN = 2,
                  FUN = quantile,
                  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
                  )
quant_phi = apply(X = contr_phi,
                  MARGIN = 2,
                  FUN = quantile,
                  probs = c(0,0.5,1)+c(1,0,-1)*(0.05)/2
                  )
par(mfrow = c(1, length(contr_mu)))
for(ct in 1:length(contr_mu) )
{
  VertHist(data = contr_mu[,ct],
            main = gsub(pattern = ' ',
                        replacement = ' → ',
                        x = names(contr_mu)[ct]),

            axes = FALSE,
            col = gray(0.75),
            ylim = c(-1, 1)*4,
            ylab = if(ct ==1){'logit mean change in accuracy'}else{''}
            )

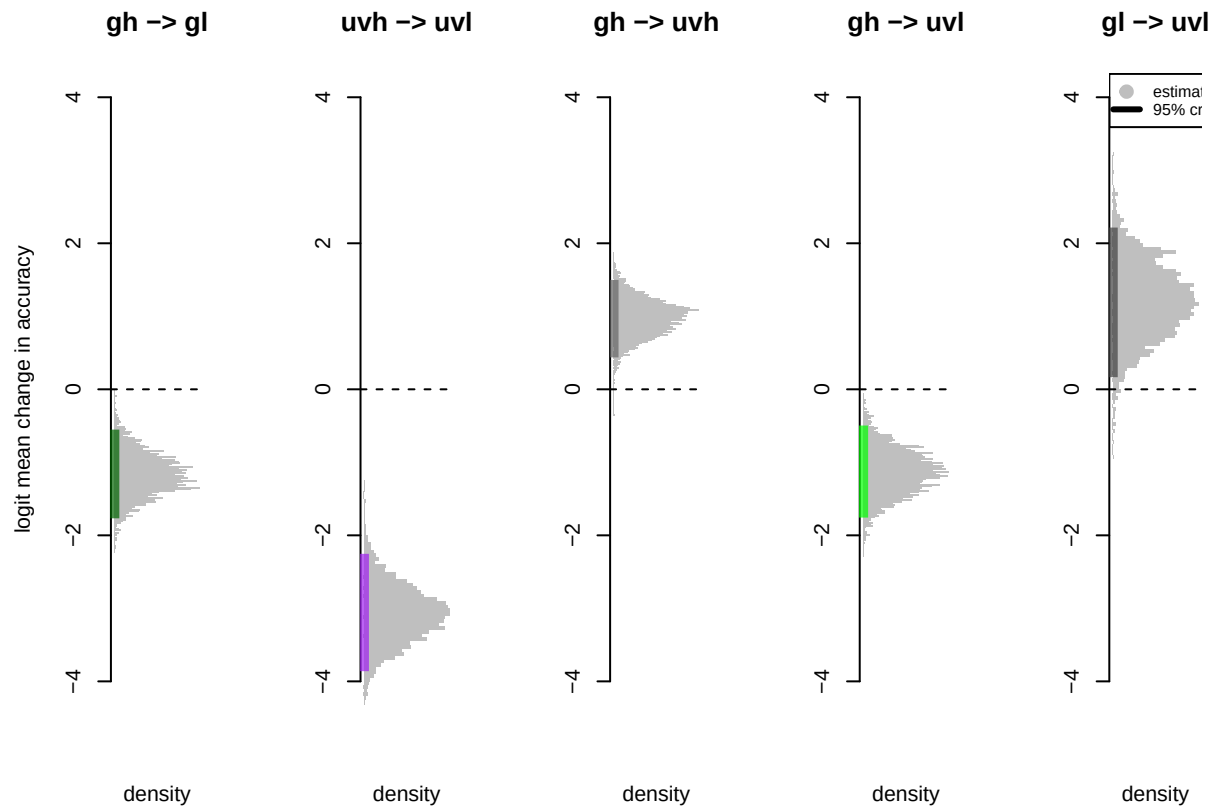
  axis(2)

  abline(h = 0,
          lty = 2)
  with(data.frame(t(quant_mu[,ct])),
        {
          arrows(x0 = 0,
                 y0 = X2.5.,
                 y1 = X97.5.,
                 length = 0,
                 angle = 90,
                 lwd = 5,
                 col = adjustcolor(c('darkgreen',
                                     'purple',
                                     'gray40',
                                     'green',
                                     'gray25'
                                     ), alpha = 0.7)[ct],
                 lend = 'butt'
                 )
        }
  )
}

legend(x = 'topleft',
       legend = c('estimates',
                  '95% credible interval'),
       pch = c(20, NA),
       lty = c(NA, 1),
       lwd = c(5, 3),
       col = c('gray', 'black'),

```

```
cex = 0.7
)
```



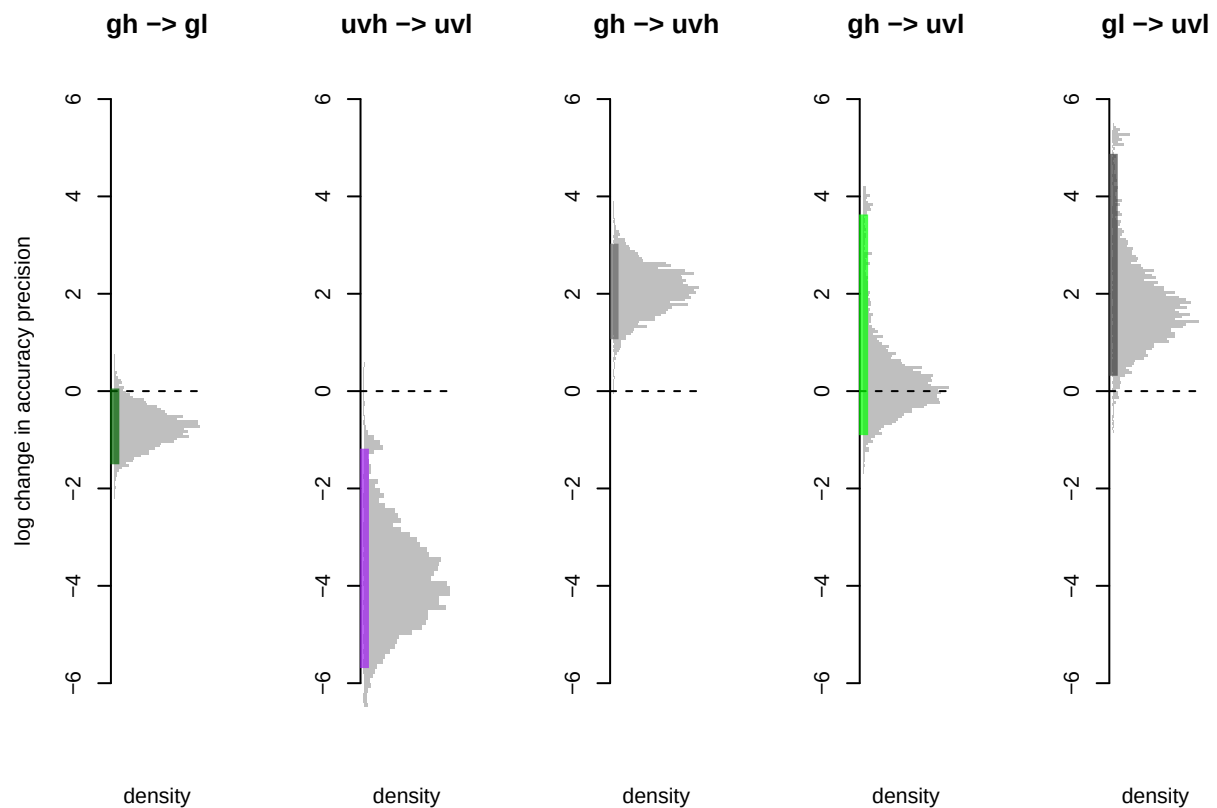
```
par(mfrow = c(1, length(contr_phi)))
for(ct in 1:length(contr_phi) )
{
  VertHist(data = contr_phi[,ct],
    main = gsub(pattern = '_',
      replacement = ' -> ',
      x = names(contr_phi)[ct]),
    axes = FALSE,
    col = gray(0.75),
    ylim = c(-1, 1)*6,
    ylab = if(ct ==1){'log change in accuracy precision'}else{''}
  )
  axis(2)

  abline(h = 0,
    lty = 2)
  with(data.frame(t(quant_phi[,ct])),
    {
      arrows(x0 = 0,
        y0 = X2.5.,
        y1 = X97.5.,
        length = 0,
```

```

    angle = 90,
    lwd = 5,
    col = adjustcolor(c('darkgreen',
                        'purple',
                        'gray40',
                        'green',
                        'gray25'
                        ), alpha = 0.7)[ct],
    lend = 'butt'
  )
}
)
}

```



```

#can also calculate these with emmeans
em_beta = emmeans(object = md_beta,
                  specs = ~brightn*colour)
ct_beta_mu = contrast(object = em_beta,
                    specs = ~brightn*colour,
                    method = 'pairwise',
                    side = c(1,-1,1,-1,-1,1)
                    )
print(ct_beta_mu, digits = 3)

```

```
## contrast estimate lower.HPD upper.HPD
```

```
## h g - l g      1.181      0.540      1.752
## h g - h u     -0.986     -1.515     -0.462
## h g - l u      1.132      0.495      1.757
## l g - h u     -2.151     -2.729     -1.599
## l g - l u     -0.033     -0.662      0.541
## h u - l u      2.107      1.637      2.605
##
## Point estimate displayed: median
## Results are given on the log odds ratio (not the response) scale.
## HPD interval probability: 0.95
```

```
#collect and estimate p values from samples
empirical.p = function(x, tails = 2){ifelse(test = median(x)>0,
                                             yes = mean(x<0)*tails,
                                             no = mean(x>0)*tails)
}

p_beta_mu = apply(X = contr_mu, MARGIN = 2, FUN = empirical.p, tails = 2)
p_beta_phi = apply(X = contr_phi, MARGIN = 2, FUN = empirical.p, tails = 2)
beta_all_res = data.frame(rho_lower = quant_rho[1,],
                          rho_median = quant_rho[2,],
                          rho_upper = quant_rho[3,],
                          mu_lower = quant_mu[1,],
                          mu_median = quant_mu[2,],
                          mu_upper = quant_mu[3,],
                          p_mu = p_beta_mu,
                          p_mu_adj = p.adjust(p = p_beta_mu,
                                                method = 'BH'),
                          phi_lower = quant_phi[1,],
                          phi_median = quant_phi[2,],
                          phi_upper = quant_phi[3,],
                          p_phi = p_beta_phi,
                          p_phi_adj = p.adjust(p = p_beta_phi,
                                                method = 'BH')
                          )
print(beta_all_res[,c(2,5,7,10)], digits = 3)
```

```
##          rho_median mu_median  p_mu phi_median
## gh_gl      -0.2077    -1.181 0.0000    -0.739
## uvh_uvl     -0.5193    -3.081 0.0000    -3.909
## gh_uvh       0.0838     0.986 0.0015     2.084
## gh_uvl      -0.1992    -1.132 0.0000     0.125
## gl_uvl       0.2117     1.209 0.0270     1.683
```

```
#set up model formula with indiv effects on light type
frm_elemod_maxRE = bf(mean_vector~ cos(sun_el_rad) + # accuracy is affected by projection distortion at
                      (1 | ID), # each individual is more or less oriented
                      phi ~ cos(sun_el_rad) + # accuracy is affected by projection distortion at high elevation
                      (1 | ID), # some individuals vary more in accuracy than others
                      family = brms::Beta())
```



```

#suggest prior values
pr_mu_Intercept = list(dist = 'normal',
                        mu = 0,
                        sd = 1)#deconstrain to beat divergences?
pr_phi_Intercept = list(dist = 'normal',
                        mu = 1,
                        sd = 1)
pr_mu_b = list(dist = 'normal',
               mu = 0,
               sd = 1)#deconstrain to beat divergences?
pr_phi_b = list(dist = 'normal',
                mu = 0,
                sd = 1)
pr_mu_sd_Intercept = list(dist = 'student_t',
                          df = 3,
                          mu = 0,
                          sd = 0.1)#constrain to beat divergences?
pr_phi_sd_Intercept = list(dist = 'student_t',
                          df = 3,
                          mu = 0,
                          sd = 0.1)#constrain to beat divergences?
pr_mu_sd = list(dist = 'student_t',
                df = 3,
                mu = 0,
                sd = 0.1)#constrain to beat divergences?
pr_phi_sd = list(dist = 'student_t',
                df = 3,
                mu = 0,
                sd = 0.1)#constrain to beat divergences?

### Assign priors -----

#set priors on:
# - all coefficients to a normal(0,1) = 95% probability of values 0.12-0.87
# - all grouping paramters to a student_t(df = 3, mu = 0, sigma = 2.5) = values < 5.88
# - all other parameters (?) to a normal(0,10)= 95% probability of values 0-1 +- 1e-7

pr_expression = expression({
  prior = sapply(class,
                 switch,
                 Intercept = with(pr_mu_Intercept,
                                   {paste0(dist, '(', mu, ', ', sd, ')'})),
                 #the general prior for coefficients (should centre on 0)
                 b = with(pr_mu_b,
                           {paste0(dist, '(', mu, ', ', sd, ')'})),
                 #the general prior for standard deviations (should be long-tailed)
                 sd = with(pr_mu_sd,
                           {paste0(dist, '(', df, ', ', mu, ', ', sd, ')'})),
                 cor = 'lkj(1)', # The only supported prior for correlation matrices is the 'lkj' prior
                 'normal(0,1)' # with this many params need more informative priors
  )
#special priors
#mu sd intercept

```

```

prior[dpar %in% '' & class %in% 'sd' & coef %in% ''] =
  with(pr_mu_sd_Intercept,
        {paste0(dist, '(', df, ',', mu, ',', sd, ')')})

#phi b
prior[dpar %in% 'phi' & class %in% 'b'] =
  with(pr_phi_b,
        {paste0(dist, '(', mu, ',', sd, ')')})

#phi intercept
prior[dpar %in% 'phi' & class %in% 'b' & coef %in% ''] =
  with(pr_phi_Intercept,
        {paste0(dist, '(', mu, ',', sd, ')')})

#phi sd
prior[dpar %in% 'phi' & class %in% 'sd'] =
  with(pr_phi_sd,
        {paste0(dist, '(', df, ',', mu, ',', sd, ')')})

#phi sd intercept
prior[dpar %in% 'phi' & class %in% 'sd' & coef %in% ''] =
  with(pr_phi_sd_Intercept,
        {paste0(dist, '(', df, ',', mu, ',', sd, ')')})
}
)

#model with all random effects
priors_elemod_mxRE = within(get_prior(formula = frm_elemod_maxRE,
                                     data = mean_vectors_full),
                           expr = eval(pr_expression))

#remove additional priors added by get_prior but not used by brms
priors_elemod_mxRE = subset(priors_elemod_mxRE, !(coef %in% ''))
print(priors_elemod_mxRE) #all parameters are accounted for

```

Investigate effect of sun elevation

##		prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
##		normal(0,1)	b	cossun_el_rad								default
##		student_t(3,0,0.1)	sd	Intercept	ID							default
##		normal(0,1)	b	cossun_el_rad			phi					default
##		student_t(3,0,0.1)	sd	Intercept	ID		phi					default

```

system.time( #takes about 2 minutes
{
md_ele = brm(formula = frm_elemod_maxRE,
              prior = priors_elemod_mxRE,
              data = within(mean_vectors_full,
                            {
mean_vector = (mean_vector * (length(mean_vector) - 1) + 0.5) / length(m
                            }
                            ),
              family = Beta(link = 'logit',
                            link_phi = 'log'),
              iter = 2e3, # only small number necessary
              control = list( adapt_delta = 0.99 ), # divergences are hard to beat, so set very high
              cores = 4,

```

```

        backend = 'cmdstanr',
        silent = 2, #don't print lots of iteration information
        refresh = 0
    )
}
)

```

```

##      user  system elapsed
## 23.866   1.816   21.120

```

```

#too many divergent transitions, check chains
rh_ele = rhat(md_ele)
sm_ele = summary(md_ele)
print(summary(rh_ele), digits= 4)#all 1.01

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9997  1.0004  1.0007  1.0009  1.0013  1.0030

```

```

print(sm_ele$fixed[, c(1, 3:4)], digits = 3)

```

```

##              Estimate 1-95% CI u-95% CI
## Intercept          1.622    0.674    2.545
## phi_Intercept       1.381    0.308    2.457
## cossun_el_rad       -0.792   -2.219    0.641
## phi_cossun_el_rad   -0.371   -1.991    1.202

```

Extract the model's predictions and compare with the original data.

```

fx_ele = sm_ele$fixed[, c(1, 3:4)]
with(mean_vectors_full,
{
  plot(x = as.numeric(deg(sun_el_rad)),
       y = mean_vector,
       xlab = 'sun elevation (°)',
       ylab = 'accuracy (mean vector length)',
       main = 'general effect of cos(sun elevation) on accuracy',
       xlim = c(20,70),
       ylim = c(0,1),
       pch = 19,
       col = sapply(X = paste0(brightn, colour),
                    FUN = switch,
                    hg = 'green',
                    lg = 'darkgreen',
                    hu = 'magenta',
                    lu = 'purple'
                    )
       )
})
}
)
abline(h = c(0,1))
polygon(x = c(0:85, 85:0),

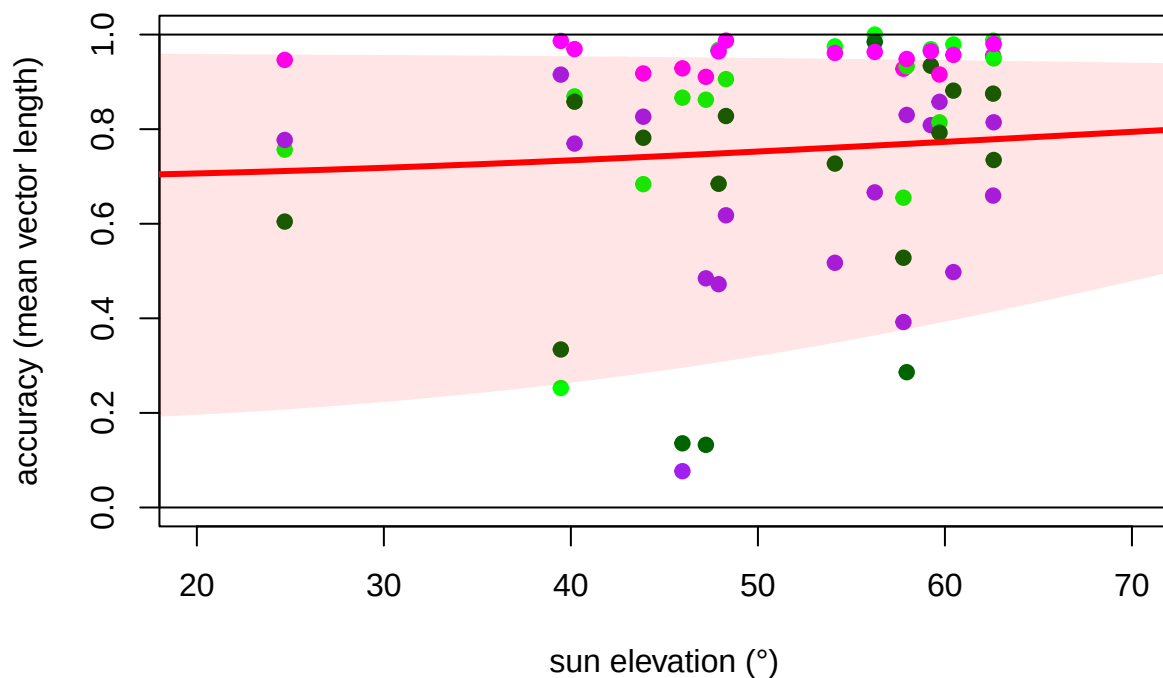
```

```

y = plogis(
  c(fx_ele$`u-95% CI`[1] + fx_ele$`u-95% CI`[3]*
      cos(rad(0:85)),
    rev( fx_ele$`l-95% CI`[1] + fx_ele$`l-95% CI`[3]*
      cos(rad(0:85)) )
  )
),
col = adjustcolor('red', alpha.f = 0.1),
border = NA)
lines(x = 0:85,
  y = plogis(
    fx_ele$Estimate[1] + fx_ele$Estimate[3]*
      cos(rad(0:85))
  ),
  col = 'red',
  lwd = 3)

```

general effect of cos(sun elevation) on accuracy



```
loo_beta = loo(md_beta)
```

```

## Warning: Found 2 observations with a pareto_k > 0.7 in model 'md_beta'. We
## recommend to set 'moment_match = TRUE' in order to perform moment matching for
## problematic observations.

```

```
loo_ele = loo(md_ele)
loo_compare(loo_beta, loo_ele)
```

```
##          elpd_diff se_diff
## md_beta   0.0       0.0
## md_ele  -24.2       7.1
```

Comparison with Edrich *et al.* (1979)

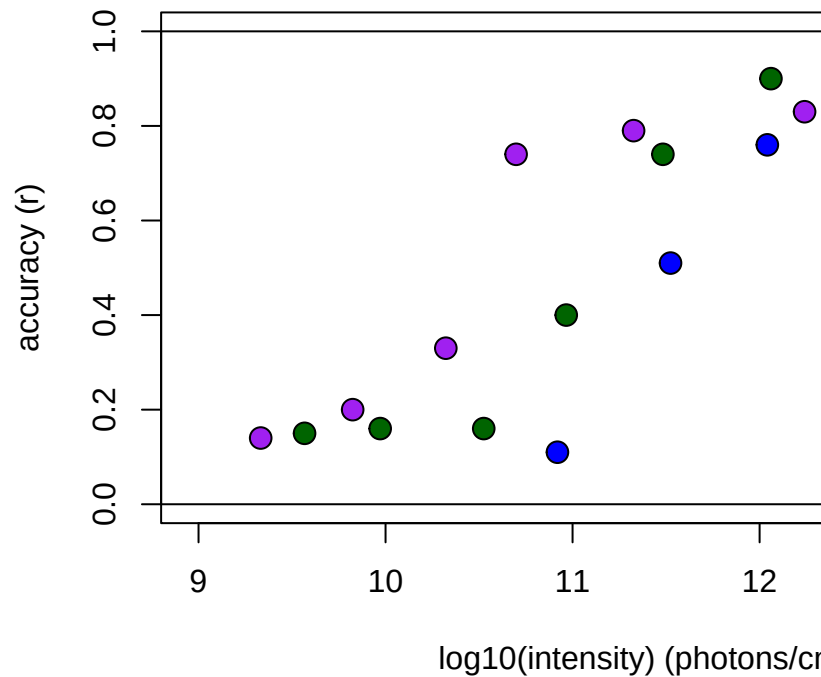
How does this compare with previous experiments in which a light spot (2.3°) was presented through bandpass filters and ND filters to produce different intensities? In that data, there was a clear sigmoidal relationship between intensity and accuracy (mean vector length). They mention that the particular relationship with light intensity was variable, but that the relative relationships remained constant, so we may also observe the same relationship but shifted.

Unfortunately, the scatter of the dance directions as a function of light intensity proved to be unstable over long periods of time; the absolute values changed from day to day. However, these changes were such that curves for adjacent wavelengths were shifted in parallel - the relative sensitivities remained constant. — Edrich et al. (1979), p154

```
with(ed,
{
  plot(x = log10(intensity),
       y = accuracy,
       xlim = c(9,14),
       ylim = c(0,1),
       main = 'Data from Edrich et al. 1979, Fig. 2',
       xlab = 'log10(intensity) (photons/cm2/s)',
       ylab = 'accuracy (r)',
       pch = 21,
       cex = 1.5,
       col = 'black',
       bg = sapply(paste(wavelength),
                   FUN = switch,
                   `354` = 'purple',
                   `429` = 'blue',
                   `535` = 'darkgreen',
                   'red'))
  )
  abline(h = c(0,1))
})

legend(x = 'bottomright',
       legend = paste(sort(unique(ed$wavelength),
                           decreasing = TRUE), 'nm'),
       col = c('darkgreen','blue', 'purple'),
       pch = c(20, 20, 20))
```

Data from Edrich et al. 1979



Plot raw data from Edrich *et al.* (1979) Fig. 2

This is best modelled with a nonlinear (psychometric model) following the beta distribution (for variables bounded between zero and one).

```
ed = within(ed,
{
  logit_accuracy = qlogis(accuracy)
  log10_intensity = log10(intensity)
  colour = sapply(paste(wavelength),
    FUN = switch,
    `354` = 'UV',
    `429` = 'blue',
    `535` = 'green',
    NA)
})
```

Reorganise data

Fit psychometric curves For each light stimulus, fit a sigmoid curve bounded between baseline accuracy (shared across stimuli) and maximum accuracy ($1 - \text{lapses}$). Inflection point and 80%-rise-region width vary between stimuli, maximising posterior probability for a beta distribution in $(0,1)$. To do this, we first define a nonlinear formula in **brms**.

```

#set up model fit
formula_beta = bf(
  formula = accuracy ~
    inv_logit(LogitBase) + (1 - inv_logit(LogitLapse) - inv_logit(LogitBase) ) *#curve region
    inv_logit( 4.39*(log10_intensity - Inflex) / exp(LogWidth) ) , #inflection-width curve
  # for each of these parameters, we can set up a separate formula
  # that describes how to predict them from the data
  #Base rate of correct choices: "
  LogitBase ~ 1, #Base rate of correct choices: "~ 1" gives the instruction "estimate the mean across a
  #Lapse rate on a log(odds) scale:
  LogitLapse ~ colour, #this is similar to the formula in our LMM example
  #inflection point of the initial curve:
  Inflex ~ colour, #N.B. this is similar to the intercept, so it does not include effects of stimulus level
  #log 80% width of the curve:
  LogWidth ~ colour, #N.B. this is similar to the slope, so all of its effects depend on stimulus level
  family = Beta(link = "identity", link_phi = 'log'),
  nl = TRUE)#the joint distribution for these parameters is undefined, and therefore the parameters the

```

Since this is a nonlinear model (there are terms both inside and outside the logistic transform) it is important to establish prior distributions for all parameters. These ensure that parameter estimates explore credible ranges during fitting. The baseline is expected to be around $r = 0.27$, while the maximum accuracy should be around $r = 0.95$, and their estimates should not converge (that would flatten the curve, or reverse its direction). The inflection points are expected around 10^{11} photons $\text{cm}^{-2} \text{s}^{-1}$, with a rise-region width of 10^3 photons $\text{cm}^{-2} \text{s}^{-1}$. All colour effects may vary by a few orders of magnitude.

```

#set up priors
prior_beta = get_prior(formula = formula_beta,
  data = ed)

#set the prior distribution
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitBase' &
    coef %in% 'Intercept'
  ] = 'normal(-1,1)' #a normal distribution centred on plogis(-1) = 0.27
  })

#this prior is automatically bounded between 0 and 1,
#Lapse rate priors -----
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitLapse' &
    coef %in% 'Intercept'
  ] = 'normal(-3,2)' #a normal distribution centred on -3
  })

#for all other fixed effects on lapse rate, we'll suggest values around 0 (no effect)
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitLapse' &
    coef %in% 'colourblue'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

```

```

    })
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogitLapse' &
    coef %in% 'colourUV'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

#Inflection point priors -----
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'Inflex' &
    coef %in% 'Intercept'
  ] = 'normal(11,2)' #a normal distribution:mean 11, sd 2
  })

#for all coefficients, we'll suggest values around 0 (no effect)
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'Inflex' &
    coef %in% 'colourblue'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 2
  })

prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'Inflex' &
    coef %in% 'colourUV'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 2
  })

#Rise region width priors -----
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogWidth' &
    coef %in% 'Intercept'
  ] = 'normal(1,2)' #a normal distribution:mean 1, sd 3
  })

#for all coefficients, we'll suggest values around 0 (no effect)
prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects
    nlpar %in% 'LogWidth' &
    coef %in% 'colourblue'
  ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

prior_beta = within(prior_beta,
  { prior[
    class %in% 'b' & #just the fixed effects

```



```

      nlpar %in% 'LogWidth' &
      coef %in% 'colourUV'
    ] = 'normal(0,2)' #a normal distribution:mean 0, sd 3
  })

```

Fit a psychometric model to the data using the nonlinear formula and specified prior distributions. 4000 iterations on 4 chains takes less than one minute.

```

#takes less than 1 minute
#parameters for the brms runs
wup = 2000
sam = 2000
refr = 0
system.time(
{
  beta_fit = brm( formula = formula_beta, # using our nonlinear formula
                  data = ed, # our data
                  prior = prior_beta, # our priors
                  iter = wup+sam, # long run for 4000 iterations
                  chains = 4, # 4 chains in parallel
                  cores = 4, # on 4 CPUs
                  refresh = refr,
                  silent = 0, # don't echo chain progress
                  control = list(adapt_delta = 0.95), #finer sampling
                  backend = 'cmdstanr') # use cmdstanr (other compilers broken)
}
)

```

```
## Start sampling
```

```

## Running MCMC with 4 parallel chains...
##
## Chain 2 finished in 2.0 seconds.
## Chain 1 finished in 2.1 seconds.
## Chain 3 finished in 2.0 seconds.
## Chain 4 finished in 2.5 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 2.2 seconds.
## Total execution time: 2.9 seconds.

```

```

##      user  system elapsed
## 23.638   1.751  19.609

```

Plot and compare data from Edrich and Kühn. Summarise the data from full cases in the Kühn data using quantiles (minimum, 25%, median, 75% and maximum).

```

#Examine quantiles
meanQ = aggregate(mean_vector ~ colour*brightn,
                  data = mean_vectors_full,
                  FUN = quantile)
#Extract different quantiles as separate columns in a data frame

```

```

meanQ = data.frame(cbind(meanQ[1:2], meanQ$mean_vector) )
#Rename
names(meanQ) = c('colour', 'brightn', paste0('q',0:4))
#Subset by condition
gh_meanQ = subset(meanQ, colour %in% 'g' & brightn %in% 'h')
gl_meanQ = subset(meanQ, colour %in% 'g' & brightn %in% 'l')
uh_meanQ = subset(meanQ, colour %in% 'u' & brightn %in% 'h')
ul_meanQ = subset(meanQ, colour %in% 'u' & brightn %in% 'l')

```

Extract the model predictions by summarising parameter estimates from the sampling phases of all chains. To identify specific combinations of parameter values, use the `conditional_effects` function, which organises all simultaneous estimates into a data frame.

```

# #N.B. for plotting we will need to also define "inv_logit" in our the R environment
inv_logit = inv_logit_scaled # this can be found in brms, but with an additional specifier

#summary of parameter estimates
full_sm_beta = summary(beta_fit,
                        robust = TRUE)#use the median estimate

#Extract fixed effects estimates
full_fix_beta = full_sm_beta$fixed
#extract rownames
full_fix_rn_beta = rownames(full_fix_beta)

# stimulus levels to add to the dataset to be able to measure this effect.
full_estimates_beta = full_fix_beta$Estimate
names(full_estimates_beta) = full_fix_rn_beta

# to predict all effects, we can use the 'posterior_epred' method
#by default 100 predictions per continuous variable (but fewer for their interactions)
#This takes <2 seconds
system.time(
{
  full_cond_beta =brms::conditional_effects(beta_fit,
                                           method = 'posterior_epred', # posterior epred not working
                                           cores = parallel::detectCores()-1,
                                           effects = c('log10_intensity:colour'),
                                           plot = FALSE
  )
}
)

```

```

##    user  system elapsed
##  2.170   0.194   2.389

```

```

#extract predictions
pred_data_beta = full_cond_beta$`log10_intensity:colour`

#Green light curve
g_curve =
with(data.frame(t(full_estimates_beta)),
{

```

```

    list(Base = plogis(LogitBase_Intercept),
          Maxima = 1- plogis(LogitLapse_Intercept),
          Inflection = Inflex_Intercept,
          Width = exp(LogWidth_Intercept)
    )
  }
)

#UV light curve
u_curve =
with(data.frame(t(full_estimates_beta)),
{
  list(Base = plogis(LogitBase_Intercept),
        Maxima = 1- plogis(LogitLapse_Intercept + LogitLapse_colourUV),
        Inflection = Inflex_Intercept + Inflex_colourUV,
        Width = exp(LogWidth_Intercept + LogWidth_colourUV)
  )
}
)

```

Plot the raw data alongside the predictions for the data from Edrich *et al.*, (1979) and a boxplot summary of the data data from Kühn (2017).

```

#Prepare a sequence for plotting
xxl = seq(from = 9, to = 16,
           length.out = 1e3)

#plot only the stimulus types also used by Kühn.
ed_ug = subset(ed, wavelength != 429)

#plot each stimulus type
with(ed_ug,
{
  plot(x = log10_intensity,
        y = accuracy,
        main =
'Comparison of data from Edrich et al. 1979
with Kühn, 2017',
        xlab = 'log10(intensity) (photons/cm2/s)',
        ylab = 'accuracy (r)',
        ylim = c(0,1),
        xlim = c(9,16),
        pch = 21,
        cex = 1.5,
        col = 'black',
        bg = sapply(paste(wavelength),
                     FUN = switch,
                     `354` = 'purple',
                     `429` = 'blue',
                     `535` = 'darkgreen',
                     'red')
        )
  abline(h = c(0,1))
}
)

```

```

    abline(v = log10(4*c(1e12,1e14)) )
  }
)
#Add the estimated range of the curve for the green stimulus
with(data.frame(t(full_estimates_beta)),
{
  abline(h = c(0,1,
    plogis(LogitBase_Intercept),
    1- plogis(LogitLapse_Intercept) ),
    lty = c(1,1,3,3)
  )
}
)

#plot total 95% prediction intervals
with(subset(pred_data_beta, colour == 'Ogreen'),
{
  polygon(x = c(sort(log10_intensity), rev(sort(log10_intensity))),
    y = c(lower__[order(log10_intensity)],
      rev(upper__[order(log10_intensity)]))
    ),
  col = adjustcolor('green', alpha.f = 25/256),
  border = NA,
  lwd = 0.1
  )
}
)
with(subset(pred_data_beta, colour == 'UV'),
{
  polygon(x = c(sort(log10_intensity), rev(sort(log10_intensity))),
    y = c(lower__[order(log10_intensity)],
      rev(upper__[order(log10_intensity)]))
    ),
  col = adjustcolor('magenta', alpha.f = 25/256),
  border = NA,
  lwd = 0.1
  )
}
)

#plot the median prediction lines
with(g_curve,
  lines(x = xx1,
    y = Base + (Maxima - Base) *
      plogis(4.39*(xx1 - Inflection)/Width),
    col = 'darkgreen',
    lwd = 3,
    lty = 3)
)
with(u_curve,
  lines(x = xx1,
    y = Base + (Maxima - Base) *
      plogis(4.39*(xx1 - Inflection)/Width),

```

```

        col = 'purple4',
        lwd = 3,
        lty = 3)
)

#plot observed data from Kühn as boxplots showing the quartiles, median and total range.
#Green High intensity
QBox(quant = subset(meanQ, colour %in% 'g' & brightn %in% 'h'),
      x = log10(4e14),
      offset = -0.1,
      cols = c('green', 'darkgreen')
)

#Green Low intensity
QBox(quant = subset(meanQ, colour %in% 'g' & brightn %in% 'l'),
      x = log10(4e12),
      offset = -0.1,
      cols = c('green', 'darkgreen')
)

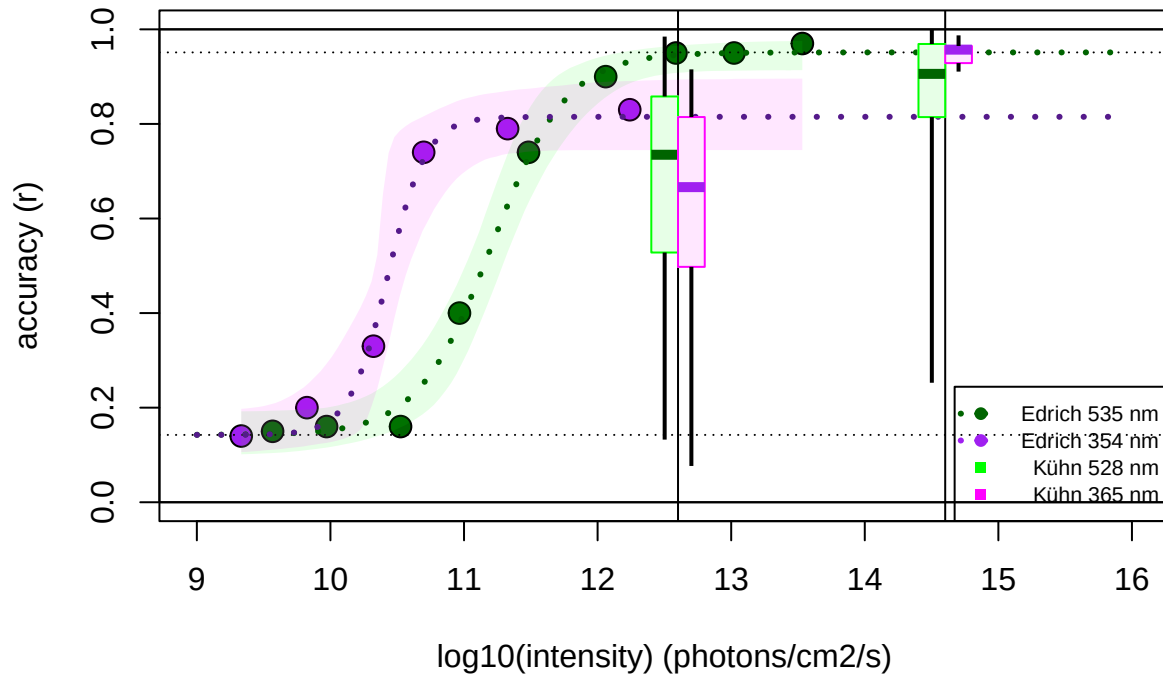
#UV High intensity
QBox(quant = subset(meanQ, colour %in% 'u' & brightn %in% 'h'),
      x = log10(4e14),
      offset = 0.1,
      cols = c('magenta', 'purple')
)

#UV Low intensity
QBox(quant = subset(meanQ, colour %in% 'u' & brightn %in% 'l'),
      x = log10(4e12),
      offset = 0.1,
      cols = c('magenta', 'purple')
)

legend(x = 'bottomright',
       legend = c(
         paste('Edrich',
              sort(unique(ed_ug$wavelength),
                     decreasing = TRUE), 'nm'),
         paste('Kühn', c(528, 365), 'nm' )
       ),
       col = c('darkgreen', 'purple', 'green', 'magenta'),
       pch = c(20, 20, 15, 15),
       lty = c(3,3,NA, NA),
       lwd = 3,
       cex = 0.7)

```

Comparison of data from Edrich et al. 1979 with Kühn, 2017



It would be difficult to explain the orientation accuracy for the high intensity (4×10^{14}) UV stimulus from the accuracy curve for Edrich's UV data, but it would fit well with the maximum accuracy observed for green light. In Edrich's study, the green light was interpreted as the sun (dances at correct angle between sun and feeder), whereas the UV light was interpreted as a patch of sky near the anti-sun. In Kühn's data the stimuli were larger (assuming 10° across) than Edrich's (2.3°), which may have resulted in them being perceived as dimmer (left shift in this graph).

Sun compass orientation

Since most dances were at least somewhat oriented ($r > 0.5$) we can assume that the mean angles of these dances typically align with an intended dance direction. If most dances are for the same geographic location, we should observe dances as the same fixed angle to the stimulus, interpreted either as the sun or the anti-sun.

```
sun_diff_gh = sapply(X = u_id,
                     FUN = SunDiff,
                     dt = mean_vectors,
                     cl = 'g',
                     br = 'h')
sun_diff_gl = sapply(X = u_id,
                     FUN = SunDiff,
                     dt = mean_vectors,
                     cl = 'g',
                     br = 'l')
sun_diff_uh = sapply(X = u_id,
                     FUN = SunDiff,
```

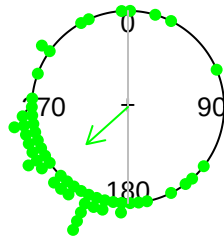
```

        dt = mean_vectors,
        cl = 'u',
        br = 'h')
sun_diff_ul = sapply(X = u_id,
                     FUN = SunDiff,
                     dt = mean_vectors,
                     cl = 'u',
                     br = 'l')

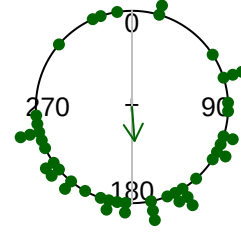
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(sun_diff_gh),
      col = 'green',
      shrink = 2.0,
      side = 3,
      title = 'Green Bright')
PCfun(angles = unlist(sun_diff_gl),
      col = 'darkgreen',
      shrink = 2.0,
      side = 3,
      title = 'Green Dim')
PCfun(angles = unlist(sun_diff_uh),
      col = 'magenta',
      shrink = 2.0,
      side = 3,
      title = 'UV Bright')
PCfun(angles = unlist(sun_diff_ul),
      col = 'purple',
      shrink = 2.0,
      side = 3,
      title = '          UV Dim')

```

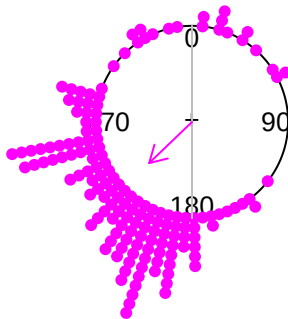
Green Bright



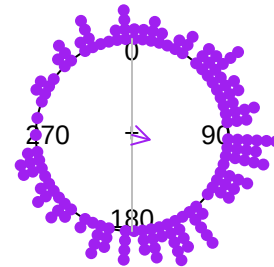
Green Dim



UV Bright



UV Dim



For the bright stimuli the dances generally indicate South, but with a very wide range. (or possibly clusters to the South-West and South-East). For the dim stimuli there is less of a clear pattern, particularly for UV dim.

We can also restrict this analysis to just the dances with all conditions.

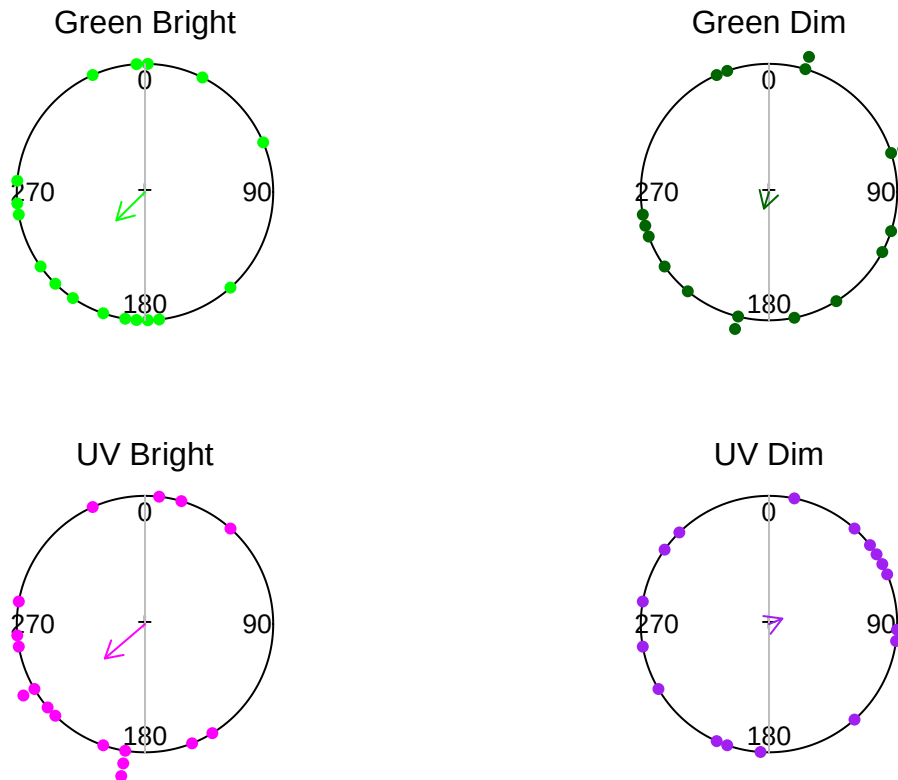
```
full_sun_diff_gh = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'g',
                          br = 'h')
full_sun_diff_gl = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'g',
                          br = 'l')
full_sun_diff_uh = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'u',
                          br = 'h')
full_sun_diff_ul = sapply(X = full_ids,
                          FUN = SunDiff,
                          dt = mean_vectors,
                          cl = 'u',
                          br = 'l')
```



```

par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(full_sun_diff_gh),
      col = 'green',
      shrink = 1.5,
      side = 3,
      title = 'Green Bright')
PCfun(angles = unlist(full_sun_diff_gl),
      col = 'darkgreen',
      shrink = 1.5,
      side = 3,
      title = 'Green Dim')
PCfun(angles = unlist(full_sun_diff_uh),
      col = 'magenta',
      shrink = 1.5,
      side = 3,
      title = 'UV Bright')
PCfun(angles = unlist(full_sun_diff_ul),
      col = 'purple',
      shrink = 1.5,
      side = 3,
      title = 'UV Dim')

```

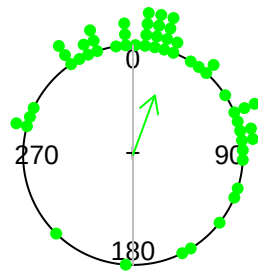


When restricting to only the balanced cases, the pattern becomes clearer. For the most oriented dances to the bright UV stimulus there are two clusters to the South-West and South-East. These are likely also present for the bright green stimulus. The dim green stimulus appears to align somewhat with the bright

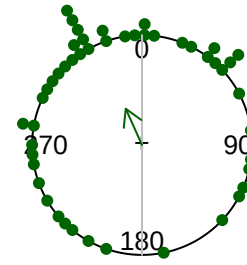
stimuli. The dim UV stimulus shows no apparent patten, though a slight southwards bias.

Inspect dance angles relative to the stimulus

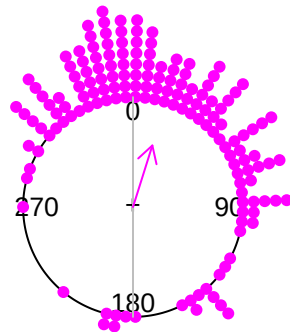
```
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'g' & brightn %in% 'h')$mu,
      col = 'green',
      shrink = 1.75,
      title = 'Green Bright')
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'g' & brightn %in% 'l')$mu,
      col = 'darkgreen',
      shrink = 1.75,
      title = 'Green Dim')
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'u' & brightn %in% 'h')$mu,
      col = 'magenta',
      shrink = 1.75,
      title = 'UV Bright')
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'u' & brightn %in% 'l')$mu,
      col = 'purple',
      shrink = 1.75,
      title = 'UV Dim')
```



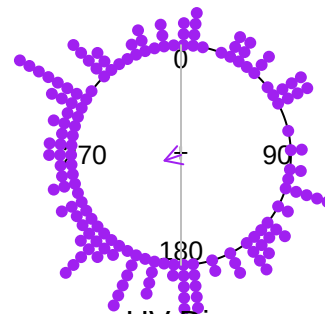
Green Bright



Green Dim



UV Bright



UV Dim

```
cd_sd = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                  FUN = sd.circular,
                  data = mean_vectors)
cd_rho = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                   FUN = rho.circular,
                   data = mean_vectors)
names(cd_sd)[3] = 'circular_sd'
names(cd_rho)[3] = 'rho'
cd_desc = merge(cd_sd, cd_rho)
cd_desc = within(cd_desc,
                 {sd_degrees = deg(circular_sd)})
print(cd_desc,digits = 2)
```

##	colour	brightn	circular_sd	rho	sd_degrees
## 1	g	h	1.0	0.58	60
## 2	g	l	1.4	0.36	82
## 3	u	h	1.0	0.59	59
## 4	u	l	1.9	0.16	110

Compare Sun-Spot & Phototaxis Hypotheses

It is likely that a plurality of dancers had visited locations in the Marburg Botanic Gardens, perhaps in the centre of the main growing area (50°48'03.3"N 8°48'24.1"E). The hive is at 50°48'22.7"N 8°48'40.8"E. That would give the following bearing:

```

# Define the coordinates

hive_coords = c(longitude = 8.811333,
                 latitude = 50.806306)

#the 150m circle intersecting with the nearest edge
garden_coords = list(left =
                      c(longitude = 8.808956,
                        latitude = 50.805509),
                      centre =
                        c(longitude = 8.810290,
                          latitude = 50.805565),
                      right =
                        c(longitude = 8.810110,
                          latitude = 50.804854)
                      )

#find the bearing from the hive
garden_bearing = sapply(garden_coords,
                        FUN = bearing,
                        p1 = hive_coords
                        )

#standard bearing
print(garden_bearing %% 360)

##      left      centre      right
## 242.1153 221.7301 208.0902

garden_bearing_rad = rad(as.numeric(garden_bearing[2]))

#save and load models instead of running them each time
load_model = FALSE
# Custom family for "unwrap" von Mises distribution
unwrap_von_mises = custom_family(
  "unwrap_von_mises", dpars = c("mu", "kappa"),
  links = c('identity', #brms cannot accept custom link functions, do via nl instead
            "softplus"),
  lb = c(-pi, 0),
  ub = c(pi, NA),
  type = "real"
)

## Stan variables

### Functions

stan_unwrap_fun = stanvar(scode = "
  real unwrap_von_mises_lpdf(real y, real mu, real kappa) {
    return von_mises_lpdf(y | mod_circular(mu), kappa);
  }
  real unwrap_von_mises_rng(real mu, real kappa) {
    return von_mises_rng( mod_circular(mu) , kappa);

```

```

}
real unwrap_von_mises_vect_lpdf(vector y, real mu, real kappa) {
  real tmp = 0;
  for(i in 1:size(y))
  {
    tmp = tmp + unwrap_von_mises_lpdf(y[i] | mu, kappa);
  }
  return tmp;
}
",
                                block = 'functions') +
stanvar(scode = "
real mod_circular(real y) {
  return fmod(y + pi(), 2*pi()) - pi();
}

real inv_mod_circular(real y) {
  return mod_circular(y);
}
",
                                block = 'functions')

### Parameters

#concentration of random effects on fixed effect on mean angle
kappamu_param = stanvar(scode = "
real kappamu;
",
                                block = "parameters") +
stanvar(scode = "
real kappa_mu = log1p_exp(kappamu);
",
                                block = 'genquant')

stanvars_unwrap = stan_unwrap_fun + kappamu_param

#subset to just gh data
cd_gh = subset(cd,
              subset = brightn %in% 'h' &
                    colour %in% 'g' &
                    !is.na(sun_az_rad) # for fair comparison only include cases where the sun azimuth is not NA
              )

dt_photo = within(cd_gh,
{
  BR = brightn
  CL = colour
  DT = date
  RN = run
  sunangle = angle + sun_az_rad
})
)

```

```

dt_sun = within(dt_photo,
                {bgangle = sunangle - garden_bearing_rad}
                )

# Model formulae
#model assuming mean direction at 0°, with individual biases
formula_photo = bf(
  formula = angle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 0, # fixed effects intercept == 0
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
  family = unwrap_von_mises, #mod mu, kappa via the softplus
  nl = TRUE) #to accept user-defined extra parameters (mui) we need to treat the formula as nonlinear

#model allowing any angle
formula_nophoto = bf(
  formula = angle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 1, # fixed effects intercept estimated
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
  family = unwrap_von_mises, #mod mu, kappa via the softplus
  nl = TRUE) #to accept user-defined extra parameters (mui) we need to treat the formula as nonlinear

formula_sunspot = bf(
  formula = bgangle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 0, # fixed effects intercept estimated
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
  family = unwrap_von_mises, #mod mu, kappa via the softplus
  nl = TRUE) #to accept user-defined extra parameters (mui) we need to treat the formula as nonlinear

formula_nosunspot = bf(
  formula = bgangle ~ mu,
  nlf(mu ~ mu0 + mui),
  mu0 ~ 1, # fixed effects intercept estimated
  mui ~ 0 + ID,
  kappa ~ 1 + (1|ID),
  family = unwrap_von_mises, #mod mu, kappa via the softplus
  nl = TRUE) #to accept user-defined extra parameters (mui) we need to treat the formula as nonlinear

# Priors
#priors for mu
pr_photo =
  set_prior(paste("target +=",
                  'unwrap_von_mises_vect_lpdpf(b_mui | 0, log1p_exp(kappamu))',
                  '+ normal_lpdpf(b_mui | 0, 2*pi())' # additional prior to keep estimates from walking a
  ),
  check = FALSE) +
  set_prior("target += lognormal_lpdpf(log1p_exp(kappamu) | log(15), 0.6)", #shrinkage prior differences

```

```

    check = FALSE) +
    prior(normal(3.0,3.0), class = Intercept, dpar = 'kappa') + # bias to oriented
    prior(student_t(3, 0, 3.0), class = sd, dpar = 'kappa') # weak bias no differences

#for models with a mean we also need an extra parameter
pr_nophoto = pr_photo +
  prior(unwrap_von_mises(0,1e-16), class = b, nlpar = 'mu0', coef = 'Intercept')

#parameters for the brms runs
wup = 1000 #number of warmup iterations used to tune the sampler
sam = 1000 #number of iterations to sample
refr = 0 #print progress every n iterations
siln = 2#print only very important messages

#these are pretty slow! About 7 minutes in total
if(load_model)
{ load(file = '2Results/phototaxis_modlist.RData')
}else
{
  cglm_photo = brm(formula = formula_photo,
    data = dt_photo,
    prior = pr_photo,
    stanvars = stanvars_unwrap,
    warmup = wup, #may be necessary
    iter = wup+sam, #doesn't take a lot of runs
    chains = 4, # 4 chains in parallel
    cores = 4, # on 4 CPUs
    # control = list(adapt_delta = 0.90),
    refresh = refr, # echo chain progress every n iterations
    silent = siln, # echo some Stan messages
    save_pars = save_pars(all = TRUE), # <--- Crucial for moment matching
    backend = 'cmdstanr')

  cglm_nophoto = brm(formula = formula_nophoto,
    data = dt_photo,
    prior = pr_nophoto,
    stanvars = stanvars_unwrap,
    warmup = wup, #may be necessary
    iter = wup+sam, #doesn't take a lot of runs
    chains = 4, # 4 chains in parallel
    cores = 4, # on 4 CPUs
    # control = list(adapt_delta = 0.90),
    refresh = refr, # echo chain progress every n iterations
    silent = siln, # echo some Stan messages
    save_pars = save_pars(all = TRUE), # <--- Crucial for moment matching
    backend = 'cmdstanr')

  cglm_sunspot = brm(formula = formula_sunspot,
    data = dt_sun,
    prior = pr_photo, #different response, same predictors
    stanvars = stanvars_unwrap,
    warmup = wup, #may be necessary
    iter = wup+sam, #doesn't take a lot of runs

```

```

        chains = 4, # 4 chains in parallel
        cores = 4, # on 4 CPUs
        control = list(adapt_delta = 0.90),
        refresh = refr, # echo chain progress every n iterations
        silent = siln, # echo some Stan messages
        save_pars = save_pars(all = TRUE), # <--- Crucial for moment matching
        backend = 'cmdstanr')

cglm_nosunspot = brm(formula = formula_nosunspot,
  data = dt_sun,
  prior = pr_nophoto, #different response, same predictors
  stanvars = stanvars_unwrap,
  warmup = wup, #may be necessary
  iter = wup+sam, #doesn't take a lot of runs
  chains = 4, # 4 chains in parallel
  cores = 4, # on 4 CPUs
  control = list(adapt_delta = 0.90),
  refresh = refr, # echo chain progress every n iterations
  silent = siln, # echo some Stan messages
  save_pars = save_pars(all = TRUE), # <--- Crucial for moment matching
  backend = 'cmdstanr')
#combine into a single list
modlist = list(phototaxis = cglm_photo,
  no_phototaxis = cglm_nophoto,
  sunspot = cglm_sunspot,
  no_sunspot = cglm_nosunspot)
}

```

```

## Warning: 2 of 4000 (0.0%) transitions ended with a divergence.
## See https://mc-stan.org/misc/warnings for details.

```

```

#summarise
suppressWarnings({ #for circular models rhats may be smaller than they appear
sm_modlist = lapply(X = modlist,
  FUN = summary)
})
Sm_only = function(x,
  type = 'fixed',
  var = 'kappa_Intercept')
{
  x[[type]][var,]
}

#average kappa
print(lapply(X = sm_modlist, Sm_only), digits = 3 )

```

```

## $phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_Intercept      8.89      1.03    6.92      11    1    1382    1990
##
## $no_phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_Intercept      8.93      1.01    7.03      11    1     998    1659

```



```
##
## $sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_Intercept      8.89      1.04      6.95      11      1      1274      1628
##
## $no_sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_Intercept      8.86      1.04      6.9      11 1.01      891      1103
```

```
#across individuals kappa
print(lapply(X = sm_modlist, Sm_only, type = 'spec_pars', var = 'kappa_mu'), digits = 3 )
```

```
## $phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.74      0.29      1.2      2.36      1      5225      2880
##
## $no_phototaxis
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.85      0.306      1.27      2.47      1      2827      2374
##
## $sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.9      0.311      1.35      2.54      1      3784      2829
##
## $no_sunspot
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_mu      1.9      0.316      1.32      2.55      1      1806      2551
```

```
#check population means
lapply(X = modlist[c(2,4)],
      FUN = UnwrapRhats,
      variable = '~b_mu0')
```

```
## $no_phototaxis
## b_mu0_Intercept
##           1.0027
##
## $no_sunspot
## b_mu0_Intercept
##           1.02284
```

```
#Summarise Rhats
lapply(X = modlist,
      FUN = SumUwRhats)
```

```
## $phototaxis
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 0.9999  1.0002  1.0007  1.0008  1.0071
##
## $no_phototaxis
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 1.0000  1.0004  1.0007  1.0012  1.0043
```

```
##
## $sunspot
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 0.9999 1.0001 1.0005 1.0006 1.0062
##
## $no_sunspot
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9998 1.0002 1.0009 1.0044 1.0060 1.0228
```

```
#calculate loo
if(!load_model)
{
modlist = lapply(modlist,
  FUN = add_criterion,
  'loo',
  moment_match = TRUE)

save(modlist,
  file = '2Results/phototaxis_modlist.RData')
}
```

```
## Recompiling the model with 'rstan'
```

```
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.
```

```
## Warning: Found 6 observations with a pareto_k > 0.7 in model 'X[[i]]'. We
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without the
## assumption that these observations are negligible. This will refit the model 6
## times to compute the ELPDs for the problematic observations directly.
```

```
## Recompiling the model with 'rstan'
```

```
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.
```

```
## Warning: Found 3 observations with a pareto_k > 0.7 in model 'X[[i]]'. We
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without the
## assumption that these observations are negligible. This will refit the model 3
## times to compute the ELPDs for the problematic observations directly.
```

```
## Recompiling the model with 'rstan'
```

```
## Recompilation done
```

```
## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.
```

```
## Warning: Found 4 observations with a pareto_k > 0.7 in model 'X[[i]]'. We
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without the
## assumption that these observations are negligible. This will refit the model 4
## times to compute the ELPDs for the problematic observations directly.
```

```
## Recompiling the model with 'rstan'
## Recompilation done

## Warning: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.

## Warning: Found 7 observations with a pareto_k > 0.7 in model 'X[[i]]'. We
## recommend to set 'reloo = TRUE' in order to calculate the ELPD without the
## assumption that these observations are negligible. This will refit the model 7
## times to compute the ELPDs for the problematic observations directly.
```

```
#compare the phototaxis models
with(modlist, loo_compare(phototaxis, no_phototaxis))
```

```
##               elpd_diff se_diff
## no_phototaxis  0.0         0.0
## phototaxis    -1.4         0.8
```

```
#compare the sun-spot orientation models
with(modlist, loo_compare(sunspot, no_sunspot))
```

```
##               elpd_diff se_diff
## sunspot       0.0         0.0
## no_sunspot   -2.8         1.1
```

```
#compare LOO IC
LooIC = function(lmod)
{
  loo(lmod)$estimates['looic', 'Estimate']
}

modLIC = lapply(X = modlist,
  FUN = LooIC)
print(sort(unlist(modLIC)), digits = 1)
```

```
## no_phototaxis    sunspot    phototaxis    no_sunspot
##              639         642         642         648
```

There is very little separation between our different hypotheses here, suggesting each would do well to explain the data. We can also look at the specific predictions to see how well each hypothesis matches.

Hypothesis test

```
#since all models are similarly credible, what about the hypotheses?
#check if the sector surrounding the expected direction is equally full

#for each individual, calculate the likelihood of foraging in the expected region

PostROPE = function(id, rope, drs, mu = 'b_mu0_Intercept')
{
```

```

mu = if(!is.na(mu)){with(drs, get(mu))}else{0}
dr_id = with(drs,
  {
    mod_circular(
      mu +
      get(paste0('b_mui_ID', id)
    )
  }
)

#centre both
circ_rope = circular(rope, template = 'none')
circ_dr = circular(dr_id, template = 'none')
mean_rope = mean.circular(circ_rope)
rope_ctr = as.numeric(mod_circular(circ_rope - mean_rope))
dr_ctr = as.numeric(mod_circular(circ_dr - mean_rope))
postp = mean( rope_ctr[1] < dr_id &
  dr_id < rope_ctr[2])
return(postp)
}

photo_id = unique(dt_photo$ID)

#width of sector
rope_width = abs(mod_circular(rad(diff(garden_bearing[-2])))) # 35°
rope_photo = c(-1,1)*rope_width/2
rope_sunspot = sort( mod_circular(rad(garden_bearing[-2] - garden_bearing[2])) )

#extract draws
dr_modlist = lapply(modlist, as_draws_df)

postl_nophoto = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_photo,
  drs = dr_modlist$no_phototaxis
)
postl_photo = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_photo,
  mu = NA,
  drs = dr_modlist$phototaxis
)

postl_nosunspot = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_sunspot, # relative to centre as in model
  drs = dr_modlist$no_sunspot
)
postl_sunspot = sapply(X = photo_id,
  FUN = PostROPE,
  rope = rope_sunspot, # relative to centre as in model
  drs = dr_modlist$sunspot,
  mu = NA

```

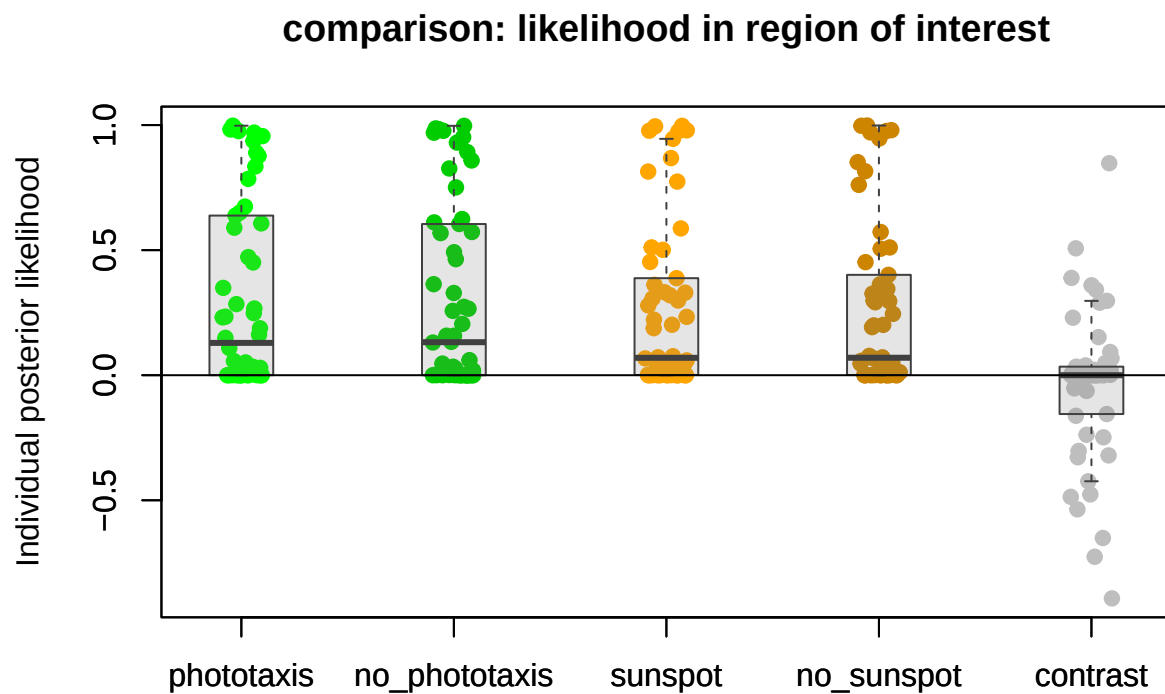
```

)

postl_dt = data.frame(phototaxis = postl_photo,
                      no_phototaxis = postl_nophoto,
                      sunspot = postl_sunspot,
                      no_sunspot = postl_nosunspot,
                      contrast = postl_sunspot -
                                postl_photo)

stripchart(x = postl_dt,
           vertical = TRUE,
           method = 'jitter',
           pch = 19,
           ylab = 'Individual posterior likelihood',
           main = 'comparison: likelihood in region of interest',
           col = c('green', 'green3', 'orange', 'orange3', 'gray'))
)
boxplot(x = postl_dt,
        add = TRUE,
        col = gray(0.5, alpha = 0.2),
        border = 'gray25',
        pars = list(boxwex = 0.3, staplewex = 0.2, outwex = 0.1),
        outline = FALSE
)
abline(h = 0)

```



It is hard to say for the individuals whether the majority orient towards the sector containing the Botanic Garden, or a similar sized sector containing the stimulus. What we can say is that the population average falls directly towards the Botanic Garden centre in real-world bearings, while the population average relative to the light source has a leftwards bias.

Plot of sun compass versus phototaxis

```
with(dr_modlist$no_phototaxis, paste0(mean(b_mu0_Intercept > 0)*100, '%') ) #nearly all estimates suggest

## [1] "98.375%"

with(dr_modlist$no_sunspot, paste0(mean(b_mu0_Intercept > 0)*100, '%') ) #there is notable overlap betw

## [1] "65.625%"

par(pty = 's')
par(mfrow = c(2,2),
    mar = c(0,0,0,0))
PCfun(angles = subset(mean_vectors,
                      subset = colour %in% 'g' &
                                brightn %in% 'h' &
                                !is.na(sun_az))$mu,

      col = 'green',
      shrink = 1.5,
      plot_rho = FALSE,
      title = '\n\n\n\nRelative to stimulus')
with(dr_modlist$phototaxis,
     arrows(x0 = 0, x1 = 0,
            y0 = 0,
            y1 = A1(softplus(median(kappamu))),
            col = 'orange',
            lwd = 5,
            length = 0.1
           )
    )

with(dr_modlist$no_phototaxis,
     arrows(x0 = 0,
            x1 = A1(softplus(median(kappamu))) *
                  sin(median(mod_circular(b_mu0_Intercept))),
            y0 = 0,
            y1 = A1(softplus(median(kappamu))) *
                  cos(median(mod_circular(b_mu0_Intercept))),
            col = 'green3',
            lwd = 5,
            length = 0.1
           )
    )
arrows(x0 = c(0,0),
      x1 = sin(c(-1,1)*rope_width/2),
      y0 = c(0,0),
```

```

    y1 = cos(c(-1,1)*rope_width/2),
    col = 'orange',
    lty = 2,
    length = 0,
    lwd = 5
  )
for(id in unique(dt_photo$ID))
{
  with(dr_modlist$no_phototaxis,
    arrows(x0 = 0,
      x1 = A1(softplus(median(
        b_kappa_Intercept +
        get(paste0('r_ID__kappa[',id,',Intercept]'))
      ))) *
      sin(median(mod_circular(
        b_mu0_Intercept +
        get(paste0('b_mui_ID', id))
      ))),
      y0 = 0,
      y1 = A1(softplus(median(
        b_kappa_Intercept +
        get(paste0('r_ID__kappa[',id,',Intercept]'))
      )))) *
      cos(median(mod_circular(
        b_mu0_Intercept +
        get(paste0('b_mui_ID', id))
      ))),
      col = adjustcolor('green4',alpha = 0.5),
      lwd = 0.5,
      length = 0
    )
  )
}
legend(x = 'topleft',
  legend = c('expected mean',
    'expected region',
    'alternative mean',
    'individual biases'
  ),
  col = c('orange',
    'orange',
    'green3',
    'green4'
  ),
  lty = c(1,3,1,1),
  lwd = c(3,3,3,1),
  cex = 0.5,
  bg = NA,
  bty = 'n',
  inset = c(0, 0)
)

```

```

PCfun(angles = with(subset(mean_vectors,
                           subset = colour %in% 'g' &
                           brightn %in% 'h' &
                           !is.na(sun_az)),
                           mu + sun_az),
      col = 'green',
      shrink = 1.5,
      plot_rho = FALSE,
      title = '\n\n\nRelative to North')
with(as_draws_df(modlist$sunspot),
     arrows(x0 = 0,
            x1 = A1(softplus(median(kappamu))) *
            sin(garden_bearing_rad),
            y0 = 0,
            y1 = A1(softplus(median(kappamu))) *
            cos(garden_bearing_rad),
            col = 'orange',
            lwd = 5,
            length = 0.1
            )
     )
with(as_draws_df(modlist$no_sunspot),
     arrows(x0 = 0,
            x1 = A1(softplus(median(kappamu))) *
            sin(median(mod_circular(b_mu0_Intercept + garden_bearing_rad
            y0 = 0,
            y1 = A1(softplus(median(kappamu))) *
            cos(median(mod_circular(b_mu0_Intercept + garden_bearing_rad
            ))),
            col = 'green3',
            lwd = 5,
            length = 0.1
            )
     )
arrows(x0 = c(0,0),
      x1 = sin(rad(garden_bearing[-2])),
      y0 = c(0,0),
      y1 = cos(rad(garden_bearing[-2])),
      col = 'orange',
      lty = 2,
      length = 0,
      lwd = 5
      )
for(id in unique(dt_photo$ID))
with(as_draws_df(modlist$no_sunspot),
     arrows(x0 = 0,
            x1 = A1(softplus(median(
            b_kappa_Intercept +
            get(paste0('r_ID__kappa[',id,',Intercept]'))))) *
            sin(median(mod_circular(
            b_mu0_Intercept +
            garden_bearing_rad +

```



```

                                get(paste0('b_mui_ID', id))
                                )),
    y0 = 0,
    y1 = A1(softplus(median(
        b_kappa_Intercept +
        get(paste0('r_ID__kappa[,id,',Intercept]')))))*
    cos(median(mod_circular(
        b_mu0_Intercept+
        garden_bearing_rad +
                                get(paste0('b_mui_ID', id))
                                )),
    col = adjustcolor('green4',alpha = 0.5),
    lwd = 0.5,
    length = 0
    )
)

par(mar = c(0,3,0,0))

with(dr_modlist$no_phototaxis,
{
  VertHist(data = deg(mod_circular(b_mu0_Intercept)),
    main = '\nPopulation mean angle',
    ylab = 'model estimates',
    ylim = c(-30, 90),
    col = adjustcolor('green3', alpha.f = 100/255),
    cex.axis = 0.7)
  abline(h = 0,
    col = 'orange',
    lwd = 7)
  arrows(x0 = 0, x1 = 0,
    y0 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 0.025),
    y1 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 1-0.025),
    col = 'green4',
    lwd = 5,
    code = 3,
    length = 0.05,
    angle = 90)
}
)

legend(x = 'bottomright',
  legend = c('stimulus angle',
    '95% CI'
    ),
  col = c('orange',
    'green4'
    ),
  lty = c(1,1),
  lwd = c(3,3),
  cex = 0.5,
  bg = NA,
  bty = 'n',

```

```

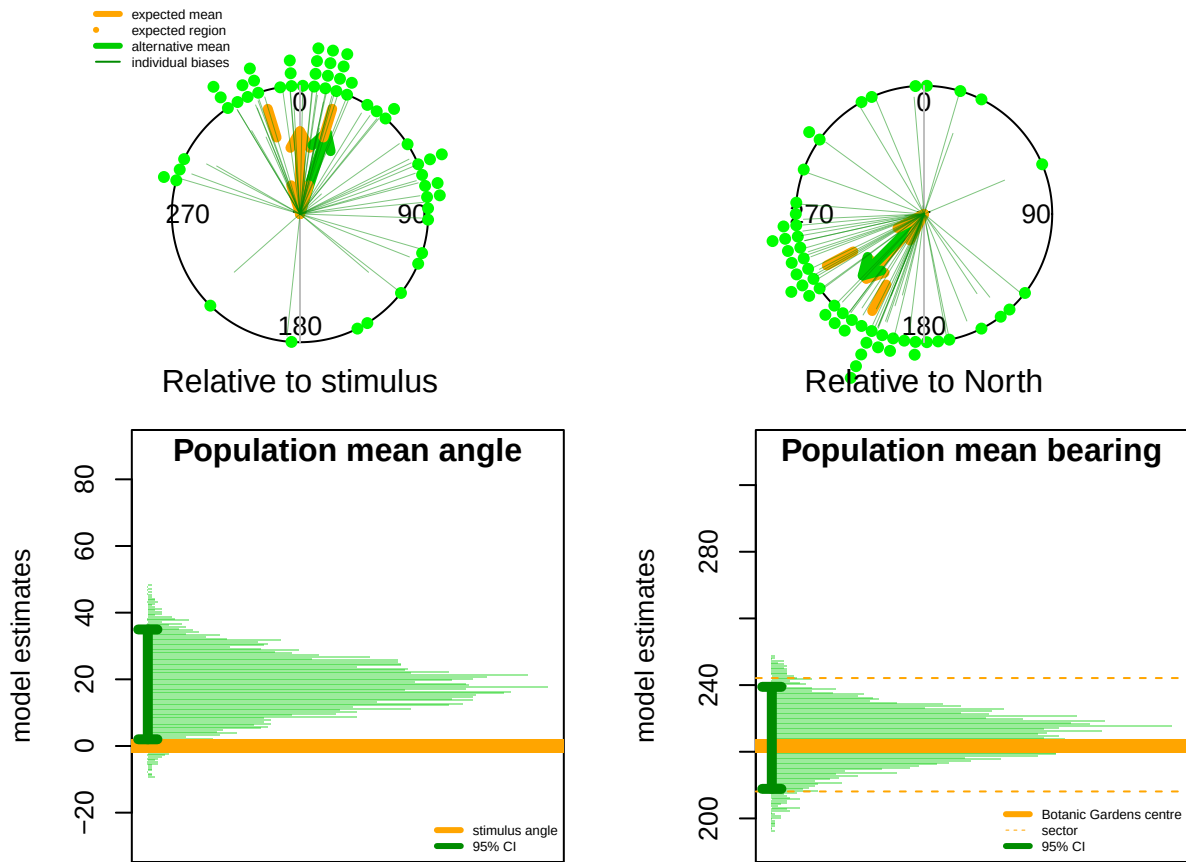
    inset = c(0, 0)
  )

gb_pos = garden_bearing[2] %% 360

with(dr_modlist$no_sunspot,
{
  VertHist(data = deg(mod_circular(b_mu0_Intercept)) + gb_pos,
    main = '\nPopulation mean bearing',
    ylab = 'model estimates',
    ylim = c(-30, 90) + gb_pos,
    col = adjustcolor('green3', alpha.f = 100/255),
    cex.axis = 0.7)
  abline(h = garden_bearing %% 360,
    col = 'orange',
    lwd = c(1,7,1),
    lty = c(2,1,2) )
  arrows(x0 = 0, x1 = 0,
    y0 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 0.025) +
      gb_pos,
    y1 = quantile(deg(mod_circular(b_mu0_Intercept)), p = 1-0.025) +
      gb_pos,
    col = 'green4',
    lwd = 5,
    code = 3,
    length = 0.05,
    angle = 90)
}
)

legend(x = 'bottomright',
  legend = c('Botanic Gardens centre',
    'sector',
    '95% CI'
  ),
  col = c('orange',
    'orange',
    'green4'
  ),
  lty = c(1,2,1),
  lwd = c(3,0.5,3),
  cex = 0.5,
  bg = NA,
  bty = 'n',
  inset = c(0, 0)
)

```



```
#How much do individual dance angles vary according to kappa_mu?
with(sm_modlist$no_sunspot,
  paste0(round(deg(MardiaSD(spec_pars['kappa_mu', 'Estimate']))), '°') )
```

```
## [1] "50°"
```

```
#Where is the population centred?
with(dr_modlist$no_sunspot,
{
  paste0(
    round(
      quantile(x = deg(b_mu0_Intercept),
        p = c(0,0.5,1)+c(1,0,-1)*(0.05/2) ) +
        gb_pos
      ) %% 360, # as a bearing
    )
  )
})
```

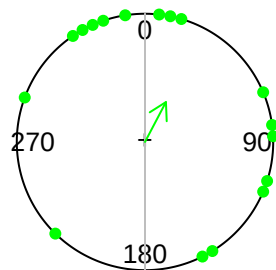
```
## [1] "209°" "225°" "239°"
```

Mean dance angles relative to the stimulus (full conditions)

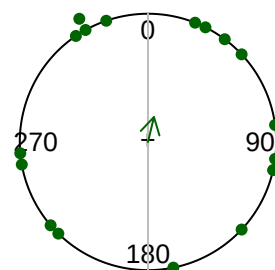
Since dances do not point to a specific geographic location, it is possible that they advertise a variety of directions relative to the sun, or are not menotactic but phototactic or uniformly oriented. We can identify

phototactic orientation from the angle relative to the stimulus.

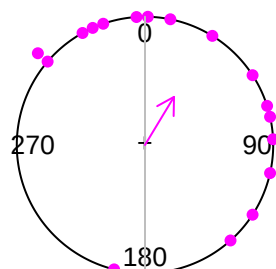
```
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'g' & brightn %in% 'h')$mu,
      col = 'green',
      shrink = 1.5,
      title = 'Green Bright')
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'g' & brightn %in% 'l')$mu,
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim')
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'u' & brightn %in% 'h')$mu,
      col = 'magenta',
      shrink = 1.5,
      title = 'UV Bright')
PCfun(angles = subset(mean_vectors_full,
                      subset = colour %in% 'u' & brightn %in% 'l')$mu,
      col = 'purple',
      shrink = 1.5,
      title = 'UV Dim')
```



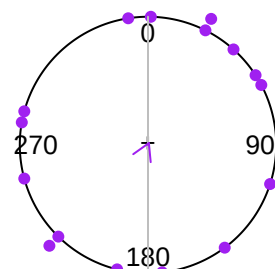
Green Bright



Green Dim



UV Bright



UV Dim

While there are some dances towards or near to the stimulus for most conditions, there is a wide spread, suggesting a factor other than phototaxis sets dance directions. N.B. for the dim stimuli where the widest spread is observed, the dances are also less accurate, which may increase variation in mean dance angle.

```

cd_sd_full = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                        FUN = sd.circular,
                        data = mean_vectors_full)
cd_rho_full = aggregate(circular(mu, units = 'degrees')~colour*brightn,
                        FUN = rho.circular,
                        data = mean_vectors_full)
names(cd_sd_full)[3] = 'circular_sd'
names(cd_rho_full)[3] = 'rho'
cd_desc_full = merge(cd_sd_full, cd_rho_full)
cd_desc_full = within(cd_desc_full,
                      {sd_degrees = deg(circular_sd)})
print(cd_desc_full,digits = 2)

```

```

##   colour brightn circular_sd   rho sd_degrees
## 1      g       h       1.4 0.351         83
## 2      g       l       1.8 0.200        103
## 3      u       h       1.3 0.438         74
## 4      u       l       2.9 0.016        164

```

Change in mean dance angle between stimuli

Change arcs

To determine whether these dances interpret the stimuli as the sun, anti-sun or something else, we would need to check the differences in dance angle between the conditions. It is also worth considering any differences in accuracy between dances to each condition, since this would influence the variation in angular change. These can be visualised as arcs.

```

par(mfrow = c(2,2), mar = c(0,0,0,0))
OpenCplot(n_sample = with(full_cd, median(run)/4) )
mtext(text = 'Green Bright to Green Dim',
      side = 1,
      line = -1,
      cex = 0.5)
#open the loop
for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
    {
      DiffArc(a1 = mu[colour %in% 'g' &
                      brightn %in% 'h'],
              a2 = mu[colour %in% 'g' &
                      brightn %in% 'l'],
              r1 = mean_vector[colour %in% 'g' &
                               brightn %in% 'h'],
              r2 = mean_vector[colour %in% 'g' &
                               brightn %in% 'l'],
              col1 = 'green',
              col2 = 'darkgreen'
            )
    }
  }
}

```

```

    )
}

OpenCplot(n_sample = with(full_cd, median(run)/4) )
mtext(text = 'UV Bright to UV Dim',
      side = 1,
      line = -1,
      cex = 0.5)
for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
      {
        DiffArc(a1 = mu[colour %in% 'u' &
                        brightn %in% 'h'],
                a2 = mu[colour %in% 'u' &
                        brightn %in% 'l'],
                r1 = mean_vector[colour %in% 'u' &
                                brightn %in% 'h'],
                r2 = mean_vector[colour %in% 'u' &
                                brightn %in% 'l'],
                col1 = 'magenta',
                col2 = 'purple'
              )
      }
    )
}

OpenCplot(n_sample = with(full_cd, median(run)/4) )
mtext(text = 'Green Bright to UV Bright',
      side = 1,
      line = -1,
      cex = 0.5)
for(ii in mean_vectors_full$ID)
{
  with(subset(mean_vectors_full,
              subset = ID %in% ii),
      {
        DiffArc(a1 = mu[colour %in% 'g' &
                        brightn %in% 'h'],
                a2 = mu[colour %in% 'u' &
                        brightn %in% 'h'],
                r1 = mean_vector[colour %in% 'g' &
                                brightn %in% 'h'],
                r2 = mean_vector[colour %in% 'u' &
                                brightn %in% 'h'],
                col1 = 'green',
                col2 = 'magenta'
              )
      }
    )
}

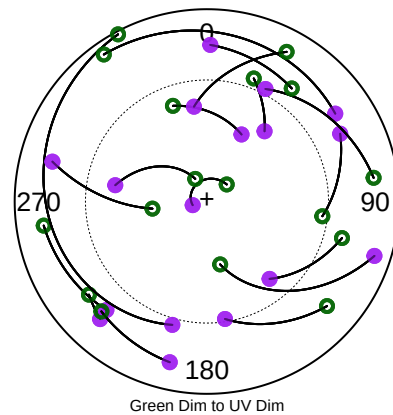
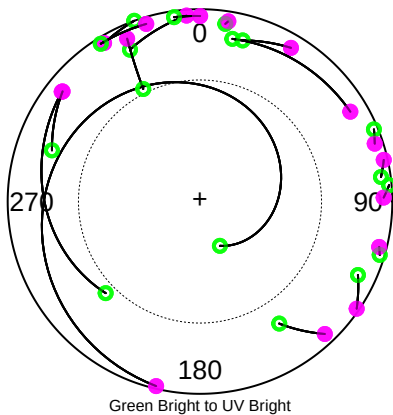
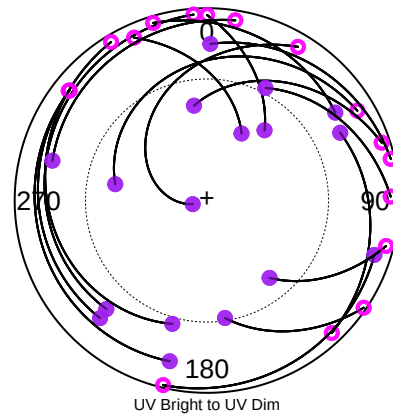
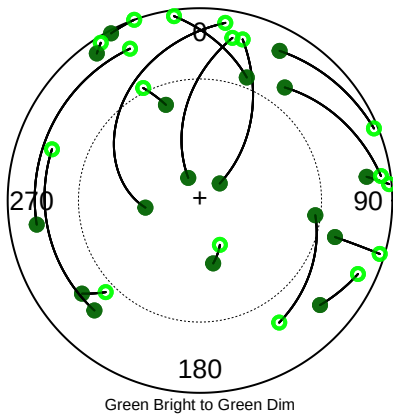
OpenCplot(n_sample = with(full_cd, median(run)/4) )
for(ii in mean_vectors_full$ID)

```

```

{
  with(subset(mean_vectors_full,
    subset = ID %in% ii),
    {
      DiffArc(a1 = mu[colour %in% 'g' &
        brightn %in% 'l'],
        a2 = mu[colour %in% 'u' &
          brightn %in% 'l'],
        r1 = mean_vector[colour %in% 'g' &
          brightn %in% 'l'],
        r2 = mean_vector[colour %in% 'u' &
          brightn %in% 'l'],
        col1 = 'darkgreen',
        col2 = 'purple'
      )
    }
  )
}
mtext(text = 'Green Dim to UV Dim',
  side = 1,
  line = -1,
  cex = 0.5)

```

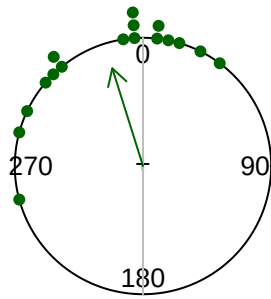


Differences in mean angle

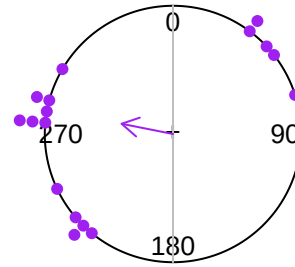
```
mu_diff_gl = sapply(X = full_ids,
  FUN = MuDiff,
  dt = mean_vectors_full,
  cl = 'g',
  br = 'l')
mu_diff_uh = sapply(X = full_ids,
  FUN = MuDiff,
  dt = mean_vectors_full,
  cl = 'u',
  br = 'h')
mu_diff_ul = sapply(X = full_ids,
  FUN = MuDiff,
  dt = mean_vectors_full,
  cl = 'u',
  br = 'l')

#Other contrasts
mu_diff_uhl = sapply(X = full_ids,
  FUN = MuDiff,
  dt = mean_vectors_full,
  cl = 'u',
  br = 'l',
  ref_cl = 'u',
  ref_br = 'h')
mu_diff_gul = sapply(X = full_ids,
  FUN = MuDiff,
  dt = mean_vectors_full,
  cl = 'u',
  br = 'l',
  ref_cl = 'g',
  ref_br = 'h')

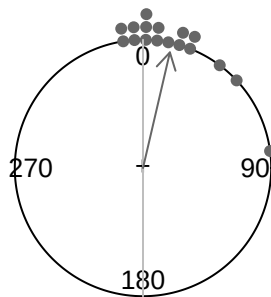
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(mu_diff_gl),
  col = 'darkgreen',
  shrink = 1.5,
  title = 'Green Dim - Green Bright')
PCfun(angles = unlist(mu_diff_uhl),
  col = 'purple',
  shrink = 1.5,
  title = 'UV Dim - UV Bright')
PCfun(angles = unlist(mu_diff_uh),
  col = 'gray40',
  shrink = 1.5,
  title = 'UV Bright - Green Bright')
PCfun(angles = unlist(mu_diff_gul),
  col = 'gray25',
  shrink = 1.5,
  title = 'UV Dim - Green Dim')
```

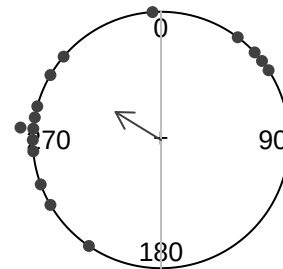
Green Dim – Green Bright



UV Dim – UV Bright



UV Bright – Green Bright



UV Dim – Green Dim

Estimate turn effects

Find fit maximum likelihood turn distributions

For each change distribution, find the maximum likelihood von Mises distribution. For the UV high to UV low change, this will be a bimodal distribution.

```
#collect all heading differences
pair_diffs_lst = lapply(X = list(#bright vs dim green
                                green_hl = mu_diff_gl,
                                #bright vs dim UV
                                uv_hl = mu_diff_uhl,
                                #bright green vs bright uv
                                uvg_h = mu_diff_uh,
                                #dim green vs bright UV
                                uvgl = mu_diff_gul ),
                        FUN = unlist)

#Provide appropriate data format
angle_unit = 'degrees'
angle_rot = 'clock'
#make sure they are in circular format
pair_diffs_lst = lapply(X = pair_diffs_lst,
                        FUN = circular,
```

```

units = angle_unit,
rotation = angle_rot)

```

Fit the maximum likelihood distribution for no difference between pairs, differences between pairs, and differences between pairs with two means. If angles shift in one direction between trials, the “difference” distribution should have significantly higher likelihood. If angles shift in two directions between pairs, then the “two means” distribution should be the best fit.

```

#TODO neaten up!
#Distribution for same means
ml_same_lst = lapply(X = pair_diffs_lst,
                      FUN = mle.vonmises,
                      bias = TRUE,
                      mu = circular(x = 0,
                                   units = angle_unit,
                                   rotation = angle_rot)
)
#Distribution for different means
ml_diff_lst = lapply(X = pair_diffs_lst,
                     FUN = mle.vonmises,
                     bias = TRUE)
#Distribution for multiple different means
mod_diff_lst = lapply(X = pair_diffs_lst,
                      FUN = circ_mle,#converts circular format data to radians
                      niter = 1e4)#some stochastic variability for green stim

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

#Forced multiple different means
bim_diff_lst = lapply(X = pair_diffs_lst,
                      FUN = circ_mle,#converts circular format data to radians
                      exclude = c('M1', 'M2A', 'M2B', 'M2C'),#no uniform or unimodal
                      niter = 1e4)#some stochastic variability for green stim

## Warning in check_data(data): Converting from degrees to radians...
## Warning in check_data(data): Converting to modulo 2pi...

```

```

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

## Warning in check_data(data): Converting from degrees to radians...

## Warning in check_data(data): Converting to modulo 2pi...

#Extract the parameters of the model with the lowest AIC
mod_diff_par = t( sapply(X = mod_diff_lst,
                        FUN = MD_extract) )
bim_diff_par = t( sapply(X = bim_diff_lst,
                        FUN = MD_extract) )

# Calculate likelihood
#calculate all in list
all_ll = mapply(FUN = LLcalc,
                ml = c(ml_same_lst, ml_diff_lst), #apply to both sets of ML estimates
                angles = pair_diffs_lst, #repeat datasets
                au = angle_unit, #important, use the same frame of reference as the original
                ar = angle_rot)

#inspect resulting parameters
mod_diff_results = do.call(rbind, c(ml_same_lst, ml_diff_lst))
#add easy to parse names
rownames(mod_diff_results) = paste( sort( #distinguish same and different mean
                                         rep(x = c( 'same', 'diff'),
                                                times = length(pair_diffs_lst) ),
                                         decreasing = TRUE
                                         ),
                                   rep(x = names(pair_diffs_lst),
                                       times = 2)
                                   )

#display version
rspar = mod_diff_results[,c('mu', 'kappa', 'se.mu', 'se.kappa')]
rspar = cbind(rspar, loglikelihood = all_ll)
rspar = apply(X = rspar, MARGIN = 2, FUN = unlist)
#reorder for easy comparison
rspar = rspar[c(rbind(1:4, 1:4+4)),]#put the 'diff' directly after 'same'
rspar[,3:4] = NA #overwrite columns for secondary parameters
rspar = cbind(rspar, weight1 = 1.0)
colnames(rspar) = colnames(bim_diff_par)[c(1:4, 6:5)]
all_res = rbind(data.frame(rspar),
                 data.frame(bim_diff_par) )
all_res = apply(all_res, MARGIN = 2, FUN = unlist)
rn_res = rownames(all_res)#NB in R4.5 need to make a copy of rownames
rn_res[9:12] = paste('multi', rownames(bim_diff_par)) #identify bimodal rows
rownames(all_res) = rn_res #

```

```

#reorder to place unimodal and bimodal distributions of the same type together
all_res = all_res[c(1:2, 9, 3:4, 10, 5:6, 11, 7:8, 12),]

all_res = within(data = data.frame(all_res),
  {
    model = c('same',
              'diff',
              'multi')
    dataset = c(sapply(X = names(pair_diffs_lst),
                       FUN = rep,
                       times = 3) )
  }
)
print(all_res, digits = 3)

```

	mu1	kappa1	mu2	kappa2	loglikelihood	weight1	dataset	model
## same green_hl	0.0	2.046	NA	NA	-19.3	1.000	green_hl	same
## diff green_hl	-17.5	2.361	NA	NA	-17.7	1.000	green_hl	diff
## multi green_hl	349.6	3.539	530	0.001	-21.8	0.500	green_hl	multi
## same uv_hl	0.0	0.000	NA	NA	-31.2	1.000	uv_hl	same
## diff uv_hl	-78.8	0.763	NA	NA	-28.3	1.000	uv_hl	diff
## multi uv_hl	49.3	20.935	260	4.796	-18.2	0.294	uv_hl	multi
## same uv_g_h	0.0	4.464	NA	NA	-10.9	1.000	uv_g_h	same
## diff uv_g_h	13.1	5.827	NA	NA	-8.4	1.000	uv_g_h	diff
## multi uv_g_h	188.7	0.001	369	15.598	-11.8	0.500	uv_g_h	multi
## same uv_g_l	0.0	0.184	NA	NA	-30.7	1.000	uv_g_l	same
## diff uv_g_l	-58.4	0.786	NA	NA	-28.2	1.000	uv_g_l	diff
## multi uv_g_l	47.3	4.979	271	4.979	-22.1	0.347	uv_g_l	multi

```

#save results
write.table(x = all_res,
  file = '2Results/MLE_paired_diffs.csv',
  sep = ',',
  row.names = TRUE)

```

As expected, the bimodal models fitted by `circ_mle` have the highest likelihood, followed by the models for differences and then the models for no difference. The difference models require at 1 more parameter, so should increase likelihood by >1 order of magnitude to be appropriate. The bimodal model require 2 more parameters, so should increase likelihood by >2 log units.

Estimate confidence intervals

For each mean, simulate a distribution of mean estimates

```

#estimate CI for unimodal von Mises models
mle_CI = with(data.frame(mod_diff_results),
  {
    mapply(FUN = CI_vM,
           angles = pair_diffs_lst,
           m1 = mu,
           k1 = kappa,
           w1 = 1,

```

```

        alternative = 'two.sided', #not a hypothesis test (symmetrical)
        n = 1e4) #takes 30s for 10 000 simulations
    }
)
#reorder for comparison like the original models
mle_CI = t(mle_CI)[c(rbind(1:4, 1:4+4)),]
rownames(mle_CI) = rownames(rspar)
colnames(mle_CI) = c('lower', 'median', 'upper')

#estimate CI for unimodal and bimodal von Mises models
mod_CI = with(data.frame(mod_diff_par),
{
    mapply(FUN = CI_vM,
           angles = pair_diffs_lst,
           m1 = mu1,
           k1 = kappa1,
           m2 = mu2,
           k2 = kappa2,
           w1 = weight1,
           alternative = 'two.sided', #not a hypothesis test (symmetrical)
           n = 1e4) #takes 15 for 10 000 simulations
    }
)
#estimate CI for unimodal and bimodal von Mises models
bim_CI = with(data.frame(bim_diff_par),
{
    mapply(FUN = CI_vM,
           angles = pair_diffs_lst,
           m1 = mu1,
           k1 = kappa1,
           m2 = mu2,
           k2 = kappa2,
           w1 = weight1,
           SIMPLIFY = FALSE,
           alternative = 'two.sided', #not a hypothesis test (symmetrical)
           n = 1e4) #takes 15 for 10 000 simulations
    }
)
#collapse to printable table
mod_CI_tb = Table_vCI(mod_CI)
bim_CI_tb = Table_vCI(bim_CI)
print(mle_CI, digits = 3)

```

```

##           lower  median  upper
## same green_hl  23.433   0.2657 -23.039
## diff green_hl   2.998 -17.4082 -38.042
## same uv_hl    -62.554 126.6761 -44.385
## diff uv_hl    -17.844 -78.5132 -139.145
## same uvg_h     13.867  -0.0656 -13.740
## diff uvg_h     24.883  12.9832   0.792
## same uvg_l    162.441   1.2226 -159.583
## diff uvg_l      0.449 -58.4850 -116.342

```

```
print(mod_CI_tb, digits = 3)
```

```
##           m1.lower m1.median m1.upper m2.lower m2.median m2.upper
## green_hl    0.964   -17.3   -35.77      NA      NA      NA
## uv_hl     -84.015  -99.8  -115.39    60.4    49.3    38.4
## uvg_h      23.802   13.2    2.36      NA      NA      NA
## uvg_l      69.133   47.2   25.36   -73.5   -89.3  -105.6
```

We see here that while the confidence interval for the difference distribution between the two green conditions and the two bright conditions nearly overlap with 0° (and the 0° distribution CI overlap with the mean difference), for differences between green lights and dim UV there is a bimodal distribution with CI that do not overlap with 0° . N.B. For *one-sided* calculations, the CI do overlap with 0° for green conditions, bright conditions and the leftmost mean for dim conditions.

Plot difference CI

```
par(mfrow = c(2,2), mar = c(0,0,0,0))
PCfun(angles = unlist(mu_diff_gl),
      col = 'darkgreen',
      shrink = 1.5,
      title = 'Green Dim - Green Bright')
PlotCI_vM(mle_CI['diff green_hl'],,
          col = 'darkgreen',
          radius = 1+0.3)

plot.circular(x = circular(unlist(mu_diff_uhl),
                          units = angle_unit,
                          rot = angle_rot
                          ),
             col = 'purple',
             stack = TRUE,
             sep = 0.1,
             bins = 355/5,
             units = angle_unit,
             rotation = angle_rot,
             zero = pi/2,
             shrink = 1.5)
mtext(text = 'UV Dim - Green Bright',
      side = 1,
      line = -2)
lines(x = c(0,0),
      y = c(-1,1),
      col = 'gray')
PlotMV_circMLE(mod_par = mod_diff_par['uv_hl'],,
              col1 = 'purple')
PlotCI_vM(mod_CI_tb['uv_hl'],,
          col = 'purple',
          radius = 1+0.3)

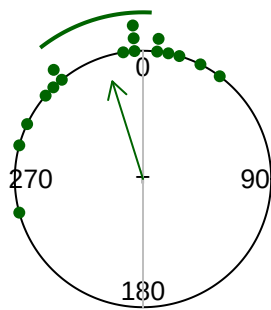
PCfun(angles = unlist(mu_diff_uh),
      col = 'gray40',
```

```

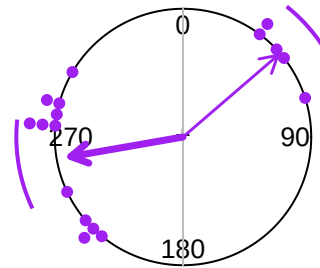
    shrink = 1.5,
    title = 'UV Bright - Green Bright')
PlotCI_vM(mod_CI_tb['uvg_h'],],
    col = 'gray40',
    radius = 1+0.3)
#according to model comparison, bim_diff_par['uvg_h',] is sometimes better
#but is not truly bimodal, kappa2 = 0

plot.circular(x = circular(unlist(mu_diff_gul),
    units = angle_unit,
    rot = angle_rot
    ),
    col = 'gray25',
    stack = TRUE,
    sep = 0.1,
    bins = 355/5,
    units = angle_unit,
    rotation = angle_rot,
    zero = pi/2,
    shrink = 1.5)
mtext(text = 'UV Dim - Green Dim',
    side = 1,
    line = -2)
lines(x = c(0,0),
    y = c(-1,1),
    col = 'gray')
PlotMV_circMLE(mod_par = mod_diff_par['uvg_l'],],
    col1 = 'gray25')
PlotCI_vM(mod_CI_tb['uvg_l'],],
    col = 'gray25',
    radius = 1+0.3)

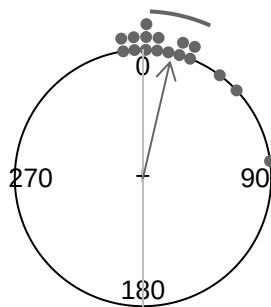
```



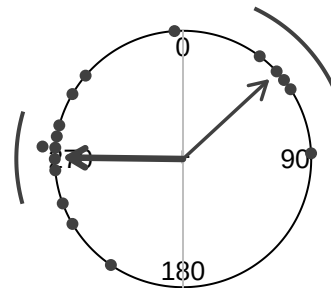
Green Dim – Green Bright



UV Dim – Green Bright



UV Bright – Green Bright



UV Dim – Green Dim

Here we see that the confidence interval for the difference between the green conditions overlaps with 0° , and the one for the two bright conditions almost does (the one sided CI does, but looks odd due to its asymmetry). To confirm whether these models better explain the data than the null hypothesis of the same mean for each condition (no difference), we can perform a likelihood-ratio test.

Likelihood ratio tests

For each condition, compare the models for same mean, different mean and multiple means. First perform all tests separately, then collect them together and finally adjust for multiple tests before reporting.

```
mod_details= lapply(X = names(pair_diffs_lst),
                    FUN = CollectDetails,
                    ll = all_res,
                    bimod = TRUE)
names(mod_details) = names(pair_diffs_lst)

#save details
save(x = mod_details,
     file = '2Results/MLEdetails_paired_diffs.RData')

#set up tests for hypotheses
lr_tests = c('pairs_diff_zero', #is the turn different from zero?
             'pairs_multi_diff', #is the turn in one or two directions?
             'pairs_multi_zero' #are the two directions different from zero?)
```



```

    )#multiple tests

#Perform collect likelihood ratio tests for each hypothesis
all_lr_results_lst = lapply(X = mod_details,
                             FUN = LR_calc_lst,
                             tst = lr_tests)
all_lr_results = do.call(what = rbind,
                          args = all_lr_results_lst
                          )
all_lr_results = within(data.frame(all_lr_results),
                        {
                          comparison = c( sapply(X = names(mod_details),
                                                  FUN = rep,
                                                  times = length(lr_tests)
                                                  ) )
                          test = rep(lr_tests,
                                      times = length(names(mod_details))
                                      )
                        }
                        )

#add labels to interpret each test according to p value and sign of comparison
all_lr_results = within(all_lr_results,
                        {
                          result = mapply(tst = lr_tests,
                                           d0 = unlist(dev0),
                                           d1 = unlist(dev1),
                                           pa = p_adjusted,
                                           FUN = H1label
                                           )
                        }
                        )

print_lr_results = all_lr_results
print_lr_results = within(print_lr_results,
                          {
                            rm(dev0, dev1, p)
                            test = sub(sub(sub(test,
                                                  pattern = '_',
                                                  replacement = ' '),
                                                  pattern = '_',
                                                  replacement = ' '),
                                                  pattern = 'pairs ',
                                                  replacement = '')
                            mu1 = round(all_res$mu1, 1)
                          }
                          )

print_lr_results = within(print_lr_results,
                          {mu2 = round(all_res$mu2, 1)}
                          )

rownames(print_lr_results) = with(print_lr_results, paste(comparison, test))#better to keep them for th
#Round numeric columns
num_col = sapply(X = print_lr_results,
                  FUN = is.numeric)

```

```
print_lr_results[,num_col] = apply(X = print_lr_results[,num_col],
                                   MARGIN = 2,
                                   FUN = round,
                                   digits = 3)

#Print the results
print(print_lr_results[,-6],
      digits = 3)
```

##		d.f.	chi_squared	p_adjusted	test	comparison	mu1	mu2
##	green_hl diff zero	1	3.31	0.103	diff zero	green_hl	0.0	NA
##	green_hl multi diff	3	8.28	0.103	multi diff	green_hl	-17.5	NA
##	green_hl multi zero	4	4.97	0.290	multi zero	green_hl	349.6	530
##	uv_hl diff zero	1	5.82	0.016	diff zero	uv_hl	0.0	NA
##	uv_hl multi diff	3	20.21	0.000	multi diff	uv_hl	-78.8	NA
##	uv_hl multi zero	4	26.03	0.000	multi zero	uv_hl	49.3	260
##	uv_g_h diff zero	1	5.04	0.074	diff zero	uv_g_h	0.0	NA
##	uv_g_h multi diff	3	6.80	0.118	multi diff	uv_g_h	13.1	NA
##	uv_g_h multi zero	4	1.76	0.780	multi zero	uv_g_h	188.7	369
##	uv_g_l diff zero	1	4.98	0.026	diff zero	uv_g_l	0.0	NA
##	uv_g_l multi diff	3	12.27	0.010	multi diff	uv_g_l	-58.4	NA
##	uv_g_l multi zero	4	17.26	0.005	multi zero	uv_g_l	47.3	271

```
#save results
write.table(x = print_lr_results,
            file = '2Results/LRT_paired_diffs.csv',
            sep = ',',
            row.names = TRUE)
```

These tests confirm, that there is no significant difference between the two green conditions, a marginally-significant 13° left turn between the bright conditions, and significant bimodal differences between the two UV conditions and the two dim conditions, driven by left and right turns for the dim UV condition.

Bayes-factor equivalent

```
# Initialize an empty list to store row segments
table_rows = list()

# Loop through each dataset in your list
for (dts in names(pair_diffs_lst)) {

  mdt_metrics = BFDDetails(all_res,
                           dts = dts,
                           bimod = TRUE,
                           n_obs = length(pair_diffs_lst[[dts]]))

  # Define the 3 tests to run per dataset to match your layout
  test_types = c("pairs_diff_zero",
                 "pairs_multi_diff",
                 "pairs_multi_zero")
  display_names = c("diff zero",
                    "multi diff",
```

```

                                "multi zero")
dfs = c(1, 3, 4) # structural degrees of freedom for the tests

for (i in 1:length(test_types)) {

  # Calculate Bayesian metrics for this specific test
  bf_out = BF_calc(tst = test_types[i],
                   mdt = mdt_metrics)
  label_out = BFlabel(test_types[i],
                      bf_out$Bayes_factor_1.0 )

  # Construct unique row name (e.g., "green_hl diff zero")
  row_id = paste(dts, display_names[i])

  # Assemble the individual row without mu1 and mu2 columns
  table_rows[[row_id]] = with(bf_out,
                              {
                                data.frame(
                                  df      = dfs[i],
                                  delta_BIC = round(delta_bic, 3),
                                  Bayes_factor_1.0 = round(Bayes_factor_1.0 , 4),
                                  test      = display_names[i],
                                  comparison = dts,
                                  result     = label_out,
                                  stringsAsFactors = FALSE
                                )
                              }
                            )
}

# Bind all collected rows into a unified final master table
bf_summary_table = do.call(rbind, table_rows)
#Print the results
print(bf_summary_table,
      digits = 3)

```

```

##              df delta_BIC Bayes_factor_1.0      test comparison
## green_hl diff zero    1    0.479      1.27e+00 diff zero  green_hl
## green_hl multi diff   3   -16.781      2.00e-04 multi diff green_hl
## green_hl multi zero   4   -16.302      3.00e-04 multi zero green_hl
## uv_hl diff zero       1    2.982      4.44e+00 diff zero  uv_hl
## uv_hl multi diff      3   11.711      3.49e+02 multi diff  uv_hl
## uv_hl multi zero      4   14.693      1.55e+03 multi zero  uv_hl
## uvg_h diff zero       1    2.210      3.02e+00 diff zero  uvg_h
## uvg_h multi diff      3   -15.301      5.00e-04 multi diff  uvg_h
## uvg_h multi zero      4   -13.091      1.40e-03 multi zero  uvg_h
## uvg_l diff zero       1    2.150      2.93e+00 diff zero  uvg_l
## uvg_l multi diff      3    3.774      6.60e+00 multi diff  uvg_l
## uvg_l multi zero      4    5.925      1.93e+01 multi zero  uvg_l
##                                     result
## green_hl diff zero Anecdotal evidence for H1
## green_hl multi diff Decisive evidence for H0

```

```
## green_hl multi zero   Decisive evidence for H0
## uv_hl diff zero      *Moderate evidence for H1
## uv_hl multi diff     Decisive evidence for H1
## uv_hl multi zero     Decisive evidence for H1
## uvg_h diff zero      *Moderate evidence for H1
## uvg_h multi diff     Decisive evidence for H0
## uvg_h multi zero     Decisive evidence for H0
## uvg_l diff zero      Anecdotal evidence for H1
## uvg_l multi diff     *Moderate evidence for H1
## uvg_l multi zero     *Strong evidence for H1
```

```
#save results
write.table(x = bf_summary_table,
            file = '2Results/BF_paired_diffs.csv',
            sep = ',',
            row.names = TRUE)
```

Are the two means for UV dim equally likely

```
wt_uv_hl = all_res['multi uv_hl', 'weight1']
est_no_m1 = round(length(full_ids)*wt_uv_hl)
binom.test(x = est_no_m1,
            n = length(full_ids),
            p = 0.5,
            alternative = 'two.sided')
```

```
##
## Exact binomial test
##
## data: est_no_m1 and length(full_ids)
## number of successes = 6, number of trials = 19, p-value = 0.1671
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.1257606 0.5655016
## sample estimates:
## probability of success
##          0.3157895
```

```
#CI for relative weight
#circMLE default from Hessians
with(mod_diff_lst,
     ci_circmle(circmle = uv_hl,
                model = uv_hl$bestmodel)
)
```

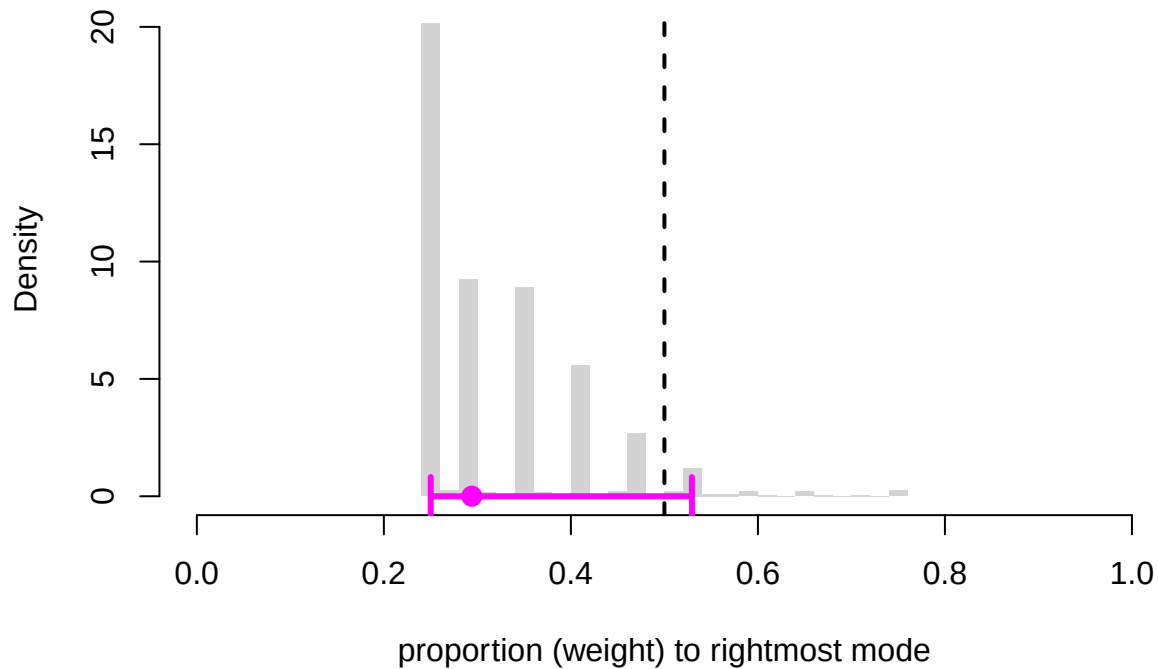
```
## Parameter      MLE      SE      CI
## 1          q1    4.54    0.14  4.266 - 4.814
## 2          k1    4.797  1.824  1.223 - 8.371
## 3          q2    0.86   0.099  0.666 - 1.054
## 4          k2   20.927 13.072 -4.694 - 46.548
## 5      lambda    0.706  0.111  0.489 - 0.923
```

```

#we can also bootstrap it
lambda_uhl = BootBimod(angles = unlist(mu_diff_uhl),
                        model = with(mod_diff_lst, uv_hl$bestmodel),
                        speedup_parallel = TRUE,
                        calc_q = FALSE,
                        n = 1e3) #save all for plotting
q_uhl = quantile(lambda_uhl,
                  probs = c(0,0.5,1)+c(1,0,-1)*((1-0.95)/2)
                  )
hist(lambda_uhl,
     breaks = 20,
     xlim = c(0,1),
     border = NA,
     probability = TRUE,
     main = 'bootstrapped M5B model',
     xlab = 'proportion (weight) to rightmost mode'
     )
abline(v = 0.5,
       lty = 2,
       col = 'black',
       lwd = 2)
arrows(x0 = q_uhl[1],
       x1 = q_uhl[3],
       y0 = 0,
       y1 = 0,
       length = 0.1,
       angle = 90,
       code = 3,
       lwd = 3,
       col = 'magenta'
       )
points(x = q_uhl[2],
       y = 0,
       pch = 20,
       col = 'magenta',
       cex = 2)

```

boostrapped M5B model



```
print(q_uhl,
      digits = 3)
```

```
## 2.5% 50% 97.5%
## 0.250 0.294 0.529
```

Angle between the means and 0°

```
abs_ang_uhl = abs(unlist(mu_diff_uhl))
dt_ang_uhl = data.frame(betaresp = abs_ang_uhl/180)
beta_ang_uhl = brm(bf(betaresp~1,
                      phi~1),
                  data = dt_ang_uhl,
                  family = Beta(link = 'logit',
                                link_phi = 'log'),
                  backend = 'cmdstanr')
```

```
## Start sampling
```

```
## Running MCMC with 4 sequential chains...
```

```
##
```

```
## Chain 1 Iteration: 1 / 2000 [ 0%] (Warmup)
```

```
## Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
```

```

## Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
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## Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
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## Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 1 finished in 0.1 seconds.
## Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
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## Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2 finished in 0.1 seconds.
## Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
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## Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)

```

```

## Chain 3 Iteration: 1000 / 2000 [ 50%] (Warmup)
## Chain 3 Iteration: 1001 / 2000 [ 50%] (Sampling)
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## Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3 finished in 0.1 seconds.
## Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
## Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
## Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
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## Chain 4 Iteration: 1700 / 2000 [ 85%] (Sampling)
## Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4 finished in 0.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.1 seconds.
## Total execution time: 0.9 seconds.

```

```

beta_ang_uhl_null = brm(bf(betaresp~0,
                           phi~1),
                        data = dt_ang_uhl,
                        family = Beta(link = 'logit',
                                      link_phi = 'log'),
                        backend = 'cmdstanr')

```

```
## Start sampling
```

```

## Running MCMC with 4 sequential chains...
##
## Chain 1 Iteration:   1 / 2000 [  0%] (Warmup)

```



```

## Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
## Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
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## Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 1 finished in 0.0 seconds.
## Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
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## Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2 finished in 0.0 seconds.
## Chain 3 Iteration: 1 / 2000 [ 0%] (Warmup)
## Chain 3 Iteration: 100 / 2000 [ 5%] (Warmup)
## Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
## Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
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```

```

## Chain 3 Iteration: 900 / 2000 [ 45%] (Warmup)
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## Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3 finished in 0.0 seconds.
## Chain 4 Iteration: 1 / 2000 [ 0%] (Warmup)
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## Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
## Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
## Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4 finished in 0.0 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 0.0 seconds.
## Total execution time: 0.9 seconds.

```

```
summary(beta_ang_uhl)
```

```

## Family: beta
## Links: mu = logit; phi = log
## Formula: betaresp ~ 1
##          phi ~ 1
## Data: dt_ang_uhl (Number of observations: 17)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##          Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS

```

```
## Intercept          -0.09      0.19    -0.46      0.29 1.00      3191      2465
## phi_Intercept      1.78       0.33      1.10      2.38 1.00      2694      2659
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
loo_beta_uhl = loo(beta_ang_uhl)
loo_beta_uhl_null = loo(beta_ang_uhl_null)
loo_compare(loo_beta_uhl, loo_beta_uhl_null)
```

```
##                elpd_diff se_diff
## beta_ang_uhl_null  0.0        0.0
## beta_ang_uhl      -0.9        0.5
```

```
#the distribution is concave (phi>2) in 99.9% of draws
mean(
  exp(
    as_draws_df(beta_ang_uhl_null,
      dpar = 'Intercept_phi')$Intercept_phi
    )<2) *100
```

```
## [1] 0.1
```

```
#estimate mean angular difference
emm_ang_uhl = emmeans(beta_ang_uhl,
  specs = '1')
print(
  summary(emm_ang_uhl,
    type = 'response')[2:4]*180,
  digits = 1)
```

```
##   response lower.HPD upper.HPD
## 1      86        69      102
```

Plot all raw data

Plot all 169 dances

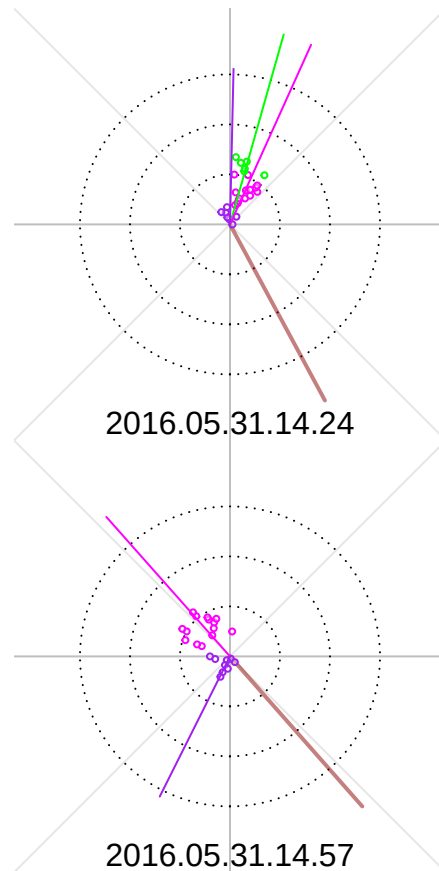
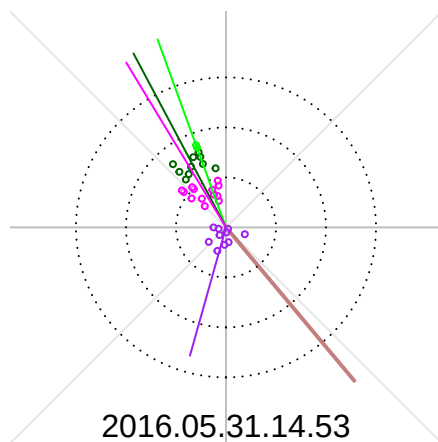
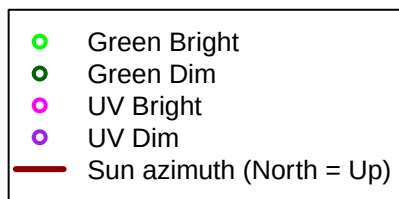
```
#set up a sequence for the axes
xc = seq(from = -pi, to = pi-1e-16, length.out = 1e3)
par(mfrow = c(2, 2),
  mar = c(0,0,0,0))
par(pty = 's')

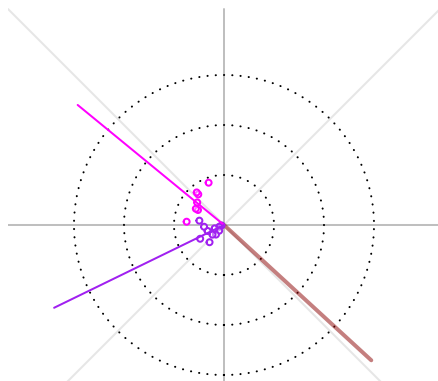
#add legend to 1st page
plot(x = NULL,
  xlim = c(-1,1),
  ylim = c(-1,1),
  pch = 19,
```

```

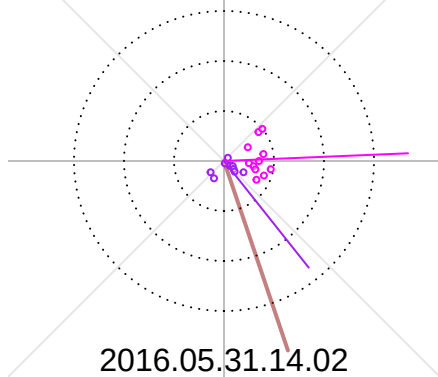
axes = FALSE,
xlab = '',
ylab = '',
main = ''
)
legend(x = 'center',
       legend = c('Green Bright',
                   'Green Dim',
                   'UV Bright',
                   'UV Dim',
                   'Sun azimuth (North = Up)'),
       col = c('green',
               'darkgreen',
               'magenta',
               'purple',
               'darkred'),
       pch = c(21, 21, 21, 21, NA),
       lty = c(NA, NA, NA, NA, 1),
       lwd = c(2, 2, 2, 2, 3)
       )
#loop through individuals
PltDance(ids = u_id,
         dtf = cd,
         cx = 1.0)

```

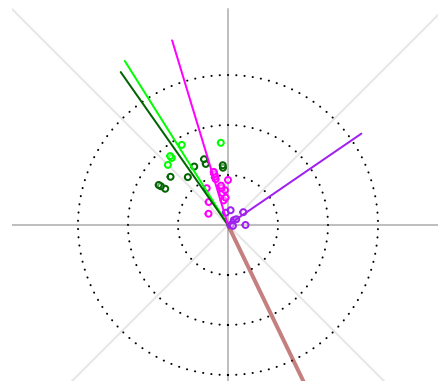




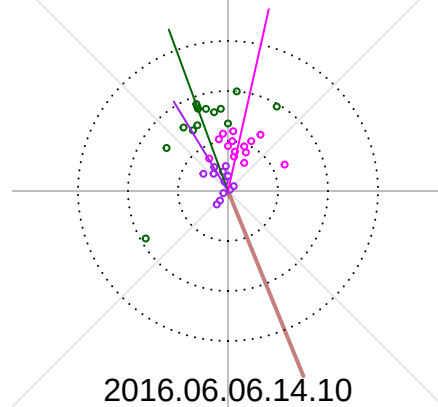
2016.05.31.15.14



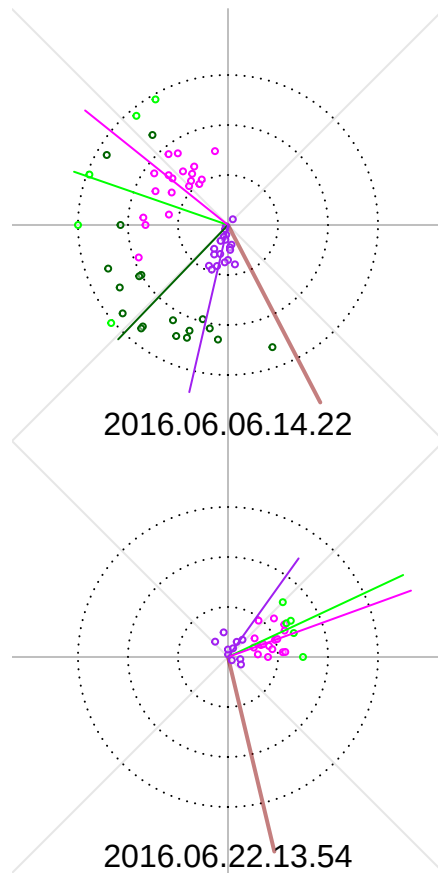
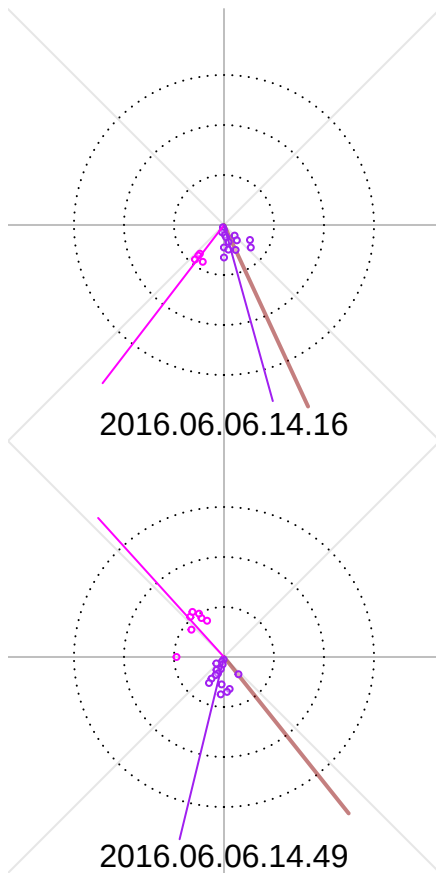
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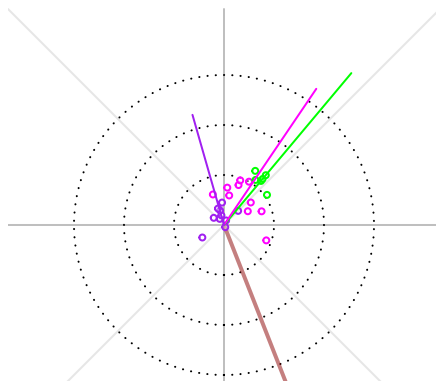


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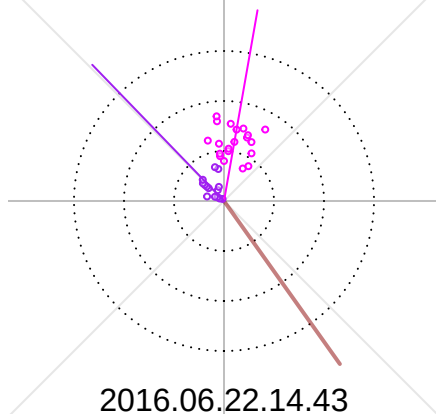


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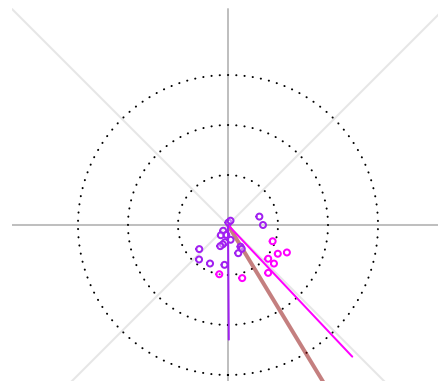




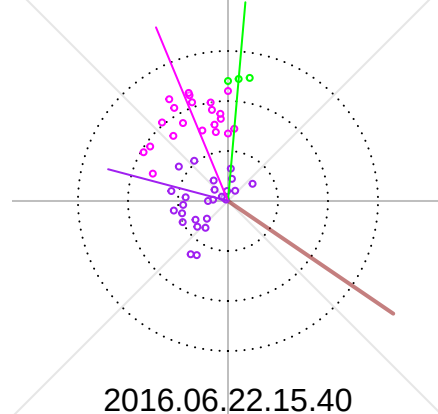
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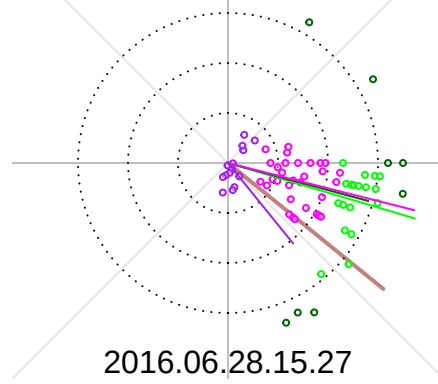
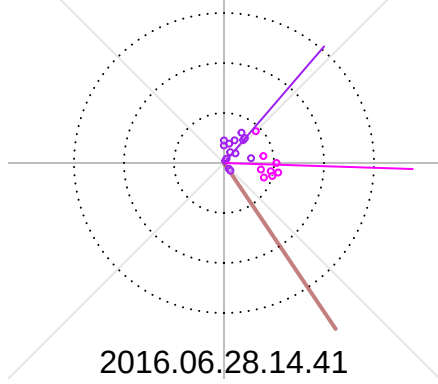
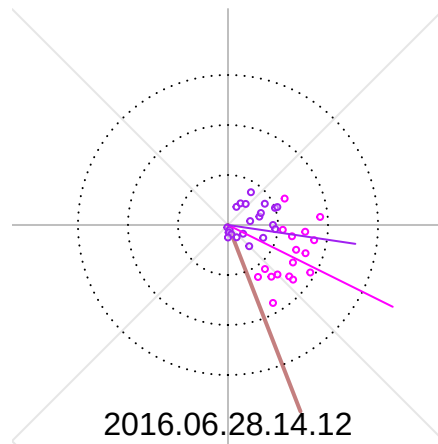
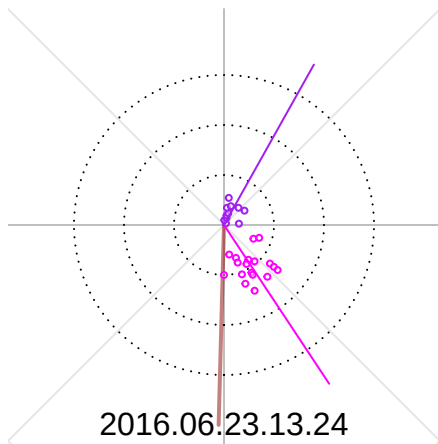
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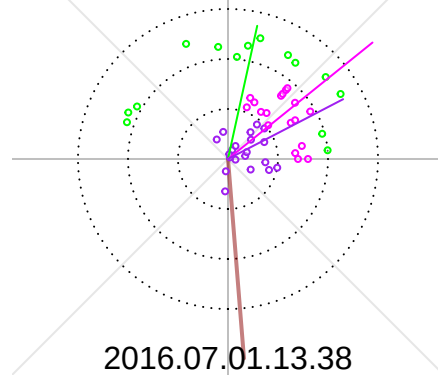
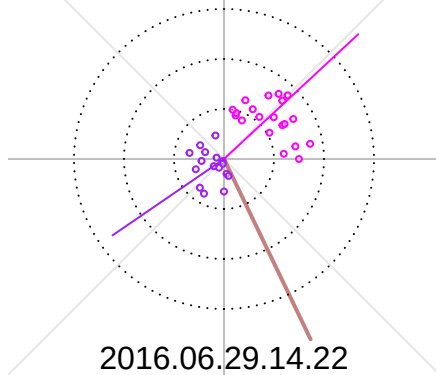
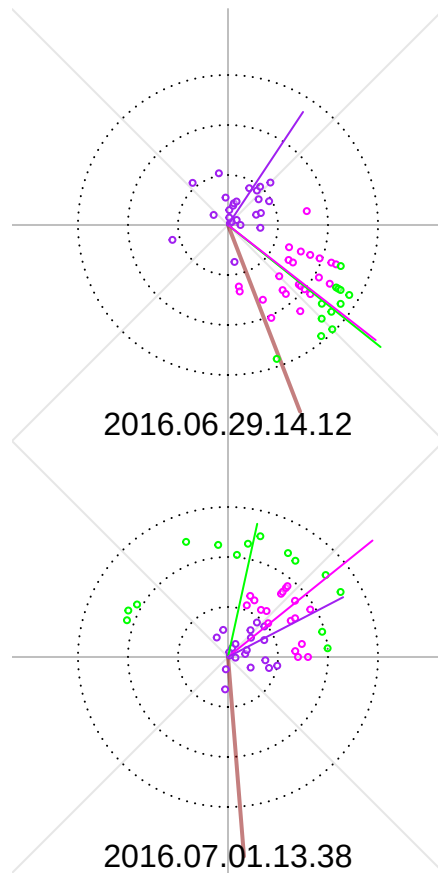
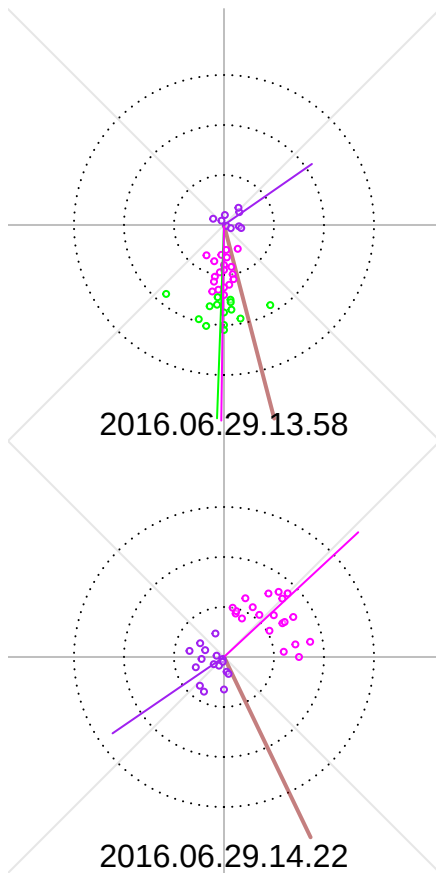


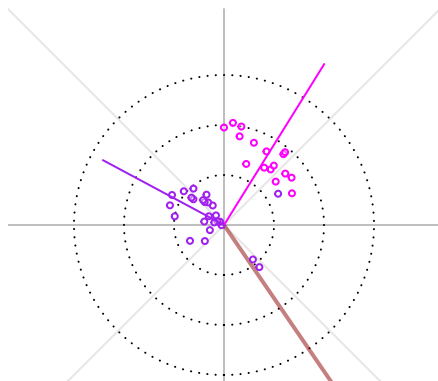
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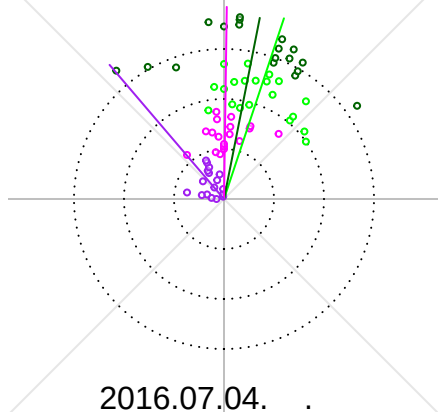
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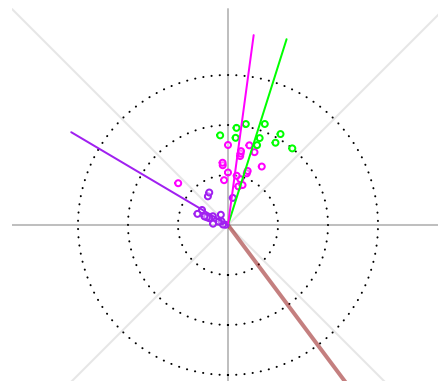




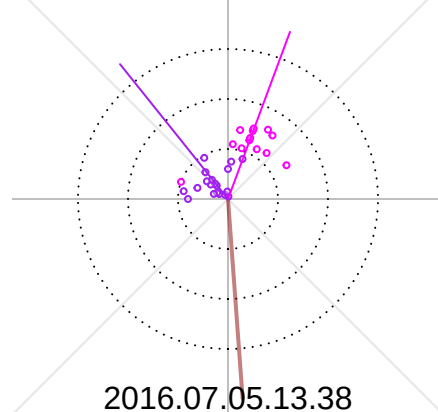
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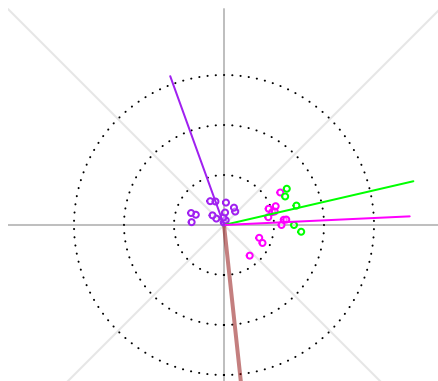
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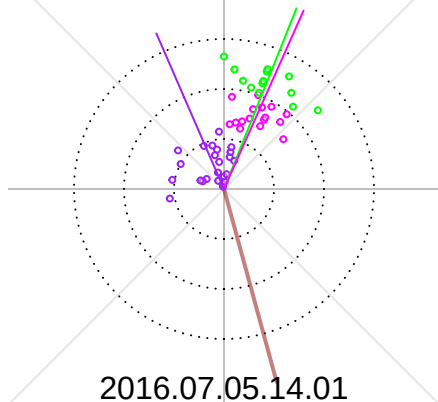
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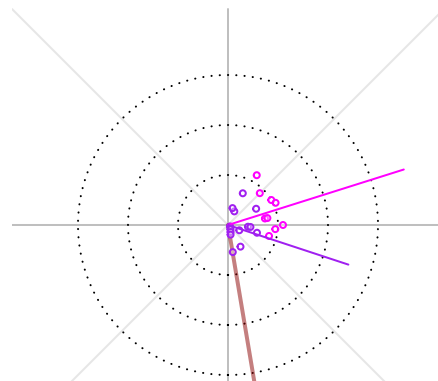
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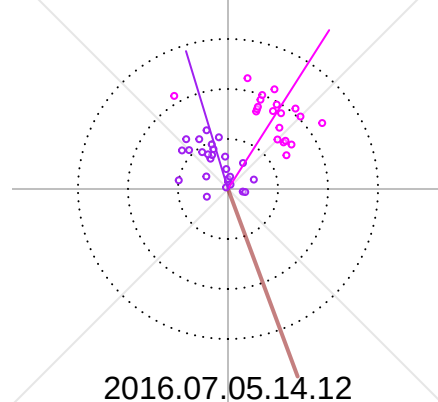
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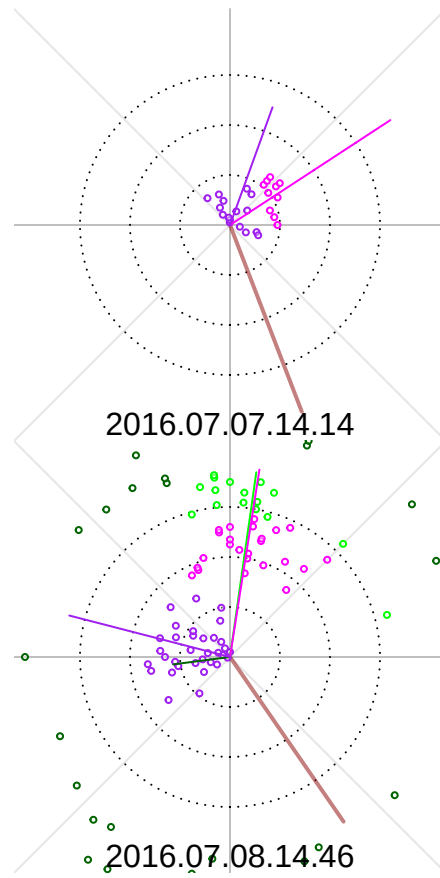
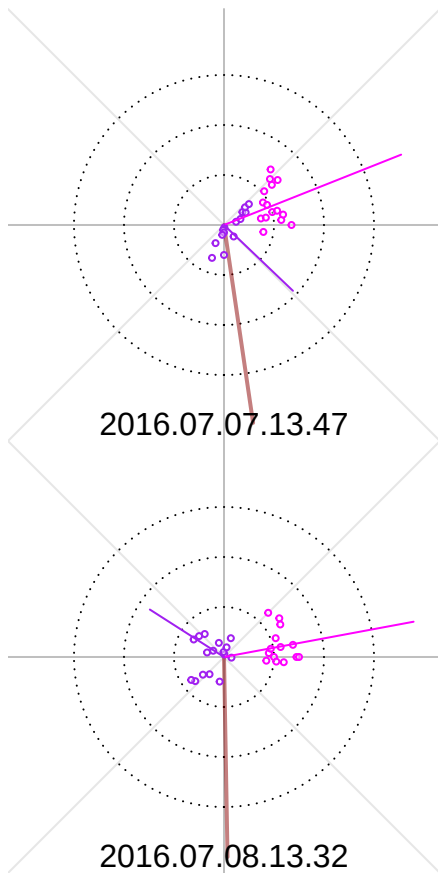
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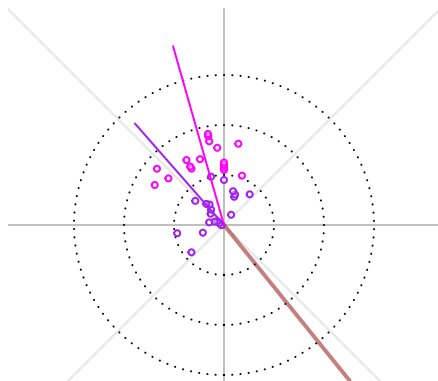


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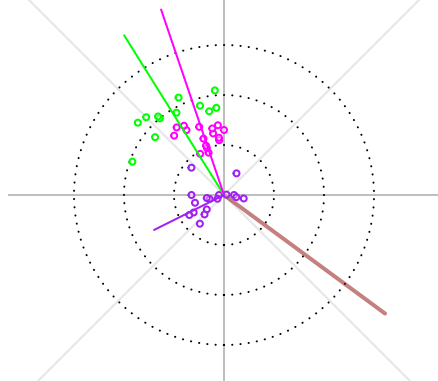


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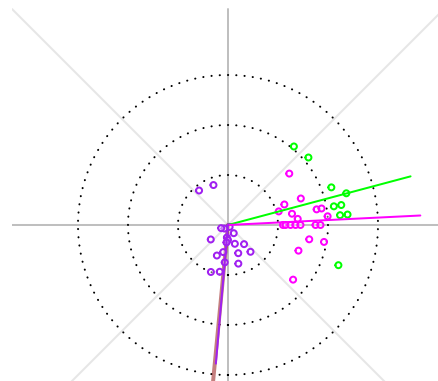




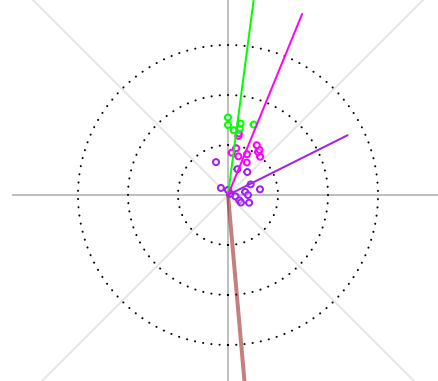
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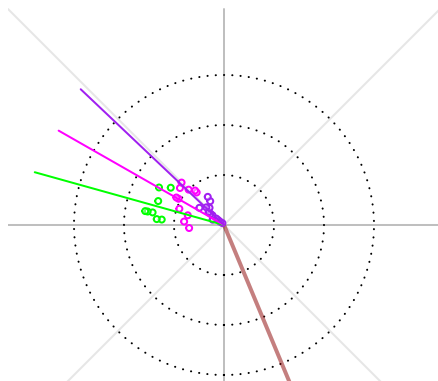
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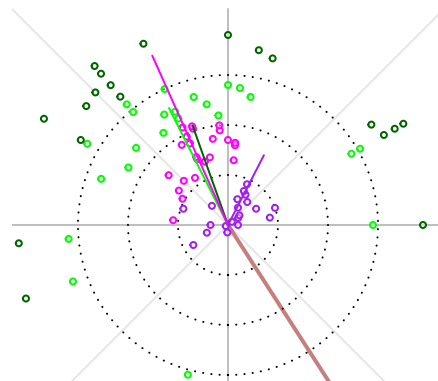
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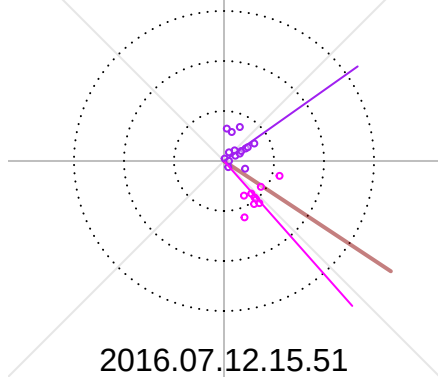
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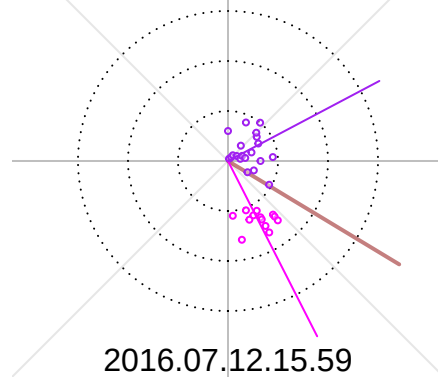
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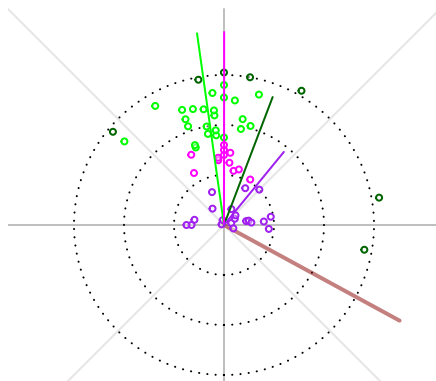
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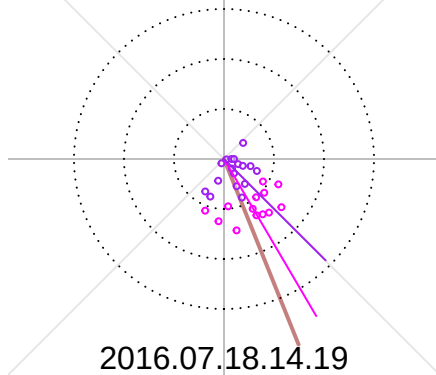
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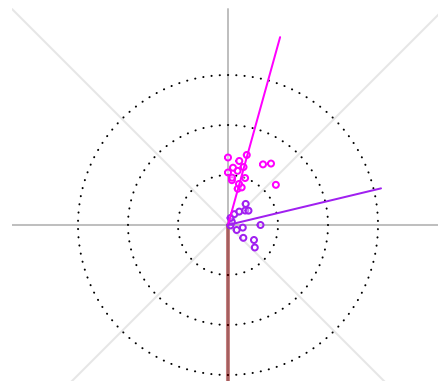
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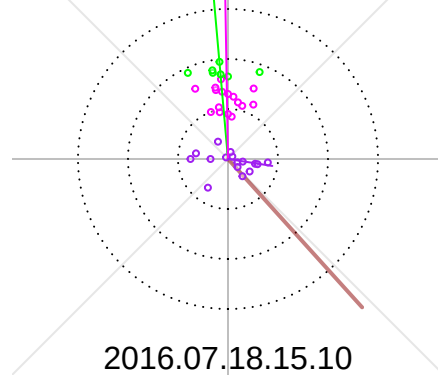
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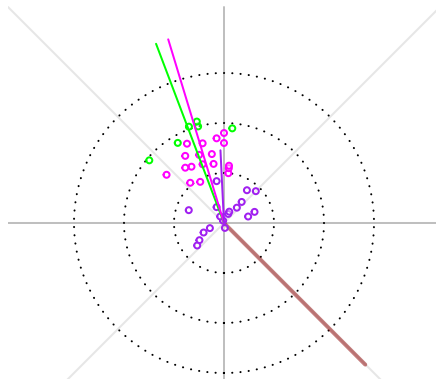
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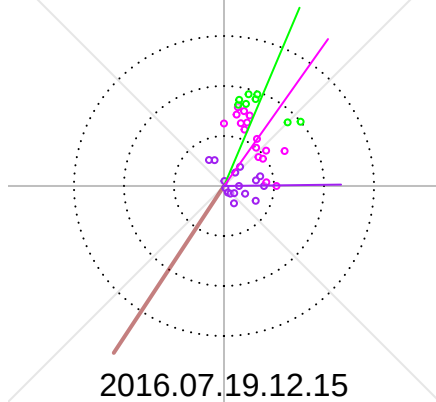
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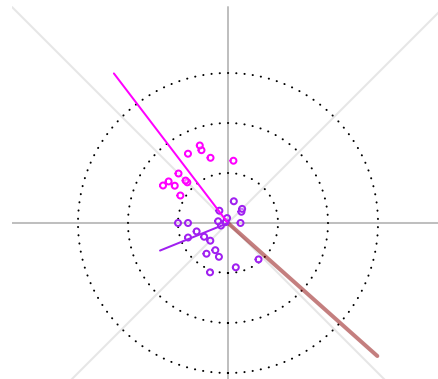
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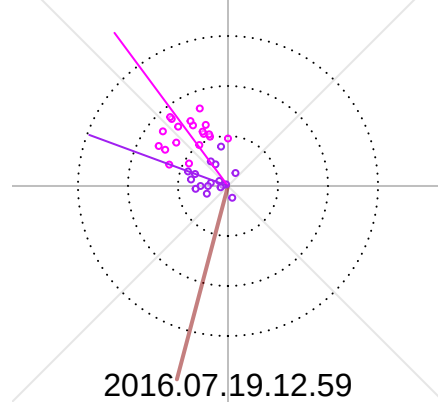
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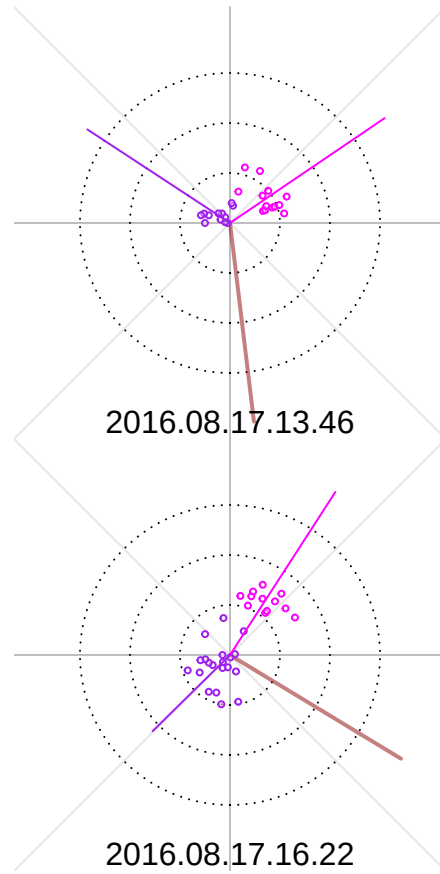
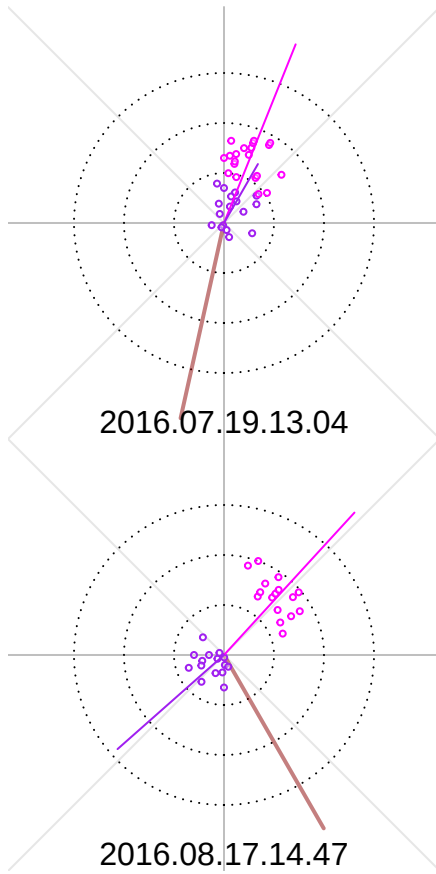
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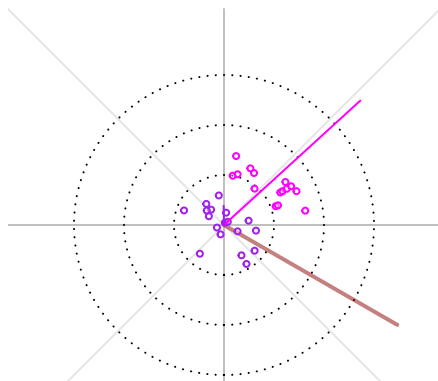


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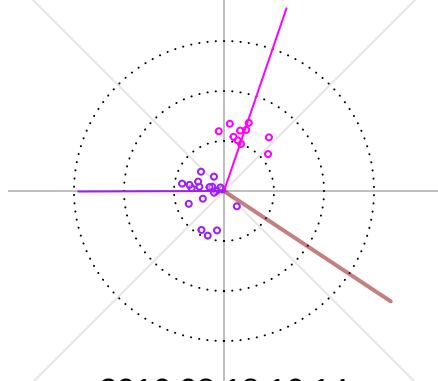


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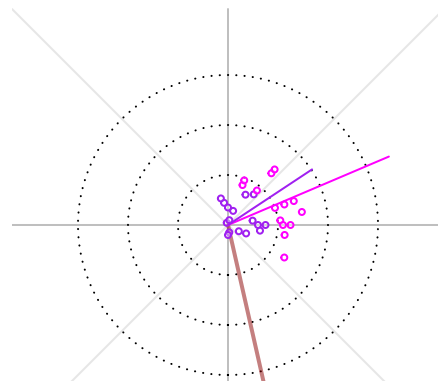




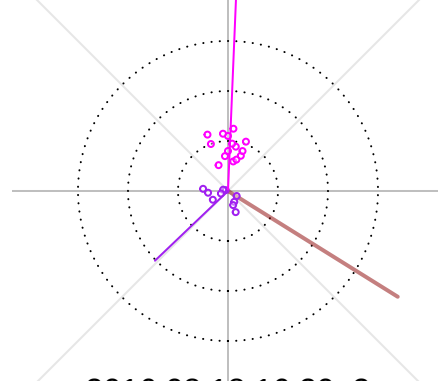
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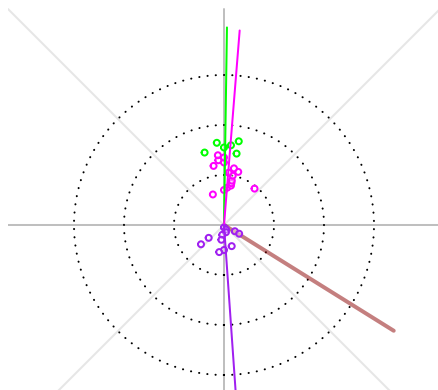
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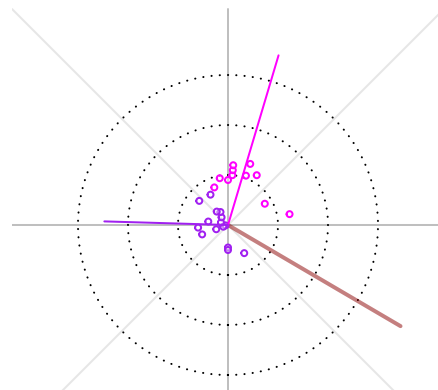
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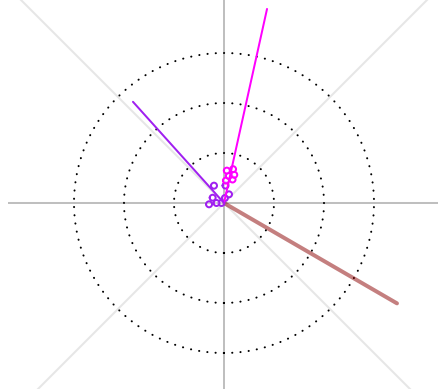
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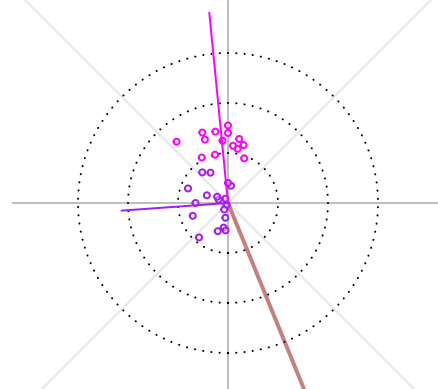
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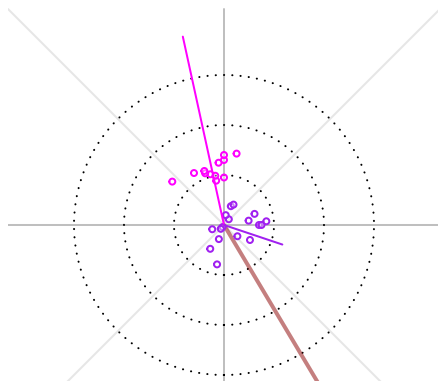
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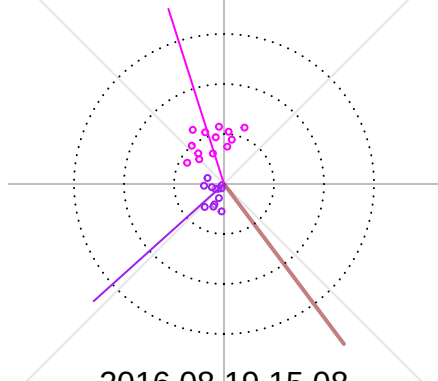
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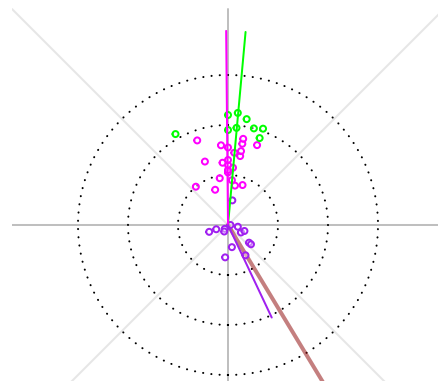
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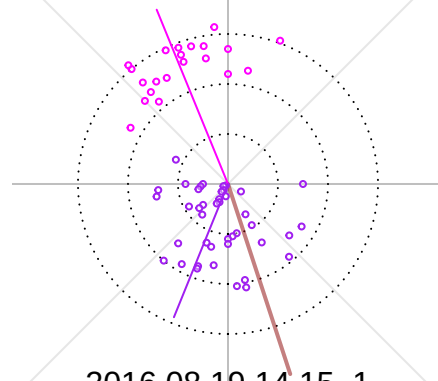
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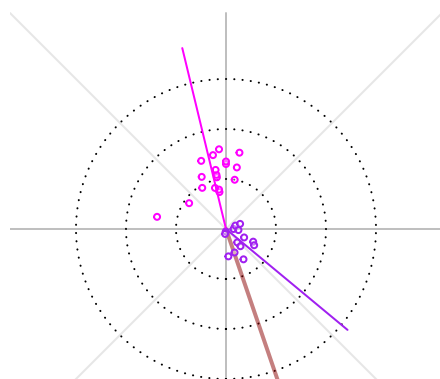
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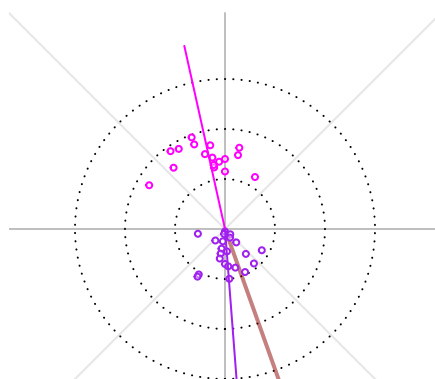
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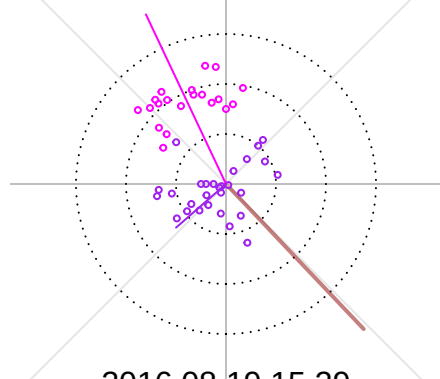
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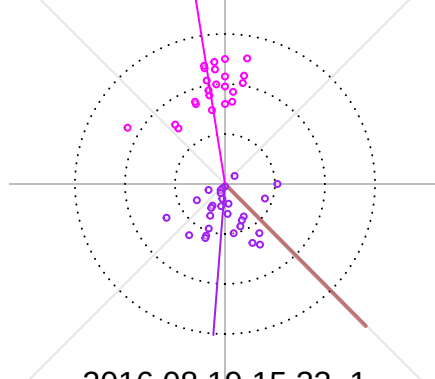
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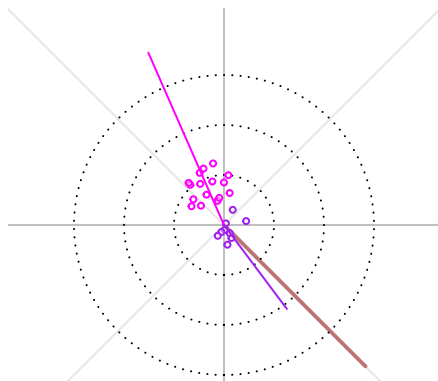
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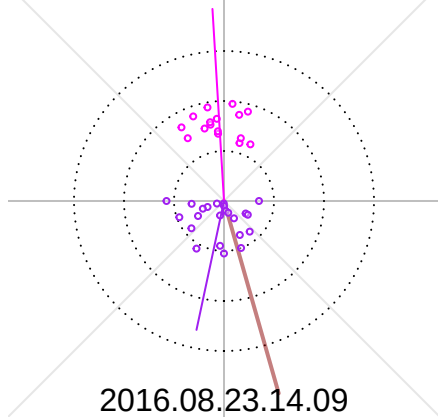
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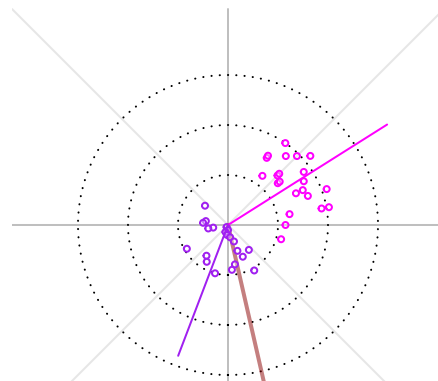
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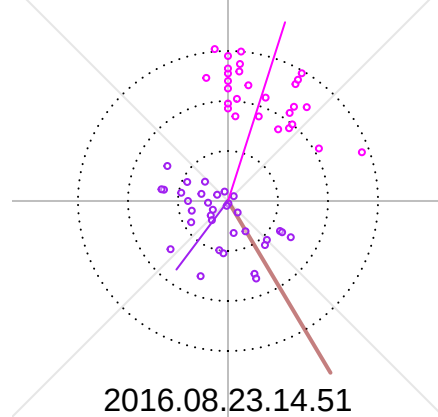
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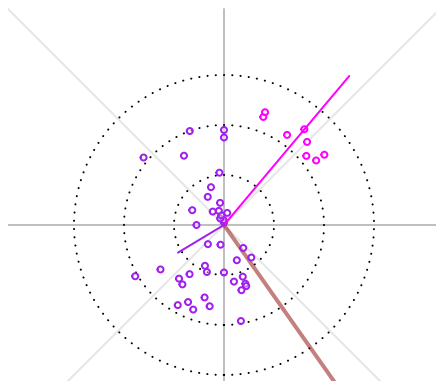
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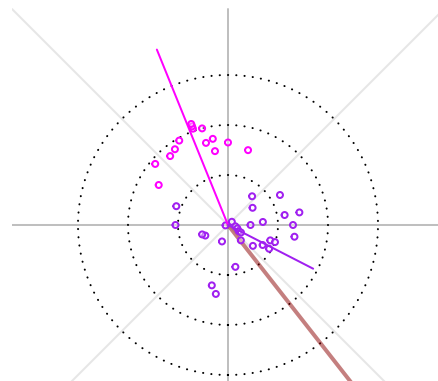
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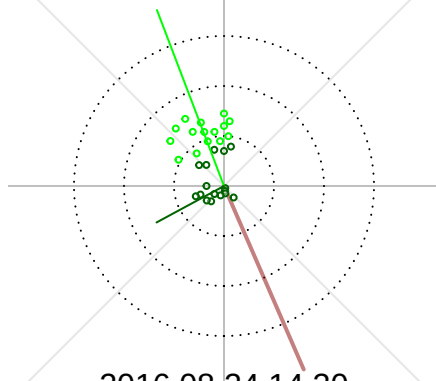
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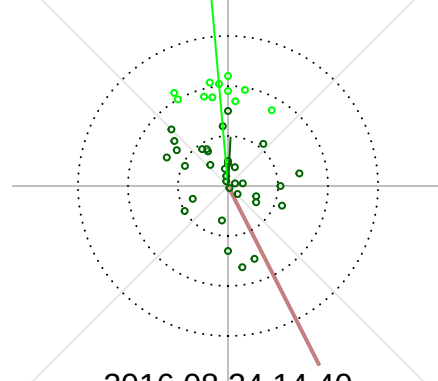
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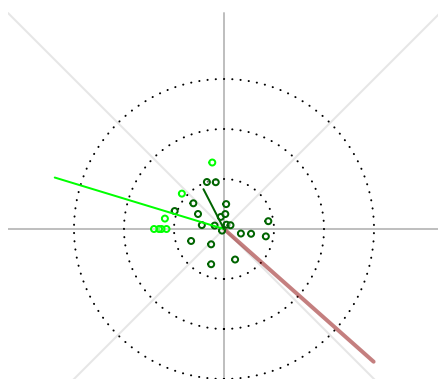
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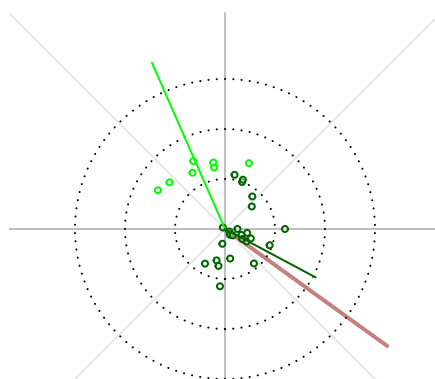
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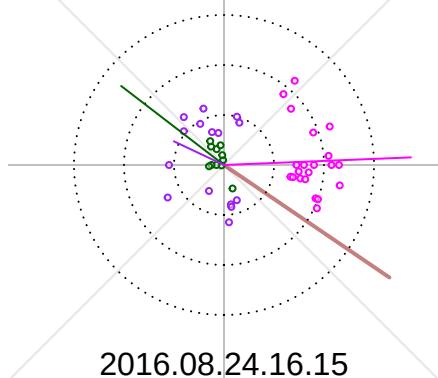
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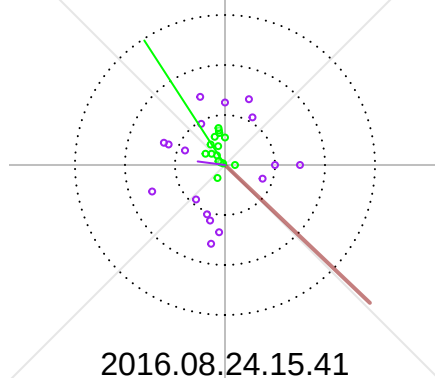
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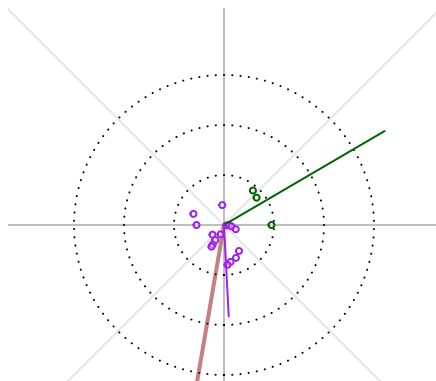
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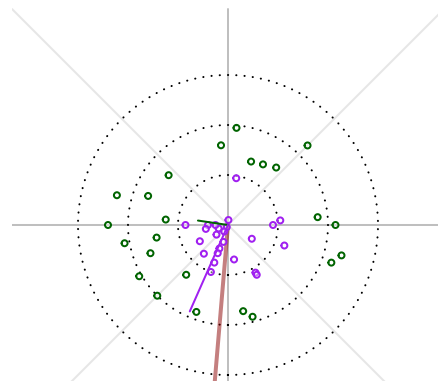
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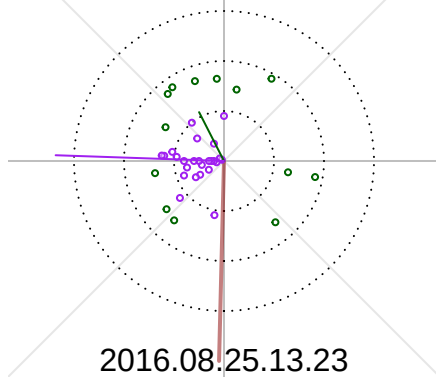
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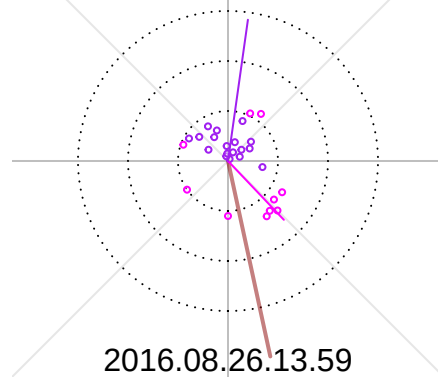
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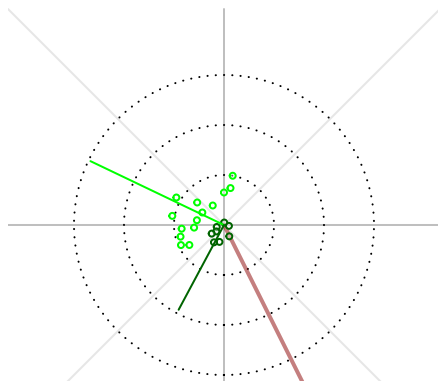
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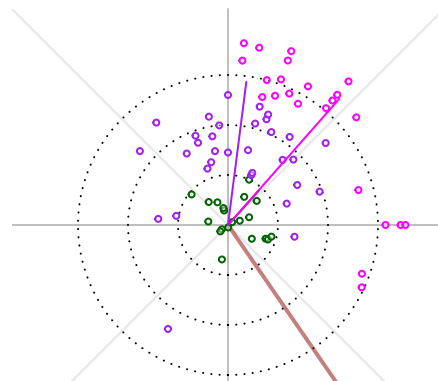
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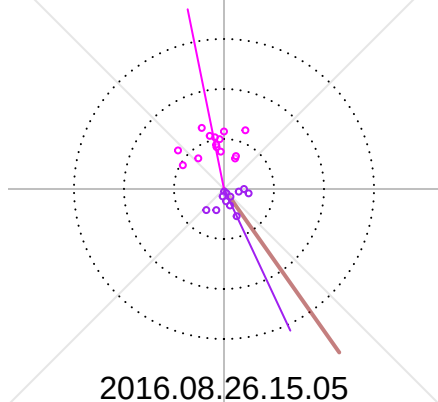
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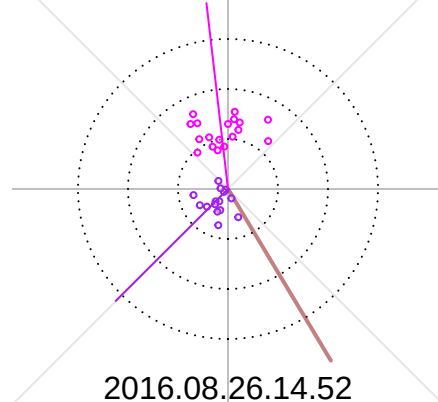
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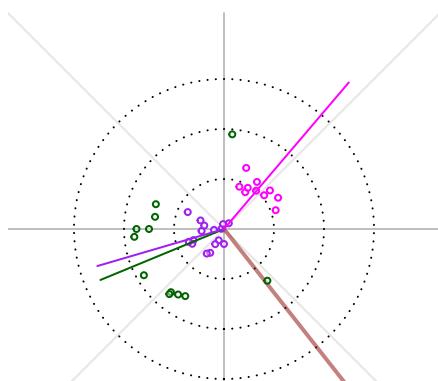
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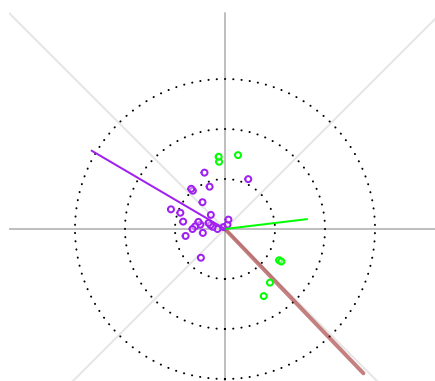
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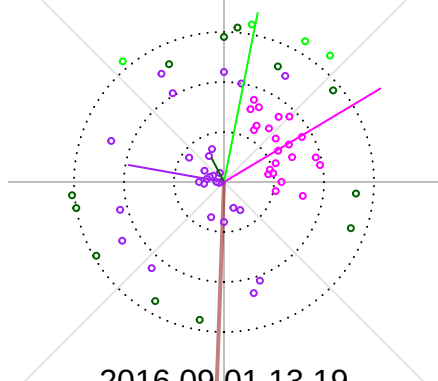
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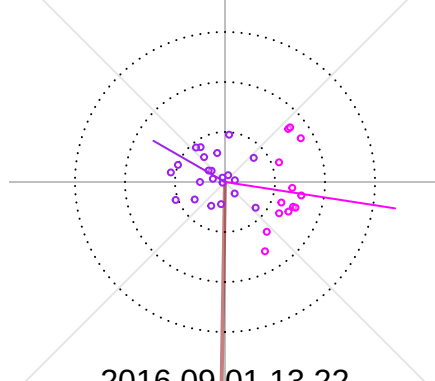
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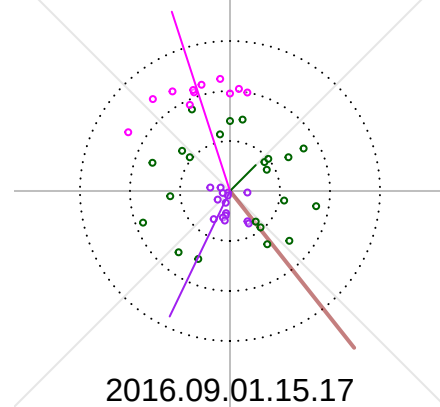
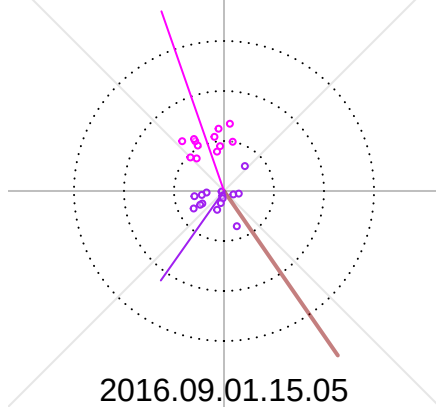
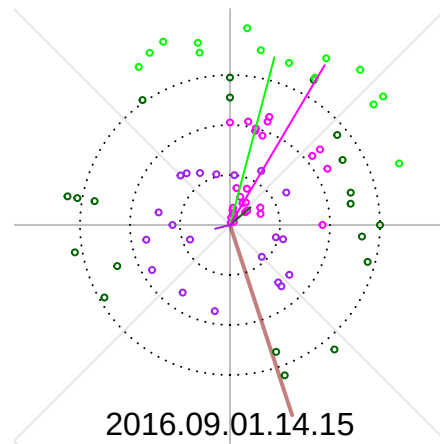
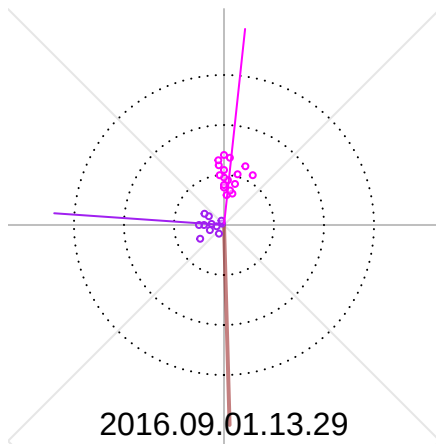
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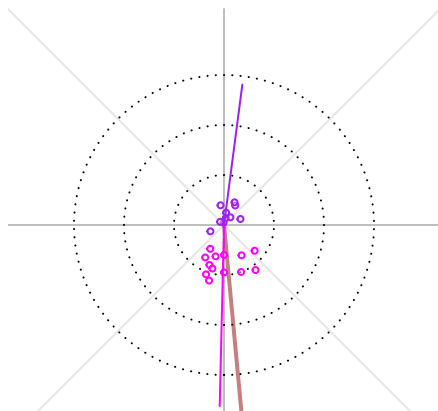


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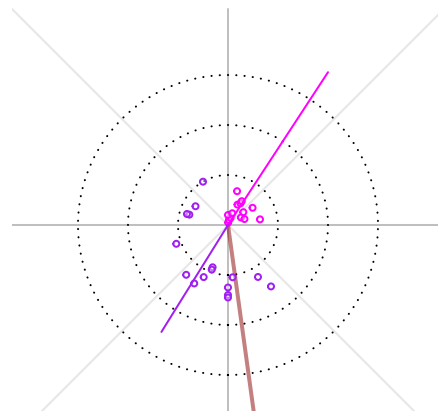


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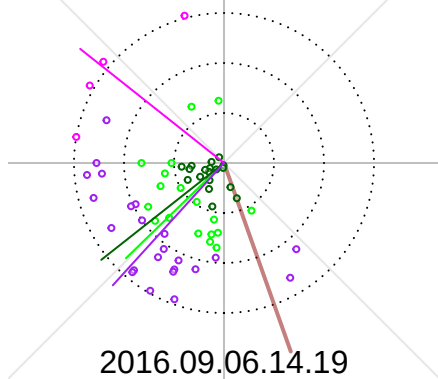




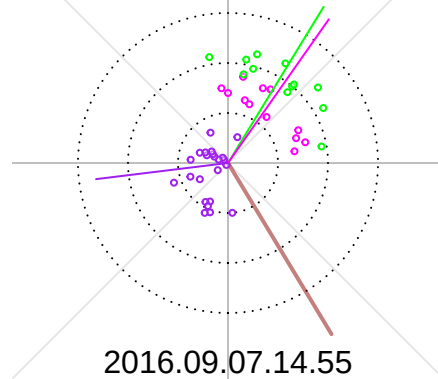
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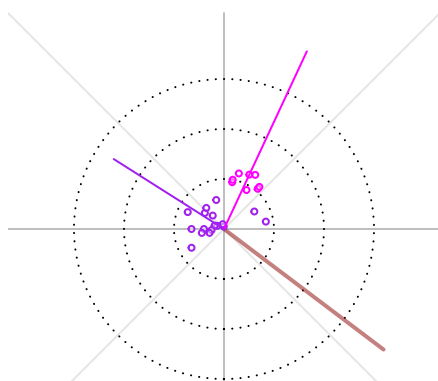
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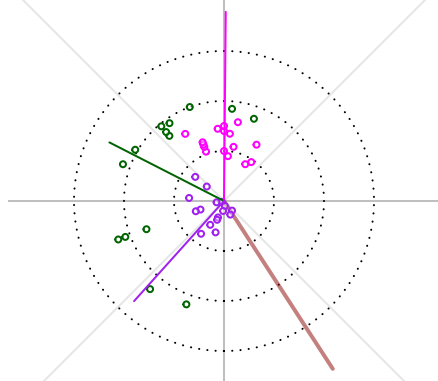
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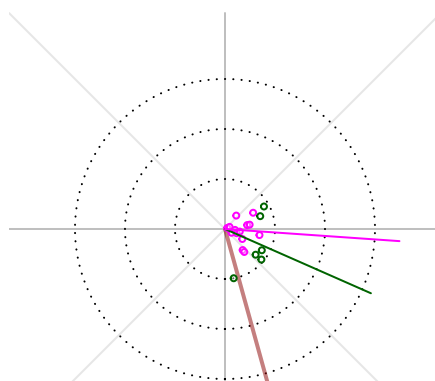
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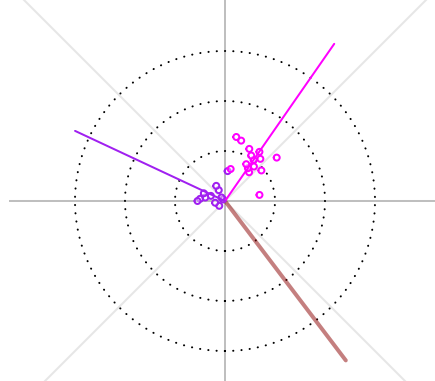
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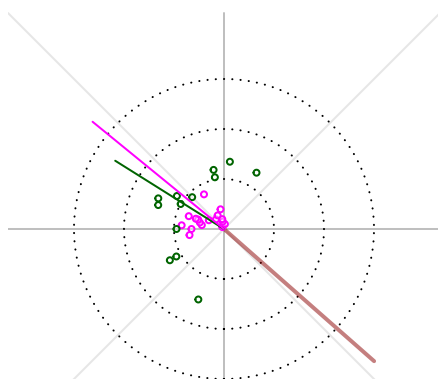
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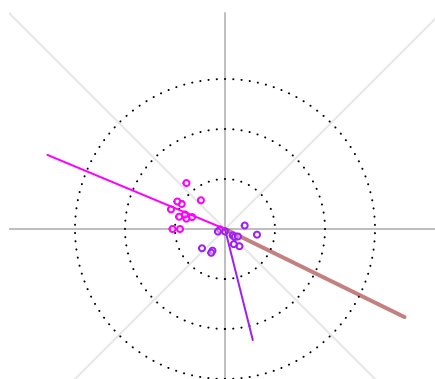
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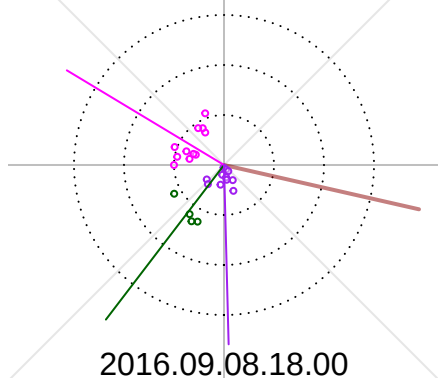
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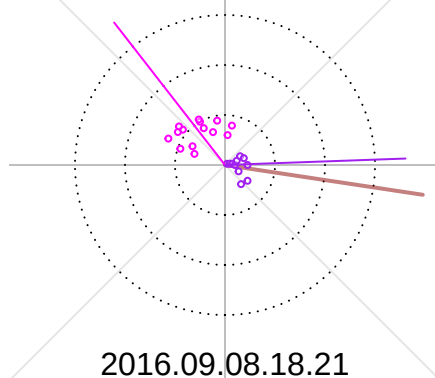
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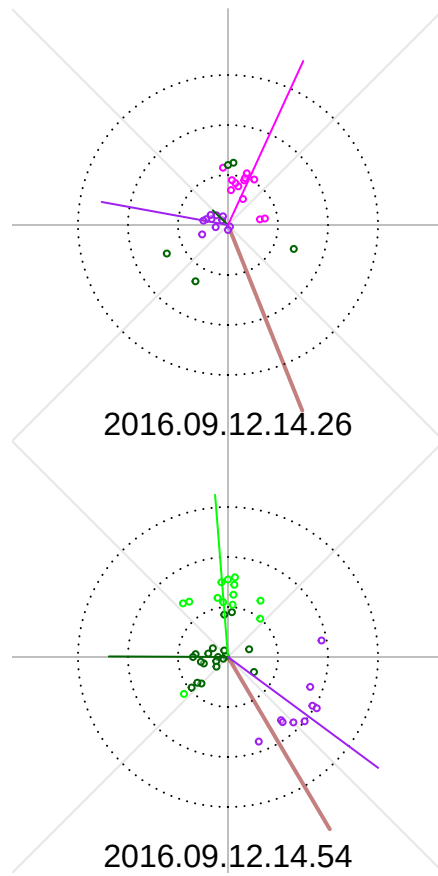
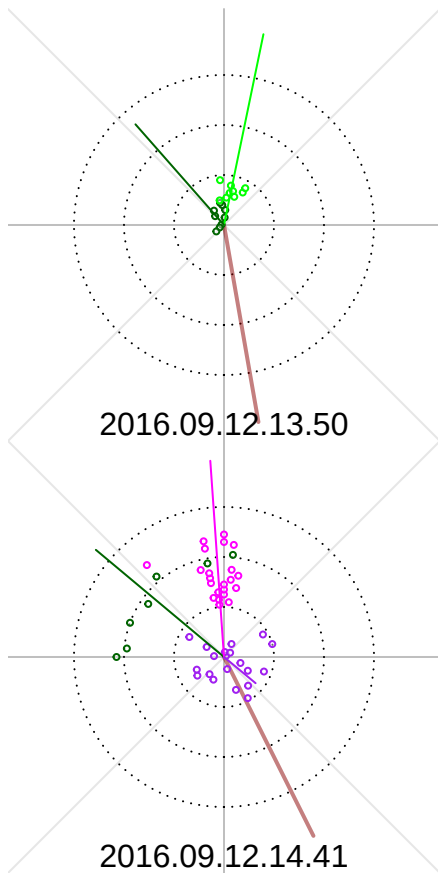
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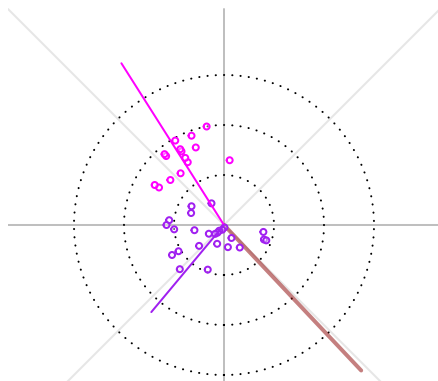


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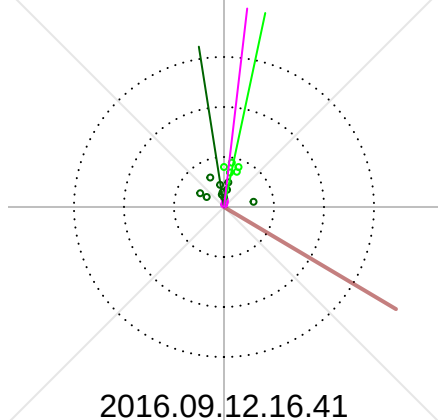


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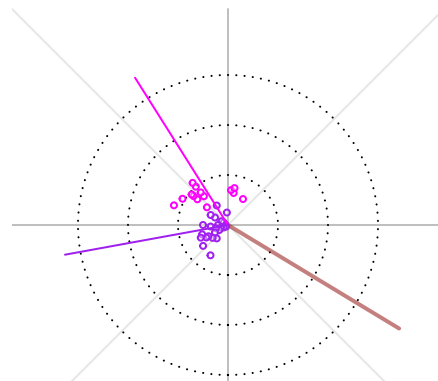




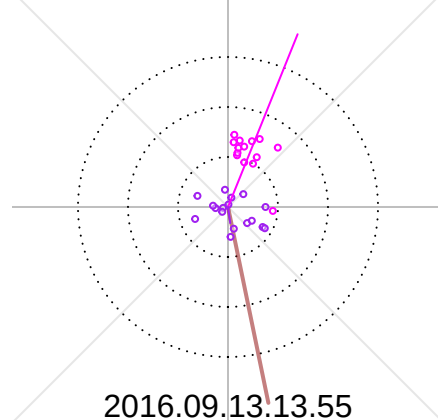
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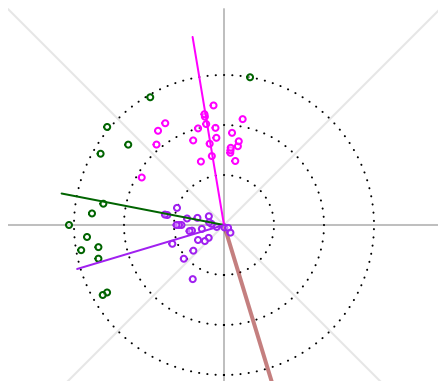
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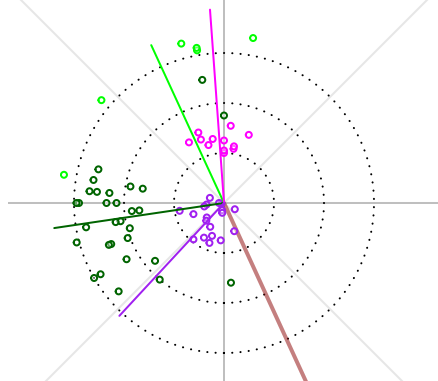
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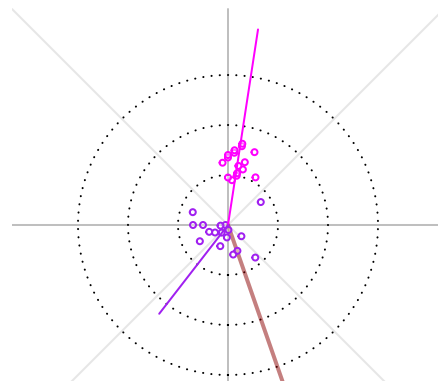
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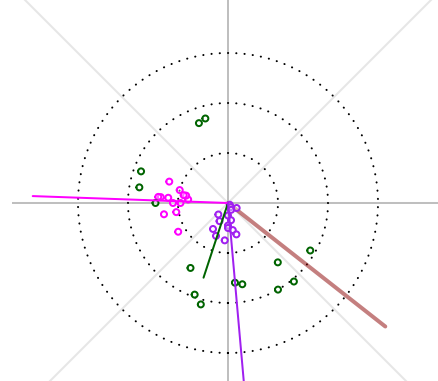
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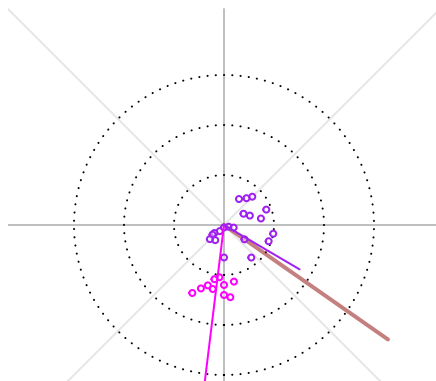
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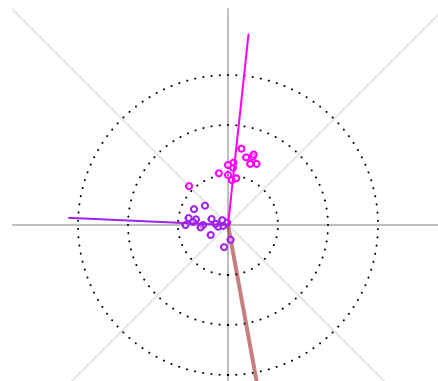
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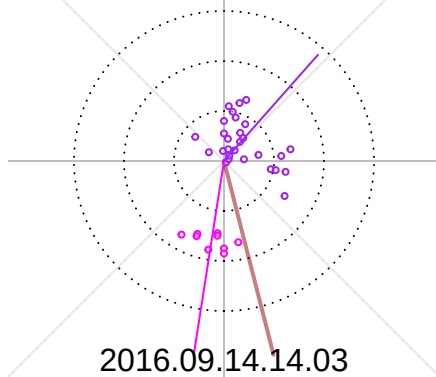
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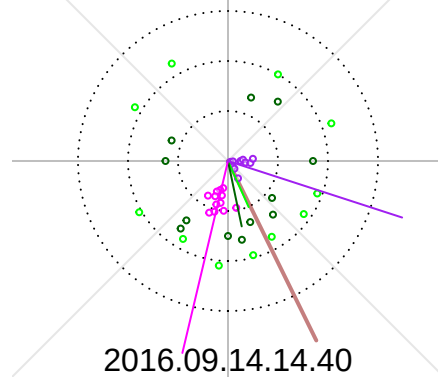
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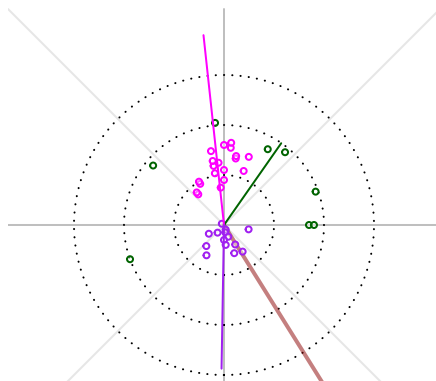
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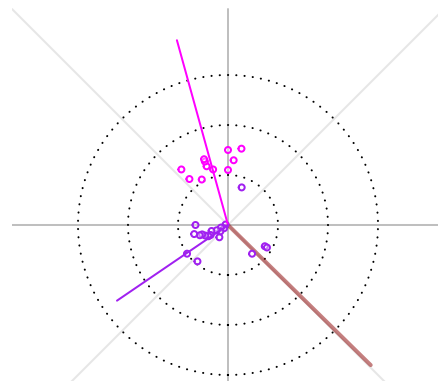
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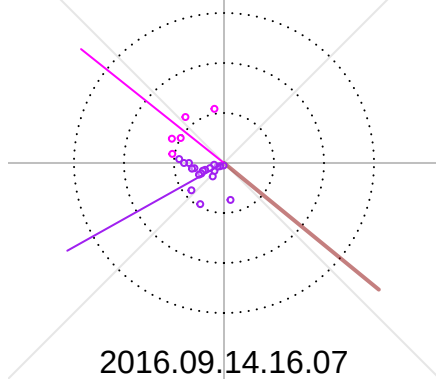
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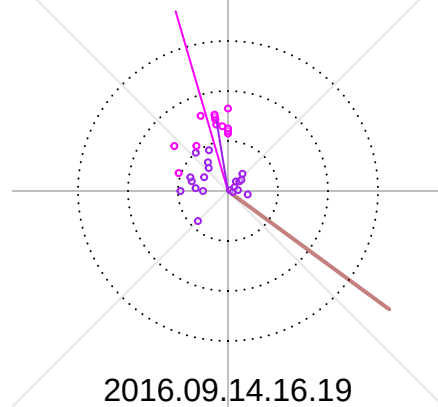
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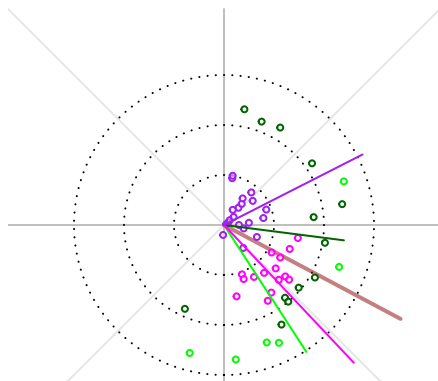
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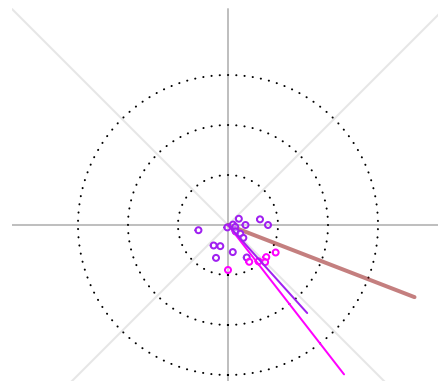
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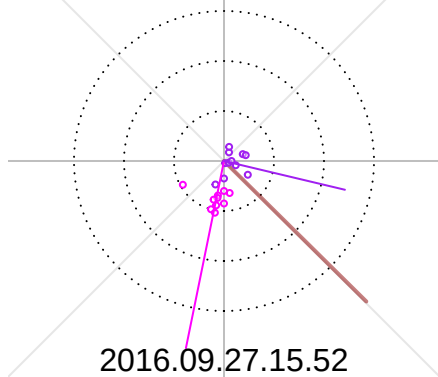
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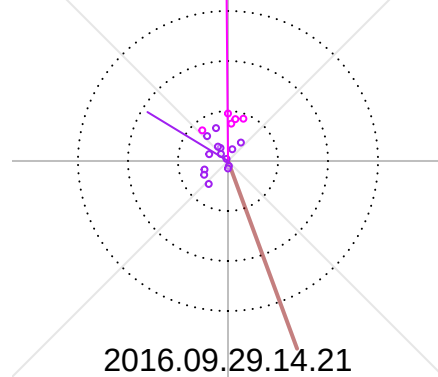
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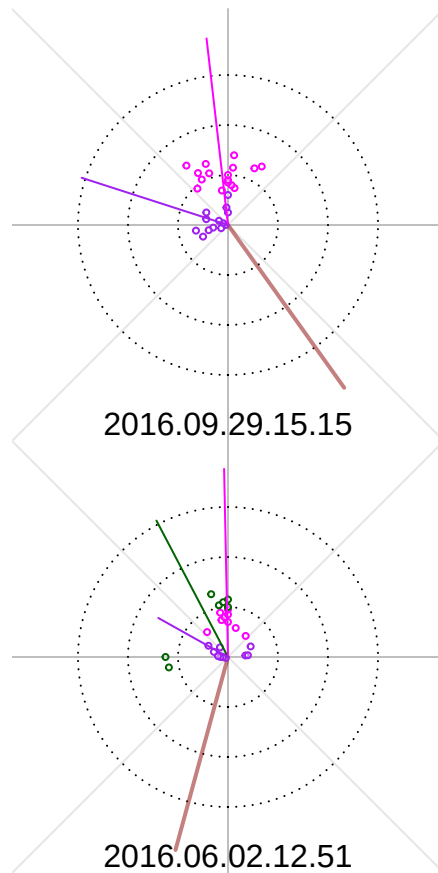
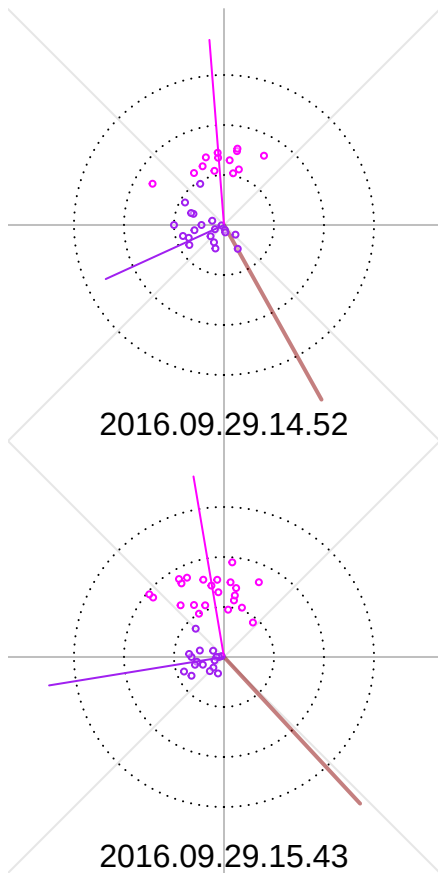
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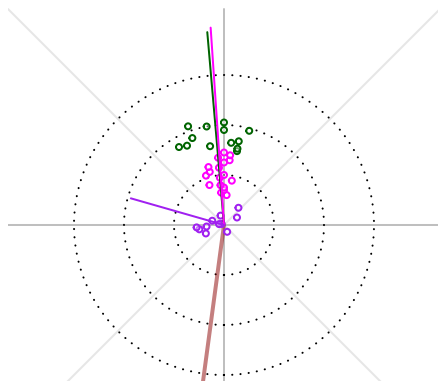


2016.09.27.15.52

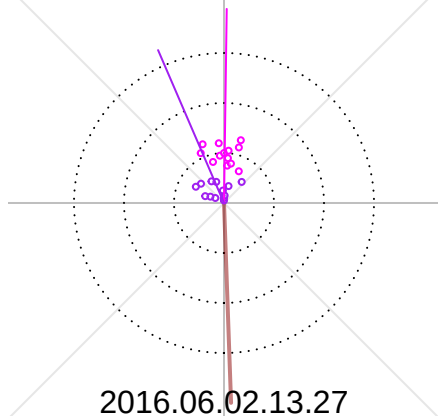


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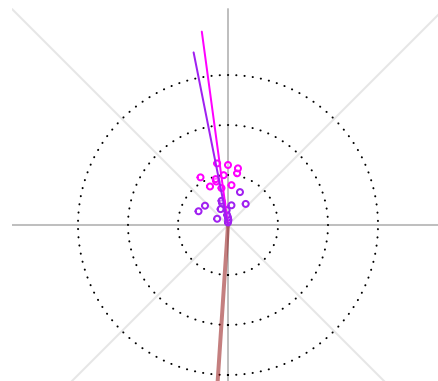




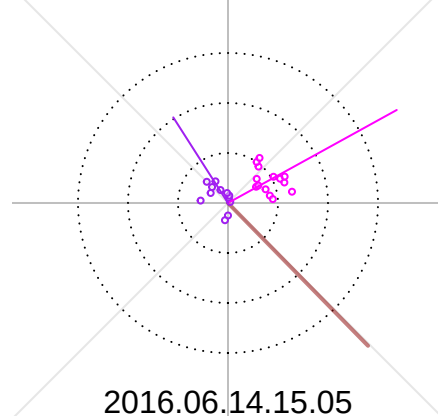
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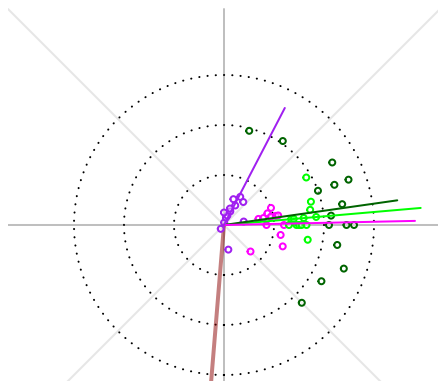
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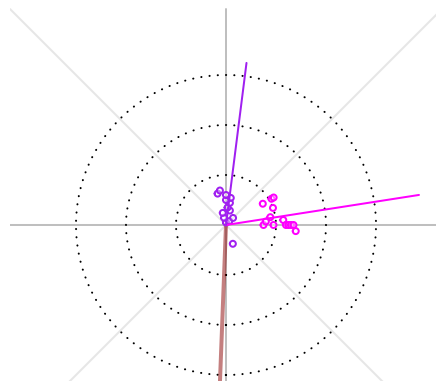
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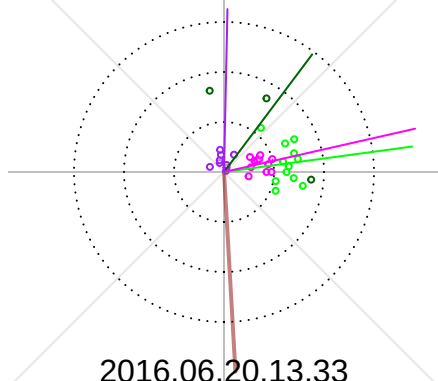
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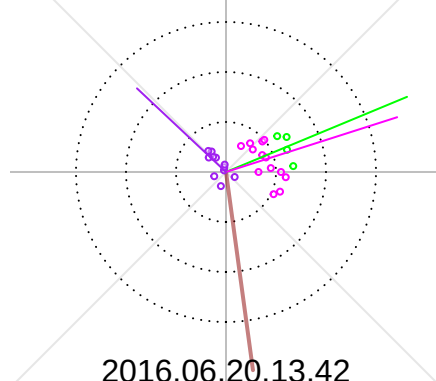
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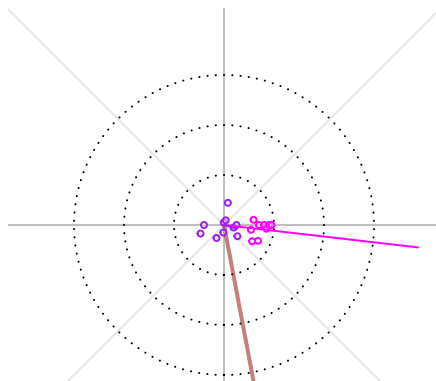
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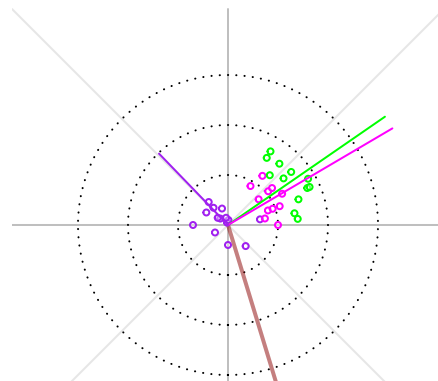
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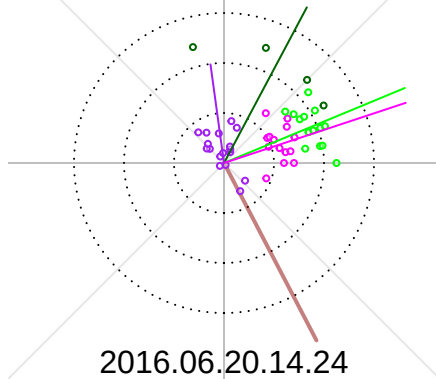
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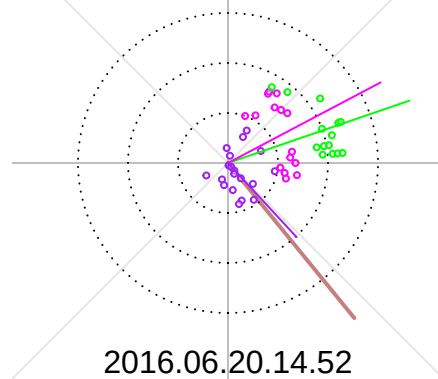
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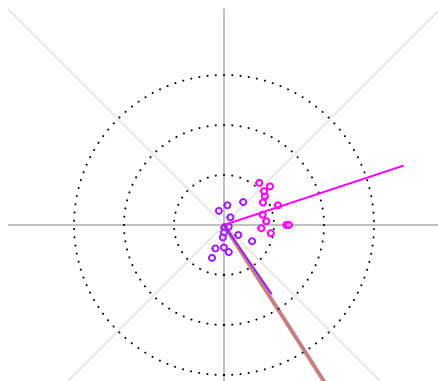
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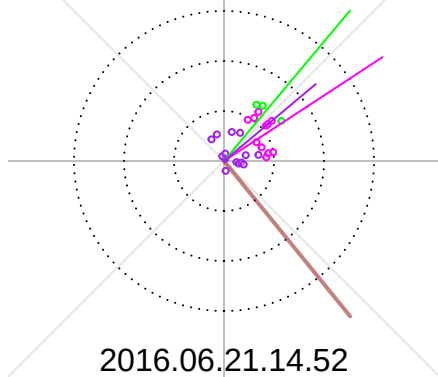
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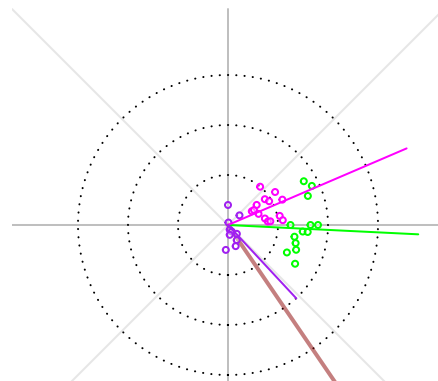
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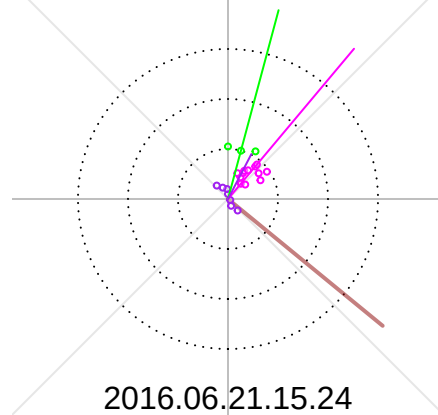
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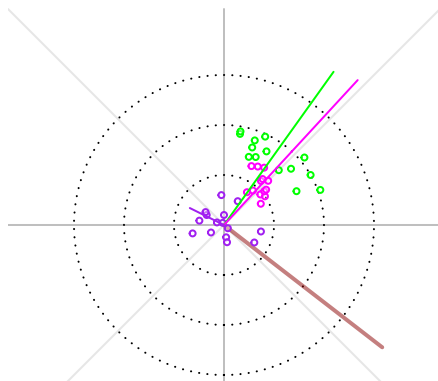
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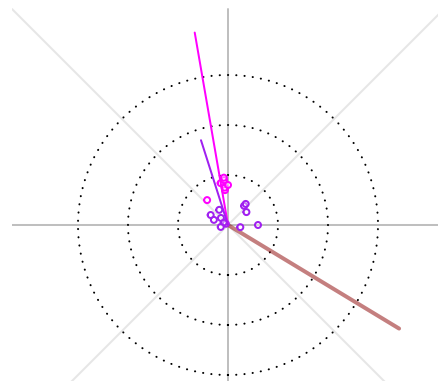
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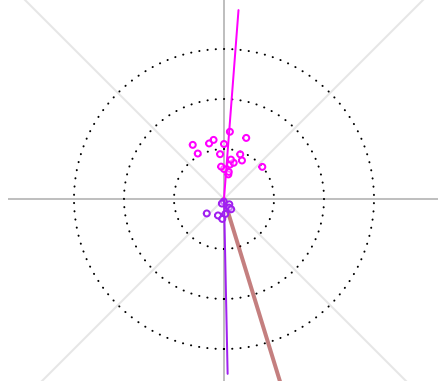
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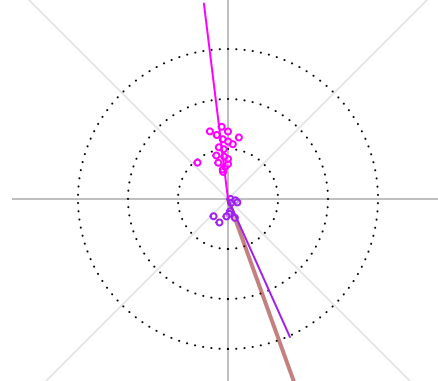
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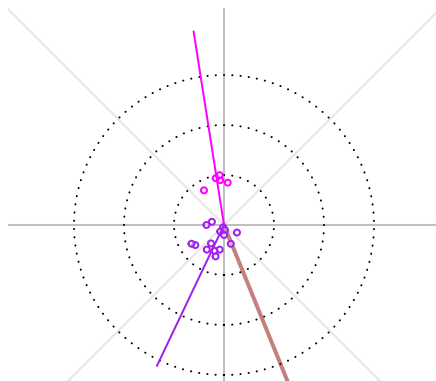
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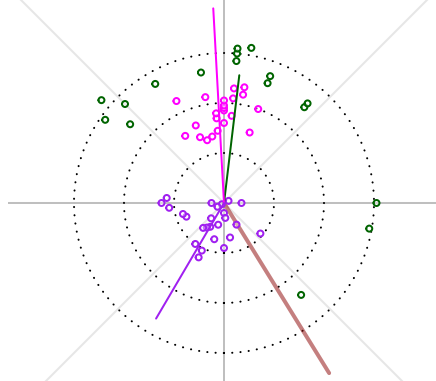
2016.06.07.13.59



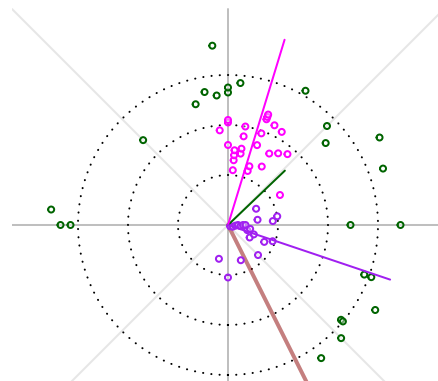
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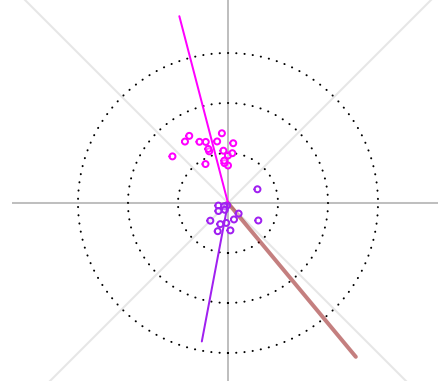
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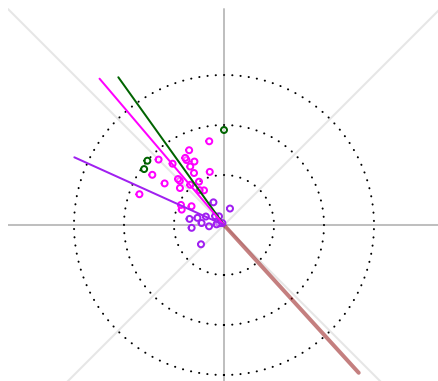
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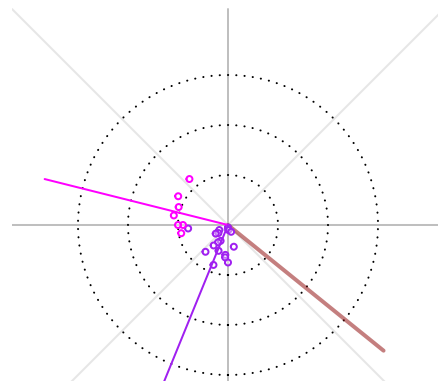
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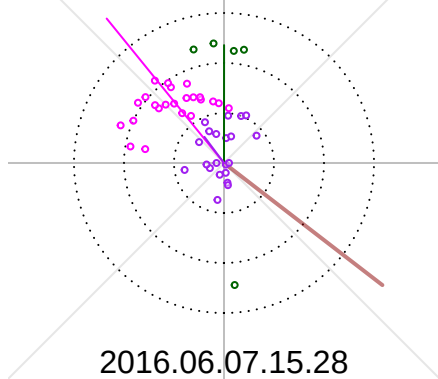
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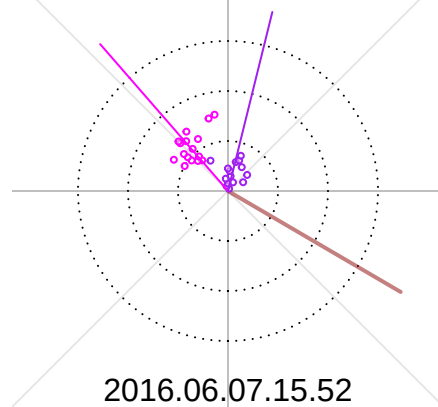
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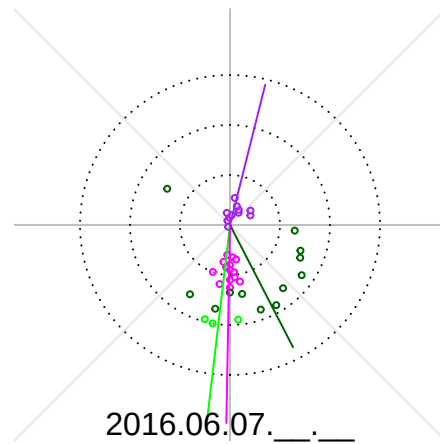
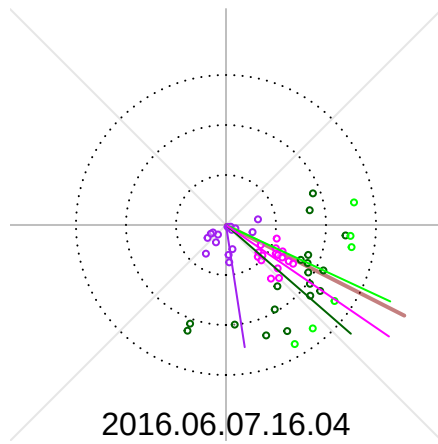
2016.06.07.15.24



2016.06.07.15.28



2016.06.07.15.52



Appendix

Print raw data

Print all dance data loaded for this analysis.

```
#our dance data
print(cd)
```

##	measured_angle	run	date	bearing	angle	ID
## 1	270	1	2016.05.31	90	1.57079633	2016.05.31.14.24
## 2	5	2	2016.05.31	355	-0.08726646	2016.05.31.14.24
## 3	20	3	2016.05.31	340	-0.34906585	2016.05.31.14.24
## 4	320	4	2016.05.31	40	0.69813170	2016.05.31.14.24
## 5	20	5	2016.05.31	340	-0.34906585	2016.05.31.14.24
## 6	35	6	2016.05.31	325	-0.61086524	2016.05.31.14.24
## 7	10	7	2016.05.31	350	-0.17453293	2016.05.31.14.24
## 8	345	8	2016.05.31	15	0.26179939	2016.05.31.14.24
## 9	340	9	2016.05.31	20	0.34906585	2016.05.31.14.24
## 10	340	10	2016.05.31	20	0.34906585	2016.05.31.14.24
## 11	340	11	2016.05.31	20	0.34906585	2016.05.31.14.24
## 12	330	12	2016.05.31	30	0.52359878	2016.05.31.14.24
## 13	350	13	2016.05.31	10	0.17453293	2016.05.31.14.24
## 14	325	14	2016.05.31	35	0.61086524	2016.05.31.14.24

## 15	335	15	2016.05.31	25	0.43633231	2016.05.31.14.24
## 16	330	16	2016.05.31	30	0.52359878	2016.05.31.14.24
## 17	320	17	2016.05.31	40	0.69813170	2016.05.31.14.24
## 18	325	18	2016.05.31	35	0.61086524	2016.05.31.14.24
## 19	325	19	2016.05.31	35	0.61086524	2016.05.31.14.24
## 20	355	20	2016.05.31	5	0.08726646	2016.05.31.14.24
## 21	340	21	2016.05.31	20	0.34906585	2016.05.31.14.24
## 22	345	22	2016.05.31	15	0.26179939	2016.05.31.14.24
## 23	345	23	2016.05.31	15	0.26179939	2016.05.31.14.24
## 24	325	24	2016.05.31	35	0.61086524	2016.05.31.14.24
## 25	350	25	2016.05.31	10	0.17453293	2016.05.31.14.24
## 26	345	26	2016.05.31	15	0.26179939	2016.05.31.14.24
## 27	355	27	2016.05.31	5	0.08726646	2016.05.31.14.24
## 28	235	1	2016.05.31	125	2.18166156	2016.05.31.14.53
## 29	185	2	2016.05.31	175	3.05432619	2016.05.31.14.53
## 30	100	3	2016.05.31	260	-1.74532925	2016.05.31.14.53
## 31	140	4	2016.05.31	220	-2.44346095	2016.05.31.14.53
## 32	90	5	2016.05.31	270	-1.57079633	2016.05.31.14.53
## 33	190	6	2016.05.31	170	2.96705973	2016.05.31.14.53
## 34	175	7	2016.05.31	185	-3.05432619	2016.05.31.14.53
## 35	250	8	2016.05.31	110	1.91986218	2016.05.31.14.53
## 36	130	9	2016.05.31	230	-2.26892803	2016.05.31.14.53
## 37	160	10	2016.05.31	200	-2.79252680	2016.05.31.14.53
## 38	15	11	2016.05.31	345	-0.26179939	2016.05.31.14.53
## 39	45	12	2016.05.31	315	-0.78539816	2016.05.31.14.53
## 40	15	13	2016.05.31	345	-0.26179939	2016.05.31.14.53
## 41	20	14	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 42	40	15	2016.05.31	320	-0.69813170	2016.05.31.14.53
## 43	20	16	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 44	10	17	2016.05.31	350	-0.17453293	2016.05.31.14.53
## 45	50	18	2016.05.31	310	-0.87266463	2016.05.31.14.53
## 46	10	19	2016.05.31	350	-0.17453293	2016.05.31.14.53
## 47	40	20	2016.05.31	320	-0.69813170	2016.05.31.14.53
## 48	40	21	2016.05.31	320	-0.69813170	2016.05.31.14.53
## 49	50	22	2016.05.31	310	-0.87266463	2016.05.31.14.53
## 50	50	23	2016.05.31	310	-0.87266463	2016.05.31.14.53
## 51	10	24	2016.05.31	350	-0.17453293	2016.05.31.14.53
## 52	40	25	2016.05.31	320	-0.69813170	2016.05.31.14.53
## 53	35	26	2016.05.31	325	-0.61086524	2016.05.31.14.53
## 54	20	27	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 55	30	28	2016.05.31	330	-0.52359878	2016.05.31.14.53
## 56	40	29	2016.05.31	320	-0.69813170	2016.05.31.14.53
## 57	20	30	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 58	25	31	2016.05.31	335	-0.43633231	2016.05.31.14.53
## 59	20	32	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 60	40	33	2016.05.31	320	-0.69813170	2016.05.31.14.53
## 61	20	34	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 62	20	35	2016.05.31	340	-0.34906585	2016.05.31.14.53
## 63	200	1	2016.05.31	160	2.79252680	2016.05.31.14.57
## 64	140	2	2016.05.31	220	-2.44346095	2016.05.31.14.57
## 65	220	3	2016.05.31	140	2.44346095	2016.05.31.14.57
## 66	150	4	2016.05.31	210	-2.61799388	2016.05.31.14.57
## 67	170	5	2016.05.31	190	-2.96705973	2016.05.31.14.57
## 68	100	6	2016.05.31	260	-1.74532925	2016.05.31.14.57

## 69	155	7	2016.05.31	205	-2.70526034	2016.05.31.14.57
## 70	90	8	2016.05.31	270	-1.57079633	2016.05.31.14.57
## 71	155	9	2016.05.31	205	-2.70526034	2016.05.31.14.57
## 72	355	10	2016.05.31	5	0.08726646	2016.05.31.14.57
## 73	40	11	2016.05.31	320	-0.69813170	2016.05.31.14.57
## 74	70	12	2016.05.31	290	-1.22173048	2016.05.31.14.57
## 75	30	13	2016.05.31	330	-0.52359878	2016.05.31.14.57
## 76	70	14	2016.05.31	290	-1.22173048	2016.05.31.14.57
## 77	25	15	2016.05.31	335	-0.43633231	2016.05.31.14.57
## 78	20	16	2016.05.31	340	-0.34906585	2016.05.31.14.57
## 79	30	17	2016.05.31	330	-0.52359878	2016.05.31.14.57
## 80	30	18	2016.05.31	330	-0.52359878	2016.05.31.14.57
## 81	70	19	2016.05.31	290	-1.22173048	2016.05.31.14.57
## 82	60	20	2016.05.31	300	-1.04719755	2016.05.31.14.57
## 83	40	21	2016.05.31	320	-0.69813170	2016.05.31.14.57
## 84	60	22	2016.05.31	300	-1.04719755	2016.05.31.14.57
## 85	40	23	2016.05.31	320	-0.69813170	2016.05.31.14.57
## 86	100	1	2016.05.31	260	-1.74532925	2016.05.31.15.14
## 87	110	2	2016.05.31	250	-1.91986218	2016.05.31.15.14
## 88	140	3	2016.05.31	220	-2.44346095	2016.05.31.15.14
## 89	110	4	2016.05.31	250	-1.91986218	2016.05.31.15.14
## 90	140	5	2016.05.31	220	-2.44346095	2016.05.31.15.14
## 91	130	6	2016.05.31	230	-2.26892803	2016.05.31.15.14
## 92	110	7	2016.05.31	250	-1.91986218	2016.05.31.15.14
## 93	95	8	2016.05.31	265	-1.65806279	2016.05.31.15.14
## 94	140	9	2016.05.31	220	-2.44346095	2016.05.31.15.14
## 95	80	10	2016.05.31	280	-1.39626340	2016.05.31.15.14
## 96	120	11	2016.05.31	240	-2.09439510	2016.05.31.15.14
## 97	60	12	2016.05.31	300	-1.04719755	2016.05.31.15.14
## 98	60	13	2016.05.31	300	-1.04719755	2016.05.31.15.14
## 99	50	14	2016.05.31	310	-0.87266463	2016.05.31.15.14
## 100	85	15	2016.05.31	275	-1.48352986	2016.05.31.15.14
## 101	40	16	2016.05.31	320	-0.69813170	2016.05.31.15.14
## 102	40	17	2016.05.31	320	-0.69813170	2016.05.31.15.14
## 103	20	18	2016.05.31	340	-0.34906585	2016.05.31.15.14
## 104	270	1	2016.05.31	90	1.57079633	2016.05.31.14.18
## 105	260	2	2016.05.31	100	1.74532925	2016.05.31.14.18
## 106	310	3	2016.05.31	50	0.87266463	2016.05.31.14.18
## 107	305	4	2016.05.31	55	0.95993109	2016.05.31.14.18
## 108	10	5	2016.05.31	350	-0.17453293	2016.05.31.14.18
## 109	350	6	2016.05.31	10	0.17453293	2016.05.31.14.18
## 110	270	7	2016.05.31	90	1.57079633	2016.05.31.14.18
## 111	310	8	2016.05.31	50	0.87266463	2016.05.31.14.18
## 112	60	9	2016.05.31	300	-1.04719755	2016.05.31.14.18
## 113	10	10	2016.05.31	350	-0.17453293	2016.05.31.14.18
## 114	5	11	2016.05.31	355	-0.08726646	2016.05.31.14.18
## 115	40	12	2016.05.31	320	-0.69813170	2016.05.31.14.18
## 116	10	13	2016.05.31	350	-0.17453293	2016.05.31.14.18
## 117	5	14	2016.05.31	355	-0.08726646	2016.05.31.14.18
## 118	10	15	2016.05.31	350	-0.17453293	2016.05.31.14.18
## 119	10	16	2016.05.31	350	-0.17453293	2016.05.31.14.18
## 120	30	17	2016.05.31	330	-0.52359878	2016.05.31.14.18
## 121	0	18	2016.05.31	0	0.00000000	2016.05.31.14.18
## 122	15	19	2016.05.31	345	-0.26179939	2016.05.31.14.18

## 123	15	20	2016.05.31	345	-0.26179939	2016.05.31.14.18
## 124	15	21	2016.05.31	345	-0.26179939	2016.05.31.14.18
## 125	15	22	2016.05.31	345	-0.26179939	2016.05.31.14.18
## 126	5	23	2016.05.31	355	-0.08726646	2016.05.31.14.18
## 127	5	24	2016.05.31	355	-0.08726646	2016.05.31.14.18
## 128	40	25	2016.05.31	320	-0.69813170	2016.05.31.14.18
## 129	20	26	2016.05.31	340	-0.34906585	2016.05.31.14.18
## 130	30	27	2016.05.31	330	-0.52359878	2016.05.31.14.18
## 131	20	28	2016.05.31	340	-0.34906585	2016.05.31.14.18
## 132	60	29	2016.05.31	300	-1.04719755	2016.05.31.14.18
## 133	50	30	2016.05.31	310	-0.87266463	2016.05.31.14.18
## 134	60	31	2016.05.31	300	-1.04719755	2016.05.31.14.18
## 135	60	32	2016.05.31	300	-1.04719755	2016.05.31.14.18
## 136	5	33	2016.05.31	355	-0.08726646	2016.05.31.14.18
## 137	45	34	2016.05.31	315	-0.78539816	2016.05.31.14.18
## 138	40	35	2016.05.31	320	-0.69813170	2016.05.31.14.18
## 139	40	36	2016.05.31	320	-0.69813170	2016.05.31.14.18
## 140	30	37	2016.05.31	330	-0.52359878	2016.05.31.14.18
## 141	200	1	2016.05.31	160	2.79252680	2016.05.31.14.02
## 142	310	2	2016.05.31	50	0.87266463	2016.05.31.14.02
## 143	230	3	2016.05.31	130	2.26892803	2016.05.31.14.02
## 144	240	4	2016.05.31	120	2.09439510	2016.05.31.14.02
## 145	230	5	2016.05.31	130	2.26892803	2016.05.31.14.02
## 146	225	6	2016.05.31	135	2.35619449	2016.05.31.14.02
## 147	130	7	2016.05.31	230	-2.26892803	2016.05.31.14.02
## 148	150	8	2016.05.31	210	-2.61799388	2016.05.31.14.02
## 149	240	9	2016.05.31	120	2.09439510	2016.05.31.14.02
## 150	265	10	2016.05.31	95	1.65806279	2016.05.31.14.02
## 151	300	11	2016.05.31	60	1.04719755	2016.05.31.14.02
## 152	260	12	2016.05.31	100	1.74532925	2016.05.31.14.02
## 153	255	13	2016.05.31	105	1.83259571	2016.05.31.14.02
## 154	270	14	2016.05.31	90	1.57079633	2016.05.31.14.02
## 155	240	15	2016.05.31	120	2.09439510	2016.05.31.14.02
## 156	280	16	2016.05.31	80	1.39626340	2016.05.31.14.02
## 157	250	17	2016.05.31	110	1.91986218	2016.05.31.14.02
## 158	310	18	2016.05.31	50	0.87266463	2016.05.31.14.02
## 159	260	19	2016.05.31	100	1.74532925	2016.05.31.14.02
## 160	310	20	2016.05.31	50	0.87266463	2016.05.31.14.02
## 161	305	1	2016.06.06	55	0.95993109	2016.06.06.14.10
## 162	115	2	2016.06.06	245	-2.00712864	2016.06.06.14.10
## 163	310	3	2016.06.06	50	0.87266463	2016.06.06.14.10
## 164	20	4	2016.06.06	340	-0.34906585	2016.06.06.14.10
## 165	140	5	2016.06.06	220	-2.44346095	2016.06.06.14.10
## 166	0	6	2016.06.06	0	0.00000000	2016.06.06.14.10
## 167	140	7	2016.06.06	220	-2.44346095	2016.06.06.14.10
## 168	15	8	2016.06.06	345	-0.26179939	2016.06.06.14.10
## 169	40	9	2016.06.06	320	-0.69813170	2016.06.06.14.10
## 170	5	10	2016.06.06	355	-0.08726646	2016.06.06.14.10
## 171	30	11	2016.06.06	330	-0.52359878	2016.06.06.14.10
## 172	55	12	2016.06.06	305	-0.95993109	2016.06.06.14.10
## 173	330	13	2016.06.06	30	0.52359878	2016.06.06.14.10
## 174	350	14	2016.06.06	10	0.17453293	2016.06.06.14.10
## 175	30	15	2016.06.06	330	-0.52359878	2016.06.06.14.10
## 176	350	16	2016.06.06	10	0.17453293	2016.06.06.14.10

## 177	335	17	2016.06.06	25	0.43633231	2016.06.06.14.10
## 178	0	18	2016.06.06	0	0.00000000	2016.06.06.14.10
## 179	340	19	2016.06.06	20	0.34906585	2016.06.06.14.10
## 180	355	20	2016.06.06	5	0.08726646	2016.06.06.14.10
## 181	10	21	2016.06.06	350	-0.17453293	2016.06.06.14.10
## 182	335	22	2016.06.06	25	0.43633231	2016.06.06.14.10
## 183	5	23	2016.06.06	355	-0.08726646	2016.06.06.14.10
## 184	355	24	2016.06.06	5	0.08726646	2016.06.06.14.10
## 185	295	25	2016.06.06	65	1.13446401	2016.06.06.14.10
## 186	330	26	2016.06.06	30	0.52359878	2016.06.06.14.10
## 187	0	27	2016.06.06	0	0.00000000	2016.06.06.14.10
## 188	30	28	2016.06.06	330	-0.52359878	2016.06.06.14.10
## 189	25	29	2016.06.06	335	-0.43633231	2016.06.06.14.10
## 190	55	30	2016.06.06	305	-0.95993109	2016.06.06.14.10
## 191	35	31	2016.06.06	325	-0.61086524	2016.06.06.14.10
## 192	10	32	2016.06.06	350	-0.17453293	2016.06.06.14.10
## 193	5	33	2016.06.06	355	-0.08726646	2016.06.06.14.10
## 194	15	34	2016.06.06	345	-0.26179939	2016.06.06.14.10
## 195	20	35	2016.06.06	340	-0.34906585	2016.06.06.14.10
## 196	20	36	2016.06.06	340	-0.34906585	2016.06.06.14.10
## 197	20	37	2016.06.06	340	-0.34906585	2016.06.06.14.10
## 198	120	38	2016.06.06	240	-2.09439510	2016.06.06.14.10
## 199	330	39	2016.06.06	30	0.52359878	2016.06.06.14.10
## 200	355	40	2016.06.06	5	0.08726646	2016.06.06.14.10
## 201	155	1	2016.06.06	205	-2.70526034	2016.06.06.14.16
## 202	180	2	2016.06.06	180	3.14159265	2016.06.06.14.16
## 203	165	3	2016.06.06	195	-2.87979327	2016.06.06.14.16
## 204	185	4	2016.06.06	175	3.05432619	2016.06.06.14.16
## 205	190	5	2016.06.06	170	2.96705973	2016.06.06.14.16
## 206	225	6	2016.06.06	135	2.35619449	2016.06.06.14.16
## 207	195	7	2016.06.06	165	2.87979327	2016.06.06.14.16
## 208	220	8	2016.06.06	140	2.44346095	2016.06.06.14.16
## 209	180	9	2016.06.06	180	3.14159265	2016.06.06.14.16
## 210	190	10	2016.06.06	170	2.96705973	2016.06.06.14.16
## 211	205	11	2016.06.06	155	2.70526034	2016.06.06.14.16
## 212	240	12	2016.06.06	120	2.09439510	2016.06.06.14.16
## 213	180	13	2016.06.06	180	3.14159265	2016.06.06.14.16
## 214	230	14	2016.06.06	130	2.26892803	2016.06.06.14.16
## 215	140	15	2016.06.06	220	-2.44346095	2016.06.06.14.16
## 216	140	16	2016.06.06	220	-2.44346095	2016.06.06.14.16
## 217	150	17	2016.06.06	210	-2.61799388	2016.06.06.14.16
## 218	140	18	2016.06.06	220	-2.44346095	2016.06.06.14.16
## 219	120	1	2016.06.06	240	-2.09439510	2016.06.06.14.22
## 220	145	2	2016.06.06	215	-2.53072742	2016.06.06.14.22
## 221	320	3	2016.06.06	40	0.69813170	2016.06.06.14.22
## 222	170	4	2016.06.06	190	-2.96705973	2016.06.06.14.22
## 223	160	5	2016.06.06	200	-2.79252680	2016.06.06.14.22
## 224	170	6	2016.06.06	190	-2.96705973	2016.06.06.14.22
## 225	155	7	2016.06.06	205	-2.70526034	2016.06.06.14.22
## 226	190	8	2016.06.06	170	2.96705973	2016.06.06.14.22
## 227	185	9	2016.06.06	175	3.05432619	2016.06.06.14.22
## 228	185	10	2016.06.06	175	3.05432619	2016.06.06.14.22
## 229	150	11	2016.06.06	210	-2.61799388	2016.06.06.14.22
## 230	165	12	2016.06.06	195	-2.87979327	2016.06.06.14.22

## 231	155	13	2016.06.06	205	-2.70526034	2016.06.06.14.22
## 232	180	14	2016.06.06	180	3.14159265	2016.06.06.14.22
## 233	175	15	2016.06.06	185	-3.05432619	2016.06.06.14.22
## 234	190	16	2016.06.06	170	2.96705973	2016.06.06.14.22
## 235	165	17	2016.06.06	195	-2.87979327	2016.06.06.14.22
## 236	155	18	2016.06.06	205	-2.70526034	2016.06.06.14.22
## 237	160	19	2016.06.06	200	-2.79252680	2016.06.06.14.22
## 238	35	20	2016.06.06	325	-0.61086524	2016.06.06.14.22
## 239	30	21	2016.06.06	330	-0.52359878	2016.06.06.14.22
## 240	45	22	2016.06.06	315	-0.78539816	2016.06.06.14.22
## 241	40	23	2016.06.06	320	-0.69813170	2016.06.06.14.22
## 242	80	24	2016.06.06	280	-1.39626340	2016.06.06.14.22
## 243	35	25	2016.06.06	325	-0.61086524	2016.06.06.14.22
## 244	60	26	2016.06.06	300	-1.04719755	2016.06.06.14.22
## 245	30	27	2016.06.06	330	-0.52359878	2016.06.06.14.22
## 246	40	28	2016.06.06	320	-0.69813170	2016.06.06.14.22
## 247	50	29	2016.06.06	310	-0.87266463	2016.06.06.14.22
## 248	10	30	2016.06.06	350	-0.17453293	2016.06.06.14.22
## 249	50	31	2016.06.06	310	-0.87266463	2016.06.06.14.22
## 250	65	32	2016.06.06	295	-1.13446401	2016.06.06.14.22
## 251	90	33	2016.06.06	270	-1.57079633	2016.06.06.14.22
## 252	85	34	2016.06.06	275	-1.48352986	2016.06.06.14.22
## 253	35	35	2016.06.06	325	-0.61086524	2016.06.06.14.22
## 254	55	36	2016.06.06	305	-0.95993109	2016.06.06.14.22
## 255	40	37	2016.06.06	320	-0.69813170	2016.06.06.14.22
## 256	110	38	2016.06.06	250	-1.91986218	2016.06.06.14.22
## 257	165	39	2016.06.06	195	-2.87979327	2016.06.06.14.22
## 258	120	40	2016.06.06	240	-2.09439510	2016.06.06.14.22
## 259	120	41	2016.06.06	240	-2.09439510	2016.06.06.14.22
## 260	170	42	2016.06.06	190	-2.96705973	2016.06.06.14.22
## 261	90	43	2016.06.06	270	-1.57079633	2016.06.06.14.22
## 262	150	44	2016.06.06	210	-2.61799388	2016.06.06.14.22
## 263	160	45	2016.06.06	200	-2.79252680	2016.06.06.14.22
## 264	175	46	2016.06.06	185	-3.05432619	2016.06.06.14.22
## 265	40	47	2016.06.06	320	-0.69813170	2016.06.06.14.22
## 266	160	48	2016.06.06	200	-2.79252680	2016.06.06.14.22
## 267	155	49	2016.06.06	205	-2.70526034	2016.06.06.14.22
## 268	120	50	2016.06.06	240	-2.09439510	2016.06.06.14.22
## 269	110	51	2016.06.06	250	-1.91986218	2016.06.06.14.22
## 270	200	52	2016.06.06	160	2.79252680	2016.06.06.14.22
## 271	140	53	2016.06.06	220	-2.44346095	2016.06.06.14.22
## 272	140	54	2016.06.06	220	-2.44346095	2016.06.06.14.22
## 273	130	55	2016.06.06	230	-2.26892803	2016.06.06.14.22
## 274	60	56	2016.06.06	300	-1.04719755	2016.06.06.14.22
## 275	40	57	2016.06.06	320	-0.69813170	2016.06.06.14.22
## 276	30	58	2016.06.06	330	-0.52359878	2016.06.06.14.22
## 277	70	59	2016.06.06	290	-1.22173048	2016.06.06.14.22
## 278	90	60	2016.06.06	270	-1.57079633	2016.06.06.14.22
## 279	130	61	2016.06.06	230	-2.26892803	2016.06.06.14.22
## 280	170	1	2016.06.06	190	-2.96705973	2016.06.06.14.49
## 281	155	2	2016.06.06	205	-2.70526034	2016.06.06.14.49
## 282	170	3	2016.06.06	190	-2.96705973	2016.06.06.14.49
## 283	130	4	2016.06.06	230	-2.26892803	2016.06.06.14.49
## 284	165	5	2016.06.06	195	-2.87979327	2016.06.06.14.49

## 285	150	6	2016.06.06	210	-2.61799388	2016.06.06.14.49
## 286	160	7	2016.06.06	200	-2.79252680	2016.06.06.14.49
## 287	155	8	2016.06.06	205	-2.70526034	2016.06.06.14.49
## 288	220	9	2016.06.06	140	2.44346095	2016.06.06.14.49
## 289	150	10	2016.06.06	210	-2.61799388	2016.06.06.14.49
## 290	175	11	2016.06.06	185	-3.05432619	2016.06.06.14.49
## 291	150	12	2016.06.06	210	-2.61799388	2016.06.06.14.49
## 292	190	13	2016.06.06	170	2.96705973	2016.06.06.14.49
## 293	185	14	2016.06.06	175	3.05432619	2016.06.06.14.49
## 294	175	15	2016.06.06	185	-3.05432619	2016.06.06.14.49
## 295	25	16	2016.06.06	335	-0.43633231	2016.06.06.14.49
## 296	50	17	2016.06.06	310	-0.87266463	2016.06.06.14.49
## 297	30	18	2016.06.06	330	-0.52359878	2016.06.06.14.49
## 298	90	19	2016.06.06	270	-1.57079633	2016.06.06.14.49
## 299	30	20	2016.06.06	330	-0.52359878	2016.06.06.14.49
## 300	40	21	2016.06.06	320	-0.69813170	2016.06.06.14.49
## 301	35	22	2016.06.06	325	-0.61086524	2016.06.06.14.49
## 302	355	1	2016.06.22	5	0.08726646	2016.06.22.13.54
## 303	230	2	2016.06.22	130	2.26892803	2016.06.22.13.54
## 304	0	3	2016.06.22	0	0.00000000	2016.06.22.13.54
## 305	330	4	2016.06.22	30	0.52359878	2016.06.22.13.54
## 306	260	5	2016.06.22	100	1.74532925	2016.06.22.13.54
## 307	240	6	2016.06.22	120	2.09439510	2016.06.22.13.54
## 308	330	7	2016.06.22	30	0.52359878	2016.06.22.13.54
## 309	40	8	2016.06.22	320	-0.69813170	2016.06.22.13.54
## 310	320	9	2016.06.22	40	0.69813170	2016.06.22.13.54
## 311	10	10	2016.06.22	350	-0.17453293	2016.06.22.13.54
## 312	290	11	2016.06.22	70	1.22173048	2016.06.22.13.54
## 313	275	12	2016.06.22	85	1.48352986	2016.06.22.13.54
## 314	305	13	2016.06.22	55	0.95993109	2016.06.22.13.54
## 315	290	14	2016.06.22	70	1.22173048	2016.06.22.13.54
## 316	290	15	2016.06.22	70	1.22173048	2016.06.22.13.54
## 317	270	16	2016.06.22	90	1.57079633	2016.06.22.13.54
## 318	285	17	2016.06.22	75	1.30899694	2016.06.22.13.54
## 319	280	18	2016.06.22	80	1.39626340	2016.06.22.13.54
## 320	320	19	2016.06.22	40	0.69813170	2016.06.22.13.54
## 321	290	20	2016.06.22	70	1.22173048	2016.06.22.13.54
## 322	290	21	2016.06.22	70	1.22173048	2016.06.22.13.54
## 323	275	22	2016.06.22	85	1.48352986	2016.06.22.13.54
## 324	275	23	2016.06.22	85	1.48352986	2016.06.22.13.54
## 325	310	24	2016.06.22	50	0.87266463	2016.06.22.13.54
## 326	295	25	2016.06.22	65	1.13446401	2016.06.22.13.54
## 327	300	26	2016.06.22	60	1.04719755	2016.06.22.13.54
## 328	300	27	2016.06.22	60	1.04719755	2016.06.22.13.54
## 329	290	28	2016.06.22	70	1.22173048	2016.06.22.13.54
## 330	300	29	2016.06.22	60	1.04719755	2016.06.22.13.54
## 331	270	30	2016.06.22	90	1.57079633	2016.06.22.13.54
## 332	315	31	2016.06.22	45	0.78539816	2016.06.22.13.54
## 333	210	1	2016.06.22	150	2.61799388	2016.06.22.14.11
## 334	335	2	2016.06.22	25	0.43633231	2016.06.22.14.11
## 335	35	3	2016.06.22	325	-0.61086524	2016.06.22.14.11
## 336	15	4	2016.06.22	345	-0.26179939	2016.06.22.14.11
## 337	55	5	2016.06.22	305	-0.95993109	2016.06.22.14.11
## 338	15	6	2016.06.22	345	-0.26179939	2016.06.22.14.11

## 339	20	7	2016.06.22	340	-0.34906585	2016.06.22.14.11
## 340	315	8	2016.06.22	45	0.78539816	2016.06.22.14.11
## 341	5	9	2016.06.22	355	-0.08726646	2016.06.22.14.11
## 342	120	10	2016.06.22	240	-2.09439510	2016.06.22.14.11
## 343	300	11	2016.06.22	60	1.04719755	2016.06.22.14.11
## 344	350	12	2016.06.22	10	0.17453293	2016.06.22.14.11
## 345	20	13	2016.06.22	340	-0.34906585	2016.06.22.14.11
## 346	310	14	2016.06.22	50	0.87266463	2016.06.22.14.11
## 347	355	15	2016.06.22	5	0.08726646	2016.06.22.14.11
## 348	290	16	2016.06.22	70	1.22173048	2016.06.22.14.11
## 349	340	17	2016.06.22	20	0.34906585	2016.06.22.14.11
## 350	250	18	2016.06.22	110	1.91986218	2016.06.22.14.11
## 351	340	19	2016.06.22	20	0.34906585	2016.06.22.14.11
## 352	330	20	2016.06.22	30	0.52359878	2016.06.22.14.11
## 353	305	21	2016.06.22	55	0.95993109	2016.06.22.14.11
## 354	325	22	2016.06.22	35	0.61086524	2016.06.22.14.11
## 355	320	23	2016.06.22	40	0.69813170	2016.06.22.14.11
## 356	320	24	2016.06.22	40	0.69813170	2016.06.22.14.11
## 357	330	25	2016.06.22	30	0.52359878	2016.06.22.14.11
## 358	320	26	2016.06.22	40	0.69813170	2016.06.22.14.11
## 359	350	1	2016.06.22	10	0.17453293	2016.06.22.14.33
## 360	330	2	2016.06.22	30	0.52359878	2016.06.22.14.33
## 361	140	3	2016.06.22	220	-2.44346095	2016.06.22.14.33
## 362	170	4	2016.06.22	190	-2.96705973	2016.06.22.14.33
## 363	145	5	2016.06.22	215	-2.53072742	2016.06.22.14.33
## 364	190	6	2016.06.22	170	2.96705973	2016.06.22.14.33
## 365	170	7	2016.06.22	190	-2.96705973	2016.06.22.14.33
## 366	165	8	2016.06.22	195	-2.87979327	2016.06.22.14.33
## 367	160	9	2016.06.22	200	-2.79252680	2016.06.22.14.33
## 368	210	10	2016.06.22	150	2.61799388	2016.06.22.14.33
## 369	210	11	2016.06.22	150	2.61799388	2016.06.22.14.33
## 370	200	12	2016.06.22	160	2.79252680	2016.06.22.14.33
## 371	285	13	2016.06.22	75	1.30899694	2016.06.22.14.33
## 372	270	14	2016.06.22	90	1.57079633	2016.06.22.14.33
## 373	130	15	2016.06.22	230	-2.26892803	2016.06.22.14.33
## 374	175	16	2016.06.22	185	-3.05432619	2016.06.22.14.33
## 375	155	17	2016.06.22	205	-2.70526034	2016.06.22.14.33
## 376	140	18	2016.06.22	220	-2.44346095	2016.06.22.14.33
## 377	250	19	2016.06.22	110	1.91986218	2016.06.22.14.33
## 378	170	20	2016.06.22	190	-2.96705973	2016.06.22.14.33
## 379	230	21	2016.06.22	130	2.26892803	2016.06.22.14.33
## 380	195	22	2016.06.22	165	2.87979327	2016.06.22.14.33
## 381	240	23	2016.06.22	120	2.09439510	2016.06.22.14.33
## 382	230	24	2016.06.22	130	2.26892803	2016.06.22.14.33
## 383	220	25	2016.06.22	140	2.44346095	2016.06.22.14.33
## 384	245	26	2016.06.22	115	2.00712864	2016.06.22.14.33
## 385	35	1	2016.06.22	325	-0.61086524	2016.06.22.14.43
## 386	60	2	2016.06.22	300	-1.04719755	2016.06.22.14.43
## 387	60	3	2016.06.22	300	-1.04719755	2016.06.22.14.43
## 388	65	4	2016.06.22	295	-1.13446401	2016.06.22.14.43
## 389	30	5	2016.06.22	330	-0.52359878	2016.06.22.14.43
## 390	20	6	2016.06.22	340	-0.34906585	2016.06.22.14.43
## 391	75	7	2016.06.22	285	-1.30899694	2016.06.22.14.43
## 392	50	8	2016.06.22	310	-0.87266463	2016.06.22.14.43

## 393	50	9	2016.06.22	310	-0.87266463	2016.06.22.14.43
## 394	50	10	2016.06.22	310	-0.87266463	2016.06.22.14.43
## 395	50	11	2016.06.22	310	-0.87266463	2016.06.22.14.43
## 396	45	12	2016.06.22	315	-0.78539816	2016.06.22.14.43
## 397	10	13	2016.06.22	350	-0.17453293	2016.06.22.14.43
## 398	15	14	2016.06.22	345	-0.26179939	2016.06.22.14.43
## 399	330	15	2016.06.22	30	0.52359878	2016.06.22.14.43
## 400	0	16	2016.06.22	0	0.00000000	2016.06.22.14.43
## 401	325	17	2016.06.22	35	0.61086524	2016.06.22.14.43
## 402	5	18	2016.06.22	355	-0.08726646	2016.06.22.14.43
## 403	5	19	2016.06.22	355	-0.08726646	2016.06.22.14.43
## 404	355	20	2016.06.22	5	0.08726646	2016.06.22.14.43
## 405	355	21	2016.06.22	5	0.08726646	2016.06.22.14.43
## 406	330	22	2016.06.22	30	0.52359878	2016.06.22.14.43
## 407	5	23	2016.06.22	355	-0.08726646	2016.06.22.14.43
## 408	350	24	2016.06.22	10	0.17453293	2016.06.22.14.43
## 409	15	25	2016.06.22	345	-0.26179939	2016.06.22.14.43
## 410	335	26	2016.06.22	25	0.43633231	2016.06.22.14.43
## 411	340	27	2016.06.22	20	0.34906585	2016.06.22.14.43
## 412	340	28	2016.06.22	20	0.34906585	2016.06.22.14.43
## 413	350	29	2016.06.22	10	0.17453293	2016.06.22.14.43
## 414	345	30	2016.06.22	15	0.26179939	2016.06.22.14.43
## 415	355	31	2016.06.22	5	0.08726646	2016.06.22.14.43
## 416	5	32	2016.06.22	355	-0.08726646	2016.06.22.14.43
## 417	330	33	2016.06.22	30	0.52359878	2016.06.22.14.43
## 418	5	34	2016.06.22	355	-0.08726646	2016.06.22.14.43
## 419	55	1	2016.06.22	305	-0.95993109	2016.06.22.15.40
## 420	50	2	2016.06.22	310	-0.87266463	2016.06.22.15.40
## 421	55	3	2016.06.22	305	-0.95993109	2016.06.22.15.40
## 422	5	4	2016.06.22	355	-0.08726646	2016.06.22.15.40
## 423	325	5	2016.06.22	35	0.61086524	2016.06.22.15.40
## 424	85	6	2016.06.22	275	-1.48352986	2016.06.22.15.40
## 425	50	7	2016.06.22	310	-0.87266463	2016.06.22.15.40
## 426	90	8	2016.06.22	270	-1.57079633	2016.06.22.15.40
## 427	350	9	2016.06.22	10	0.17453293	2016.06.22.15.40
## 428	35	10	2016.06.22	325	-0.61086524	2016.06.22.15.40
## 429	130	11	2016.06.22	230	-2.26892803	2016.06.22.15.40
## 430	305	12	2016.06.22	55	0.95993109	2016.06.22.15.40
## 431	355	13	2016.06.22	5	0.08726646	2016.06.22.15.40
## 432	140	14	2016.06.22	220	-2.44346095	2016.06.22.15.40
## 433	120	15	2016.06.22	240	-2.09439510	2016.06.22.15.40
## 434	130	16	2016.06.22	230	-2.26892803	2016.06.22.15.40
## 435	85	17	2016.06.22	275	-1.48352986	2016.06.22.15.40
## 436	95	18	2016.06.22	265	-1.65806279	2016.06.22.15.40
## 437	105	19	2016.06.22	255	-1.83259571	2016.06.22.15.40
## 438	115	20	2016.06.22	245	-2.00712864	2016.06.22.15.40
## 439	40	21	2016.06.22	320	-0.69813170	2016.06.22.15.40
## 440	100	22	2016.06.22	260	-1.74532925	2016.06.22.15.40
## 441	80	23	2016.06.22	280	-1.39626340	2016.06.22.15.40
## 442	55	24	2016.06.22	305	-0.95993109	2016.06.22.15.40
## 443	150	25	2016.06.22	210	-2.61799388	2016.06.22.15.40
## 444	145	26	2016.06.22	215	-2.53072742	2016.06.22.15.40
## 445	0	27	2016.06.22	0	0.00000000	2016.06.22.15.40
## 446	10	28	2016.06.22	350	-0.17453293	2016.06.22.15.40

## 447	355	29	2016.06.22	5	0.08726646	2016.06.22.15.40
## 448	20	30	2016.06.22	340	-0.34906585	2016.06.22.15.40
## 449	10	31	2016.06.22	350	-0.17453293	2016.06.22.15.40
## 450	70	32	2016.06.22	290	-1.22173048	2016.06.22.15.40
## 451	5	33	2016.06.22	355	-0.08726646	2016.06.22.15.40
## 452	40	34	2016.06.22	320	-0.69813170	2016.06.22.15.40
## 453	5	35	2016.06.22	355	-0.08726646	2016.06.22.15.40
## 454	30	36	2016.06.22	330	-0.52359878	2016.06.22.15.40
## 455	10	37	2016.06.22	350	-0.17453293	2016.06.22.15.40
## 456	55	38	2016.06.22	305	-0.95993109	2016.06.22.15.40
## 457	60	39	2016.06.22	300	-1.04719755	2016.06.22.15.40
## 458	10	40	2016.06.22	350	-0.17453293	2016.06.22.15.40
## 459	40	41	2016.06.22	320	-0.69813170	2016.06.22.15.40
## 460	20	42	2016.06.22	340	-0.34906585	2016.06.22.15.40
## 461	30	43	2016.06.22	330	-0.52359878	2016.06.22.15.40
## 462	0	44	2016.06.22	0	0.00000000	2016.06.22.15.40
## 463	20	45	2016.06.22	340	-0.34906585	2016.06.22.15.40
## 464	20	46	2016.06.22	340	-0.34906585	2016.06.22.15.40
## 465	30	47	2016.06.22	330	-0.52359878	2016.06.22.15.40
## 466	0	48	2016.06.22	0	0.00000000	2016.06.22.15.40
## 467	355	49	2016.06.22	5	0.08726646	2016.06.22.15.40
## 468	350	50	2016.06.22	10	0.17453293	2016.06.22.15.40
## 469	310	1	2016.06.23	50	0.87266463	2016.06.23.13.24
## 470	355	2	2016.06.23	5	0.08726646	2016.06.23.13.24
## 471	340	3	2016.06.23	20	0.34906585	2016.06.23.13.24
## 472	345	4	2016.06.23	15	0.26179939	2016.06.23.13.24
## 473	340	5	2016.06.23	20	0.34906585	2016.06.23.13.24
## 474	275	6	2016.06.23	85	1.48352986	2016.06.23.13.24
## 475	350	7	2016.06.23	10	0.17453293	2016.06.23.13.24
## 476	340	8	2016.06.23	20	0.34906585	2016.06.23.13.24
## 477	320	9	2016.06.23	40	0.69813170	2016.06.23.13.24
## 478	305	10	2016.06.23	55	0.95993109	2016.06.23.13.24
## 479	350	11	2016.06.23	10	0.17453293	2016.06.23.13.24
## 480	190	12	2016.06.23	170	2.96705973	2016.06.23.13.24
## 481	245	13	2016.06.23	115	2.00712864	2016.06.23.13.24
## 482	200	14	2016.06.23	160	2.79252680	2016.06.23.13.24
## 483	250	15	2016.06.23	110	1.91986218	2016.06.23.13.24
## 484	200	16	2016.06.23	160	2.79252680	2016.06.23.13.24
## 485	215	17	2016.06.23	145	2.53072742	2016.06.23.13.24
## 486	210	18	2016.06.23	150	2.61799388	2016.06.23.13.24
## 487	220	19	2016.06.23	140	2.44346095	2016.06.23.13.24
## 488	180	20	2016.06.23	180	3.14159265	2016.06.23.13.24
## 489	200	21	2016.06.23	160	2.79252680	2016.06.23.13.24
## 490	210	22	2016.06.23	150	2.61799388	2016.06.23.13.24
## 491	210	23	2016.06.23	150	2.61799388	2016.06.23.13.24
## 492	230	24	2016.06.23	130	2.26892803	2016.06.23.13.24
## 493	200	25	2016.06.23	160	2.79252680	2016.06.23.13.24
## 494	230	26	2016.06.23	130	2.26892803	2016.06.23.13.24
## 495	220	27	2016.06.23	140	2.44346095	2016.06.23.13.24
## 496	230	28	2016.06.23	130	2.26892803	2016.06.23.13.24
## 497	205	29	2016.06.23	155	2.70526034	2016.06.23.13.24
## 498	165	1	2016.06.28	195	-2.87979327	2016.06.28.14.12
## 499	200	2	2016.06.28	160	2.79252680	2016.06.28.14.12
## 500	185	3	2016.06.28	175	3.05432619	2016.06.28.14.12

## 501	200	4	2016.06.28	160	2.79252680	2016.06.28.14.12
## 502	180	5	2016.06.28	180	3.14159265	2016.06.28.14.12
## 503	215	6	2016.06.28	145	2.53072742	2016.06.28.14.12
## 504	240	7	2016.06.28	120	2.09439510	2016.06.28.14.12
## 505	335	8	2016.06.28	25	0.43633231	2016.06.28.14.12
## 506	280	9	2016.06.28	80	1.39626340	2016.06.28.14.12
## 507	330	10	2016.06.28	30	0.52359878	2016.06.28.14.12
## 508	320	11	2016.06.28	40	0.69813170	2016.06.28.14.12
## 509	225	12	2016.06.28	135	2.35619449	2016.06.28.14.12
## 510	285	13	2016.06.28	75	1.30899694	2016.06.28.14.12
## 511	290	14	2016.06.28	70	1.22173048	2016.06.28.14.12
## 512	250	15	2016.06.28	110	1.91986218	2016.06.28.14.12
## 513	325	16	2016.06.28	35	0.61086524	2016.06.28.14.12
## 514	300	17	2016.06.28	60	1.04719755	2016.06.28.14.12
## 515	270	18	2016.06.28	90	1.57079633	2016.06.28.14.12
## 516	265	19	2016.06.28	95	1.65806279	2016.06.28.14.12
## 517	290	20	2016.06.28	70	1.22173048	2016.06.28.14.12
## 518	290	21	2016.06.28	70	1.22173048	2016.06.28.14.12
## 519	265	22	2016.06.28	95	1.65806279	2016.06.28.14.12
## 520	220	23	2016.06.28	140	2.44346095	2016.06.28.14.12
## 521	210	24	2016.06.28	150	2.61799388	2016.06.28.14.12
## 522	295	25	2016.06.28	65	1.13446401	2016.06.28.14.12
## 523	260	26	2016.06.28	100	1.74532925	2016.06.28.14.12
## 524	220	27	2016.06.28	140	2.44346095	2016.06.28.14.12
## 525	225	28	2016.06.28	135	2.35619449	2016.06.28.14.12
## 526	250	29	2016.06.28	110	1.91986218	2016.06.28.14.12
## 527	240	30	2016.06.28	120	2.09439510	2016.06.28.14.12
## 528	265	31	2016.06.28	95	1.65806279	2016.06.28.14.12
## 529	230	32	2016.06.28	130	2.26892803	2016.06.28.14.12
## 530	250	33	2016.06.28	110	1.91986218	2016.06.28.14.12
## 531	230	34	2016.06.28	130	2.26892803	2016.06.28.14.12
## 532	260	35	2016.06.28	100	1.74532925	2016.06.28.14.12
## 533	210	36	2016.06.28	150	2.61799388	2016.06.28.14.12
## 534	275	37	2016.06.28	85	1.48352986	2016.06.28.14.12
## 535	240	38	2016.06.28	120	2.09439510	2016.06.28.14.12
## 536	330	1	2016.06.28	30	0.52359878	2016.06.28.14.41
## 537	330	2	2016.06.28	30	0.52359878	2016.06.28.14.41
## 538	220	3	2016.06.28	140	2.44346095	2016.06.28.14.41
## 539	220	4	2016.06.28	140	2.44346095	2016.06.28.14.41
## 540	330	5	2016.06.28	30	0.52359878	2016.06.28.14.41
## 541	310	6	2016.06.28	50	0.87266463	2016.06.28.14.41
## 542	0	7	2016.06.28	0	0.00000000	2016.06.28.14.41
## 543	345	8	2016.06.28	15	0.26179939	2016.06.28.14.41
## 544	0	9	2016.06.28	0	0.00000000	2016.06.28.14.41
## 545	335	10	2016.06.28	25	0.43633231	2016.06.28.14.41
## 546	280	11	2016.06.28	80	1.39626340	2016.06.28.14.41
## 547	320	12	2016.06.28	40	0.69813170	2016.06.28.14.41
## 548	320	13	2016.06.28	40	0.69813170	2016.06.28.14.41
## 549	330	14	2016.06.28	30	0.52359878	2016.06.28.14.41
## 550	260	15	2016.06.28	100	1.74532925	2016.06.28.14.41
## 551	280	16	2016.06.28	80	1.39626340	2016.06.28.14.41
## 552	250	17	2016.06.28	110	1.91986218	2016.06.28.14.41
## 553	315	18	2016.06.28	45	0.78539816	2016.06.28.14.41
## 554	260	19	2016.06.28	100	1.74532925	2016.06.28.14.41

## 555	255	20	2016.06.28	105	1.83259571	2016.06.28.14.41
## 556	270	21	2016.06.28	90	1.57079633	2016.06.28.14.41
## 557	260	22	2016.06.28	100	1.74532925	2016.06.28.14.41
## 558	170	1	2016.06.28	190	-2.96705973	2016.06.28.15.27
## 559	265	2	2016.06.28	95	1.65806279	2016.06.28.15.27
## 560	210	3	2016.06.28	150	2.61799388	2016.06.28.15.27
## 561	190	4	2016.06.28	170	2.96705973	2016.06.28.15.27
## 562	170	5	2016.06.28	190	-2.96705973	2016.06.28.15.27
## 563	160	6	2016.06.28	200	-2.79252680	2016.06.28.15.27
## 564	220	7	2016.06.28	140	2.44346095	2016.06.28.15.27
## 565	310	8	2016.06.28	50	0.87266463	2016.06.28.15.27
## 566	320	9	2016.06.28	40	0.69813170	2016.06.28.15.27
## 567	195	10	2016.06.28	165	2.87979327	2016.06.28.15.27
## 568	190	11	2016.06.28	170	2.96705973	2016.06.28.15.27
## 569	170	12	2016.06.28	190	-2.96705973	2016.06.28.15.27
## 570	330	13	2016.06.28	30	0.52359878	2016.06.28.15.27
## 571	310	14	2016.06.28	50	0.87266463	2016.06.28.15.27
## 572	240	15	2016.06.28	120	2.09439510	2016.06.28.15.27
## 573	290	16	2016.06.28	70	1.22173048	2016.06.28.15.27
## 574	270	17	2016.06.28	90	1.57079633	2016.06.28.15.27
## 575	240	18	2016.06.28	120	2.09439510	2016.06.28.15.27
## 576	250	19	2016.06.28	110	1.91986218	2016.06.28.15.27
## 577	265	20	2016.06.28	95	1.65806279	2016.06.28.15.27
## 578	250	21	2016.06.28	110	1.91986218	2016.06.28.15.27
## 579	260	22	2016.06.28	100	1.74532925	2016.06.28.15.27
## 580	270	23	2016.06.28	90	1.57079633	2016.06.28.15.27
## 581	280	24	2016.06.28	80	1.39626340	2016.06.28.15.27
## 582	285	25	2016.06.28	75	1.30899694	2016.06.28.15.27
## 583	250	26	2016.06.28	110	1.91986218	2016.06.28.15.27
## 584	255	27	2016.06.28	105	1.83259571	2016.06.28.15.27
## 585	270	28	2016.06.28	90	1.57079633	2016.06.28.15.27
## 586	240	29	2016.06.28	120	2.09439510	2016.06.28.15.27
## 587	255	30	2016.06.28	105	1.83259571	2016.06.28.15.27
## 588	260	31	2016.06.28	100	1.74532925	2016.06.28.15.27
## 589	230	32	2016.06.28	130	2.26892803	2016.06.28.15.27
## 590	270	33	2016.06.28	90	1.57079633	2016.06.28.15.27
## 591	230	34	2016.06.28	130	2.26892803	2016.06.28.15.27
## 592	230	35	2016.06.28	130	2.26892803	2016.06.28.15.27
## 593	240	36	2016.06.28	120	2.09439510	2016.06.28.15.27
## 594	270	37	2016.06.28	90	1.57079633	2016.06.28.15.27
## 595	265	38	2016.06.28	95	1.65806279	2016.06.28.15.27
## 596	270	39	2016.06.28	90	1.57079633	2016.06.28.15.27
## 597	250	40	2016.06.28	110	1.91986218	2016.06.28.15.27
## 598	240	41	2016.06.28	120	2.09439510	2016.06.28.15.27
## 599	240	42	2016.06.28	120	2.09439510	2016.06.28.15.27
## 600	240	43	2016.06.28	120	2.09439510	2016.06.28.15.27
## 601	260	44	2016.06.28	100	1.74532925	2016.06.28.15.27
## 602	265	45	2016.06.28	95	1.65806279	2016.06.28.15.27
## 603	270	46	2016.06.28	90	1.57079633	2016.06.28.15.27
## 604	250	47	2016.06.28	110	1.91986218	2016.06.28.15.27
## 605	260	48	2016.06.28	100	1.74532925	2016.06.28.15.27
## 606	250	49	2016.06.28	110	1.91986218	2016.06.28.15.27
## 607	260	50	2016.06.28	100	1.74532925	2016.06.28.15.27
## 608	260	51	2016.06.28	100	1.74532925	2016.06.28.15.27

## 609	250	52	2016.06.28	110	1.91986218	2016.06.28.15.27
## 610	260	53	2016.06.28	100	1.74532925	2016.06.28.15.27
## 611	240	54	2016.06.28	120	2.09439510	2016.06.28.15.27
## 612	265	55	2016.06.28	95	1.65806279	2016.06.28.15.27
## 613	260	56	2016.06.28	100	1.74532925	2016.06.28.15.27
## 614	240	57	2016.06.28	120	2.09439510	2016.06.28.15.27
## 615	220	58	2016.06.28	140	2.44346095	2016.06.28.15.27
## 616	265	59	2016.06.28	95	1.65806279	2016.06.28.15.27
## 617	260	60	2016.06.28	100	1.74532925	2016.06.28.15.27
## 618	265	61	2016.06.28	95	1.65806279	2016.06.28.15.27
## 619	255	62	2016.06.28	105	1.83259571	2016.06.28.15.27
## 620	230	63	2016.06.28	130	2.26892803	2016.06.28.15.27
## 621	270	64	2016.06.28	90	1.57079633	2016.06.28.15.27
## 622	330	65	2016.06.28	30	0.52359878	2016.06.28.15.27
## 623	205	66	2016.06.28	155	2.70526034	2016.06.28.15.27
## 624	300	67	2016.06.28	60	1.04719755	2016.06.28.15.27
## 625	200	68	2016.06.28	160	2.79252680	2016.06.28.15.27
## 626	210	69	2016.06.28	150	2.61799388	2016.06.28.15.27
## 627	270	70	2016.06.28	90	1.57079633	2016.06.28.15.27
## 628	260	71	2016.06.28	100	1.74532925	2016.06.28.15.27
## 629	250	1	2016.06.29	110	1.91986218	2016.06.29.13.58
## 630	30	2	2016.06.29	330	-0.52359878	2016.06.29.13.58
## 631	245	3	2016.06.29	115	2.00712864	2016.06.29.13.58
## 632	355	4	2016.06.29	5	0.08726646	2016.06.29.13.58
## 633	60	5	2016.06.29	300	-1.04719755	2016.06.29.13.58
## 634	265	6	2016.06.29	95	1.65806279	2016.06.29.13.58
## 635	260	7	2016.06.29	100	1.74532925	2016.06.29.13.58
## 636	310	8	2016.06.29	50	0.87266463	2016.06.29.13.58
## 637	320	9	2016.06.29	40	0.69813170	2016.06.29.13.58
## 638	185	10	2016.06.29	175	3.05432619	2016.06.29.13.58
## 639	210	11	2016.06.29	150	2.61799388	2016.06.29.13.58
## 640	175	12	2016.06.29	185	-3.05432619	2016.06.29.13.58
## 641	185	13	2016.06.29	175	3.05432619	2016.06.29.13.58
## 642	150	14	2016.06.29	210	-2.61799388	2016.06.29.13.58
## 643	165	15	2016.06.29	195	-2.87979327	2016.06.29.13.58
## 644	180	16	2016.06.29	180	3.14159265	2016.06.29.13.58
## 645	190	17	2016.06.29	170	2.96705973	2016.06.29.13.58
## 646	180	18	2016.06.29	180	3.14159265	2016.06.29.13.58
## 647	175	19	2016.06.29	185	-3.05432619	2016.06.29.13.58
## 648	190	20	2016.06.29	170	2.96705973	2016.06.29.13.58
## 649	170	21	2016.06.29	190	-2.96705973	2016.06.29.13.58
## 650	190	22	2016.06.29	170	2.96705973	2016.06.29.13.58
## 651	170	23	2016.06.29	190	-2.96705973	2016.06.29.13.58
## 652	185	24	2016.06.29	175	3.05432619	2016.06.29.13.58
## 653	180	25	2016.06.29	180	3.14159265	2016.06.29.13.58
## 654	175	26	2016.06.29	185	-3.05432619	2016.06.29.13.58
## 655	170	27	2016.06.29	190	-2.96705973	2016.06.29.13.58
## 656	180	28	2016.06.29	180	3.14159265	2016.06.29.13.58
## 657	175	29	2016.06.29	185	-3.05432619	2016.06.29.13.58
## 658	185	30	2016.06.29	175	3.05432619	2016.06.29.13.58
## 659	185	31	2016.06.29	175	3.05432619	2016.06.29.13.58
## 660	175	32	2016.06.29	185	-3.05432619	2016.06.29.13.58
## 661	170	33	2016.06.29	190	-2.96705973	2016.06.29.13.58
## 662	185	34	2016.06.29	175	3.05432619	2016.06.29.13.58

## 663	180	35	2016.06.29	180	3.14159265	2016.06.29.13.58
## 664	140	36	2016.06.29	220	-2.44346095	2016.06.29.13.58
## 665	210	37	2016.06.29	150	2.61799388	2016.06.29.13.58
## 666	190	38	2016.06.29	170	2.96705973	2016.06.29.13.58
## 667	165	39	2016.06.29	195	-2.87979327	2016.06.29.13.58
## 668	180	40	2016.06.29	180	3.14159265	2016.06.29.13.58
## 669	170	41	2016.06.29	190	-2.96705973	2016.06.29.13.58
## 670	180	42	2016.06.29	180	3.14159265	2016.06.29.13.58
## 671	310	1	2016.06.29	50	0.87266463	2016.06.29.14.12
## 672	310	2	2016.06.29	50	0.87266463	2016.06.29.14.12
## 673	350	3	2016.06.29	10	0.17453293	2016.06.29.14.12
## 674	300	4	2016.06.29	60	1.04719755	2016.06.29.14.12
## 675	270	5	2016.06.29	90	1.57079633	2016.06.29.14.12
## 676	355	6	2016.06.29	5	0.08726646	2016.06.29.14.12
## 677	55	7	2016.06.29	305	-0.95993109	2016.06.29.14.12
## 678	345	8	2016.06.29	15	0.26179939	2016.06.29.14.12
## 679	345	9	2016.06.29	15	0.26179939	2016.06.29.14.12
## 680	340	10	2016.06.29	20	0.34906585	2016.06.29.14.12
## 681	5	11	2016.06.29	355	-0.08726646	2016.06.29.14.12
## 682	290	12	2016.06.29	70	1.22173048	2016.06.29.14.12
## 683	265	13	2016.06.29	95	1.65806279	2016.06.29.14.12
## 684	290	14	2016.06.29	70	1.22173048	2016.06.29.14.12
## 685	190	15	2016.06.29	170	2.96705973	2016.06.29.14.12
## 686	310	16	2016.06.29	50	0.87266463	2016.06.29.14.12
## 687	330	17	2016.06.29	30	0.52359878	2016.06.29.14.12
## 688	320	18	2016.06.29	40	0.69813170	2016.06.29.14.12
## 689	300	19	2016.06.29	60	1.04719755	2016.06.29.14.12
## 690	320	20	2016.06.29	40	0.69813170	2016.06.29.14.12
## 691	10	21	2016.06.29	350	-0.17453293	2016.06.29.14.12
## 692	40	22	2016.06.29	320	-0.69813170	2016.06.29.14.12
## 693	105	23	2016.06.29	255	-1.83259571	2016.06.29.14.12
## 694	315	24	2016.06.29	45	0.78539816	2016.06.29.14.12
## 695	190	25	2016.06.29	170	2.96705973	2016.06.29.14.12
## 696	250	26	2016.06.29	110	1.91986218	2016.06.29.14.12
## 697	190	27	2016.06.29	170	2.96705973	2016.06.29.14.12
## 698	240	28	2016.06.29	120	2.09439510	2016.06.29.14.12
## 699	225	29	2016.06.29	135	2.35619449	2016.06.29.14.12
## 700	240	30	2016.06.29	120	2.09439510	2016.06.29.14.12
## 701	250	31	2016.06.29	110	1.91986218	2016.06.29.14.12
## 702	280	32	2016.06.29	80	1.39626340	2016.06.29.14.12
## 703	205	33	2016.06.29	155	2.70526034	2016.06.29.14.12
## 704	220	34	2016.06.29	140	2.44346095	2016.06.29.14.12
## 705	250	35	2016.06.29	110	1.91986218	2016.06.29.14.12
## 706	220	36	2016.06.29	140	2.44346095	2016.06.29.14.12
## 707	230	37	2016.06.29	130	2.26892803	2016.06.29.14.12
## 708	230	38	2016.06.29	130	2.26892803	2016.06.29.14.12
## 709	250	39	2016.06.29	110	1.91986218	2016.06.29.14.12
## 710	230	40	2016.06.29	130	2.26892803	2016.06.29.14.12
## 711	205	41	2016.06.29	155	2.70526034	2016.06.29.14.12
## 712	240	42	2016.06.29	120	2.09439510	2016.06.29.14.12
## 713	230	43	2016.06.29	130	2.26892803	2016.06.29.14.12
## 714	250	44	2016.06.29	110	1.91986218	2016.06.29.14.12
## 715	220	45	2016.06.29	140	2.44346095	2016.06.29.14.12
## 716	250	46	2016.06.29	110	1.91986218	2016.06.29.14.12

## 717	240	47	2016.06.29	120	2.09439510	2016.06.29.14.12
## 718	250	48	2016.06.29	110	1.91986218	2016.06.29.14.12
## 719	230	49	2016.06.29	130	2.26892803	2016.06.29.14.12
## 720	240	50	2016.06.29	120	2.09439510	2016.06.29.14.12
## 721	240	51	2016.06.29	120	2.09439510	2016.06.29.14.12
## 722	240	52	2016.06.29	120	2.09439510	2016.06.29.14.12
## 723	225	53	2016.06.29	135	2.35619449	2016.06.29.14.12
## 724	230	54	2016.06.29	130	2.26892803	2016.06.29.14.12
## 725	235	55	2016.06.29	125	2.18166156	2016.06.29.14.12
## 726	240	56	2016.06.29	120	2.09439510	2016.06.29.14.12
## 727	200	57	2016.06.29	160	2.79252680	2016.06.29.14.12
## 728	220	58	2016.06.29	140	2.44346095	2016.06.29.14.12
## 729	225	59	2016.06.29	135	2.35619449	2016.06.29.14.12
## 730	145	1	2016.06.29	215	-2.53072742	2016.06.29.14.22
## 731	165	2	2016.06.29	195	-2.87979327	2016.06.29.14.22
## 732	80	3	2016.06.29	280	-1.39626340	2016.06.29.14.22
## 733	150	4	2016.06.29	210	-2.61799388	2016.06.29.14.22
## 734	125	5	2016.06.29	235	-2.18166156	2016.06.29.14.22
## 735	190	6	2016.06.29	170	2.96705973	2016.06.29.14.22
## 736	195	7	2016.06.29	165	2.87979327	2016.06.29.14.22
## 737	70	8	2016.06.29	290	-1.22173048	2016.06.29.14.22
## 738	95	9	2016.06.29	265	-1.65806279	2016.06.29.14.22
## 739	20	10	2016.06.29	340	-0.34906585	2016.06.29.14.22
## 740	60	11	2016.06.29	300	-1.04719755	2016.06.29.14.22
## 741	110	12	2016.06.29	250	-1.91986218	2016.06.29.14.22
## 742	180	13	2016.06.29	180	3.14159265	2016.06.29.14.22
## 743	80	14	2016.06.29	280	-1.39626340	2016.06.29.14.22
## 744	140	15	2016.06.29	220	-2.44346095	2016.06.29.14.22
## 745	150	16	2016.06.29	210	-2.61799388	2016.06.29.14.22
## 746	335	17	2016.06.29	25	0.43633231	2016.06.29.14.22
## 747	345	18	2016.06.29	15	0.26179939	2016.06.29.14.22
## 748	345	19	2016.06.29	15	0.26179939	2016.06.29.14.22
## 749	350	20	2016.06.29	10	0.17453293	2016.06.29.14.22
## 750	300	21	2016.06.29	60	1.04719755	2016.06.29.14.22
## 751	320	22	2016.06.29	40	0.69813170	2016.06.29.14.22
## 752	330	23	2016.06.29	30	0.52359878	2016.06.29.14.22
## 753	275	24	2016.06.29	85	1.48352986	2016.06.29.14.22
## 754	340	25	2016.06.29	20	0.34906585	2016.06.29.14.22
## 755	310	26	2016.06.29	50	0.87266463	2016.06.29.14.22
## 756	300	27	2016.06.29	60	1.04719755	2016.06.29.14.22
## 757	300	28	2016.06.29	60	1.04719755	2016.06.29.14.22
## 758	280	29	2016.06.29	80	1.39626340	2016.06.29.14.22
## 759	270	30	2016.06.29	90	1.57079633	2016.06.29.14.22
## 760	325	31	2016.06.29	35	0.61086524	2016.06.29.14.22
## 761	300	32	2016.06.29	60	1.04719755	2016.06.29.14.22
## 762	315	33	2016.06.29	45	0.78539816	2016.06.29.14.22
## 763	320	34	2016.06.29	40	0.69813170	2016.06.29.14.22
## 764	280	35	2016.06.29	80	1.39626340	2016.06.29.14.22
## 765	315	36	2016.06.29	45	0.78539816	2016.06.29.14.22
## 766	310	1	2016.07.01	50	0.87266463	2016.07.01.13.38
## 767	345	2	2016.07.01	15	0.26179939	2016.07.01.13.38
## 768	265	3	2016.07.01	95	1.65806279	2016.07.01.13.38
## 769	335	4	2016.07.01	25	0.43633231	2016.07.01.13.38
## 770	170	5	2016.07.01	190	-2.96705973	2016.07.01.13.38

## 771	330	6	2016.07.01	30	0.52359878	2016.07.01.13.38
## 772	280	7	2016.07.01	80	1.39626340	2016.07.01.13.38
## 773	290	8	2016.07.01	70	1.22173048	2016.07.01.13.38
## 774	30	9	2016.07.01	330	-0.52359878	2016.07.01.13.38
## 775	245	10	2016.07.01	115	2.00712864	2016.07.01.13.38
## 776	10	11	2016.07.01	350	-0.17453293	2016.07.01.13.38
## 777	310	12	2016.07.01	50	0.87266463	2016.07.01.13.38
## 778	175	13	2016.07.01	185	-3.05432619	2016.07.01.13.38
## 779	320	14	2016.07.01	40	0.69813170	2016.07.01.13.38
## 780	265	15	2016.07.01	95	1.65806279	2016.07.01.13.38
## 781	295	16	2016.07.01	65	1.13446401	2016.07.01.13.38
## 782	255	17	2016.07.01	105	1.83259571	2016.07.01.13.38
## 783	320	18	2016.07.01	40	0.69813170	2016.07.01.13.38
## 784	310	19	2016.07.01	50	0.87266463	2016.07.01.13.38
## 785	260	20	2016.07.01	100	1.74532925	2016.07.01.13.38
## 786	310	21	2016.07.01	50	0.87266463	2016.07.01.13.38
## 787	340	22	2016.07.01	20	0.34906585	2016.07.01.13.38
## 788	325	23	2016.07.01	35	0.61086524	2016.07.01.13.38
## 789	320	24	2016.07.01	40	0.69813170	2016.07.01.13.38
## 790	335	25	2016.07.01	25	0.43633231	2016.07.01.13.38
## 791	340	26	2016.07.01	20	0.34906585	2016.07.01.13.38
## 792	275	27	2016.07.01	85	1.48352986	2016.07.01.13.38
## 793	270	28	2016.07.01	90	1.57079633	2016.07.01.13.38
## 794	300	29	2016.07.01	60	1.04719755	2016.07.01.13.38
## 795	280	30	2016.07.01	80	1.39626340	2016.07.01.13.38
## 796	300	31	2016.07.01	60	1.04719755	2016.07.01.13.38
## 797	270	32	2016.07.01	90	1.57079633	2016.07.01.13.38
## 798	320	33	2016.07.01	40	0.69813170	2016.07.01.13.38
## 799	320	34	2016.07.01	40	0.69813170	2016.07.01.13.38
## 800	310	35	2016.07.01	50	0.87266463	2016.07.01.13.38
## 801	320	36	2016.07.01	40	0.69813170	2016.07.01.13.38
## 802	320	37	2016.07.01	40	0.69813170	2016.07.01.13.38
## 803	300	38	2016.07.01	60	1.04719755	2016.07.01.13.38
## 804	285	39	2016.07.01	75	1.30899694	2016.07.01.13.38
## 805	275	40	2016.07.01	85	1.48352986	2016.07.01.13.38
## 806	355	41	2016.07.01	5	0.08726646	2016.07.01.13.38
## 807	60	42	2016.07.01	300	-1.04719755	2016.07.01.13.38
## 808	70	43	2016.07.01	290	-1.22173048	2016.07.01.13.38
## 809	65	44	2016.07.01	295	-1.13446401	2016.07.01.13.38
## 810	5	45	2016.07.01	355	-0.08726646	2016.07.01.13.38
## 811	350	46	2016.07.01	10	0.17453293	2016.07.01.13.38
## 812	325	47	2016.07.01	35	0.61086524	2016.07.01.13.38
## 813	330	48	2016.07.01	30	0.52359878	2016.07.01.13.38
## 814	20	49	2016.07.01	340	-0.34906585	2016.07.01.13.38
## 815	345	50	2016.07.01	15	0.26179939	2016.07.01.13.38
## 816	310	51	2016.07.01	50	0.87266463	2016.07.01.13.38
## 817	300	52	2016.07.01	60	1.04719755	2016.07.01.13.38
## 818	95	1	2016.07.01	265	-1.65806279	2016.07.01.14.43
## 819	50	2	2016.07.01	310	-0.87266463	2016.07.01.14.43
## 820	60	3	2016.07.01	300	-1.04719755	2016.07.01.14.43
## 821	75	4	2016.07.01	285	-1.30899694	2016.07.01.14.43
## 822	40	5	2016.07.01	320	-0.69813170	2016.07.01.14.43
## 823	110	6	2016.07.01	250	-1.91986218	2016.07.01.14.43
## 824	60	7	2016.07.01	300	-1.04719755	2016.07.01.14.43

## 825	80	8	2016.07.01	280	-1.39626340	2016.07.01.14.43
## 826	30	9	2016.07.01	330	-0.52359878	2016.07.01.14.43
## 827	130	10	2016.07.01	230	-2.26892803	2016.07.01.14.43
## 828	35	11	2016.07.01	325	-0.61086524	2016.07.01.14.43
## 829	40	12	2016.07.01	320	-0.69813170	2016.07.01.14.43
## 830	40	13	2016.07.01	320	-0.69813170	2016.07.01.14.43
## 831	30	14	2016.07.01	330	-0.52359878	2016.07.01.14.43
## 832	115	15	2016.07.01	245	-2.00712864	2016.07.01.14.43
## 833	50	16	2016.07.01	310	-0.87266463	2016.07.01.14.43
## 834	50	17	2016.07.01	310	-0.87266463	2016.07.01.14.43
## 835	220	18	2016.07.01	140	2.44346095	2016.07.01.14.43
## 836	40	19	2016.07.01	320	-0.69813170	2016.07.01.14.43
## 837	80	20	2016.07.01	280	-1.39626340	2016.07.01.14.43
## 838	50	21	2016.07.01	310	-0.87266463	2016.07.01.14.43
## 839	220	22	2016.07.01	140	2.44346095	2016.07.01.14.43
## 840	70	23	2016.07.01	290	-1.22173048	2016.07.01.14.43
## 841	60	24	2016.07.01	300	-1.04719755	2016.07.01.14.43
## 842	300	25	2016.07.01	60	1.04719755	2016.07.01.14.43
## 843	340	26	2016.07.01	20	0.34906585	2016.07.01.14.43
## 844	310	27	2016.07.01	50	0.87266463	2016.07.01.14.43
## 845	325	28	2016.07.01	35	0.61086524	2016.07.01.14.43
## 846	320	29	2016.07.01	40	0.69813170	2016.07.01.14.43
## 847	295	30	2016.07.01	65	1.13446401	2016.07.01.14.43
## 848	320	31	2016.07.01	40	0.69813170	2016.07.01.14.43
## 849	310	32	2016.07.01	50	0.87266463	2016.07.01.14.43
## 850	305	33	2016.07.01	55	0.95993109	2016.07.01.14.43
## 851	330	34	2016.07.01	30	0.52359878	2016.07.01.14.43
## 852	340	35	2016.07.01	20	0.34906585	2016.07.01.14.43
## 853	350	36	2016.07.01	10	0.17453293	2016.07.01.14.43
## 854	320	37	2016.07.01	40	0.69813170	2016.07.01.14.43
## 855	320	38	2016.07.01	40	0.69813170	2016.07.01.14.43
## 856	0	39	2016.07.01	0	0.00000000	2016.07.01.14.43
## 857	350	40	2016.07.01	10	0.17453293	2016.07.01.14.43
## 858	355	41	2016.07.01	5	0.08726646	2016.07.01.14.43
## 859	85	1	2016.07.01	275	-1.48352986	2016.07.01.14.49
## 860	85	2	2016.07.01	275	-1.48352986	2016.07.01.14.49
## 861	60	3	2016.07.01	300	-1.04719755	2016.07.01.14.49
## 862	70	4	2016.07.01	290	-1.22173048	2016.07.01.14.49
## 863	35	5	2016.07.01	325	-0.61086524	2016.07.01.14.49
## 864	85	6	2016.07.01	275	-1.48352986	2016.07.01.14.49
## 865	60	7	2016.07.01	300	-1.04719755	2016.07.01.14.49
## 866	70	8	2016.07.01	290	-1.22173048	2016.07.01.14.49
## 867	70	9	2016.07.01	290	-1.22173048	2016.07.01.14.49
## 868	70	10	2016.07.01	290	-1.22173048	2016.07.01.14.49
## 869	350	11	2016.07.01	10	0.17453293	2016.07.01.14.49
## 870	60	12	2016.07.01	300	-1.04719755	2016.07.01.14.49
## 871	70	13	2016.07.01	290	-1.22173048	2016.07.01.14.49
## 872	35	14	2016.07.01	325	-0.61086524	2016.07.01.14.49
## 873	30	15	2016.07.01	330	-0.52359878	2016.07.01.14.49
## 874	345	16	2016.07.01	15	0.26179939	2016.07.01.14.49
## 875	340	17	2016.07.01	20	0.34906585	2016.07.01.14.49
## 876	5	18	2016.07.01	355	-0.08726646	2016.07.01.14.49
## 877	345	19	2016.07.01	15	0.26179939	2016.07.01.14.49
## 878	350	20	2016.07.01	10	0.17453293	2016.07.01.14.49

## 879	0	21	2016.07.01	0	0.00000000	2016.07.01.14.49
## 880	340	22	2016.07.01	20	0.34906585	2016.07.01.14.49
## 881	340	23	2016.07.01	20	0.34906585	2016.07.01.14.49
## 882	5	24	2016.07.01	355	-0.08726646	2016.07.01.14.49
## 883	5	25	2016.07.01	355	-0.08726646	2016.07.01.14.49
## 884	50	26	2016.07.01	310	-0.87266463	2016.07.01.14.49
## 885	330	27	2016.07.01	30	0.52359878	2016.07.01.14.49
## 886	350	28	2016.07.01	10	0.17453293	2016.07.01.14.49
## 887	350	29	2016.07.01	10	0.17453293	2016.07.01.14.49
## 888	350	30	2016.07.01	10	0.17453293	2016.07.01.14.49
## 889	340	31	2016.07.01	20	0.34906585	2016.07.01.14.49
## 890	0	32	2016.07.01	0	0.00000000	2016.07.01.14.49
## 891	345	33	2016.07.01	15	0.26179939	2016.07.01.14.49
## 892	340	34	2016.07.01	20	0.34906585	2016.07.01.14.49
## 893	355	35	2016.07.01	5	0.08726646	2016.07.01.14.49
## 894	5	36	2016.07.01	355	-0.08726646	2016.07.01.14.49
## 895	340	37	2016.07.01	20	0.34906585	2016.07.01.14.49
## 896	330	38	2016.07.01	30	0.52359878	2016.07.01.14.49
## 897	355	39	2016.07.01	5	0.08726646	2016.07.01.14.49
## 898	320	40	2016.07.01	40	0.69813170	2016.07.01.14.49
## 899	350	41	2016.07.01	10	0.17453293	2016.07.01.14.49
## 900	330	42	2016.07.01	30	0.52359878	2016.07.01.14.49
## 901	340	43	2016.07.01	20	0.34906585	2016.07.01.14.49
## 902	20	1	2016.07.04	340	-0.34906585	2016.07.04.____
## 903	10	2	2016.07.04	350	-0.17453293	2016.07.04.____
## 904	90	3	2016.07.04	270	-1.57079633	2016.07.04.____
## 905	5	4	2016.07.04	355	-0.08726646	2016.07.04.____
## 906	85	5	2016.07.04	275	-1.48352986	2016.07.04.____
## 907	40	6	2016.07.04	320	-0.69813170	2016.07.04.____
## 908	75	7	2016.07.04	285	-1.30899694	2016.07.04.____
## 909	20	8	2016.07.04	340	-0.34906585	2016.07.04.____
## 910	80	9	2016.07.04	280	-1.39626340	2016.07.04.____
## 911	10	10	2016.07.04	350	-0.17453293	2016.07.04.____
## 912	50	11	2016.07.04	310	-0.87266463	2016.07.04.____
## 913	30	12	2016.07.04	330	-0.52359878	2016.07.04.____
## 914	30	13	2016.07.04	330	-0.52359878	2016.07.04.____
## 915	25	14	2016.07.04	335	-0.43633231	2016.07.04.____
## 916	80	15	2016.07.04	280	-1.39626340	2016.07.04.____
## 917	25	16	2016.07.04	335	-0.43633231	2016.07.04.____
## 918	25	17	2016.07.04	335	-0.43633231	2016.07.04.____
## 919	5	18	2016.07.04	355	-0.08726646	2016.07.04.____
## 920	10	19	2016.07.04	350	-0.17453293	2016.07.04.____
## 921	0	20	2016.07.04	0	0.00000000	2016.07.04.____
## 922	0	21	2016.07.04	0	0.00000000	2016.07.04.____
## 923	0	22	2016.07.04	0	0.00000000	2016.07.04.____
## 924	40	23	2016.07.04	320	-0.69813170	2016.07.04.____
## 925	345	24	2016.07.04	15	0.26179939	2016.07.04.____
## 926	5	25	2016.07.04	355	-0.08726646	2016.07.04.____
## 927	355	26	2016.07.04	5	0.08726646	2016.07.04.____
## 928	10	27	2016.07.04	350	-0.17453293	2016.07.04.____
## 929	15	28	2016.07.04	345	-0.26179939	2016.07.04.____
## 930	355	29	2016.07.04	5	0.08726646	2016.07.04.____
## 931	340	30	2016.07.04	20	0.34906585	2016.07.04.____
## 932	340	31	2016.07.04	20	0.34906585	2016.07.04.____

## 933	5	32	2016.07.04	355	-0.08726646	2016.07.04.____
## 934	355	33	2016.07.04	5	0.08726646	2016.07.04.____
## 935	320	34	2016.07.04	40	0.69813170	2016.07.04.____
## 936	5	35	2016.07.04	355	-0.08726646	2016.07.04.____
## 937	10	36	2016.07.04	350	-0.17453293	2016.07.04.____
## 938	350	37	2016.07.04	10	0.17453293	2016.07.04.____
## 939	355	38	2016.07.04	5	0.08726646	2016.07.04.____
## 940	345	39	2016.07.04	15	0.26179939	2016.07.04.____
## 941	305	40	2016.07.04	55	0.95993109	2016.07.04.____
## 942	320	41	2016.07.04	40	0.69813170	2016.07.04.____
## 943	310	42	2016.07.04	50	0.87266463	2016.07.04.____
## 944	320	43	2016.07.04	40	0.69813170	2016.07.04.____
## 945	0	44	2016.07.04	0	0.00000000	2016.07.04.____
## 946	5	45	2016.07.04	355	-0.08726646	2016.07.04.____
## 947	335	46	2016.07.04	25	0.43633231	2016.07.04.____
## 948	355	47	2016.07.04	5	0.08726646	2016.07.04.____
## 949	350	48	2016.07.04	10	0.17453293	2016.07.04.____
## 950	345	49	2016.07.04	15	0.26179939	2016.07.04.____
## 951	340	50	2016.07.04	20	0.34906585	2016.07.04.____
## 952	320	51	2016.07.04	40	0.69813170	2016.07.04.____
## 953	335	52	2016.07.04	25	0.43633231	2016.07.04.____
## 954	340	53	2016.07.04	20	0.34906585	2016.07.04.____
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## 957	20	56	2016.07.04	340	-0.34906585	2016.07.04.____
## 958	330	57	2016.07.04	30	0.52359878	2016.07.04.____
## 959	340	58	2016.07.04	20	0.34906585	2016.07.04.____
## 960	330	59	2016.07.04	30	0.52359878	2016.07.04.____
## 961	340	60	2016.07.04	20	0.34906585	2016.07.04.____
## 962	30	61	2016.07.04	330	-0.52359878	2016.07.04.____
## 963	335	62	2016.07.04	25	0.43633231	2016.07.04.____
## 964	330	63	2016.07.04	30	0.52359878	2016.07.04.____
## 965	340	64	2016.07.04	20	0.34906585	2016.07.04.____
## 966	305	65	2016.07.04	55	0.95993109	2016.07.04.____
## 967	335	66	2016.07.04	25	0.43633231	2016.07.04.____
## 968	40	67	2016.07.04	320	-0.69813170	2016.07.04.____
## 969	340	68	2016.07.04	20	0.34906585	2016.07.04.____
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## 971	355	70	2016.07.04	5	0.08726646	2016.07.04.____
## 972	5	71	2016.07.04	355	-0.08726646	2016.07.04.____
## 973	355	72	2016.07.04	5	0.08726646	2016.07.04.____
## 974	355	73	2016.07.04	5	0.08726646	2016.07.04.____
## 975	350	1	2016.07.05	10	0.17453293	2016.07.05.13.38
## 976	35	2	2016.07.05	325	-0.61086524	2016.07.05.13.38
## 977	5	3	2016.07.05	355	-0.08726646	2016.07.05.13.38
## 978	60	4	2016.07.05	300	-1.04719755	2016.07.05.13.38
## 979	60	5	2016.07.05	300	-1.04719755	2016.07.05.13.38
## 980	70	6	2016.07.05	290	-1.22173048	2016.07.05.13.38
## 981	40	7	2016.07.05	320	-0.69813170	2016.07.05.13.38
## 982	40	8	2016.07.05	320	-0.69813170	2016.07.05.13.38
## 983	50	9	2016.07.05	310	-0.87266463	2016.07.05.13.38
## 984	40	10	2016.07.05	320	-0.69813170	2016.07.05.13.38
## 985	50	11	2016.07.05	310	-0.87266463	2016.07.05.13.38
## 986	0	12	2016.07.05	0	0.00000000	2016.07.05.13.38

## 987	70	13	2016.07.05	290	-1.22173048	2016.07.05.13.38
## 988	40	14	2016.07.05	320	-0.69813170	2016.07.05.13.38
## 989	355	15	2016.07.05	5	0.08726646	2016.07.05.13.38
## 990	90	16	2016.07.05	270	-1.57079633	2016.07.05.13.38
## 991	340	17	2016.07.05	20	0.34906585	2016.07.05.13.38
## 992	80	18	2016.07.05	280	-1.39626340	2016.07.05.13.38
## 993	30	19	2016.07.05	330	-0.52359878	2016.07.05.13.38
## 994	70	20	2016.07.05	290	-1.22173048	2016.07.05.13.38
## 995	345	21	2016.07.05	15	0.26179939	2016.07.05.13.38
## 996	355	22	2016.07.05	5	0.08726646	2016.07.05.13.38
## 997	330	23	2016.07.05	30	0.52359878	2016.07.05.13.38
## 998	320	24	2016.07.05	40	0.69813170	2016.07.05.13.38
## 999	340	25	2016.07.05	20	0.34906585	2016.07.05.13.38
## 1000	340	26	2016.07.05	20	0.34906585	2016.07.05.13.38
## 1001	300	27	2016.07.05	60	1.04719755	2016.07.05.13.38
## 1002	350	28	2016.07.05	10	0.17453293	2016.07.05.13.38
## 1003	340	29	2016.07.05	20	0.34906585	2016.07.05.13.38
## 1004	340	30	2016.07.05	20	0.34906585	2016.07.05.13.38
## 1005	325	31	2016.07.05	35	0.61086524	2016.07.05.13.38
## 1006	330	32	2016.07.05	30	0.52359878	2016.07.05.13.38
## 1007	5	1	2016.07.05	355	-0.08726646	2016.07.05.13.42
## 1008	340	2	2016.07.05	20	0.34906585	2016.07.05.13.42
## 1009	5	3	2016.07.05	355	-0.08726646	2016.07.05.13.42
## 1010	50	4	2016.07.05	310	-0.87266463	2016.07.05.13.42
## 1011	355	5	2016.07.05	5	0.08726646	2016.07.05.13.42
## 1012	50	6	2016.07.05	310	-0.87266463	2016.07.05.13.42
## 1013	320	7	2016.07.05	40	0.69813170	2016.07.05.13.42
## 1014	330	8	2016.07.05	30	0.52359878	2016.07.05.13.42
## 1015	355	9	2016.07.05	5	0.08726646	2016.07.05.13.42
## 1016	20	10	2016.07.05	340	-0.34906585	2016.07.05.13.42
## 1017	30	11	2016.07.05	330	-0.52359878	2016.07.05.13.42
## 1018	70	12	2016.07.05	290	-1.22173048	2016.07.05.13.42
## 1019	85	13	2016.07.05	275	-1.48352986	2016.07.05.13.42
## 1020	70	14	2016.07.05	290	-1.22173048	2016.07.05.13.42
## 1021	250	15	2016.07.05	110	1.91986218	2016.07.05.13.42
## 1022	220	16	2016.07.05	140	2.44346095	2016.07.05.13.42
## 1023	245	17	2016.07.05	115	2.00712864	2016.07.05.13.42
## 1024	280	18	2016.07.05	80	1.39626340	2016.07.05.13.42
## 1025	290	19	2016.07.05	70	1.22173048	2016.07.05.13.42
## 1026	285	20	2016.07.05	75	1.30899694	2016.07.05.13.42
## 1027	285	21	2016.07.05	75	1.30899694	2016.07.05.13.42
## 1028	290	22	2016.07.05	70	1.22173048	2016.07.05.13.42
## 1029	270	23	2016.07.05	90	1.57079633	2016.07.05.13.42
## 1030	275	24	2016.07.05	85	1.48352986	2016.07.05.13.42
## 1031	275	25	2016.07.05	85	1.48352986	2016.07.05.13.42
## 1032	300	26	2016.07.05	60	1.04719755	2016.07.05.13.42
## 1033	295	27	2016.07.05	65	1.13446401	2016.07.05.13.42
## 1034	270	28	2016.07.05	90	1.57079633	2016.07.05.13.42
## 1035	300	29	2016.07.05	60	1.04719755	2016.07.05.13.42
## 1036	285	30	2016.07.05	75	1.30899694	2016.07.05.13.42
## 1037	265	31	2016.07.05	95	1.65806279	2016.07.05.13.42
## 1038	235	1	2016.07.05	125	2.18166156	2016.07.05.13.49
## 1039	210	2	2016.07.05	150	2.61799388	2016.07.05.13.49
## 1040	200	3	2016.07.05	160	2.79252680	2016.07.05.13.49

## 1041	195	4	2016.07.05	165	2.87979327	2016.07.05.13.49
## 1042	245	5	2016.07.05	115	2.00712864	2016.07.05.13.49
## 1043	335	6	2016.07.05	25	0.43633231	2016.07.05.13.49
## 1044	345	7	2016.07.05	15	0.26179939	2016.07.05.13.49
## 1045	265	8	2016.07.05	95	1.65806279	2016.07.05.13.49
## 1046	265	9	2016.07.05	95	1.65806279	2016.07.05.13.49
## 1047	210	10	2016.07.05	150	2.61799388	2016.07.05.13.49
## 1048	190	11	2016.07.05	170	2.96705973	2016.07.05.13.49
## 1049	255	12	2016.07.05	105	1.83259571	2016.07.05.13.49
## 1050	300	13	2016.07.05	60	1.04719755	2016.07.05.13.49
## 1051	335	14	2016.07.05	25	0.43633231	2016.07.05.13.49
## 1052	280	15	2016.07.05	80	1.39626340	2016.07.05.13.49
## 1053	280	16	2016.07.05	80	1.39626340	2016.07.05.13.49
## 1054	255	17	2016.07.05	105	1.83259571	2016.07.05.13.49
## 1055	315	18	2016.07.05	45	0.78539816	2016.07.05.13.49
## 1056	265	19	2016.07.05	95	1.65806279	2016.07.05.13.49
## 1057	300	20	2016.07.05	60	1.04719755	2016.07.05.13.49
## 1058	295	21	2016.07.05	65	1.13446401	2016.07.05.13.49
## 1059	270	22	2016.07.05	90	1.57079633	2016.07.05.13.49
## 1060	330	23	2016.07.05	30	0.52359878	2016.07.05.13.49
## 1061	20	1	2016.07.05	340	-0.34906585	2016.07.05.14.01
## 1062	15	2	2016.07.05	345	-0.26179939	2016.07.05.14.01
## 1063	355	3	2016.07.05	5	0.08726646	2016.07.05.14.01
## 1064	35	4	2016.07.05	325	-0.61086524	2016.07.05.14.01
## 1065	0	5	2016.07.05	0	0.00000000	2016.07.05.14.01
## 1066	350	6	2016.07.05	10	0.17453293	2016.07.05.14.01
## 1067	20	7	2016.07.05	340	-0.34906585	2016.07.05.14.01
## 1068	60	8	2016.07.05	300	-1.04719755	2016.07.05.14.01
## 1069	70	9	2016.07.05	290	-1.22173048	2016.07.05.14.01
## 1070	70	10	2016.07.05	290	-1.22173048	2016.07.05.14.01
## 1071	10	11	2016.07.05	350	-0.17453293	2016.07.05.14.01
## 1072	340	12	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1073	350	13	2016.07.05	10	0.17453293	2016.07.05.14.01
## 1074	15	14	2016.07.05	345	-0.26179939	2016.07.05.14.01
## 1075	350	15	2016.07.05	10	0.17453293	2016.07.05.14.01
## 1076	10	16	2016.07.05	350	-0.17453293	2016.07.05.14.01
## 1077	350	17	2016.07.05	10	0.17453293	2016.07.05.14.01
## 1078	15	18	2016.07.05	345	-0.26179939	2016.07.05.14.01
## 1079	25	19	2016.07.05	335	-0.43633231	2016.07.05.14.01
## 1080	60	20	2016.07.05	300	-1.04719755	2016.07.05.14.01
## 1081	80	21	2016.07.05	280	-1.39626340	2016.07.05.14.01
## 1082	100	22	2016.07.05	260	-1.74532925	2016.07.05.14.01
## 1083	5	23	2016.07.05	355	-0.08726646	2016.07.05.14.01
## 1084	50	24	2016.07.05	310	-0.87266463	2016.07.05.14.01
## 1085	345	25	2016.07.05	15	0.26179939	2016.07.05.14.01
## 1086	355	26	2016.07.05	5	0.08726646	2016.07.05.14.01
## 1087	350	27	2016.07.05	10	0.17453293	2016.07.05.14.01
## 1088	345	28	2016.07.05	15	0.26179939	2016.07.05.14.01
## 1089	330	29	2016.07.05	30	0.52359878	2016.07.05.14.01
## 1090	340	30	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1091	310	31	2016.07.05	50	0.87266463	2016.07.05.14.01
## 1092	330	32	2016.07.05	30	0.52359878	2016.07.05.14.01
## 1093	330	33	2016.07.05	30	0.52359878	2016.07.05.14.01
## 1094	340	34	2016.07.05	20	0.34906585	2016.07.05.14.01

## 1095	320	35	2016.07.05	40	0.69813170	2016.07.05.14.01
## 1096	335	36	2016.07.05	25	0.43633231	2016.07.05.14.01
## 1097	355	37	2016.07.05	5	0.08726646	2016.07.05.14.01
## 1098	330	38	2016.07.05	30	0.52359878	2016.07.05.14.01
## 1099	320	39	2016.07.05	40	0.69813170	2016.07.05.14.01
## 1100	340	40	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1101	340	41	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1102	345	42	2016.07.05	15	0.26179939	2016.07.05.14.01
## 1103	320	43	2016.07.05	40	0.69813170	2016.07.05.14.01
## 1104	350	44	2016.07.05	10	0.17453293	2016.07.05.14.01
## 1105	340	45	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1106	340	46	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1107	325	47	2016.07.05	35	0.61086524	2016.07.05.14.01
## 1108	355	48	2016.07.05	5	0.08726646	2016.07.05.14.01
## 1109	310	49	2016.07.05	50	0.87266463	2016.07.05.14.01
## 1110	340	50	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1111	340	51	2016.07.05	20	0.34906585	2016.07.05.14.01
## 1112	330	52	2016.07.05	30	0.52359878	2016.07.05.14.01
## 1113	0	53	2016.07.05	0	0.00000000	2016.07.05.14.01
## 1114	50	1	2016.07.05	310	-0.87266463	2016.07.05.14.12
## 1115	330	2	2016.07.05	30	0.52359878	2016.07.05.14.12
## 1116	0	3	2016.07.05	0	0.00000000	2016.07.05.14.12
## 1117	0	4	2016.07.05	0	0.00000000	2016.07.05.14.12
## 1118	350	5	2016.07.05	10	0.17453293	2016.07.05.14.12
## 1119	260	6	2016.07.05	100	1.74532925	2016.07.05.14.12
## 1120	260	7	2016.07.05	100	1.74532925	2016.07.05.14.12
## 1121	5	8	2016.07.05	355	-0.08726646	2016.07.05.14.12
## 1122	110	9	2016.07.05	250	-1.91986218	2016.07.05.14.12
## 1123	60	10	2016.07.05	300	-1.04719755	2016.07.05.14.12
## 1124	290	11	2016.07.05	70	1.22173048	2016.07.05.14.12
## 1125	330	12	2016.07.05	30	0.52359878	2016.07.05.14.12
## 1126	5	13	2016.07.05	355	-0.08726646	2016.07.05.14.12
## 1127	30	14	2016.07.05	330	-0.52359878	2016.07.05.14.12
## 1128	25	15	2016.07.05	335	-0.43633231	2016.07.05.14.12
## 1129	30	16	2016.07.05	330	-0.52359878	2016.07.05.14.12
## 1130	20	17	2016.07.05	340	-0.34906585	2016.07.05.14.12
## 1131	35	18	2016.07.05	325	-0.61086524	2016.07.05.14.12
## 1132	20	19	2016.07.05	340	-0.34906585	2016.07.05.14.12
## 1133	80	20	2016.07.05	280	-1.39626340	2016.07.05.14.12
## 1134	10	21	2016.07.05	350	-0.17453293	2016.07.05.14.12
## 1135	45	22	2016.07.05	315	-0.78539816	2016.07.05.14.12
## 1136	30	23	2016.07.05	330	-0.52359878	2016.07.05.14.12
## 1137	50	24	2016.07.05	310	-0.87266463	2016.07.05.14.12
## 1138	20	25	2016.07.05	340	-0.34906585	2016.07.05.14.12
## 1139	40	26	2016.07.05	320	-0.69813170	2016.07.05.14.12
## 1140	300	27	2016.07.05	60	1.04719755	2016.07.05.14.12
## 1141	315	28	2016.07.05	45	0.78539816	2016.07.05.14.12
## 1142	310	29	2016.07.05	50	0.87266463	2016.07.05.14.12
## 1143	310	30	2016.07.05	50	0.87266463	2016.07.05.14.12
## 1144	305	31	2016.07.05	55	0.95993109	2016.07.05.14.12
## 1145	320	32	2016.07.05	40	0.69813170	2016.07.05.14.12
## 1146	340	33	2016.07.05	20	0.34906585	2016.07.05.14.12
## 1147	340	34	2016.07.05	20	0.34906585	2016.07.05.14.12
## 1148	340	35	2016.07.05	20	0.34906585	2016.07.05.14.12

## 1149	330	36	2016.07.05	30	0.52359878	2016.07.05.14.12
## 1150	325	37	2016.07.05	35	0.61086524	2016.07.05.14.12
## 1151	340	38	2016.07.05	20	0.34906585	2016.07.05.14.12
## 1152	330	39	2016.07.05	30	0.52359878	2016.07.05.14.12
## 1153	340	40	2016.07.05	20	0.34906585	2016.07.05.14.12
## 1154	315	41	2016.07.05	45	0.78539816	2016.07.05.14.12
## 1155	320	42	2016.07.05	40	0.69813170	2016.07.05.14.12
## 1156	30	43	2016.07.05	330	-0.52359878	2016.07.05.14.12
## 1157	335	44	2016.07.05	25	0.43633231	2016.07.05.14.12
## 1158	350	45	2016.07.05	10	0.17453293	2016.07.05.14.12
## 1159	305	46	2016.07.05	55	0.95993109	2016.07.05.14.12
## 1160	190	1	2016.07.07	170	2.96705973	2016.07.07.13.47
## 1161	170	2	2016.07.07	190	-2.96705973	2016.07.07.13.47
## 1162	180	3	2016.07.07	180	3.14159265	2016.07.07.13.47
## 1163	170	4	2016.07.07	190	-2.96705973	2016.07.07.13.47
## 1164	285	5	2016.07.07	75	1.30899694	2016.07.07.13.47
## 1165	220	6	2016.07.07	140	2.44346095	2016.07.07.13.47
## 1166	290	7	2016.07.07	70	1.22173048	2016.07.07.13.47
## 1167	155	8	2016.07.07	205	-2.70526034	2016.07.07.13.47
## 1168	305	9	2016.07.07	55	0.95993109	2016.07.07.13.47
## 1169	300	10	2016.07.07	60	1.04719755	2016.07.07.13.47
## 1170	310	11	2016.07.07	50	0.87266463	2016.07.07.13.47
## 1171	180	12	2016.07.07	180	3.14159265	2016.07.07.13.47
## 1172	310	13	2016.07.07	50	0.87266463	2016.07.07.13.47
## 1173	160	14	2016.07.07	200	-2.79252680	2016.07.07.13.47
## 1174	280	15	2016.07.07	80	1.39626340	2016.07.07.13.47
## 1175	260	16	2016.07.07	100	1.74532925	2016.07.07.13.47
## 1176	280	17	2016.07.07	80	1.39626340	2016.07.07.13.47
## 1177	300	18	2016.07.07	60	1.04719755	2016.07.07.13.47
## 1178	295	19	2016.07.07	65	1.13446401	2016.07.07.13.47
## 1179	285	20	2016.07.07	75	1.30899694	2016.07.07.13.47
## 1180	310	21	2016.07.07	50	0.87266463	2016.07.07.13.47
## 1181	285	22	2016.07.07	75	1.30899694	2016.07.07.13.47
## 1182	275	23	2016.07.07	85	1.48352986	2016.07.07.13.47
## 1183	280	24	2016.07.07	80	1.39626340	2016.07.07.13.47
## 1184	310	25	2016.07.07	50	0.87266463	2016.07.07.13.47
## 1185	315	26	2016.07.07	45	0.78539816	2016.07.07.13.47
## 1186	270	27	2016.07.07	90	1.57079633	2016.07.07.13.47
## 1187	310	28	2016.07.07	50	0.87266463	2016.07.07.13.47
## 1188	320	29	2016.07.07	40	0.69813170	2016.07.07.13.47
## 1189	5	1	2016.07.07	355	-0.08726646	2016.07.07.14.14
## 1190	0	2	2016.07.07	0	0.00000000	2016.07.07.14.14
## 1191	10	3	2016.07.07	350	-0.17453293	2016.07.07.14.14
## 1192	260	4	2016.07.07	100	1.74532925	2016.07.07.14.14
## 1193	35	5	2016.07.07	325	-0.61086524	2016.07.07.14.14
## 1194	335	6	2016.07.07	25	0.43633231	2016.07.07.14.14
## 1195	245	7	2016.07.07	115	2.00712864	2016.07.07.14.14
## 1196	30	8	2016.07.07	330	-0.52359878	2016.07.07.14.14
## 1197	310	9	2016.07.07	50	0.87266463	2016.07.07.14.14
## 1198	15	10	2016.07.07	345	-0.26179939	2016.07.07.14.14
## 1199	255	11	2016.07.07	105	1.83259571	2016.07.07.14.14
## 1200	250	12	2016.07.07	110	1.91986218	2016.07.07.14.14
## 1201	20	13	2016.07.07	340	-0.34906585	2016.07.07.14.14
## 1202	40	14	2016.07.07	320	-0.69813170	2016.07.07.14.14

## 1203	325	15	2016.07.07	35	0.61086524	2016.07.07.14.14
## 1204	335	16	2016.07.07	25	0.43633231	2016.07.07.14.14
## 1205	290	17	2016.07.07	70	1.22173048	2016.07.07.14.14
## 1206	280	18	2016.07.07	80	1.39626340	2016.07.07.14.14
## 1207	270	19	2016.07.07	90	1.57079633	2016.07.07.14.14
## 1208	310	20	2016.07.07	50	0.87266463	2016.07.07.14.14
## 1209	320	21	2016.07.07	40	0.69813170	2016.07.07.14.14
## 1210	300	22	2016.07.07	60	1.04719755	2016.07.07.14.14
## 1211	320	23	2016.07.07	40	0.69813170	2016.07.07.14.14
## 1212	310	24	2016.07.07	50	0.87266463	2016.07.07.14.14
## 1213	320	25	2016.07.07	40	0.69813170	2016.07.07.14.14
## 1214	310	26	2016.07.07	50	0.87266463	2016.07.07.14.14
## 1215	350	1	2016.07.08	10	0.17453293	2016.07.08.13.32
## 1216	5	2	2016.07.08	355	-0.08726646	2016.07.08.13.32
## 1217	265	3	2016.07.08	95	1.65806279	2016.07.08.13.32
## 1218	345	4	2016.07.08	15	0.26179939	2016.07.08.13.32
## 1219	60	5	2016.07.08	300	-1.04719755	2016.07.08.13.32
## 1220	20	6	2016.07.08	340	-0.34906585	2016.07.08.13.32
## 1221	75	7	2016.07.08	285	-1.30899694	2016.07.08.13.32
## 1222	340	8	2016.07.08	20	0.34906585	2016.07.08.13.32
## 1223	140	9	2016.07.08	220	-2.44346095	2016.07.08.13.32
## 1224	170	10	2016.07.08	190	-2.96705973	2016.07.08.13.32
## 1225	130	11	2016.07.08	230	-2.26892803	2016.07.08.13.32
## 1226	40	12	2016.07.08	320	-0.69813170	2016.07.08.13.32
## 1227	50	13	2016.07.08	310	-0.87266463	2016.07.08.13.32
## 1228	60	14	2016.07.08	300	-1.04719755	2016.07.08.13.32
## 1229	130	15	2016.07.08	230	-2.26892803	2016.07.08.13.32
## 1230	125	16	2016.07.08	235	-2.18166156	2016.07.08.13.32
## 1231	265	17	2016.07.08	95	1.65806279	2016.07.08.13.32
## 1232	275	18	2016.07.08	85	1.48352986	2016.07.08.13.32
## 1233	280	19	2016.07.08	80	1.39626340	2016.07.08.13.32
## 1234	270	20	2016.07.08	90	1.57079633	2016.07.08.13.32
## 1235	265	21	2016.07.08	95	1.65806279	2016.07.08.13.32
## 1236	290	22	2016.07.08	70	1.22173048	2016.07.08.13.32
## 1237	280	23	2016.07.08	80	1.39626340	2016.07.08.13.32
## 1238	265	24	2016.07.08	95	1.65806279	2016.07.08.13.32
## 1239	315	25	2016.07.08	45	0.78539816	2016.07.08.13.32
## 1240	300	26	2016.07.08	60	1.04719755	2016.07.08.13.32
## 1241	305	27	2016.07.08	55	0.95993109	2016.07.08.13.32
## 1242	280	28	2016.07.08	80	1.39626340	2016.07.08.13.32
## 1243	270	29	2016.07.08	90	1.57079633	2016.07.08.13.32
## 1244	270	30	2016.07.08	90	1.57079633	2016.07.08.13.32
## 1245	105	1	2016.07.08	255	-1.83259571	2016.07.08.14.46
## 1246	0	2	2016.07.08	0	0.00000000	2016.07.08.14.46
## 1247	75	3	2016.07.08	285	-1.30899694	2016.07.08.14.46
## 1248	30	4	2016.07.08	330	-0.52359878	2016.07.08.14.46
## 1249	70	5	2016.07.08	290	-1.22173048	2016.07.08.14.46
## 1250	120	6	2016.07.08	240	-2.09439510	2016.07.08.14.46
## 1251	30	7	2016.07.08	330	-0.52359878	2016.07.08.14.46
## 1252	105	8	2016.07.08	255	-1.83259571	2016.07.08.14.46
## 1253	80	9	2016.07.08	280	-1.39626340	2016.07.08.14.46
## 1254	40	10	2016.07.08	320	-0.69813170	2016.07.08.14.46
## 1255	95	11	2016.07.08	265	-1.65806279	2016.07.08.14.46
## 1256	120	12	2016.07.08	240	-2.09439510	2016.07.08.14.46

## 1257	55	13	2016.07.08	305	-0.95993109	2016.07.08.14.46
## 1258	100	14	2016.07.08	260	-1.74532925	2016.07.08.14.46
## 1259	15	15	2016.07.08	345	-0.26179939	2016.07.08.14.46
## 1260	80	16	2016.07.08	280	-1.39626340	2016.07.08.14.46
## 1261	60	17	2016.07.08	300	-1.04719755	2016.07.08.14.46
## 1262	55	18	2016.07.08	305	-0.95993109	2016.07.08.14.46
## 1263	140	19	2016.07.08	220	-2.44346095	2016.07.08.14.46
## 1264	10	20	2016.07.08	350	-0.17453293	2016.07.08.14.46
## 1265	100	21	2016.07.08	260	-1.74532925	2016.07.08.14.46
## 1266	95	22	2016.07.08	265	-1.65806279	2016.07.08.14.46
## 1267	70	23	2016.07.08	290	-1.22173048	2016.07.08.14.46
## 1268	105	24	2016.07.08	255	-1.83259571	2016.07.08.14.46
## 1269	60	25	2016.07.08	300	-1.04719755	2016.07.08.14.46
## 1270	90	26	2016.07.08	270	-1.57079633	2016.07.08.14.46
## 1271	30	27	2016.07.08	330	-0.52359878	2016.07.08.14.46
## 1272	85	28	2016.07.08	275	-1.48352986	2016.07.08.14.46
## 1273	75	29	2016.07.08	285	-1.30899694	2016.07.08.14.46
## 1274	125	30	2016.07.08	235	-2.18166156	2016.07.08.14.46
## 1275	50	31	2016.07.08	310	-0.87266463	2016.07.08.14.46
## 1276	100	32	2016.07.08	260	-1.74532925	2016.07.08.14.46
## 1277	95	33	2016.07.08	265	-1.65806279	2016.07.08.14.46
## 1278	350	34	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1279	320	35	2016.07.08	40	0.69813170	2016.07.08.14.46
## 1280	25	36	2016.07.08	335	-0.43633231	2016.07.08.14.46
## 1281	20	37	2016.07.08	340	-0.34906585	2016.07.08.14.46
## 1282	20	38	2016.07.08	340	-0.34906585	2016.07.08.14.46
## 1283	340	39	2016.07.08	20	0.34906585	2016.07.08.14.46
## 1284	350	40	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1285	15	41	2016.07.08	345	-0.26179939	2016.07.08.14.46
## 1286	350	42	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1287	355	43	2016.07.08	5	0.08726646	2016.07.08.14.46
## 1288	330	44	2016.07.08	30	0.52359878	2016.07.08.14.46
## 1289	0	45	2016.07.08	0	0.00000000	2016.07.08.14.46
## 1290	320	46	2016.07.08	40	0.69813170	2016.07.08.14.46
## 1291	0	47	2016.07.08	0	0.00000000	2016.07.08.14.46
## 1292	345	48	2016.07.08	15	0.26179939	2016.07.08.14.46
## 1293	345	49	2016.07.08	15	0.26179939	2016.07.08.14.46
## 1294	5	50	2016.07.08	355	-0.08726646	2016.07.08.14.46
## 1295	5	51	2016.07.08	355	-0.08726646	2016.07.08.14.46
## 1296	0	52	2016.07.08	0	0.00000000	2016.07.08.14.46
## 1297	350	53	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1298	340	54	2016.07.08	20	0.34906585	2016.07.08.14.46
## 1299	315	55	2016.07.08	45	0.78539816	2016.07.08.14.46
## 1300	350	56	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1301	335	57	2016.07.08	25	0.43633231	2016.07.08.14.46
## 1302	345	58	2016.07.08	15	0.26179939	2016.07.08.14.46
## 1303	15	59	2016.07.08	345	-0.26179939	2016.07.08.14.46
## 1304	350	60	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1305	5	61	2016.07.08	355	-0.08726646	2016.07.08.14.46
## 1306	355	62	2016.07.08	5	0.08726646	2016.07.08.14.46
## 1307	350	63	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1308	315	64	2016.07.08	45	0.78539816	2016.07.08.14.46
## 1309	285	65	2016.07.08	75	1.30899694	2016.07.08.14.46
## 1310	355	66	2016.07.08	5	0.08726646	2016.07.08.14.46

## 1311	5	67	2016.07.08	355	-0.08726646	2016.07.08.14.46
## 1312	345	68	2016.07.08	15	0.26179939	2016.07.08.14.46
## 1313	10	69	2016.07.08	350	-0.17453293	2016.07.08.14.46
## 1314	0	70	2016.07.08	0	0.00000000	2016.07.08.14.46
## 1315	350	71	2016.07.08	10	0.17453293	2016.07.08.14.46
## 1316	5	72	2016.07.08	355	-0.08726646	2016.07.08.14.46
## 1317	5	73	2016.07.08	355	-0.08726646	2016.07.08.14.46
## 1318	20	74	2016.07.08	340	-0.34906585	2016.07.08.14.46
## 1319	115	75	2016.07.08	245	-2.00712864	2016.07.08.14.46
## 1320	20	76	2016.07.08	340	-0.34906585	2016.07.08.14.46
## 1321	40	77	2016.07.08	320	-0.69813170	2016.07.08.14.46
## 1322	30	78	2016.07.08	330	-0.52359878	2016.07.08.14.46
## 1323	50	79	2016.07.08	310	-0.87266463	2016.07.08.14.46
## 1324	130	80	2016.07.08	230	-2.26892803	2016.07.08.14.46
## 1325	175	81	2016.07.08	185	-3.05432619	2016.07.08.14.46
## 1326	90	82	2016.07.08	270	-1.57079633	2016.07.08.14.46
## 1327	145	83	2016.07.08	215	-2.53072742	2016.07.08.14.46
## 1328	205	84	2016.07.08	155	2.70526034	2016.07.08.14.46
## 1329	140	85	2016.07.08	220	-2.44346095	2016.07.08.14.46
## 1330	230	86	2016.07.08	130	2.26892803	2016.07.08.14.46
## 1331	200	87	2016.07.08	160	2.79252680	2016.07.08.14.46
## 1332	170	88	2016.07.08	190	-2.96705973	2016.07.08.14.46
## 1333	25	89	2016.07.08	335	-0.43633231	2016.07.08.14.46
## 1334	340	90	2016.07.08	20	0.34906585	2016.07.08.14.46
## 1335	295	91	2016.07.08	65	1.13446401	2016.07.08.14.46
## 1336	340	92	2016.07.08	20	0.34906585	2016.07.08.14.46
## 1337	90	93	2016.07.08	270	-1.57079633	2016.07.08.14.46
## 1338	160	94	2016.07.08	200	-2.79252680	2016.07.08.14.46
## 1339	310	95	2016.07.08	50	0.87266463	2016.07.08.14.46
## 1340	105	96	2016.07.08	255	-1.83259571	2016.07.08.14.46
## 1341	90	97	2016.07.08	270	-1.57079633	2016.07.08.14.46
## 1342	150	98	2016.07.08	210	-2.61799388	2016.07.08.14.46
## 1343	145	99	2016.07.08	215	-2.53072742	2016.07.08.14.46
## 1344	0	100	2016.07.08	0	0.00000000	2016.07.08.14.46
## 1345	95	1	2016.07.08	265	-1.65806279	2016.07.08.14.57
## 1346	70	2	2016.07.08	290	-1.22173048	2016.07.08.14.57
## 1347	60	3	2016.07.08	300	-1.04719755	2016.07.08.14.57
## 1348	70	4	2016.07.08	290	-1.22173048	2016.07.08.14.57
## 1349	325	5	2016.07.08	35	0.61086524	2016.07.08.14.57
## 1350	80	6	2016.07.08	280	-1.39626340	2016.07.08.14.57
## 1351	50	7	2016.07.08	310	-0.87266463	2016.07.08.14.57
## 1352	40	8	2016.07.08	320	-0.69813170	2016.07.08.14.57
## 1353	110	9	2016.07.08	250	-1.91986218	2016.07.08.14.57
## 1354	35	10	2016.07.08	325	-0.61086524	2016.07.08.14.57
## 1355	40	11	2016.07.08	320	-0.69813170	2016.07.08.14.57
## 1356	340	12	2016.07.08	20	0.34906585	2016.07.08.14.57
## 1357	340	13	2016.07.08	20	0.34906585	2016.07.08.14.57
## 1358	345	14	2016.07.08	15	0.26179939	2016.07.08.14.57
## 1359	50	15	2016.07.08	310	-0.87266463	2016.07.08.14.57
## 1360	320	16	2016.07.08	40	0.69813170	2016.07.08.14.57
## 1361	130	17	2016.07.08	230	-2.26892803	2016.07.08.14.57
## 1362	0	18	2016.07.08	0	0.00000000	2016.07.08.14.57
## 1363	100	19	2016.07.08	260	-1.74532925	2016.07.08.14.57
## 1364	15	20	2016.07.08	345	-0.26179939	2016.07.08.14.57

## 1365	340	21	2016.07.08	20	0.34906585	2016.07.08.14.57
## 1366	0	22	2016.07.08	0	0.00000000	2016.07.08.14.57
## 1367	0	23	2016.07.08	0	0.00000000	2016.07.08.14.57
## 1368	0	24	2016.07.08	0	0.00000000	2016.07.08.14.57
## 1369	0	25	2016.07.08	0	0.00000000	2016.07.08.14.57
## 1370	30	26	2016.07.08	330	-0.52359878	2016.07.08.14.57
## 1371	30	27	2016.07.08	330	-0.52359878	2016.07.08.14.57
## 1372	20	28	2016.07.08	340	-0.34906585	2016.07.08.14.57
## 1373	50	29	2016.07.08	310	-0.87266463	2016.07.08.14.57
## 1374	30	30	2016.07.08	330	-0.52359878	2016.07.08.14.57
## 1375	5	31	2016.07.08	355	-0.08726646	2016.07.08.14.57
## 1376	60	32	2016.07.08	300	-1.04719755	2016.07.08.14.57
## 1377	350	33	2016.07.08	10	0.17453293	2016.07.08.14.57
## 1378	10	34	2016.07.08	350	-0.17453293	2016.07.08.14.57
## 1379	50	35	2016.07.08	310	-0.87266463	2016.07.08.14.57
## 1380	10	36	2016.07.08	350	-0.17453293	2016.07.08.14.57
## 1381	10	37	2016.07.08	350	-0.17453293	2016.07.08.14.57
## 1382	220	1	2016.07.11	140	2.44346095	2016.07.11.13.19
## 1383	135	2	2016.07.11	225	-2.35619449	2016.07.11.13.19
## 1384	115	3	2016.07.11	245	-2.00712864	2016.07.11.13.19
## 1385	215	4	2016.07.11	145	2.53072742	2016.07.11.13.19
## 1386	175	5	2016.07.11	185	-3.05432619	2016.07.11.13.19
## 1387	180	6	2016.07.11	180	3.14159265	2016.07.11.13.19
## 1388	175	7	2016.07.11	185	-3.05432619	2016.07.11.13.19
## 1389	200	8	2016.07.11	160	2.79252680	2016.07.11.13.19
## 1390	130	9	2016.07.11	230	-2.26892803	2016.07.11.13.19
## 1391	220	10	2016.07.11	140	2.44346095	2016.07.11.13.19
## 1392	170	11	2016.07.11	190	-2.96705973	2016.07.11.13.19
## 1393	200	12	2016.07.11	160	2.79252680	2016.07.11.13.19
## 1394	160	13	2016.07.11	200	-2.79252680	2016.07.11.13.19
## 1395	220	14	2016.07.11	140	2.44346095	2016.07.11.13.19
## 1396	175	15	2016.07.11	185	-3.05432619	2016.07.11.13.19
## 1397	195	16	2016.07.11	165	2.87979327	2016.07.11.13.19
## 1398	20	17	2016.07.11	340	-0.34906585	2016.07.11.13.19
## 1399	40	18	2016.07.11	320	-0.69813170	2016.07.11.13.19
## 1400	170	19	2016.07.11	190	-2.96705973	2016.07.11.13.19
## 1401	160	20	2016.07.11	200	-2.79252680	2016.07.11.13.19
## 1402	285	21	2016.07.11	75	1.30899694	2016.07.11.13.19
## 1403	270	22	2016.07.11	90	1.57079633	2016.07.11.13.19
## 1404	270	23	2016.07.11	90	1.57079633	2016.07.11.13.19
## 1405	290	24	2016.07.11	70	1.22173048	2016.07.11.13.19
## 1406	270	25	2016.07.11	90	1.57079633	2016.07.11.13.19
## 1407	280	26	2016.07.11	80	1.39626340	2016.07.11.13.19
## 1408	270	27	2016.07.11	90	1.57079633	2016.07.11.13.19
## 1409	275	28	2016.07.11	85	1.48352986	2016.07.11.13.19
## 1410	270	29	2016.07.11	90	1.57079633	2016.07.11.13.19
## 1411	250	30	2016.07.11	110	1.91986218	2016.07.11.13.19
## 1412	290	31	2016.07.11	70	1.22173048	2016.07.11.13.19
## 1413	310	32	2016.07.11	50	0.87266463	2016.07.11.13.19
## 1414	260	33	2016.07.11	100	1.74532925	2016.07.11.13.19
## 1415	230	34	2016.07.11	130	2.26892803	2016.07.11.13.19
## 1416	270	35	2016.07.11	90	1.57079633	2016.07.11.13.19
## 1417	280	36	2016.07.11	80	1.39626340	2016.07.11.13.19
## 1418	270	37	2016.07.11	90	1.57079633	2016.07.11.13.19

## 1419	280	38	2016.07.11	80	1.39626340	2016.07.11.13.19
## 1420	260	39	2016.07.11	100	1.74532925	2016.07.11.13.19
## 1421	275	40	2016.07.11	85	1.48352986	2016.07.11.13.19
## 1422	320	41	2016.07.11	40	0.69813170	2016.07.11.13.19
## 1423	310	42	2016.07.11	50	0.87266463	2016.07.11.13.19
## 1424	280	43	2016.07.11	80	1.39626340	2016.07.11.13.19
## 1425	290	44	2016.07.11	70	1.22173048	2016.07.11.13.19
## 1426	275	45	2016.07.11	85	1.48352986	2016.07.11.13.19
## 1427	280	46	2016.07.11	80	1.39626340	2016.07.11.13.19
## 1428	250	47	2016.07.11	110	1.91986218	2016.07.11.13.19
## 1429	275	48	2016.07.11	85	1.48352986	2016.07.11.13.19
## 1430	285	49	2016.07.11	75	1.30899694	2016.07.11.13.19
## 1431	280	1	2016.07.11	80	1.39626340	2016.07.11.15.41
## 1432	90	2	2016.07.11	270	-1.57079633	2016.07.11.15.41
## 1433	120	3	2016.07.11	240	-2.09439510	2016.07.11.15.41
## 1434	270	4	2016.07.11	90	1.57079633	2016.07.11.15.41
## 1435	260	5	2016.07.11	100	1.74532925	2016.07.11.15.41
## 1436	105	6	2016.07.11	255	-1.83259571	2016.07.11.15.41
## 1437	100	7	2016.07.11	260	-1.74532925	2016.07.11.15.41
## 1438	260	8	2016.07.11	100	1.74532925	2016.07.11.15.41
## 1439	130	9	2016.07.11	230	-2.26892803	2016.07.11.15.41
## 1440	330	10	2016.07.11	30	0.52359878	2016.07.11.15.41
## 1441	135	11	2016.07.11	225	-2.35619449	2016.07.11.15.41
## 1442	105	12	2016.07.11	255	-1.83259571	2016.07.11.15.41
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## 1444	120	14	2016.07.11	240	-2.09439510	2016.07.11.15.41
## 1445	140	15	2016.07.11	220	-2.44346095	2016.07.11.15.41
## 1446	120	16	2016.07.11	240	-2.09439510	2016.07.11.15.41
## 1447	50	17	2016.07.11	310	-0.87266463	2016.07.11.15.41
## 1448	20	18	2016.07.11	340	-0.34906585	2016.07.11.15.41
## 1449	30	19	2016.07.11	330	-0.52359878	2016.07.11.15.41
## 1450	20	20	2016.07.11	340	-0.34906585	2016.07.11.15.41
## 1451	20	21	2016.07.11	340	-0.34906585	2016.07.11.15.41
## 1452	5	22	2016.07.11	355	-0.08726646	2016.07.11.15.41
## 1453	5	23	2016.07.11	355	-0.08726646	2016.07.11.15.41
## 1454	20	24	2016.07.11	340	-0.34906585	2016.07.11.15.41
## 1455	10	25	2016.07.11	350	-0.17453293	2016.07.11.15.41
## 1456	0	26	2016.07.11	0	0.00000000	2016.07.11.15.41
## 1457	10	27	2016.07.11	350	-0.17453293	2016.07.11.15.41
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## 1459	20	29	2016.07.11	340	-0.34906585	2016.07.11.15.41
## 1460	30	30	2016.07.11	330	-0.52359878	2016.07.11.15.41
## 1461	40	31	2016.07.11	320	-0.69813170	2016.07.11.15.41
## 1462	30	32	2016.07.11	330	-0.52359878	2016.07.11.15.41
## 1463	35	33	2016.07.11	325	-0.61086524	2016.07.11.15.41
## 1464	10	34	2016.07.11	350	-0.17453293	2016.07.11.15.41
## 1465	5	35	2016.07.11	355	-0.08726646	2016.07.11.15.41
## 1466	50	36	2016.07.11	310	-0.87266463	2016.07.11.15.41
## 1467	15	37	2016.07.11	345	-0.26179939	2016.07.11.15.41
## 1468	30	38	2016.07.11	330	-0.52359878	2016.07.11.15.41
## 1469	70	39	2016.07.11	290	-1.22173048	2016.07.11.15.41
## 1470	40	40	2016.07.11	320	-0.69813170	2016.07.11.15.41
## 1471	40	41	2016.07.11	320	-0.69813170	2016.07.11.15.41
## 1472	5	42	2016.07.11	355	-0.08726646	2016.07.11.15.41

## 1473	25	43	2016.07.11	335	-0.43633231	2016.07.11.15.41
## 1474	45	44	2016.07.11	315	-0.78539816	2016.07.11.15.41
## 1475	50	45	2016.07.11	310	-0.87266463	2016.07.11.15.41
## 1476	300	1	2016.07.12	60	1.04719755	2016.07.12.13.41
## 1477	0	2	2016.07.12	0	0.00000000	2016.07.12.13.41
## 1478	260	3	2016.07.12	100	1.74532925	2016.07.12.13.41
## 1479	45	4	2016.07.12	315	-0.78539816	2016.07.12.13.41
## 1480	245	5	2016.07.12	115	2.00712864	2016.07.12.13.41
## 1481	240	6	2016.07.12	120	2.09439510	2016.07.12.13.41
## 1482	280	7	2016.07.12	80	1.39626340	2016.07.12.13.41
## 1483	270	8	2016.07.12	90	1.57079633	2016.07.12.13.41
## 1484	250	9	2016.07.12	110	1.91986218	2016.07.12.13.41
## 1485	295	10	2016.07.12	65	1.13446401	2016.07.12.13.41
## 1486	340	11	2016.07.12	20	0.34906585	2016.07.12.13.41
## 1487	320	12	2016.07.12	40	0.69813170	2016.07.12.13.41
## 1488	280	13	2016.07.12	80	1.39626340	2016.07.12.13.41
## 1489	20	14	2016.07.12	340	-0.34906585	2016.07.12.13.41
## 1490	330	15	2016.07.12	30	0.52359878	2016.07.12.13.41
## 1491	345	16	2016.07.12	15	0.26179939	2016.07.12.13.41
## 1492	355	17	2016.07.12	5	0.08726646	2016.07.12.13.41
## 1493	335	18	2016.07.12	25	0.43633231	2016.07.12.13.41
## 1494	350	19	2016.07.12	10	0.17453293	2016.07.12.13.41
## 1495	320	20	2016.07.12	40	0.69813170	2016.07.12.13.41
## 1496	325	21	2016.07.12	35	0.61086524	2016.07.12.13.41
## 1497	325	22	2016.07.12	35	0.61086524	2016.07.12.13.41
## 1498	330	23	2016.07.12	30	0.52359878	2016.07.12.13.41
## 1499	350	24	2016.07.12	10	0.17453293	2016.07.12.13.41
## 1500	350	25	2016.07.12	10	0.17453293	2016.07.12.13.41
## 1501	355	26	2016.07.12	5	0.08726646	2016.07.12.13.41
## 1502	350	27	2016.07.12	10	0.17453293	2016.07.12.13.41
## 1503	0	28	2016.07.12	0	0.00000000	2016.07.12.13.41
## 1504	350	29	2016.07.12	10	0.17453293	2016.07.12.13.41
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## 1507	40	1	2016.07.12	320	-0.69813170	2016.07.12.14.19
## 1508	50	2	2016.07.12	310	-0.87266463	2016.07.12.14.19
## 1509	50	3	2016.07.12	310	-0.87266463	2016.07.12.14.19
## 1510	50	4	2016.07.12	310	-0.87266463	2016.07.12.14.19
## 1511	65	5	2016.07.12	295	-1.13446401	2016.07.12.14.19
## 1512	50	6	2016.07.12	310	-0.87266463	2016.07.12.14.19
## 1513	50	7	2016.07.12	310	-0.87266463	2016.07.12.14.19
## 1514	50	8	2016.07.12	310	-0.87266463	2016.07.12.14.19
## 1515	40	9	2016.07.12	320	-0.69813170	2016.07.12.14.19
## 1516	45	10	2016.07.12	315	-0.78539816	2016.07.12.14.19
## 1517	30	11	2016.07.12	330	-0.52359878	2016.07.12.14.19
## 1518	55	12	2016.07.12	305	-0.95993109	2016.07.12.14.19
## 1519	30	13	2016.07.12	330	-0.52359878	2016.07.12.14.19
## 1520	95	14	2016.07.12	265	-1.65806279	2016.07.12.14.19
## 1521	75	15	2016.07.12	285	-1.30899694	2016.07.12.14.19
## 1522	85	16	2016.07.12	275	-1.48352986	2016.07.12.14.19
## 1523	40	17	2016.07.12	320	-0.69813170	2016.07.12.14.19
## 1524	40	18	2016.07.12	320	-0.69813170	2016.07.12.14.19
## 1525	70	19	2016.07.12	290	-1.22173048	2016.07.12.14.19
## 1526	45	20	2016.07.12	315	-0.78539816	2016.07.12.14.19

## 1527	60	21	2016.07.12	300	-1.04719755	2016.07.12.14.19
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## 1530	45	24	2016.07.12	315	-0.78539816	2016.07.12.14.19
## 1531	85	25	2016.07.12	275	-1.48352986	2016.07.12.14.19
## 1532	55	26	2016.07.12	305	-0.95993109	2016.07.12.14.19
## 1533	85	27	2016.07.12	275	-1.48352986	2016.07.12.14.19
## 1534	70	28	2016.07.12	290	-1.22173048	2016.07.12.14.19
## 1535	80	29	2016.07.12	280	-1.39626340	2016.07.12.14.19
## 1536	60	30	2016.07.12	300	-1.04719755	2016.07.12.14.19
## 1537	80	31	2016.07.12	280	-1.39626340	2016.07.12.14.19
## 1538	80	32	2016.07.12	280	-1.39626340	2016.07.12.14.19
## 1539	115	1	2016.07.12	245	-2.00712864	2016.07.12.14.43
## 1540	310	2	2016.07.12	50	0.87266463	2016.07.12.14.43
## 1541	175	3	2016.07.12	185	-3.05432619	2016.07.12.14.43
## 1542	270	4	2016.07.12	90	1.57079633	2016.07.12.14.43
## 1543	305	5	2016.07.12	55	0.95993109	2016.07.12.14.43
## 1544	310	6	2016.07.12	50	0.87266463	2016.07.12.14.43
## 1545	90	7	2016.07.12	270	-1.57079633	2016.07.12.14.43
## 1546	330	8	2016.07.12	30	0.52359878	2016.07.12.14.43
## 1547	110	9	2016.07.12	250	-1.91986218	2016.07.12.14.43
## 1548	40	10	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1549	340	11	2016.07.12	20	0.34906585	2016.07.12.14.43
## 1550	320	12	2016.07.12	40	0.69813170	2016.07.12.14.43
## 1551	300	13	2016.07.12	60	1.04719755	2016.07.12.14.43
## 1552	330	14	2016.07.12	30	0.52359878	2016.07.12.14.43
## 1553	335	15	2016.07.12	25	0.43633231	2016.07.12.14.43
## 1554	120	16	2016.07.12	240	-2.09439510	2016.07.12.14.43
## 1555	280	17	2016.07.12	80	1.39626340	2016.07.12.14.43
## 1556	335	18	2016.07.12	25	0.43633231	2016.07.12.14.43
## 1557	70	19	2016.07.12	290	-1.22173048	2016.07.12.14.43
## 1558	290	20	2016.07.12	70	1.22173048	2016.07.12.14.43
## 1559	60	21	2016.07.12	300	-1.04719755	2016.07.12.14.43
## 1560	85	22	2016.07.12	275	-1.48352986	2016.07.12.14.43
## 1561	35	23	2016.07.12	325	-0.61086524	2016.07.12.14.43
## 1562	55	24	2016.07.12	305	-0.95993109	2016.07.12.14.43
## 1563	45	25	2016.07.12	315	-0.78539816	2016.07.12.14.43
## 1564	355	26	2016.07.12	5	0.08726646	2016.07.12.14.43
## 1565	20	27	2016.07.12	340	-0.34906585	2016.07.12.14.43
## 1566	15	28	2016.07.12	345	-0.26179939	2016.07.12.14.43
## 1567	25	29	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1568	25	30	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1569	50	31	2016.07.12	310	-0.87266463	2016.07.12.14.43
## 1570	355	32	2016.07.12	5	0.08726646	2016.07.12.14.43
## 1571	355	33	2016.07.12	5	0.08726646	2016.07.12.14.43
## 1572	0	34	2016.07.12	0	0.00000000	2016.07.12.14.43
## 1573	10	35	2016.07.12	350	-0.17453293	2016.07.12.14.43
## 1574	25	36	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1575	30	37	2016.07.12	330	-0.52359878	2016.07.12.14.43
## 1576	5	38	2016.07.12	355	-0.08726646	2016.07.12.14.43
## 1577	25	39	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1578	5	40	2016.07.12	355	-0.08726646	2016.07.12.14.43
## 1579	20	41	2016.07.12	340	-0.34906585	2016.07.12.14.43
## 1580	20	42	2016.07.12	340	-0.34906585	2016.07.12.14.43

## 1581	25	43	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1582	5	44	2016.07.12	355	-0.08726646	2016.07.12.14.43
## 1583	35	45	2016.07.12	325	-0.61086524	2016.07.12.14.43
## 1584	60	46	2016.07.12	300	-1.04719755	2016.07.12.14.43
## 1585	25	47	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1586	50	48	2016.07.12	310	-0.87266463	2016.07.12.14.43
## 1587	10	49	2016.07.12	350	-0.17453293	2016.07.12.14.43
## 1588	25	50	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1589	30	51	2016.07.12	330	-0.52359878	2016.07.12.14.43
## 1590	350	52	2016.07.12	10	0.17453293	2016.07.12.14.43
## 1591	15	53	2016.07.12	345	-0.26179939	2016.07.12.14.43
## 1592	70	54	2016.07.12	290	-1.22173048	2016.07.12.14.43
## 1593	355	55	2016.07.12	5	0.08726646	2016.07.12.14.43
## 1594	0	56	2016.07.12	0	0.00000000	2016.07.12.14.43
## 1595	300	57	2016.07.12	60	1.04719755	2016.07.12.14.43
## 1596	270	58	2016.07.12	90	1.57079633	2016.07.12.14.43
## 1597	40	59	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1598	25	60	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1599	300	61	2016.07.12	60	1.04719755	2016.07.12.14.43
## 1600	165	62	2016.07.12	195	-2.87979327	2016.07.12.14.43
## 1601	40	63	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1602	95	64	2016.07.12	265	-1.65806279	2016.07.12.14.43
## 1603	60	65	2016.07.12	300	-1.04719755	2016.07.12.14.43
## 1604	110	66	2016.07.12	250	-1.91986218	2016.07.12.14.43
## 1605	40	67	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1606	60	68	2016.07.12	300	-1.04719755	2016.07.12.14.43
## 1607	345	69	2016.07.12	15	0.26179939	2016.07.12.14.43
## 1608	305	70	2016.07.12	55	0.95993109	2016.07.12.14.43
## 1609	350	71	2016.07.12	10	0.17453293	2016.07.12.14.43
## 1610	300	72	2016.07.12	60	1.04719755	2016.07.12.14.43
## 1611	40	73	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1612	50	74	2016.07.12	310	-0.87266463	2016.07.12.14.43
## 1613	45	75	2016.07.12	315	-0.78539816	2016.07.12.14.43
## 1614	0	76	2016.07.12	0	0.00000000	2016.07.12.14.43
## 1615	300	77	2016.07.12	60	1.04719755	2016.07.12.14.43
## 1616	270	78	2016.07.12	90	1.57079633	2016.07.12.14.43
## 1617	40	79	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1618	25	80	2016.07.12	335	-0.43633231	2016.07.12.14.43
## 1619	300	81	2016.07.12	60	1.04719755	2016.07.12.14.43
## 1620	165	82	2016.07.12	195	-2.87979327	2016.07.12.14.43
## 1621	40	83	2016.07.12	320	-0.69813170	2016.07.12.14.43
## 1622	95	84	2016.07.12	265	-1.65806279	2016.07.12.14.43
## 1623	60	85	2016.07.12	300	-1.04719755	2016.07.12.14.43
## 1624	110	86	2016.07.12	250	-1.91986218	2016.07.12.14.43
## 1625	345	1	2016.07.12	15	0.26179939	2016.07.12.15.51
## 1626	270	2	2016.07.12	90	1.57079633	2016.07.12.15.51
## 1627	215	3	2016.07.12	145	2.53072742	2016.07.12.15.51
## 1628	330	4	2016.07.12	30	0.52359878	2016.07.12.15.51
## 1629	295	5	2016.07.12	65	1.13446401	2016.07.12.15.51
## 1630	315	6	2016.07.12	45	0.78539816	2016.07.12.15.51
## 1631	295	7	2016.07.12	65	1.13446401	2016.07.12.15.51
## 1632	300	8	2016.07.12	60	1.04719755	2016.07.12.15.51
## 1633	250	9	2016.07.12	110	1.91986218	2016.07.12.15.51
## 1634	300	10	2016.07.12	60	1.04719755	2016.07.12.15.51

## 1635	300	11	2016.07.12	60	1.04719755	2016.07.12.15.51
## 1636	345	12	2016.07.12	15	0.26179939	2016.07.12.15.51
## 1637	355	13	2016.07.12	5	0.08726646	2016.07.12.15.51
## 1638	300	14	2016.07.12	60	1.04719755	2016.07.12.15.51
## 1639	335	15	2016.07.12	25	0.43633231	2016.07.12.15.51
## 1640	210	16	2016.07.12	150	2.61799388	2016.07.12.15.51
## 1641	220	17	2016.07.12	140	2.44346095	2016.07.12.15.51
## 1642	235	18	2016.07.12	125	2.18166156	2016.07.12.15.51
## 1643	220	19	2016.07.12	140	2.44346095	2016.07.12.15.51
## 1644	220	20	2016.07.12	140	2.44346095	2016.07.12.15.51
## 1645	215	21	2016.07.12	145	2.53072742	2016.07.12.15.51
## 1646	220	22	2016.07.12	140	2.44346095	2016.07.12.15.51
## 1647	255	23	2016.07.12	105	1.83259571	2016.07.12.15.51
## 1648	200	24	2016.07.12	160	2.79252680	2016.07.12.15.51
## 1649	335	1	2016.07.12	25	0.43633231	2016.07.12.15.59
## 1650	320	2	2016.07.12	40	0.69813170	2016.07.12.15.59
## 1651	320	3	2016.07.12	40	0.69813170	2016.07.12.15.59
## 1652	300	4	2016.07.12	60	1.04719755	2016.07.12.15.59
## 1653	280	5	2016.07.12	80	1.39626340	2016.07.12.15.59
## 1654	290	6	2016.07.12	70	1.22173048	2016.07.12.15.59
## 1655	280	7	2016.07.12	80	1.39626340	2016.07.12.15.59
## 1656	320	8	2016.07.12	40	0.69813170	2016.07.12.15.59
## 1657	240	9	2016.07.12	120	2.09439510	2016.07.12.15.59
## 1658	290	10	2016.07.12	70	1.22173048	2016.07.12.15.59
## 1659	250	11	2016.07.12	110	1.91986218	2016.07.12.15.59
## 1660	0	12	2016.07.12	0	0.00000000	2016.07.12.15.59
## 1661	270	13	2016.07.12	90	1.57079633	2016.07.12.15.59
## 1662	300	14	2016.07.12	60	1.04719755	2016.07.12.15.59
## 1663	310	15	2016.07.12	50	0.87266463	2016.07.12.15.59
## 1664	315	16	2016.07.12	45	0.78539816	2016.07.12.15.59
## 1665	335	17	2016.07.12	25	0.43633231	2016.07.12.15.59
## 1666	275	18	2016.07.12	85	1.48352986	2016.07.12.15.59
## 1667	240	19	2016.07.12	120	2.09439510	2016.07.12.15.59
## 1668	320	20	2016.07.12	40	0.69813170	2016.07.12.15.59
## 1669	200	21	2016.07.12	160	2.79252680	2016.07.12.15.59
## 1670	185	22	2016.07.12	175	3.05432619	2016.07.12.15.59
## 1671	210	23	2016.07.12	150	2.61799388	2016.07.12.15.59
## 1672	205	24	2016.07.12	155	2.70526034	2016.07.12.15.59
## 1673	200	25	2016.07.12	160	2.79252680	2016.07.12.15.59
## 1674	210	26	2016.07.12	150	2.61799388	2016.07.12.15.59
## 1675	210	27	2016.07.12	150	2.61799388	2016.07.12.15.59
## 1676	220	28	2016.07.12	140	2.44346095	2016.07.12.15.59
## 1677	220	29	2016.07.12	140	2.44346095	2016.07.12.15.59
## 1678	210	30	2016.07.12	150	2.61799388	2016.07.12.15.59
## 1679	220	31	2016.07.12	140	2.44346095	2016.07.12.15.59
## 1680	190	32	2016.07.12	170	2.96705973	2016.07.12.15.59
## 1681	210	33	2016.07.12	150	2.61799388	2016.07.12.15.59
## 1682	75	1	2016.07.12	285	-1.30899694	2016.07.12.16.08
## 1683	15	2	2016.07.12	345	-0.26179939	2016.07.12.16.08
## 1684	285	3	2016.07.12	75	1.30899694	2016.07.12.16.08
## 1685	250	4	2016.07.12	110	1.91986218	2016.07.12.16.08
## 1686	300	5	2016.07.12	60	1.04719755	2016.07.12.16.08
## 1687	310	6	2016.07.12	50	0.87266463	2016.07.12.16.08
## 1688	335	7	2016.07.12	25	0.43633231	2016.07.12.16.08

## 1689	35	8	2016.07.12	325	-0.61086524	2016.07.12.16.08
## 1690	280	9	2016.07.12	80	1.39626340	2016.07.12.16.08
## 1691	280	10	2016.07.12	80	1.39626340	2016.07.12.16.08
## 1692	275	11	2016.07.12	85	1.48352986	2016.07.12.16.08
## 1693	80	12	2016.07.12	280	-1.39626340	2016.07.12.16.08
## 1694	90	13	2016.07.12	270	-1.57079633	2016.07.12.16.08
## 1695	20	14	2016.07.12	340	-0.34906585	2016.07.12.16.08
## 1696	90	15	2016.07.12	270	-1.57079633	2016.07.12.16.08
## 1697	275	16	2016.07.12	85	1.48352986	2016.07.12.16.08
## 1698	330	17	2016.07.12	30	0.52359878	2016.07.12.16.08
## 1699	265	18	2016.07.12	95	1.65806279	2016.07.12.16.08
## 1700	280	19	2016.07.12	80	1.39626340	2016.07.12.16.08
## 1701	315	20	2016.07.12	45	0.78539816	2016.07.12.16.08
## 1702	330	21	2016.07.12	30	0.52359878	2016.07.12.16.08
## 1703	350	22	2016.07.12	10	0.17453293	2016.07.12.16.08
## 1704	345	23	2016.07.12	15	0.26179939	2016.07.12.16.08
## 1705	30	24	2016.07.12	330	-0.52359878	2016.07.12.16.08
## 1706	355	25	2016.07.12	5	0.08726646	2016.07.12.16.08
## 1707	5	26	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1708	5	27	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1709	0	28	2016.07.12	0	0.00000000	2016.07.12.16.08
## 1710	355	29	2016.07.12	5	0.08726646	2016.07.12.16.08
## 1711	0	30	2016.07.12	0	0.00000000	2016.07.12.16.08
## 1712	25	31	2016.07.12	335	-0.43633231	2016.07.12.16.08
## 1713	0	32	2016.07.12	0	0.00000000	2016.07.12.16.08
## 1714	20	33	2016.07.12	340	-0.34906585	2016.07.12.16.08
## 1715	20	34	2016.07.12	340	-0.34906585	2016.07.12.16.08
## 1716	0	35	2016.07.12	0	0.00000000	2016.07.12.16.08
## 1717	5	36	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1718	10	37	2016.07.12	350	-0.17453293	2016.07.12.16.08
## 1719	5	38	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1720	350	39	2016.07.12	10	0.17453293	2016.07.12.16.08
## 1721	10	40	2016.07.12	350	-0.17453293	2016.07.12.16.08
## 1722	345	41	2016.07.12	15	0.26179939	2016.07.12.16.08
## 1723	20	42	2016.07.12	340	-0.34906585	2016.07.12.16.08
## 1724	350	43	2016.07.12	10	0.17453293	2016.07.12.16.08
## 1725	5	44	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1726	20	45	2016.07.12	340	-0.34906585	2016.07.12.16.08
## 1727	5	46	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1728	10	47	2016.07.12	350	-0.17453293	2016.07.12.16.08
## 1729	15	48	2016.07.12	345	-0.26179939	2016.07.12.16.08
## 1730	20	49	2016.07.12	340	-0.34906585	2016.07.12.16.08
## 1731	355	50	2016.07.12	5	0.08726646	2016.07.12.16.08
## 1732	0	51	2016.07.12	0	0.00000000	2016.07.12.16.08
## 1733	50	52	2016.07.12	310	-0.87266463	2016.07.12.16.08
## 1734	5	53	2016.07.12	355	-0.08726646	2016.07.12.16.08
## 1735	345	54	2016.07.12	15	0.26179939	2016.07.12.16.08
## 1736	30	55	2016.07.12	330	-0.52359878	2016.07.12.16.08
## 1737	0	56	2016.07.12	0	0.00000000	2016.07.12.16.08
## 1738	260	57	2016.07.12	100	1.74532925	2016.07.12.16.08
## 1739	50	58	2016.07.12	310	-0.87266463	2016.07.12.16.08
## 1740	10	59	2016.07.12	350	-0.17453293	2016.07.12.16.08
## 1741	350	60	2016.07.12	10	0.17453293	2016.07.12.16.08
## 1742	0	61	2016.07.12	0	0.00000000	2016.07.12.16.08

## 1743	330	62	2016.07.12	30	0.52359878	2016.07.12.16.08
## 1744	280	63	2016.07.12	80	1.39626340	2016.07.12.16.08
## 1745	260	1	2016.07.18	100	1.74532925	2016.07.18.13.31
## 1746	310	2	2016.07.18	50	0.87266463	2016.07.18.13.31
## 1747	340	3	2016.07.18	20	0.34906585	2016.07.18.13.31
## 1748	240	4	2016.07.18	120	2.09439510	2016.07.18.13.31
## 1749	330	5	2016.07.18	30	0.52359878	2016.07.18.13.31
## 1750	260	6	2016.07.18	100	1.74532925	2016.07.18.13.31
## 1751	320	7	2016.07.18	40	0.69813170	2016.07.18.13.31
## 1752	230	8	2016.07.18	130	2.26892803	2016.07.18.13.31
## 1753	310	9	2016.07.18	50	0.87266463	2016.07.18.13.31
## 1754	305	10	2016.07.18	55	0.95993109	2016.07.18.13.31
## 1755	320	11	2016.07.18	40	0.69813170	2016.07.18.13.31
## 1756	240	12	2016.07.18	120	2.09439510	2016.07.18.13.31
## 1757	270	13	2016.07.18	90	1.57079633	2016.07.18.13.31
## 1758	230	14	2016.07.18	130	2.26892803	2016.07.18.13.31
## 1759	345	15	2016.07.18	15	0.26179939	2016.07.18.13.31
## 1760	340	16	2016.07.18	20	0.34906585	2016.07.18.13.31
## 1761	345	17	2016.07.18	15	0.26179939	2016.07.18.13.31
## 1762	355	18	2016.07.18	5	0.08726646	2016.07.18.13.31
## 1763	355	19	2016.07.18	5	0.08726646	2016.07.18.13.31
## 1764	340	20	2016.07.18	20	0.34906585	2016.07.18.13.31
## 1765	0	21	2016.07.18	0	0.00000000	2016.07.18.13.31
## 1766	350	22	2016.07.18	10	0.17453293	2016.07.18.13.31
## 1767	355	23	2016.07.18	5	0.08726646	2016.07.18.13.31
## 1768	345	24	2016.07.18	15	0.26179939	2016.07.18.13.31
## 1769	310	25	2016.07.18	50	0.87266463	2016.07.18.13.31
## 1770	350	26	2016.07.18	10	0.17453293	2016.07.18.13.31
## 1771	0	27	2016.07.18	0	0.00000000	2016.07.18.13.31
## 1772	330	28	2016.07.18	30	0.52359878	2016.07.18.13.31
## 1773	345	29	2016.07.18	15	0.26179939	2016.07.18.13.31
## 1774	325	30	2016.07.18	35	0.61086524	2016.07.18.13.31
## 1775	260	1	2016.07.18	100	1.74532925	2016.07.18.14.19
## 1776	150	2	2016.07.18	210	-2.61799388	2016.07.18.14.19
## 1777	270	3	2016.07.18	90	1.57079633	2016.07.18.14.19
## 1778	270	4	2016.07.18	90	1.57079633	2016.07.18.14.19
## 1779	220	5	2016.07.18	140	2.44346095	2016.07.18.14.19
## 1780	250	6	2016.07.18	110	1.91986218	2016.07.18.14.19
## 1781	215	7	2016.07.18	145	2.53072742	2016.07.18.14.19
## 1782	250	8	2016.07.18	110	1.91986218	2016.07.18.14.19
## 1783	165	9	2016.07.18	195	-2.87979327	2016.07.18.14.19
## 1784	310	10	2016.07.18	50	0.87266463	2016.07.18.14.19
## 1785	255	11	2016.07.18	105	1.83259571	2016.07.18.14.19
## 1786	205	12	2016.07.18	155	2.70526034	2016.07.18.14.19
## 1787	220	13	2016.07.18	140	2.44346095	2016.07.18.14.19
## 1788	250	14	2016.07.18	110	1.91986218	2016.07.18.14.19
## 1789	150	15	2016.07.18	210	-2.61799388	2016.07.18.14.19
## 1790	160	16	2016.07.18	200	-2.79252680	2016.07.18.14.19
## 1791	205	17	2016.07.18	155	2.70526034	2016.07.18.14.19
## 1792	240	18	2016.07.18	120	2.09439510	2016.07.18.14.19
## 1793	185	19	2016.07.18	175	3.05432619	2016.07.18.14.19
## 1794	220	20	2016.07.18	140	2.44346095	2016.07.18.14.19
## 1795	230	21	2016.07.18	130	2.26892803	2016.07.18.14.19
## 1796	160	22	2016.07.18	200	-2.79252680	2016.07.18.14.19

## 1797	210	23	2016.07.18	150	2.61799388	2016.07.18.14.19
## 1798	245	24	2016.07.18	115	2.00712864	2016.07.18.14.19
## 1799	175	25	2016.07.18	185	-3.05432619	2016.07.18.14.19
## 1800	210	26	2016.07.18	150	2.61799388	2016.07.18.14.19
## 1801	215	27	2016.07.18	145	2.53072742	2016.07.18.14.19
## 1802	220	28	2016.07.18	140	2.44346095	2016.07.18.14.19
## 1803	190	29	2016.07.18	170	2.96705973	2016.07.18.14.19
## 1804	230	30	2016.07.18	130	2.26892803	2016.07.18.14.19
## 1805	50	1	2016.07.18	310	-0.87266463	2016.07.18.15.10
## 1806	300	2	2016.07.18	60	1.04719755	2016.07.18.15.10
## 1807	340	3	2016.07.18	20	0.34906585	2016.07.18.15.10
## 1808	250	4	2016.07.18	110	1.91986218	2016.07.18.15.10
## 1809	230	5	2016.07.18	130	2.26892803	2016.07.18.15.10
## 1810	260	6	2016.07.18	100	1.74532925	2016.07.18.15.10
## 1811	90	7	2016.07.18	270	-1.57079633	2016.07.18.15.10
## 1812	30	8	2016.07.18	330	-0.52359878	2016.07.18.15.10
## 1813	220	9	2016.07.18	140	2.44346095	2016.07.18.15.10
## 1814	240	10	2016.07.18	120	2.09439510	2016.07.18.15.10
## 1815	260	11	2016.07.18	100	1.74532925	2016.07.18.15.10
## 1816	260	12	2016.07.18	100	1.74532925	2016.07.18.15.10
## 1817	80	13	2016.07.18	280	-1.39626340	2016.07.18.15.10
## 1818	145	14	2016.07.18	215	-2.53072742	2016.07.18.15.10
## 1819	90	15	2016.07.18	270	-1.57079633	2016.07.18.15.10
## 1820	265	16	2016.07.18	95	1.65806279	2016.07.18.15.10
## 1821	355	17	2016.07.18	5	0.08726646	2016.07.18.15.10
## 1822	0	18	2016.07.18	0	0.00000000	2016.07.18.15.10
## 1823	10	19	2016.07.18	350	-0.17453293	2016.07.18.15.10
## 1824	20	20	2016.07.18	340	-0.34906585	2016.07.18.15.10
## 1825	10	21	2016.07.18	350	-0.17453293	2016.07.18.15.10
## 1826	345	22	2016.07.18	15	0.26179939	2016.07.18.15.10
## 1827	350	23	2016.07.18	10	0.17453293	2016.07.18.15.10
## 1828	335	24	2016.07.18	25	0.43633231	2016.07.18.15.10
## 1829	355	25	2016.07.18	5	0.08726646	2016.07.18.15.10
## 1830	0	26	2016.07.18	0	0.00000000	2016.07.18.15.10
## 1831	5	27	2016.07.18	355	-0.08726646	2016.07.18.15.10
## 1832	10	28	2016.07.18	350	-0.17453293	2016.07.18.15.10
## 1833	10	29	2016.07.18	350	-0.17453293	2016.07.18.15.10
## 1834	340	30	2016.07.18	20	0.34906585	2016.07.18.15.10
## 1835	25	31	2016.07.18	335	-0.43633231	2016.07.18.15.10
## 1836	5	32	2016.07.18	355	-0.08726646	2016.07.18.15.10
## 1837	0	33	2016.07.18	0	0.00000000	2016.07.18.15.10
## 1838	5	34	2016.07.18	355	-0.08726646	2016.07.18.15.10
## 1839	10	35	2016.07.18	350	-0.17453293	2016.07.18.15.10
## 1840	10	36	2016.07.18	350	-0.17453293	2016.07.18.15.10
## 1841	340	37	2016.07.18	20	0.34906585	2016.07.18.15.10
## 1842	25	38	2016.07.18	335	-0.43633231	2016.07.18.15.10
## 1843	5	39	2016.07.18	355	-0.08726646	2016.07.18.15.10
## 1844	20	1	2016.07.18	340	-0.34906585	2016.07.18.15.18
## 1845	190	2	2016.07.18	170	2.96705973	2016.07.18.15.18
## 1846	30	3	2016.07.18	330	-0.52359878	2016.07.18.15.18
## 1847	335	4	2016.07.18	25	0.43633231	2016.07.18.15.18
## 1848	335	5	2016.07.18	25	0.43633231	2016.07.18.15.18
## 1849	110	6	2016.07.18	250	-1.91986218	2016.07.18.15.18
## 1850	25	7	2016.07.18	335	-0.43633231	2016.07.18.15.18

## 1851	320	8	2016.07.18	40	0.69813170	2016.07.18.15.18
## 1852	115	9	2016.07.18	245	-2.00712864	2016.07.18.15.18
## 1853	285	10	2016.07.18	75	1.30899694	2016.07.18.15.18
## 1854	320	11	2016.07.18	40	0.69813170	2016.07.18.15.18
## 1855	125	12	2016.07.18	235	-2.18166156	2016.07.18.15.18
## 1856	290	13	2016.07.18	70	1.22173048	2016.07.18.15.18
## 1857	130	14	2016.07.18	230	-2.26892803	2016.07.18.15.18
## 1858	70	15	2016.07.18	290	-1.22173048	2016.07.18.15.18
## 1859	325	16	2016.07.18	35	0.61086524	2016.07.18.15.18
## 1860	10	17	2016.07.18	350	-0.17453293	2016.07.18.15.18
## 1861	315	18	2016.07.18	45	0.78539816	2016.07.18.15.18
## 1862	30	19	2016.07.18	330	-0.52359878	2016.07.18.15.18
## 1863	355	20	2016.07.18	5	0.08726646	2016.07.18.15.18
## 1864	40	21	2016.07.18	320	-0.69813170	2016.07.18.15.18
## 1865	355	22	2016.07.18	5	0.08726646	2016.07.18.15.18
## 1866	355	23	2016.07.18	5	0.08726646	2016.07.18.15.18
## 1867	10	24	2016.07.18	350	-0.17453293	2016.07.18.15.18
## 1868	20	25	2016.07.18	340	-0.34906585	2016.07.18.15.18
## 1869	30	26	2016.07.18	330	-0.52359878	2016.07.18.15.18
## 1870	35	27	2016.07.18	325	-0.61086524	2016.07.18.15.18
## 1871	10	28	2016.07.18	350	-0.17453293	2016.07.18.15.18
## 1872	20	29	2016.07.18	340	-0.34906585	2016.07.18.15.18
## 1873	50	30	2016.07.18	310	-0.87266463	2016.07.18.15.18
## 1874	30	31	2016.07.18	330	-0.52359878	2016.07.18.15.18
## 1875	0	32	2016.07.18	0	0.00000000	2016.07.18.15.18
## 1876	15	33	2016.07.18	345	-0.26179939	2016.07.18.15.18
## 1877	5	34	2016.07.18	355	-0.08726646	2016.07.18.15.18
## 1878	25	35	2016.07.18	335	-0.43633231	2016.07.18.15.18
## 1879	0	36	2016.07.18	0	0.00000000	2016.07.18.15.18
## 1880	30	37	2016.07.18	330	-0.52359878	2016.07.18.15.18
## 1881	355	38	2016.07.18	5	0.08726646	2016.07.18.15.18
## 1882	50	39	2016.07.18	310	-0.87266463	2016.07.18.15.18
## 1883	15	40	2016.07.18	345	-0.26179939	2016.07.18.15.18
## 1884	20	41	2016.07.18	340	-0.34906585	2016.07.18.15.18
## 1885	15	42	2016.07.18	345	-0.26179939	2016.07.18.15.18
## 1886	70	1	2016.07.18	290	-1.22173048	2016.07.18.15.28
## 1887	10	2	2016.07.18	350	-0.17453293	2016.07.18.15.28
## 1888	110	3	2016.07.18	250	-1.91986218	2016.07.18.15.28
## 1889	80	4	2016.07.18	280	-1.39626340	2016.07.18.15.28
## 1890	270	5	2016.07.18	90	1.57079633	2016.07.18.15.28
## 1891	35	6	2016.07.18	325	-0.61086524	2016.07.18.15.28
## 1892	310	7	2016.07.18	50	0.87266463	2016.07.18.15.28
## 1893	315	8	2016.07.18	45	0.78539816	2016.07.18.15.28
## 1894	345	9	2016.07.18	15	0.26179939	2016.07.18.15.28
## 1895	135	10	2016.07.18	225	-2.35619449	2016.07.18.15.28
## 1896	120	11	2016.07.18	240	-2.09439510	2016.07.18.15.28
## 1897	155	12	2016.07.18	205	-2.70526034	2016.07.18.15.28
## 1898	105	13	2016.07.18	255	-1.83259571	2016.07.18.15.28
## 1899	165	14	2016.07.18	195	-2.87979327	2016.07.18.15.28
## 1900	145	15	2016.07.18	215	-2.53072742	2016.07.18.15.28
## 1901	90	16	2016.07.18	270	-1.57079633	2016.07.18.15.28
## 1902	110	17	2016.07.18	250	-1.91986218	2016.07.18.15.28
## 1903	190	18	2016.07.18	170	2.96705973	2016.07.18.15.28
## 1904	220	19	2016.07.18	140	2.44346095	2016.07.18.15.28

## 1905	90	20	2016.07.18	270	-1.57079633	2016.07.18.15.28
## 1906	160	21	2016.07.18	200	-2.79252680	2016.07.18.15.28
## 1907	60	22	2016.07.18	300	-1.04719755	2016.07.18.15.28
## 1908	45	23	2016.07.18	315	-0.78539816	2016.07.18.15.28
## 1909	45	24	2016.07.18	315	-0.78539816	2016.07.18.15.28
## 1910	355	25	2016.07.18	5	0.08726646	2016.07.18.15.28
## 1911	55	26	2016.07.18	305	-0.95993109	2016.07.18.15.28
## 1912	15	27	2016.07.18	345	-0.26179939	2016.07.18.15.28
## 1913	45	28	2016.07.18	315	-0.78539816	2016.07.18.15.28
## 1914	55	29	2016.07.18	305	-0.95993109	2016.07.18.15.28
## 1915	60	30	2016.07.18	300	-1.04719755	2016.07.18.15.28
## 1916	20	31	2016.07.18	340	-0.34906585	2016.07.18.15.28
## 1917	30	32	2016.07.18	330	-0.52359878	2016.07.18.15.28
## 1918	20	33	2016.07.18	340	-0.34906585	2016.07.18.15.28
## 1919	215	1	2016.07.19	145	2.53072742	2016.07.19.12.15
## 1920	355	2	2016.07.19	5	0.08726646	2016.07.19.12.15
## 1921	210	3	2016.07.19	150	2.61799388	2016.07.19.12.15
## 1922	220	4	2016.07.19	140	2.44346095	2016.07.19.12.15
## 1923	235	5	2016.07.19	125	2.18166156	2016.07.19.12.15
## 1924	270	6	2016.07.19	90	1.57079633	2016.07.19.12.15
## 1925	320	7	2016.07.19	40	0.69813170	2016.07.19.12.15
## 1926	210	8	2016.07.19	150	2.61799388	2016.07.19.12.15
## 1927	250	9	2016.07.19	110	1.91986218	2016.07.19.12.15
## 1928	320	10	2016.07.19	40	0.69813170	2016.07.19.12.15
## 1929	20	11	2016.07.19	340	-0.34906585	2016.07.19.12.15
## 1930	30	12	2016.07.19	330	-0.52359878	2016.07.19.12.15
## 1931	280	13	2016.07.19	80	1.39626340	2016.07.19.12.15
## 1932	245	14	2016.07.19	115	2.00712864	2016.07.19.12.15
## 1933	285	15	2016.07.19	75	1.30899694	2016.07.19.12.15
## 1934	270	16	2016.07.19	90	1.57079633	2016.07.19.12.15
## 1935	275	17	2016.07.19	85	1.48352986	2016.07.19.12.15
## 1936	310	18	2016.07.19	50	0.87266463	2016.07.19.12.15
## 1937	305	19	2016.07.19	55	0.95993109	2016.07.19.12.15
## 1938	320	20	2016.07.19	40	0.69813170	2016.07.19.12.15
## 1939	270	21	2016.07.19	90	1.57079633	2016.07.19.12.15
## 1940	310	22	2016.07.19	50	0.87266463	2016.07.19.12.15
## 1941	325	23	2016.07.19	35	0.61086524	2016.07.19.12.15
## 1942	340	24	2016.07.19	20	0.34906585	2016.07.19.12.15
## 1943	0	25	2016.07.19	0	0.00000000	2016.07.19.12.15
## 1944	345	26	2016.07.19	15	0.26179939	2016.07.19.12.15
## 1945	340	27	2016.07.19	20	0.34906585	2016.07.19.12.15
## 1946	300	28	2016.07.19	60	1.04719755	2016.07.19.12.15
## 1947	350	29	2016.07.19	10	0.17453293	2016.07.19.12.15
## 1948	340	30	2016.07.19	20	0.34906585	2016.07.19.12.15
## 1949	345	31	2016.07.19	15	0.26179939	2016.07.19.12.15
## 1950	350	32	2016.07.19	10	0.17453293	2016.07.19.12.15
## 1951	350	33	2016.07.19	10	0.17453293	2016.07.19.12.15
## 1952	345	34	2016.07.19	15	0.26179939	2016.07.19.12.15
## 1953	350	35	2016.07.19	10	0.17453293	2016.07.19.12.15
## 1954	315	36	2016.07.19	45	0.78539816	2016.07.19.12.15
## 1955	340	37	2016.07.19	20	0.34906585	2016.07.19.12.15
## 1956	345	38	2016.07.19	15	0.26179939	2016.07.19.12.15
## 1957	340	39	2016.07.19	20	0.34906585	2016.07.19.12.15
## 1958	310	40	2016.07.19	50	0.87266463	2016.07.19.12.15

## 1959	50	1	2016.07.19	310	-0.87266463	2016.07.19.12.59
## 1960	80	2	2016.07.19	280	-1.39626340	2016.07.19.12.59
## 1961	100	3	2016.07.19	260	-1.74532925	2016.07.19.12.59
## 1962	60	4	2016.07.19	300	-1.04719755	2016.07.19.12.59
## 1963	200	5	2016.07.19	160	2.79252680	2016.07.19.12.59
## 1964	330	6	2016.07.19	30	0.52359878	2016.07.19.12.59
## 1965	80	7	2016.07.19	280	-1.39626340	2016.07.19.12.59
## 1966	90	8	2016.07.19	270	-1.57079633	2016.07.19.12.59
## 1967	110	9	2016.07.19	250	-1.91986218	2016.07.19.12.59
## 1968	30	10	2016.07.19	330	-0.52359878	2016.07.19.12.59
## 1969	90	11	2016.07.19	270	-1.57079633	2016.07.19.12.59
## 1970	35	12	2016.07.19	325	-0.61086524	2016.07.19.12.59
## 1971	95	13	2016.07.19	265	-1.65806279	2016.07.19.12.59
## 1972	70	14	2016.07.19	290	-1.22173048	2016.07.19.12.59
## 1973	80	15	2016.07.19	280	-1.39626340	2016.07.19.12.59
## 1974	10	16	2016.07.19	350	-0.17453293	2016.07.19.12.59
## 1975	70	17	2016.07.19	290	-1.22173048	2016.07.19.12.59
## 1976	60	18	2016.07.19	300	-1.04719755	2016.07.19.12.59
## 1977	0	19	2016.07.19	0	0.00000000	2016.07.19.12.59
## 1978	35	20	2016.07.19	325	-0.61086524	2016.07.19.12.59
## 1979	20	21	2016.07.19	340	-0.34906585	2016.07.19.12.59
## 1980	20	22	2016.07.19	340	-0.34906585	2016.07.19.12.59
## 1981	25	23	2016.07.19	335	-0.43633231	2016.07.19.12.59
## 1982	25	24	2016.07.19	335	-0.43633231	2016.07.19.12.59
## 1983	70	25	2016.07.19	290	-1.22173048	2016.07.19.12.59
## 1984	20	26	2016.07.19	340	-0.34906585	2016.07.19.12.59
## 1985	50	27	2016.07.19	310	-0.87266463	2016.07.19.12.59
## 1986	30	28	2016.07.19	330	-0.52359878	2016.07.19.12.59
## 1987	60	29	2016.07.19	300	-1.04719755	2016.07.19.12.59
## 1988	30	30	2016.07.19	330	-0.52359878	2016.07.19.12.59
## 1989	40	31	2016.07.19	320	-0.69813170	2016.07.19.12.59
## 1990	60	32	2016.07.19	300	-1.04719755	2016.07.19.12.59
## 1991	20	33	2016.07.19	340	-0.34906585	2016.07.19.12.59
## 1992	50	34	2016.07.19	310	-0.87266463	2016.07.19.12.59
## 1993	40	35	2016.07.19	320	-0.69813170	2016.07.19.12.59
## 1994	40	36	2016.07.19	320	-0.69813170	2016.07.19.12.59
## 1995	160	1	2016.07.19	200	-2.79252680	2016.07.19.13.04
## 1996	150	2	2016.07.19	210	-2.61799388	2016.07.19.13.04
## 1997	200	3	2016.07.19	160	2.79252680	2016.07.19.13.04
## 1998	25	4	2016.07.19	335	-0.43633231	2016.07.19.13.04
## 1999	100	5	2016.07.19	260	-1.74532925	2016.07.19.13.04
## 2000	200	6	2016.07.19	160	2.79252680	2016.07.19.13.04
## 2001	340	7	2016.07.19	20	0.34906585	2016.07.19.13.04
## 2002	15	8	2016.07.19	345	-0.26179939	2016.07.19.13.04
## 2003	300	9	2016.07.19	60	1.04719755	2016.07.19.13.04
## 2004	330	10	2016.07.19	30	0.52359878	2016.07.19.13.04
## 2005	345	11	2016.07.19	15	0.26179939	2016.07.19.13.04
## 2006	250	12	2016.07.19	110	1.91986218	2016.07.19.13.04
## 2007	340	13	2016.07.19	20	0.34906585	2016.07.19.13.04
## 2008	0	14	2016.07.19	0	0.00000000	2016.07.19.13.04
## 2009	300	15	2016.07.19	60	1.04719755	2016.07.19.13.04
## 2010	10	16	2016.07.19	350	-0.17453293	2016.07.19.13.04
## 2011	310	17	2016.07.19	50	0.87266463	2016.07.19.13.04
## 2012	310	18	2016.07.19	50	0.87266463	2016.07.19.13.04

## 2013	345	19	2016.07.19	15	0.26179939	2016.07.19.13.04
## 2014	355	20	2016.07.19	5	0.08726646	2016.07.19.13.04
## 2015	305	21	2016.07.19	55	0.95993109	2016.07.19.13.04
## 2016	325	22	2016.07.19	35	0.61086524	2016.07.19.13.04
## 2017	325	23	2016.07.19	35	0.61086524	2016.07.19.13.04
## 2018	350	24	2016.07.19	10	0.17453293	2016.07.19.13.04
## 2019	350	25	2016.07.19	10	0.17453293	2016.07.19.13.04
## 2020	0	26	2016.07.19	0	0.00000000	2016.07.19.13.04
## 2021	355	27	2016.07.19	5	0.08726646	2016.07.19.13.04
## 2022	350	28	2016.07.19	10	0.17453293	2016.07.19.13.04
## 2023	340	29	2016.07.19	20	0.34906585	2016.07.19.13.04
## 2024	310	30	2016.07.19	50	0.87266463	2016.07.19.13.04
## 2025	345	31	2016.07.19	15	0.26179939	2016.07.19.13.04
## 2026	340	32	2016.07.19	20	0.34906585	2016.07.19.13.04
## 2027	355	33	2016.07.19	5	0.08726646	2016.07.19.13.04
## 2028	340	34	2016.07.19	20	0.34906585	2016.07.19.13.04
## 2029	340	35	2016.07.19	20	0.34906585	2016.07.19.13.04
## 2030	330	36	2016.07.19	30	0.52359878	2016.07.19.13.04
## 2031	330	37	2016.07.19	30	0.52359878	2016.07.19.13.04
## 2032	90	1	2016.08.17	270	-1.57079633	2016.08.17.13.46
## 2033	80	2	2016.08.17	280	-1.39626340	2016.08.17.13.46
## 2034	40	3	2016.08.17	320	-0.69813170	2016.08.17.13.46
## 2035	70	4	2016.08.17	290	-1.22173048	2016.08.17.13.46
## 2036	40	5	2016.08.17	320	-0.69813170	2016.08.17.13.46
## 2037	50	6	2016.08.17	310	-0.87266463	2016.08.17.13.46
## 2038	350	7	2016.08.17	10	0.17453293	2016.08.17.13.46
## 2039	355	8	2016.08.17	5	0.08726646	2016.08.17.13.46
## 2040	70	9	2016.08.17	290	-1.22173048	2016.08.17.13.46
## 2041	90	10	2016.08.17	270	-1.57079633	2016.08.17.13.46
## 2042	70	11	2016.08.17	290	-1.22173048	2016.08.17.13.46
## 2043	75	12	2016.08.17	285	-1.30899694	2016.08.17.13.46
## 2044	345	13	2016.08.17	15	0.26179939	2016.08.17.13.46
## 2045	290	14	2016.08.17	70	1.22173048	2016.08.17.13.46
## 2046	290	15	2016.08.17	70	1.22173048	2016.08.17.13.46
## 2047	295	16	2016.08.17	65	1.13446401	2016.08.17.13.46
## 2048	310	17	2016.08.17	50	0.87266463	2016.08.17.13.46
## 2049	290	18	2016.08.17	70	1.22173048	2016.08.17.13.46
## 2050	290	19	2016.08.17	70	1.22173048	2016.08.17.13.46
## 2051	310	20	2016.08.17	50	0.87266463	2016.08.17.13.46
## 2052	290	21	2016.08.17	70	1.22173048	2016.08.17.13.46
## 2053	280	22	2016.08.17	80	1.39626340	2016.08.17.13.46
## 2054	345	23	2016.08.17	15	0.26179939	2016.08.17.13.46
## 2055	330	24	2016.08.17	30	0.52359878	2016.08.17.13.46
## 2056	295	25	2016.08.17	65	1.13446401	2016.08.17.13.46
## 2057	185	1	2016.08.17	175	3.05432619	2016.08.17.14.47
## 2058	65	2	2016.08.17	295	-1.13446401	2016.08.17.14.47
## 2059	120	3	2016.08.17	240	-2.09439510	2016.08.17.14.47
## 2060	185	4	2016.08.17	175	3.05432619	2016.08.17.14.47
## 2061	200	5	2016.08.17	160	2.79252680	2016.08.17.14.47
## 2062	90	6	2016.08.17	270	-1.57079633	2016.08.17.14.47
## 2063	175	7	2016.08.17	185	-3.05432619	2016.08.17.14.47
## 2064	155	8	2016.08.17	205	-2.70526034	2016.08.17.14.47
## 2065	105	9	2016.08.17	255	-1.83259571	2016.08.17.14.47
## 2066	115	10	2016.08.17	245	-2.00712864	2016.08.17.14.47

## 2067	50	11	2016.08.17	310	-0.87266463	2016.08.17.14.47
## 2068	90	12	2016.08.17	270	-1.57079633	2016.08.17.14.47
## 2069	180	13	2016.08.17	180	3.14159265	2016.08.17.14.47
## 2070	140	14	2016.08.17	220	-2.44346095	2016.08.17.14.47
## 2071	110	15	2016.08.17	250	-1.91986218	2016.08.17.14.47
## 2072	290	25	2016.08.17	70	1.22173048	2016.08.17.14.47
## 2073	300	26	2016.08.17	60	1.04719755	2016.08.17.14.47
## 2074	330	27	2016.08.17	30	0.52359878	2016.08.17.14.47
## 2075	310	28	2016.08.17	50	0.87266463	2016.08.17.14.47
## 2076	330	29	2016.08.17	30	0.52359878	2016.08.17.14.47
## 2077	320	30	2016.08.17	40	0.69813170	2016.08.17.14.47
## 2078	300	31	2016.08.17	60	1.04719755	2016.08.17.14.47
## 2079	320	32	2016.08.17	40	0.69813170	2016.08.17.14.47
## 2080	330	33	2016.08.17	30	0.52359878	2016.08.17.14.47
## 2081	320	34	2016.08.17	40	0.69813170	2016.08.17.14.47
## 2082	300	35	2016.08.17	60	1.04719755	2016.08.17.14.47
## 2083	310	36	2016.08.17	50	0.87266463	2016.08.17.14.47
## 2084	345	37	2016.08.17	15	0.26179939	2016.08.17.14.47
## 2085	325	38	2016.08.17	35	0.61086524	2016.08.17.14.47
## 2086	310	39	2016.08.17	50	0.87266463	2016.08.17.14.47
## 2087	340	40	2016.08.17	20	0.34906585	2016.08.17.14.47
## 2088	190	1	2016.08.17	170	2.96705973	2016.08.17.16.22
## 2089	280	2	2016.08.17	80	1.39626340	2016.08.17.16.22
## 2090	90	3	2016.08.17	270	-1.57079633	2016.08.17.16.22
## 2091	135	4	2016.08.17	225	-2.35619449	2016.08.17.16.22
## 2092	170	5	2016.08.17	190	-2.96705973	2016.08.17.16.22
## 2093	150	6	2016.08.17	210	-2.61799388	2016.08.17.16.22
## 2094	200	7	2016.08.17	160	2.79252680	2016.08.17.16.22
## 2095	120	8	2016.08.17	240	-2.09439510	2016.08.17.16.22
## 2096	110	9	2016.08.17	250	-1.91986218	2016.08.17.16.22
## 2097	100	10	2016.08.17	260	-1.74532925	2016.08.17.16.22
## 2098	330	11	2016.08.17	30	0.52359878	2016.08.17.16.22
## 2099	100	12	2016.08.17	260	-1.74532925	2016.08.17.16.22
## 2100	50	13	2016.08.17	310	-0.87266463	2016.08.17.16.22
## 2101	120	14	2016.08.17	240	-2.09439510	2016.08.17.16.22
## 2102	10	15	2016.08.17	350	-0.17453293	2016.08.17.16.22
## 2103	160	16	2016.08.17	200	-2.79252680	2016.08.17.16.22
## 2104	150	17	2016.08.17	210	-2.61799388	2016.08.17.16.22
## 2105	110	18	2016.08.17	250	-1.91986218	2016.08.17.16.22
## 2106	190	19	2016.08.17	170	2.96705973	2016.08.17.16.22
## 2107	170	20	2016.08.17	190	-2.96705973	2016.08.17.16.22
## 2108	340	21	2016.08.17	20	0.34906585	2016.08.17.16.22
## 2109	320	22	2016.08.17	40	0.69813170	2016.08.17.16.22
## 2110	320	23	2016.08.17	40	0.69813170	2016.08.17.16.22
## 2111	350	24	2016.08.17	10	0.17453293	2016.08.17.16.22
## 2112	340	25	2016.08.17	20	0.34906585	2016.08.17.16.22
## 2113	330	26	2016.08.17	30	0.52359878	2016.08.17.16.22
## 2114	340	27	2016.08.17	20	0.34906585	2016.08.17.16.22
## 2115	320	28	2016.08.17	40	0.69813170	2016.08.17.16.22
## 2116	310	29	2016.08.17	50	0.87266463	2016.08.17.16.22
## 2117	300	30	2016.08.17	60	1.04719755	2016.08.17.16.22
## 2118	335	31	2016.08.17	25	0.43633231	2016.08.17.16.22
## 2119	320	32	2016.08.17	40	0.69813170	2016.08.17.16.22
## 2120	340	1	2016.08.17	20	0.34906585	2016.08.17.16.27

## 2121	310	2	2016.08.17	50	0.87266463	2016.08.17.16.27
## 2122	110	3	2016.08.17	250	-1.91986218	2016.08.17.16.27
## 2123	160	4	2016.08.17	200	-2.79252680	2016.08.17.16.27
## 2124	350	5	2016.08.17	10	0.17453293	2016.08.17.16.27
## 2125	245	6	2016.08.17	115	2.00712864	2016.08.17.16.27
## 2126	60	7	2016.08.17	300	-1.04719755	2016.08.17.16.27
## 2127	40	8	2016.08.17	320	-0.69813170	2016.08.17.16.27
## 2128	50	9	2016.08.17	310	-0.87266463	2016.08.17.16.27
## 2129	280	10	2016.08.17	80	1.39626340	2016.08.17.16.27
## 2130	40	11	2016.08.17	320	-0.69813170	2016.08.17.16.27
## 2131	10	12	2016.08.17	350	-0.17453293	2016.08.17.16.27
## 2132	260	13	2016.08.17	100	1.74532925	2016.08.17.16.27
## 2133	210	14	2016.08.17	150	2.61799388	2016.08.17.16.27
## 2134	140	15	2016.08.17	220	-2.44346095	2016.08.17.16.27
## 2135	230	16	2016.08.17	130	2.26892803	2016.08.17.16.27
## 2136	70	17	2016.08.17	290	-1.22173048	2016.08.17.16.27
## 2137	210	18	2016.08.17	150	2.61799388	2016.08.17.16.27
## 2138	320	19	2016.08.17	40	0.69813170	2016.08.17.16.27
## 2139	350	20	2016.08.17	10	0.17453293	2016.08.17.16.27
## 2140	345	21	2016.08.17	15	0.26179939	2016.08.17.16.27
## 2141	290	22	2016.08.17	70	1.22173048	2016.08.17.16.27
## 2142	290	23	2016.08.17	70	1.22173048	2016.08.17.16.27
## 2143	330	24	2016.08.17	30	0.52359878	2016.08.17.16.27
## 2144	335	25	2016.08.17	25	0.43633231	2016.08.17.16.27
## 2145	300	26	2016.08.17	60	1.04719755	2016.08.17.16.27
## 2146	300	27	2016.08.17	60	1.04719755	2016.08.17.16.27
## 2147	350	28	2016.08.17	10	0.17453293	2016.08.17.16.27
## 2148	300	29	2016.08.17	60	1.04719755	2016.08.17.16.27
## 2149	305	30	2016.08.17	55	0.95993109	2016.08.17.16.27
## 2150	300	31	2016.08.17	60	1.04719755	2016.08.17.16.27
## 2151	295	32	2016.08.17	65	1.13446401	2016.08.17.16.27
## 2152	280	33	2016.08.17	80	1.39626340	2016.08.17.16.27
## 2153	30	1	2016.08.18	330	-0.52359878	2016.08.18.14.01
## 2154	345	2	2016.08.18	15	0.26179939	2016.08.18.14.01
## 2155	190	3	2016.08.18	170	2.96705973	2016.08.18.14.01
## 2156	180	4	2016.08.18	180	3.14159265	2016.08.18.14.01
## 2157	240	5	2016.08.18	120	2.09439510	2016.08.18.14.01
## 2158	340	6	2016.08.18	20	0.34906585	2016.08.18.14.01
## 2159	0	7	2016.08.18	0	0.00000000	2016.08.18.14.01
## 2160	245	8	2016.08.18	115	2.00712864	2016.08.18.14.01
## 2161	10	9	2016.08.18	350	-0.17453293	2016.08.18.14.01
## 2162	280	10	2016.08.18	80	1.39626340	2016.08.18.14.01
## 2163	15	11	2016.08.18	345	-0.26179939	2016.08.18.14.01
## 2164	270	12	2016.08.18	90	1.57079633	2016.08.18.14.01
## 2165	260	13	2016.08.18	100	1.74532925	2016.08.18.14.01
## 2166	330	14	2016.08.18	30	0.52359878	2016.08.18.14.01
## 2167	270	15	2016.08.18	90	1.57079633	2016.08.18.14.01
## 2168	320	16	2016.08.18	40	0.69813170	2016.08.18.14.01
## 2169	340	17	2016.08.18	20	0.34906585	2016.08.18.14.01
## 2170	320	18	2016.08.18	40	0.69813170	2016.08.18.14.01
## 2171	340	19	2016.08.18	20	0.34906585	2016.08.18.14.01
## 2172	290	20	2016.08.18	70	1.22173048	2016.08.18.14.01
## 2173	275	21	2016.08.18	85	1.48352986	2016.08.18.14.01
## 2174	270	22	2016.08.18	90	1.57079633	2016.08.18.14.01

## 2175	260	23	2016.08.18	100	1.74532925	2016.08.18.14.01
## 2176	290	24	2016.08.18	70	1.22173048	2016.08.18.14.01
## 2177	270	25	2016.08.18	90	1.57079633	2016.08.18.14.01
## 2178	240	26	2016.08.18	120	2.09439510	2016.08.18.14.01
## 2179	320	27	2016.08.18	40	0.69813170	2016.08.18.14.01
## 2180	290	28	2016.08.18	70	1.22173048	2016.08.18.14.01
## 2181	320	29	2016.08.18	40	0.69813170	2016.08.18.14.01
## 2182	280	30	2016.08.18	80	1.39626340	2016.08.18.14.01
## 2183	50	1	2016.08.18	310	-0.87266463	2016.08.18.16.14
## 2184	45	2	2016.08.18	315	-0.78539816	2016.08.18.16.14
## 2185	90	3	2016.08.18	270	-1.57079633	2016.08.18.16.14
## 2186	100	4	2016.08.18	260	-1.74532925	2016.08.18.16.14
## 2187	70	5	2016.08.18	290	-1.22173048	2016.08.18.16.14
## 2188	75	6	2016.08.18	285	-1.30899694	2016.08.18.16.14
## 2189	35	7	2016.08.18	325	-0.61086524	2016.08.18.16.14
## 2190	220	8	2016.08.18	140	2.44346095	2016.08.18.16.14
## 2191	110	9	2016.08.18	250	-1.91986218	2016.08.18.16.14
## 2192	80	10	2016.08.18	280	-1.39626340	2016.08.18.16.14
## 2193	70	11	2016.08.18	290	-1.22173048	2016.08.18.16.14
## 2194	50	12	2016.08.18	310	-0.87266463	2016.08.18.16.14
## 2195	85	13	2016.08.18	275	-1.48352986	2016.08.18.16.14
## 2196	80	14	2016.08.18	280	-1.39626340	2016.08.18.16.14
## 2197	110	15	2016.08.18	250	-1.91986218	2016.08.18.16.14
## 2198	170	16	2016.08.18	190	-2.96705973	2016.08.18.16.14
## 2199	80	17	2016.08.18	280	-1.39626340	2016.08.18.16.14
## 2200	150	18	2016.08.18	210	-2.61799388	2016.08.18.16.14
## 2201	160	19	2016.08.18	200	-2.79252680	2016.08.18.16.14
## 2202	340	20	2016.08.18	20	0.34906585	2016.08.18.16.14
## 2203	345	21	2016.08.18	15	0.26179939	2016.08.18.16.14
## 2204	350	22	2016.08.18	10	0.17453293	2016.08.18.16.14
## 2205	310	23	2016.08.18	50	0.87266463	2016.08.18.16.14
## 2206	5	24	2016.08.18	355	-0.08726646	2016.08.18.16.14
## 2207	345	25	2016.08.18	15	0.26179939	2016.08.18.16.14
## 2208	340	26	2016.08.18	20	0.34906585	2016.08.18.16.14
## 2209	355	27	2016.08.18	5	0.08726646	2016.08.18.16.14
## 2210	320	28	2016.08.18	40	0.69813170	2016.08.18.16.14
## 2211	340	29	2016.08.18	20	0.34906585	2016.08.18.16.14
## 2212	70	1	2016.08.18	290	-1.22173048	2016.08.18.16.20_2
## 2213	75	2	2016.08.18	285	-1.30899694	2016.08.18.16.20_2
## 2214	110	3	2016.08.18	250	-1.91986218	2016.08.18.16.20_2
## 2215	240	4	2016.08.18	120	2.09439510	2016.08.18.16.20_2
## 2216	210	5	2016.08.18	150	2.61799388	2016.08.18.16.20_2
## 2217	200	6	2016.08.18	160	2.79252680	2016.08.18.16.20_2
## 2218	120	7	2016.08.18	240	-2.09439510	2016.08.18.16.20_2
## 2219	95	8	2016.08.18	265	-1.65806279	2016.08.18.16.20_2
## 2220	200	9	2016.08.18	160	2.79252680	2016.08.18.16.20_2
## 2221	85	10	2016.08.18	275	-1.48352986	2016.08.18.16.20_2
## 2222	20	11	2016.08.18	340	-0.34906585	2016.08.18.16.20_2
## 2223	350	12	2016.08.18	10	0.17453293	2016.08.18.16.20_2
## 2224	345	13	2016.08.18	15	0.26179939	2016.08.18.16.20_2
## 2225	5	14	2016.08.18	355	-0.08726646	2016.08.18.16.20_2
## 2226	340	15	2016.08.18	20	0.34906585	2016.08.18.16.20_2
## 2227	0	16	2016.08.18	0	0.00000000	2016.08.18.16.20_2
## 2228	340	17	2016.08.18	20	0.34906585	2016.08.18.16.20_2

## 2229	350	18	2016.08.18	10	0.17453293	2016.08.18.16.20_2
## 2230	355	19	2016.08.18	5	0.08726646	2016.08.18.16.20_2
## 2231	20	20	2016.08.18	340	-0.34906585	2016.08.18.16.20_2
## 2232	340	21	2016.08.18	20	0.34906585	2016.08.18.16.20_2
## 2233	0	22	2016.08.18	0	0.00000000	2016.08.18.16.20_2
## 2234	5	23	2016.08.18	355	-0.08726646	2016.08.18.16.20_2
## 2235	20	24	2016.08.18	340	-0.34906585	2016.08.18.16.20_2
## 2236	355	25	2016.08.18	5	0.08726646	2016.08.18.16.20_2
## 2237	180	1	2016.08.18	180	3.14159265	2016.08.18.16.20_1
## 2238	205	2	2016.08.18	155	2.70526034	2016.08.18.16.20_1
## 2239	195	3	2016.08.18	165	2.87979327	2016.08.18.16.20_1
## 2240	170	4	2016.08.18	190	-2.96705973	2016.08.18.16.20_1
## 2241	240	5	2016.08.18	120	2.09439510	2016.08.18.16.20_1
## 2242	170	6	2016.08.18	190	-2.96705973	2016.08.18.16.20_1
## 2243	240	7	2016.08.18	120	2.09439510	2016.08.18.16.20_1
## 2244	130	8	2016.08.18	230	-2.26892803	2016.08.18.16.20_1
## 2245	200	9	2016.08.18	160	2.79252680	2016.08.18.16.20_1
## 2246	180	10	2016.08.18	180	3.14159265	2016.08.18.16.20_1
## 2247	170	11	2016.08.18	190	-2.96705973	2016.08.18.16.20_1
## 2248	130	12	2016.08.18	230	-2.26892803	2016.08.18.16.20_1
## 2249	20	13	2016.08.18	340	-0.34906585	2016.08.18.16.20_1
## 2250	0	14	2016.08.18	0	0.00000000	2016.08.18.16.20_1
## 2251	355	15	2016.08.18	5	0.08726646	2016.08.18.16.20_1
## 2252	350	16	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2253	350	17	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2254	350	18	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2255	320	19	2016.08.18	40	0.69813170	2016.08.18.16.20_1
## 2256	350	20	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2257	355	21	2016.08.18	5	0.08726646	2016.08.18.16.20_1
## 2258	345	22	2016.08.18	15	0.26179939	2016.08.18.16.20_1
## 2259	350	23	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2260	10	24	2016.08.18	350	-0.17453293	2016.08.18.16.20_1
## 2261	0	25	2016.08.18	0	0.00000000	2016.08.18.16.20_1
## 2262	5	26	2016.08.18	355	-0.08726646	2016.08.18.16.20_1
## 2263	0	27	2016.08.18	0	0.00000000	2016.08.18.16.20_1
## 2264	5	28	2016.08.18	355	-0.08726646	2016.08.18.16.20_1
## 2265	350	29	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2266	15	30	2016.08.18	345	-0.26179939	2016.08.18.16.20_1
## 2267	0	31	2016.08.18	0	0.00000000	2016.08.18.16.20_1
## 2268	355	32	2016.08.18	5	0.08726646	2016.08.18.16.20_1
## 2269	5	33	2016.08.18	355	-0.08726646	2016.08.18.16.20_1
## 2270	350	34	2016.08.18	10	0.17453293	2016.08.18.16.20_1
## 2271	100	1	2016.08.18	260	-1.74532925	2016.08.18.16.26_1
## 2272	110	2	2016.08.18	250	-1.91986218	2016.08.18.16.26_1
## 2273	70	3	2016.08.18	290	-1.22173048	2016.08.18.16.26_1
## 2274	40	4	2016.08.18	320	-0.69813170	2016.08.18.16.26_1
## 2275	110	5	2016.08.18	250	-1.91986218	2016.08.18.16.26_1
## 2276	30	6	2016.08.18	330	-0.52359878	2016.08.18.16.26_1
## 2277	40	7	2016.08.18	320	-0.69813170	2016.08.18.16.26_1
## 2278	80	8	2016.08.18	280	-1.39626340	2016.08.18.16.26_1
## 2279	180	9	2016.08.18	180	3.14159265	2016.08.18.16.26_1
## 2280	180	10	2016.08.18	180	3.14159265	2016.08.18.16.26_1
## 2281	110	11	2016.08.18	250	-1.91986218	2016.08.18.16.26_1
## 2282	95	12	2016.08.18	265	-1.65806279	2016.08.18.16.26_1

## 2283	210	13	2016.08.18	150	2.61799388	2016.08.18.16.26_1
## 2284	30	14	2016.08.18	330	-0.52359878	2016.08.18.16.26_1
## 2285	50	15	2016.08.18	310	-0.87266463	2016.08.18.16.26_1
## 2286	20	16	2016.08.18	340	-0.34906585	2016.08.18.16.26_1
## 2287	300	17	2016.08.18	60	1.04719755	2016.08.18.16.26_1
## 2288	0	18	2016.08.18	0	0.00000000	2016.08.18.16.26_1
## 2289	10	19	2016.08.18	350	-0.17453293	2016.08.18.16.26_1
## 2290	355	20	2016.08.18	5	0.08726646	2016.08.18.16.26_1
## 2291	340	21	2016.08.18	20	0.34906585	2016.08.18.16.26_1
## 2292	355	22	2016.08.18	5	0.08726646	2016.08.18.16.26_1
## 2293	330	23	2016.08.18	30	0.52359878	2016.08.18.16.26_1
## 2294	355	24	2016.08.18	5	0.08726646	2016.08.18.16.26_1
## 2295	280	25	2016.08.18	80	1.39626340	2016.08.18.16.26_1
## 2296	340	26	2016.08.18	20	0.34906585	2016.08.18.16.26_1
## 2297	90	1	2016.08.18	270	-1.57079633	2016.08.18.16.27_2
## 2298	350	2	2016.08.18	10	0.17453293	2016.08.18.16.27_2
## 2299	90	3	2016.08.18	270	-1.57079633	2016.08.18.16.27_2
## 2300	330	4	2016.08.18	30	0.52359878	2016.08.18.16.27_2
## 2301	65	5	2016.08.18	295	-1.13446401	2016.08.18.16.27_2
## 2302	95	6	2016.08.18	265	-1.65806279	2016.08.18.16.27_2
## 2303	355	7	2016.08.18	5	0.08726646	2016.08.18.16.27_2
## 2304	30	8	2016.08.18	330	-0.52359878	2016.08.18.16.27_2
## 2305	355	9	2016.08.18	5	0.08726646	2016.08.18.16.27_2
## 2306	340	10	2016.08.18	20	0.34906585	2016.08.18.16.27_2
## 2307	350	11	2016.08.18	10	0.17453293	2016.08.18.16.27_2
## 2308	340	12	2016.08.18	20	0.34906585	2016.08.18.16.27_2
## 2309	355	13	2016.08.18	5	0.08726646	2016.08.18.16.27_2
## 2310	345	14	2016.08.18	15	0.26179939	2016.08.18.16.27_2
## 2311	145	1	2016.08.19	215	-2.53072742	2016.08.19.14.26
## 2312	30	2	2016.08.19	330	-0.52359878	2016.08.19.14.26
## 2313	150	3	2016.08.19	210	-2.61799388	2016.08.19.14.26
## 2314	70	4	2016.08.19	290	-1.22173048	2016.08.19.14.26
## 2315	60	5	2016.08.19	300	-1.04719755	2016.08.19.14.26
## 2316	170	6	2016.08.19	190	-2.96705973	2016.08.19.14.26
## 2317	350	7	2016.08.19	10	0.17453293	2016.08.19.14.26
## 2318	0	8	2016.08.19	0	0.00000000	2016.08.19.14.26
## 2319	70	9	2016.08.19	290	-1.22173048	2016.08.19.14.26
## 2320	170	10	2016.08.19	190	-2.96705973	2016.08.19.14.26
## 2321	175	11	2016.08.19	185	-3.05432619	2016.08.19.14.26
## 2322	160	12	2016.08.19	200	-2.79252680	2016.08.19.14.26
## 2323	90	13	2016.08.19	270	-1.57079633	2016.08.19.14.26
## 2324	30	14	2016.08.19	330	-0.52359878	2016.08.19.14.26
## 2325	110	15	2016.08.19	250	-1.91986218	2016.08.19.14.26
## 2326	40	16	2016.08.19	320	-0.69813170	2016.08.19.14.26
## 2327	70	17	2016.08.19	290	-1.22173048	2016.08.19.14.26
## 2328	140	18	2016.08.19	220	-2.44346095	2016.08.19.14.26
## 2329	340	19	2016.08.19	20	0.34906585	2016.08.19.14.26
## 2330	15	20	2016.08.19	345	-0.26179939	2016.08.19.14.26
## 2331	30	21	2016.08.19	330	-0.52359878	2016.08.19.14.26
## 2332	350	22	2016.08.19	10	0.17453293	2016.08.19.14.26
## 2333	355	23	2016.08.19	5	0.08726646	2016.08.19.14.26
## 2334	345	24	2016.08.19	15	0.26179939	2016.08.19.14.26
## 2335	5	25	2016.08.19	355	-0.08726646	2016.08.19.14.26
## 2336	350	26	2016.08.19	10	0.17453293	2016.08.19.14.26

## 2337	20	27	2016.08.19	340	-0.34906585	2016.08.19.14.26
## 2338	0	28	2016.08.19	0	0.00000000	2016.08.19.14.26
## 2339	10	29	2016.08.19	350	-0.17453293	2016.08.19.14.26
## 2340	20	30	2016.08.19	340	-0.34906585	2016.08.19.14.26
## 2341	0	31	2016.08.19	0	0.00000000	2016.08.19.14.26
## 2342	40	32	2016.08.19	320	-0.69813170	2016.08.19.14.26
## 2343	145	1	2016.08.19	215	-2.53072742	2016.08.19.14.50_1
## 2344	140	2	2016.08.19	220	-2.44346095	2016.08.19.14.50_1
## 2345	320	3	2016.08.19	40	0.69813170	2016.08.19.14.50_1
## 2346	350	4	2016.08.19	10	0.17453293	2016.08.19.14.50_1
## 2347	110	5	2016.08.19	250	-1.91986218	2016.08.19.14.50_1
## 2348	160	6	2016.08.19	200	-2.79252680	2016.08.19.14.50_1
## 2349	230	7	2016.08.19	130	2.26892803	2016.08.19.14.50_1
## 2350	340	8	2016.08.19	20	0.34906585	2016.08.19.14.50_1
## 2351	335	9	2016.08.19	25	0.43633231	2016.08.19.14.50_1
## 2352	280	10	2016.08.19	80	1.39626340	2016.08.19.14.50_1
## 2353	150	11	2016.08.19	210	-2.61799388	2016.08.19.14.50_1
## 2354	240	12	2016.08.19	120	2.09439510	2016.08.19.14.50_1
## 2355	290	13	2016.08.19	70	1.22173048	2016.08.19.14.50_1
## 2356	270	14	2016.08.19	90	1.57079633	2016.08.19.14.50_1
## 2357	270	15	2016.08.19	90	1.57079633	2016.08.19.14.50_1
## 2358	170	16	2016.08.19	190	-2.96705973	2016.08.19.14.50_1
## 2359	275	17	2016.08.19	85	1.48352986	2016.08.19.14.50_1
## 2360	10	18	2016.08.19	350	-0.17453293	2016.08.19.14.50_1
## 2361	0	19	2016.08.19	0	0.00000000	2016.08.19.14.50_1
## 2362	10	20	2016.08.19	350	-0.17453293	2016.08.19.14.50_1
## 2363	15	21	2016.08.19	345	-0.26179939	2016.08.19.14.50_1
## 2364	20	22	2016.08.19	340	-0.34906585	2016.08.19.14.50_1
## 2365	20	23	2016.08.19	340	-0.34906585	2016.08.19.14.50_1
## 2366	30	24	2016.08.19	330	-0.52359878	2016.08.19.14.50_1
## 2367	5	25	2016.08.19	355	-0.08726646	2016.08.19.14.50_1
## 2368	0	26	2016.08.19	0	0.00000000	2016.08.19.14.50_1
## 2369	50	27	2016.08.19	310	-0.87266463	2016.08.19.14.50_1
## 2370	0	28	2016.08.19	0	0.00000000	2016.08.19.14.50_1
## 2371	350	29	2016.08.19	10	0.17453293	2016.08.19.14.50_1
## 2372	270	1	2016.08.19	90	1.57079633	2016.08.19.14.51_2
## 2373	140	2	2016.08.19	220	-2.44346095	2016.08.19.14.51_2
## 2374	150	3	2016.08.19	210	-2.61799388	2016.08.19.14.51_2
## 2375	260	4	2016.08.19	100	1.74532925	2016.08.19.14.51_2
## 2376	110	5	2016.08.19	250	-1.91986218	2016.08.19.14.51_2
## 2377	240	6	2016.08.19	120	2.09439510	2016.08.19.14.51_2
## 2378	250	7	2016.08.19	110	1.91986218	2016.08.19.14.51_2
## 2379	110	8	2016.08.19	250	-1.91986218	2016.08.19.14.51_2
## 2380	190	9	2016.08.19	170	2.96705973	2016.08.19.14.51_2
## 2381	350	10	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2382	230	11	2016.08.19	130	2.26892803	2016.08.19.14.51_2
## 2383	230	12	2016.08.19	130	2.26892803	2016.08.19.14.51_2
## 2384	175	13	2016.08.19	185	-3.05432619	2016.08.19.14.51_2
## 2385	210	14	2016.08.19	150	2.61799388	2016.08.19.14.51_2
## 2386	20	15	2016.08.19	340	-0.34906585	2016.08.19.14.51_2
## 2387	350	16	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2388	340	17	2016.08.19	20	0.34906585	2016.08.19.14.51_2
## 2389	355	18	2016.08.19	5	0.08726646	2016.08.19.14.51_2
## 2390	10	19	2016.08.19	350	-0.17453293	2016.08.19.14.51_2

## 2391	40	20	2016.08.19	320	-0.69813170	2016.08.19.14.51_2
## 2392	0	21	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2393	0	22	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2394	355	23	2016.08.19	5	0.08726646	2016.08.19.14.51_2
## 2395	0	24	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2396	5	25	2016.08.19	355	-0.08726646	2016.08.19.14.51_2
## 2397	0	26	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2398	20	27	2016.08.19	340	-0.34906585	2016.08.19.14.51_2
## 2399	350	28	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2400	355	29	2016.08.19	5	0.08726646	2016.08.19.14.51_2
## 2401	350	30	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2402	0	31	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2403	5	32	2016.08.19	355	-0.08726646	2016.08.19.14.51_2
## 2404	350	33	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2405	340	34	2016.08.19	20	0.34906585	2016.08.19.14.51_2
## 2406	350	35	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2407	20	36	2016.08.19	340	-0.34906585	2016.08.19.14.51_2
## 2408	340	37	2016.08.19	20	0.34906585	2016.08.19.14.51_2
## 2409	0	38	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2410	355	39	2016.08.19	5	0.08726646	2016.08.19.14.51_2
## 2411	345	40	2016.08.19	15	0.26179939	2016.08.19.14.51_2
## 2412	340	41	2016.08.19	20	0.34906585	2016.08.19.14.51_2
## 2413	30	42	2016.08.19	330	-0.52359878	2016.08.19.14.51_2
## 2414	350	43	2016.08.19	10	0.17453293	2016.08.19.14.51_2
## 2415	0	44	2016.08.19	0	0.00000000	2016.08.19.14.51_2
## 2416	355	45	2016.08.19	5	0.08726646	2016.08.19.14.51_2
## 2417	120	1	2016.08.19	240	-2.09439510	2016.08.19.15.08
## 2418	150	2	2016.08.19	210	-2.61799388	2016.08.19.15.08
## 2419	130	3	2016.08.19	230	-2.26892803	2016.08.19.15.08
## 2420	120	4	2016.08.19	240	-2.09439510	2016.08.19.15.08
## 2421	105	5	2016.08.19	255	-1.83259571	2016.08.19.15.08
## 2422	160	6	2016.08.19	200	-2.79252680	2016.08.19.15.08
## 2423	70	7	2016.08.19	290	-1.22173048	2016.08.19.15.08
## 2424	95	8	2016.08.19	265	-1.65806279	2016.08.19.15.08
## 2425	155	9	2016.08.19	205	-2.70526034	2016.08.19.15.08
## 2426	155	10	2016.08.19	205	-2.70526034	2016.08.19.15.08
## 2427	175	11	2016.08.19	185	-3.05432619	2016.08.19.15.08
## 2428	140	12	2016.08.19	220	-2.44346095	2016.08.19.15.08
## 2429	20	13	2016.08.19	340	-0.34906585	2016.08.19.15.08
## 2430	45	14	2016.08.19	315	-0.78539816	2016.08.19.15.08
## 2431	355	15	2016.08.19	5	0.08726646	2016.08.19.15.08
## 2432	40	16	2016.08.19	320	-0.69813170	2016.08.19.15.08
## 2433	60	17	2016.08.19	300	-1.04719755	2016.08.19.15.08
## 2434	350	18	2016.08.19	10	0.17453293	2016.08.19.15.08
## 2435	10	19	2016.08.19	350	-0.17453293	2016.08.19.15.08
## 2436	40	20	2016.08.19	320	-0.69813170	2016.08.19.15.08
## 2437	355	21	2016.08.19	5	0.08726646	2016.08.19.15.08
## 2438	20	22	2016.08.19	340	-0.34906585	2016.08.19.15.08
## 2439	5	23	2016.08.19	355	-0.08726646	2016.08.19.15.08
## 2440	340	24	2016.08.19	20	0.34906585	2016.08.19.15.08
## 2441	30	25	2016.08.19	330	-0.52359878	2016.08.19.15.08
## 2442	125	1	2016.08.19	235	-2.18166156	2016.08.19.14.15_1
## 2443	115	2	2016.08.19	245	-2.00712864	2016.08.19.14.15_1
## 2444	145	3	2016.08.19	215	-2.53072742	2016.08.19.14.15_1

## 2445	140	4	2016.08.19	220	-2.44346095	2016.08.19.14.15_1
## 2446	170	5	2016.08.19	190	-2.96705973	2016.08.19.14.15_1
## 2447	240	6	2016.08.19	120	2.09439510	2016.08.19.14.15_1
## 2448	150	7	2016.08.19	210	-2.61799388	2016.08.19.14.15_1
## 2449	155	8	2016.08.19	205	-2.70526034	2016.08.19.14.15_1
## 2450	150	9	2016.08.19	210	-2.61799388	2016.08.19.14.15_1
## 2451	90	10	2016.08.19	270	-1.57079633	2016.08.19.14.15_1
## 2452	95	11	2016.08.19	265	-1.65806279	2016.08.19.14.15_1
## 2453	100	12	2016.08.19	260	-1.74532925	2016.08.19.14.15_1
## 2454	130	13	2016.08.19	230	-2.26892803	2016.08.19.14.15_1
## 2455	210	14	2016.08.19	150	2.61799388	2016.08.19.14.15_1
## 2456	130	15	2016.08.19	230	-2.26892803	2016.08.19.14.15_1
## 2457	140	16	2016.08.19	220	-2.44346095	2016.08.19.14.15_1
## 2458	90	17	2016.08.19	270	-1.57079633	2016.08.19.14.15_1
## 2459	120	18	2016.08.19	240	-2.09439510	2016.08.19.14.15_1
## 2460	210	19	2016.08.19	150	2.61799388	2016.08.19.14.15_1
## 2461	190	20	2016.08.19	170	2.96705973	2016.08.19.14.15_1
## 2462	185	21	2016.08.19	175	3.05432619	2016.08.19.14.15_1
## 2463	180	22	2016.08.19	180	3.14159265	2016.08.19.14.15_1
## 2464	65	23	2016.08.19	295	-1.13446401	2016.08.19.14.15_1
## 2465	180	24	2016.08.19	180	3.14159265	2016.08.19.14.15_1
## 2466	160	25	2016.08.19	200	-2.79252680	2016.08.19.14.15_1
## 2467	165	26	2016.08.19	195	-2.87979327	2016.08.19.14.15_1
## 2468	210	27	2016.08.19	150	2.61799388	2016.08.19.14.15_1
## 2469	95	28	2016.08.19	265	-1.65806279	2016.08.19.14.15_1
## 2470	100	29	2016.08.19	260	-1.74532925	2016.08.19.14.15_1
## 2471	270	30	2016.08.19	90	1.57079633	2016.08.19.14.15_1
## 2472	140	31	2016.08.19	220	-2.44346095	2016.08.19.14.15_1
## 2473	230	32	2016.08.19	130	2.26892803	2016.08.19.14.15_1
## 2474	170	33	2016.08.19	190	-2.96705973	2016.08.19.14.15_1
## 2475	240	34	2016.08.19	120	2.09439510	2016.08.19.14.15_1
## 2476	160	35	2016.08.19	200	-2.79252680	2016.08.19.14.15_1
## 2477	160	36	2016.08.19	200	-2.79252680	2016.08.19.14.15_1
## 2478	150	37	2016.08.19	210	-2.61799388	2016.08.19.14.15_1
## 2479	220	38	2016.08.19	140	2.44346095	2016.08.19.14.15_1
## 2480	190	39	2016.08.19	170	2.96705973	2016.08.19.14.15_1
## 2481	140	40	2016.08.19	220	-2.44346095	2016.08.19.14.15_1
## 2482	185	41	2016.08.19	175	3.05432619	2016.08.19.14.15_1
## 2483	190	42	2016.08.19	170	2.96705973	2016.08.19.14.15_1
## 2484	40	43	2016.08.19	320	-0.69813170	2016.08.19.14.15_1
## 2485	0	44	2016.08.19	0	0.00000000	2016.08.19.14.15_1
## 2486	60	45	2016.08.19	300	-1.04719755	2016.08.19.14.15_1
## 2487	350	46	2016.08.19	10	0.17453293	2016.08.19.14.15_1
## 2488	45	47	2016.08.19	315	-0.78539816	2016.08.19.14.15_1
## 2489	40	48	2016.08.19	320	-0.69813170	2016.08.19.14.15_1
## 2490	30	49	2016.08.19	330	-0.52359878	2016.08.19.14.15_1
## 2491	35	50	2016.08.19	325	-0.61086524	2016.08.19.14.15_1
## 2492	10	51	2016.08.19	350	-0.17453293	2016.08.19.14.15_1
## 2493	20	52	2016.08.19	340	-0.34906585	2016.08.19.14.15_1
## 2494	40	53	2016.08.19	320	-0.69813170	2016.08.19.14.15_1
## 2495	0	54	2016.08.19	0	0.00000000	2016.08.19.14.15_1
## 2496	20	55	2016.08.19	340	-0.34906585	2016.08.19.14.15_1
## 2497	10	56	2016.08.19	350	-0.17453293	2016.08.19.14.15_1
## 2498	15	57	2016.08.19	345	-0.26179939	2016.08.19.14.15_1

## 2499	20	58	2016.08.19	340	-0.34906585	2016.08.19.14.15_1
## 2500	25	59	2016.08.19	335	-0.43633231	2016.08.19.14.15_1
## 2501	40	60	2016.08.19	320	-0.69813170	2016.08.19.14.15_1
## 2502	340	61	2016.08.19	20	0.34906585	2016.08.19.14.15_1
## 2503	40	62	2016.08.19	320	-0.69813170	2016.08.19.14.15_1
## 2504	5	63	2016.08.19	355	-0.08726646	2016.08.19.14.15_1
## 2505	175	1	2016.08.19	185	-3.05432619	2016.08.19.14.17_2
## 2506	170	2	2016.08.19	190	-2.96705973	2016.08.19.14.17_2
## 2507	270	3	2016.08.19	90	1.57079633	2016.08.19.14.17_2
## 2508	290	4	2016.08.19	70	1.22173048	2016.08.19.14.17_2
## 2509	265	5	2016.08.19	95	1.65806279	2016.08.19.14.17_2
## 2510	290	6	2016.08.19	70	1.22173048	2016.08.19.14.17_2
## 2511	220	7	2016.08.19	140	2.44346095	2016.08.19.14.17_2
## 2512	245	8	2016.08.19	115	2.00712864	2016.08.19.14.17_2
## 2513	220	9	2016.08.19	140	2.44346095	2016.08.19.14.17_2
## 2514	200	10	2016.08.19	160	2.79252680	2016.08.19.14.17_2
## 2515	185	11	2016.08.19	175	3.05432619	2016.08.19.14.17_2
## 2516	245	12	2016.08.19	115	2.00712864	2016.08.19.14.17_2
## 2517	240	13	2016.08.19	120	2.09439510	2016.08.19.14.17_2
## 2518	210	14	2016.08.19	150	2.61799388	2016.08.19.14.17_2
## 2519	10	15	2016.08.19	350	-0.17453293	2016.08.19.14.17_2
## 2520	10	16	2016.08.19	350	-0.17453293	2016.08.19.14.17_2
## 2521	15	17	2016.08.19	345	-0.26179939	2016.08.19.14.17_2
## 2522	55	18	2016.08.19	305	-0.95993109	2016.08.19.14.17_2
## 2523	30	19	2016.08.19	330	-0.52359878	2016.08.19.14.17_2
## 2524	350	20	2016.08.19	10	0.17453293	2016.08.19.14.17_2
## 2525	10	21	2016.08.19	350	-0.17453293	2016.08.19.14.17_2
## 2526	10	22	2016.08.19	350	-0.17453293	2016.08.19.14.17_2
## 2527	25	23	2016.08.19	335	-0.43633231	2016.08.19.14.17_2
## 2528	10	24	2016.08.19	350	-0.17453293	2016.08.19.14.17_2
## 2529	350	25	2016.08.19	10	0.17453293	2016.08.19.14.17_2
## 2530	0	26	2016.08.19	0	0.00000000	2016.08.19.14.17_2
## 2531	0	27	2016.08.19	0	0.00000000	2016.08.19.14.17_2
## 2532	80	28	2016.08.19	280	-1.39626340	2016.08.19.14.17_2
## 2533	20	29	2016.08.19	340	-0.34906585	2016.08.19.14.17_2
## 2534	10	30	2016.08.19	350	-0.17453293	2016.08.19.14.17_2
## 2535	350	31	2016.08.19	10	0.17453293	2016.08.19.14.17_2
## 2536	5	32	2016.08.19	355	-0.08726646	2016.08.19.14.17_2
## 2537	180	1	2016.08.19	180	3.14159265	2016.08.19.14.19
## 2538	170	2	2016.08.19	190	-2.96705973	2016.08.19.14.19
## 2539	225	3	2016.08.19	135	2.35619449	2016.08.19.14.19
## 2540	210	4	2016.08.19	150	2.61799388	2016.08.19.14.19
## 2541	170	5	2016.08.19	190	-2.96705973	2016.08.19.14.19
## 2542	140	6	2016.08.19	220	-2.44346095	2016.08.19.14.19
## 2543	220	7	2016.08.19	140	2.44346095	2016.08.19.14.19
## 2544	170	8	2016.08.19	190	-2.96705973	2016.08.19.14.19
## 2545	185	9	2016.08.19	175	3.05432619	2016.08.19.14.19
## 2546	170	10	2016.08.19	190	-2.96705973	2016.08.19.14.19
## 2547	100	11	2016.08.19	260	-1.74532925	2016.08.19.14.19
## 2548	170	12	2016.08.19	190	-2.96705973	2016.08.19.14.19
## 2549	220	13	2016.08.19	140	2.44346095	2016.08.19.14.19
## 2550	180	14	2016.08.19	180	3.14159265	2016.08.19.14.19
## 2551	185	15	2016.08.19	175	3.05432619	2016.08.19.14.19
## 2552	195	16	2016.08.19	165	2.87979327	2016.08.19.14.19

## 2553	240	17	2016.08.19	120	2.09439510	2016.08.19.14.19
## 2554	220	18	2016.08.19	140	2.44346095	2016.08.19.14.19
## 2555	205	19	2016.08.19	155	2.70526034	2016.08.19.14.19
## 2556	185	20	2016.08.19	175	3.05432619	2016.08.19.14.19
## 2557	150	21	2016.08.19	210	-2.61799388	2016.08.19.14.19
## 2558	150	22	2016.08.19	210	-2.61799388	2016.08.19.14.19
## 2559	0	23	2016.08.19	0	0.00000000	2016.08.19.14.19
## 2560	330	24	2016.08.19	30	0.52359878	2016.08.19.14.19
## 2561	10	25	2016.08.19	350	-0.17453293	2016.08.19.14.19
## 2562	10	26	2016.08.19	350	-0.17453293	2016.08.19.14.19
## 2563	5	27	2016.08.19	355	-0.08726646	2016.08.19.14.19
## 2564	0	28	2016.08.19	0	0.00000000	2016.08.19.14.19
## 2565	10	29	2016.08.19	350	-0.17453293	2016.08.19.14.19
## 2566	350	30	2016.08.19	10	0.17453293	2016.08.19.14.19
## 2567	15	31	2016.08.19	345	-0.26179939	2016.08.19.14.19
## 2568	40	32	2016.08.19	320	-0.69813170	2016.08.19.14.19
## 2569	350	33	2016.08.19	10	0.17453293	2016.08.19.14.19
## 2570	10	34	2016.08.19	350	-0.17453293	2016.08.19.14.19
## 2571	60	35	2016.08.19	300	-1.04719755	2016.08.19.14.19
## 2572	20	36	2016.08.19	340	-0.34906585	2016.08.19.14.19
## 2573	30	37	2016.08.19	330	-0.52359878	2016.08.19.14.19
## 2574	35	38	2016.08.19	325	-0.61086524	2016.08.19.14.19
## 2575	20	39	2016.08.19	340	-0.34906585	2016.08.19.14.19
## 2576	245	1	2016.08.19	115	2.00712864	2016.08.19.15.29
## 2577	115	2	2016.08.19	245	-2.00712864	2016.08.19.15.29
## 2578	120	3	2016.08.19	240	-2.09439510	2016.08.19.15.29
## 2579	150	4	2016.08.19	210	-2.61799388	2016.08.19.15.29
## 2580	90	5	2016.08.19	270	-1.57079633	2016.08.19.15.29
## 2581	330	6	2016.08.19	30	0.52359878	2016.08.19.15.29
## 2582	240	7	2016.08.19	120	2.09439510	2016.08.19.15.29
## 2583	90	8	2016.08.19	270	-1.57079633	2016.08.19.15.29
## 2584	120	9	2016.08.19	240	-2.09439510	2016.08.19.15.29
## 2585	90	10	2016.08.19	270	-1.57079633	2016.08.19.15.29
## 2586	140	11	2016.08.19	220	-2.44346095	2016.08.19.15.29
## 2587	170	12	2016.08.19	190	-2.96705973	2016.08.19.15.29
## 2588	320	13	2016.08.19	40	0.69813170	2016.08.19.15.29
## 2589	205	14	2016.08.19	155	2.70526034	2016.08.19.15.29
## 2590	135	15	2016.08.19	225	-2.35619449	2016.08.19.15.29
## 2591	120	16	2016.08.19	240	-2.09439510	2016.08.19.15.29
## 2592	185	17	2016.08.19	175	3.05432619	2016.08.19.15.29
## 2593	300	18	2016.08.19	60	1.04719755	2016.08.19.15.29
## 2594	125	19	2016.08.19	235	-2.18166156	2016.08.19.15.29
## 2595	320	20	2016.08.19	40	0.69813170	2016.08.19.15.29
## 2596	280	21	2016.08.19	80	1.39626340	2016.08.19.15.29
## 2597	100	22	2016.08.19	260	-1.74532925	2016.08.19.15.29
## 2598	320	23	2016.08.19	40	0.69813170	2016.08.19.15.29
## 2599	125	24	2016.08.19	235	-2.18166156	2016.08.19.15.29
## 2600	200	25	2016.08.19	160	2.79252680	2016.08.19.15.29
## 2601	50	26	2016.08.19	310	-0.87266463	2016.08.19.15.29
## 2602	95	27	2016.08.19	265	-1.65806279	2016.08.19.15.29
## 2603	100	28	2016.08.19	260	-1.74532925	2016.08.19.15.29
## 2604	60	29	2016.08.19	300	-1.04719755	2016.08.19.15.29
## 2605	0	30	2016.08.19	0	0.00000000	2016.08.19.15.29
## 2606	50	31	2016.08.19	310	-0.87266463	2016.08.19.15.29

## 2607	355	32	2016.08.19	5	0.08726646	2016.08.19.15.29
## 2608	10	33	2016.08.19	350	-0.17453293	2016.08.19.15.29
## 2609	5	34	2016.08.19	355	-0.08726646	2016.08.19.15.29
## 2610	50	35	2016.08.19	310	-0.87266463	2016.08.19.15.29
## 2611	30	36	2016.08.19	330	-0.52359878	2016.08.19.15.29
## 2612	15	37	2016.08.19	345	-0.26179939	2016.08.19.15.29
## 2613	20	38	2016.08.19	340	-0.34906585	2016.08.19.15.29
## 2614	350	39	2016.08.19	10	0.17453293	2016.08.19.15.29
## 2615	20	40	2016.08.19	340	-0.34906585	2016.08.19.15.29
## 2616	35	41	2016.08.19	325	-0.61086524	2016.08.19.15.29
## 2617	40	42	2016.08.19	320	-0.69813170	2016.08.19.15.29
## 2618	45	43	2016.08.19	315	-0.78539816	2016.08.19.15.29
## 2619	40	44	2016.08.19	320	-0.69813170	2016.08.19.15.29
## 2620	35	45	2016.08.19	325	-0.61086524	2016.08.19.15.29
## 2621	50	46	2016.08.19	310	-0.87266463	2016.08.19.15.29
## 2622	5	47	2016.08.19	355	-0.08726646	2016.08.19.15.29
## 2623	10	48	2016.08.19	350	-0.17453293	2016.08.19.15.29
## 2624	180	1	2016.08.19	180	3.14159265	2016.08.19.15.33_1
## 2625	150	2	2016.08.19	210	-2.61799388	2016.08.19.15.33_1
## 2626	145	3	2016.08.19	215	-2.53072742	2016.08.19.15.33_1
## 2627	155	4	2016.08.19	205	-2.70526034	2016.08.19.15.33_1
## 2628	310	5	2016.08.19	50	0.87266463	2016.08.19.15.33_1
## 2629	170	6	2016.08.19	190	-2.96705973	2016.08.19.15.33_1
## 2630	110	7	2016.08.19	250	-1.91986218	2016.08.19.15.33_1
## 2631	190	8	2016.08.19	170	2.96705973	2016.08.19.15.33_1
## 2632	170	9	2016.08.19	190	-2.96705973	2016.08.19.15.33_1
## 2633	150	10	2016.08.19	210	-2.61799388	2016.08.19.15.33_1
## 2634	150	11	2016.08.19	210	-2.61799388	2016.08.19.15.33_1
## 2635	185	12	2016.08.19	175	3.05432619	2016.08.19.15.33_1
## 2636	120	13	2016.08.19	240	-2.09439510	2016.08.19.15.33_1
## 2637	155	14	2016.08.19	205	-2.70526034	2016.08.19.15.33_1
## 2638	210	15	2016.08.19	150	2.61799388	2016.08.19.15.33_1
## 2639	205	16	2016.08.19	155	2.70526034	2016.08.19.15.33_1
## 2640	250	17	2016.08.19	110	1.91986218	2016.08.19.15.33_1
## 2641	200	18	2016.08.19	160	2.79252680	2016.08.19.15.33_1
## 2642	160	19	2016.08.19	200	-2.79252680	2016.08.19.15.33_1
## 2643	190	20	2016.08.19	170	2.96705973	2016.08.19.15.33_1
## 2644	270	21	2016.08.19	90	1.57079633	2016.08.19.15.33_1
## 2645	160	22	2016.08.19	200	-2.79252680	2016.08.19.15.33_1
## 2646	160	23	2016.08.19	200	-2.79252680	2016.08.19.15.33_1
## 2647	215	24	2016.08.19	145	2.53072742	2016.08.19.15.33_1
## 2648	145	25	2016.08.19	215	-2.53072742	2016.08.19.15.33_1
## 2649	205	26	2016.08.19	155	2.70526034	2016.08.19.15.33_1
## 2650	120	27	2016.08.19	240	-2.09439510	2016.08.19.15.33_1
## 2651	210	28	2016.08.19	150	2.61799388	2016.08.19.15.33_1
## 2652	40	29	2016.08.19	320	-0.69813170	2016.08.19.15.33_1
## 2653	10	30	2016.08.19	350	-0.17453293	2016.08.19.15.33_1
## 2654	40	31	2016.08.19	320	-0.69813170	2016.08.19.15.33_1
## 2655	0	32	2016.08.19	0	0.00000000	2016.08.19.15.33_1
## 2656	355	33	2016.08.19	5	0.08726646	2016.08.19.15.33_1
## 2657	20	34	2016.08.19	340	-0.34906585	2016.08.19.15.33_1
## 2658	20	35	2016.08.19	340	-0.34906585	2016.08.19.15.33_1
## 2659	10	36	2016.08.19	350	-0.17453293	2016.08.19.15.33_1
## 2660	355	37	2016.08.19	5	0.08726646	2016.08.19.15.33_1

## 2661	10	38	2016.08.19	350	-0.17453293	2016.08.19.15.33_1
## 2662	0	39	2016.08.19	0	0.00000000	2016.08.19.15.33_1
## 2663	5	40	2016.08.19	355	-0.08726646	2016.08.19.15.33_1
## 2664	350	41	2016.08.19	10	0.17453293	2016.08.19.15.33_1
## 2665	10	42	2016.08.19	350	-0.17453293	2016.08.19.15.33_1
## 2666	0	43	2016.08.19	0	0.00000000	2016.08.19.15.33_1
## 2667	350	44	2016.08.19	10	0.17453293	2016.08.19.15.33_1
## 2668	60	45	2016.08.19	300	-1.04719755	2016.08.19.15.33_1
## 2669	5	46	2016.08.19	355	-0.08726646	2016.08.19.15.33_1
## 2670	10	47	2016.08.19	350	-0.17453293	2016.08.19.15.33_1
## 2671	10	48	2016.08.19	350	-0.17453293	2016.08.19.15.33_1
## 2672	5	49	2016.08.19	355	-0.08726646	2016.08.19.15.33_1
## 2673	0	50	2016.08.19	0	0.00000000	2016.08.19.15.33_1
## 2674	350	51	2016.08.19	10	0.17453293	2016.08.19.15.33_1
## 2675	310	1	2016.08.19	50	0.87266463	2016.08.19.15.34_2
## 2676	190	2	2016.08.19	170	2.96705973	2016.08.19.15.34_2
## 2677	160	3	2016.08.19	200	-2.79252680	2016.08.19.15.34_2
## 2678	215	4	2016.08.19	145	2.53072742	2016.08.19.15.34_2
## 2679	150	5	2016.08.19	210	-2.61799388	2016.08.19.15.34_2
## 2680	210	6	2016.08.19	150	2.61799388	2016.08.19.15.34_2
## 2681	330	7	2016.08.19	30	0.52359878	2016.08.19.15.34_2
## 2682	190	8	2016.08.19	170	2.96705973	2016.08.19.15.34_2
## 2683	280	9	2016.08.19	80	1.39626340	2016.08.19.15.34_2
## 2684	15	10	2016.08.19	345	-0.26179939	2016.08.19.15.34_2
## 2685	10	11	2016.08.19	350	-0.17453293	2016.08.19.15.34_2
## 2686	50	12	2016.08.19	310	-0.87266463	2016.08.19.15.34_2
## 2687	350	13	2016.08.19	10	0.17453293	2016.08.19.15.34_2
## 2688	30	14	2016.08.19	330	-0.52359878	2016.08.19.15.34_2
## 2689	60	15	2016.08.19	300	-1.04719755	2016.08.19.15.34_2
## 2690	50	16	2016.08.19	310	-0.87266463	2016.08.19.15.34_2
## 2691	0	17	2016.08.19	0	0.00000000	2016.08.19.15.34_2
## 2692	15	18	2016.08.19	345	-0.26179939	2016.08.19.15.34_2
## 2693	30	19	2016.08.19	330	-0.52359878	2016.08.19.15.34_2
## 2694	355	20	2016.08.19	5	0.08726646	2016.08.19.15.34_2
## 2695	40	21	2016.08.19	320	-0.69813170	2016.08.19.15.34_2
## 2696	40	22	2016.08.19	320	-0.69813170	2016.08.19.15.34_2
## 2697	25	23	2016.08.19	335	-0.43633231	2016.08.19.15.34_2
## 2698	20	24	2016.08.19	340	-0.34906585	2016.08.19.15.34_2
## 2699	10	25	2016.08.19	350	-0.17453293	2016.08.19.15.34_2
## 2700	145	1	2016.08.23	215	-2.53072742	2016.08.23.14.01
## 2701	180	2	2016.08.23	180	3.14159265	2016.08.23.14.01
## 2702	160	3	2016.08.23	200	-2.79252680	2016.08.23.14.01
## 2703	175	4	2016.08.23	185	-3.05432619	2016.08.23.14.01
## 2704	190	5	2016.08.23	170	2.96705973	2016.08.23.14.01
## 2705	100	6	2016.08.23	260	-1.74532925	2016.08.23.14.01
## 2706	200	7	2016.08.23	160	2.79252680	2016.08.23.14.01
## 2707	100	8	2016.08.23	260	-1.74532925	2016.08.23.14.01
## 2708	80	9	2016.08.23	280	-1.39626340	2016.08.23.14.01
## 2709	85	10	2016.08.23	275	-1.48352986	2016.08.23.14.01
## 2710	200	11	2016.08.23	160	2.79252680	2016.08.23.14.01
## 2711	50	12	2016.08.23	310	-0.87266463	2016.08.23.14.01
## 2712	220	13	2016.08.23	140	2.44346095	2016.08.23.14.01
## 2713	205	14	2016.08.23	155	2.70526034	2016.08.23.14.01
## 2714	145	15	2016.08.23	215	-2.53072742	2016.08.23.14.01

## 2715	190	16	2016.08.23	170	2.96705973	2016.08.23.14.01
## 2716	150	17	2016.08.23	210	-2.61799388	2016.08.23.14.01
## 2717	185	18	2016.08.23	175	3.05432619	2016.08.23.14.01
## 2718	120	19	2016.08.23	240	-2.09439510	2016.08.23.14.01
## 2719	165	20	2016.08.23	195	-2.87979327	2016.08.23.14.01
## 2720	210	21	2016.08.23	150	2.61799388	2016.08.23.14.01
## 2721	255	22	2016.08.23	105	1.83259571	2016.08.23.14.01
## 2722	270	23	2016.08.23	90	1.57079633	2016.08.23.14.01
## 2723	325	24	2016.08.23	35	0.61086524	2016.08.23.14.01
## 2724	280	25	2016.08.23	80	1.39626340	2016.08.23.14.01
## 2725	310	26	2016.08.23	50	0.87266463	2016.08.23.14.01
## 2726	310	27	2016.08.23	50	0.87266463	2016.08.23.14.01
## 2727	315	28	2016.08.23	45	0.78539816	2016.08.23.14.01
## 2728	315	29	2016.08.23	45	0.78539816	2016.08.23.14.01
## 2729	295	30	2016.08.23	65	1.13446401	2016.08.23.14.01
## 2730	330	31	2016.08.23	30	0.52359878	2016.08.23.14.01
## 2731	330	32	2016.08.23	30	0.52359878	2016.08.23.14.01
## 2732	295	33	2016.08.23	65	1.13446401	2016.08.23.14.01
## 2733	290	34	2016.08.23	70	1.22173048	2016.08.23.14.01
## 2734	300	35	2016.08.23	60	1.04719755	2016.08.23.14.01
## 2735	320	36	2016.08.23	40	0.69813170	2016.08.23.14.01
## 2736	305	37	2016.08.23	55	0.95993109	2016.08.23.14.01
## 2737	280	38	2016.08.23	80	1.39626340	2016.08.23.14.01
## 2738	315	39	2016.08.23	45	0.78539816	2016.08.23.14.01
## 2739	325	40	2016.08.23	35	0.61086524	2016.08.23.14.01
## 2740	280	41	2016.08.23	80	1.39626340	2016.08.23.14.01
## 2741	290	42	2016.08.23	70	1.22173048	2016.08.23.14.01
## 2742	310	43	2016.08.23	50	0.87266463	2016.08.23.14.01
## 2743	175	1	2016.08.23	185	-3.05432619	2016.08.23.14.09
## 2744	180	2	2016.08.23	180	3.14159265	2016.08.23.14.09
## 2745	110	3	2016.08.23	250	-1.91986218	2016.08.23.14.09
## 2746	190	4	2016.08.23	170	2.96705973	2016.08.23.14.09
## 2747	200	5	2016.08.23	160	2.79252680	2016.08.23.14.09
## 2748	165	6	2016.08.23	195	-2.87979327	2016.08.23.14.09
## 2749	110	7	2016.08.23	250	-1.91986218	2016.08.23.14.09
## 2750	210	8	2016.08.23	150	2.61799388	2016.08.23.14.09
## 2751	110	9	2016.08.23	250	-1.91986218	2016.08.23.14.09
## 2752	240	10	2016.08.23	120	2.09439510	2016.08.23.14.09
## 2753	240	11	2016.08.23	120	2.09439510	2016.08.23.14.09
## 2754	120	12	2016.08.23	240	-2.09439510	2016.08.23.14.09
## 2755	95	13	2016.08.23	265	-1.65806279	2016.08.23.14.09
## 2756	270	14	2016.08.23	90	1.57079633	2016.08.23.14.09
## 2757	205	15	2016.08.23	155	2.70526034	2016.08.23.14.09
## 2758	220	16	2016.08.23	140	2.44346095	2016.08.23.14.09
## 2759	130	17	2016.08.23	230	-2.26892803	2016.08.23.14.09
## 2760	175	18	2016.08.23	185	-3.05432619	2016.08.23.14.09
## 2761	110	19	2016.08.23	250	-1.91986218	2016.08.23.14.09
## 2762	200	20	2016.08.23	160	2.79252680	2016.08.23.14.09
## 2763	180	21	2016.08.23	180	3.14159265	2016.08.23.14.09
## 2764	150	22	2016.08.23	210	-2.61799388	2016.08.23.14.09
## 2765	90	23	2016.08.23	270	-1.57079633	2016.08.23.14.09
## 2766	345	24	2016.08.23	15	0.26179939	2016.08.23.14.09
## 2767	335	25	2016.08.23	25	0.43633231	2016.08.23.14.09
## 2768	345	26	2016.08.23	15	0.26179939	2016.08.23.14.09

## 2769	5	27	2016.08.23	355	-0.08726646	2016.08.23.14.09
## 2770	5	28	2016.08.23	355	-0.08726646	2016.08.23.14.09
## 2771	30	29	2016.08.23	330	-0.52359878	2016.08.23.14.09
## 2772	15	30	2016.08.23	345	-0.26179939	2016.08.23.14.09
## 2773	10	31	2016.08.23	350	-0.17453293	2016.08.23.14.09
## 2774	10	32	2016.08.23	350	-0.17453293	2016.08.23.14.09
## 2775	5	33	2016.08.23	355	-0.08726646	2016.08.23.14.09
## 2776	30	34	2016.08.23	330	-0.52359878	2016.08.23.14.09
## 2777	350	35	2016.08.23	10	0.17453293	2016.08.23.14.09
## 2778	20	36	2016.08.23	340	-0.34906585	2016.08.23.14.09
## 2779	345	37	2016.08.23	15	0.26179939	2016.08.23.14.09
## 2780	10	38	2016.08.23	350	-0.17453293	2016.08.23.14.09
## 2781	355	39	2016.08.23	5	0.08726646	2016.08.23.14.09
## 2782	190	1	2016.08.23	170	2.96705973	2016.08.23.14.51
## 2783	165	2	2016.08.23	195	-2.87979327	2016.08.23.14.51
## 2784	310	3	2016.08.23	50	0.87266463	2016.08.23.14.51
## 2785	20	4	2016.08.23	340	-0.34906585	2016.08.23.14.51
## 2786	60	5	2016.08.23	300	-1.04719755	2016.08.23.14.51
## 2787	220	6	2016.08.23	140	2.44346095	2016.08.23.14.51
## 2788	120	7	2016.08.23	240	-2.09439510	2016.08.23.14.51
## 2789	95	8	2016.08.23	265	-1.65806279	2016.08.23.14.51
## 2790	130	9	2016.08.23	230	-2.26892803	2016.08.23.14.51
## 2791	140	10	2016.08.23	220	-2.44346095	2016.08.23.14.51
## 2792	75	11	2016.08.23	285	-1.30899694	2016.08.23.14.51
## 2793	50	12	2016.08.23	310	-0.87266463	2016.08.23.14.51
## 2794	190	13	2016.08.23	170	2.96705973	2016.08.23.14.51
## 2795	210	14	2016.08.23	150	2.61799388	2016.08.23.14.51
## 2796	105	15	2016.08.23	255	-1.83259571	2016.08.23.14.51
## 2797	90	16	2016.08.23	270	-1.57079633	2016.08.23.14.51
## 2798	120	17	2016.08.23	240	-2.09439510	2016.08.23.14.51
## 2799	65	18	2016.08.23	295	-1.13446401	2016.08.23.14.51
## 2800	80	19	2016.08.23	280	-1.39626340	2016.08.23.14.51
## 2801	170	20	2016.08.23	190	-2.96705973	2016.08.23.14.51
## 2802	175	21	2016.08.23	185	-3.05432619	2016.08.23.14.51
## 2803	225	22	2016.08.23	135	2.35619449	2016.08.23.14.51
## 2804	220	23	2016.08.23	140	2.44346095	2016.08.23.14.51
## 2805	240	24	2016.08.23	120	2.09439510	2016.08.23.14.51
## 2806	240	25	2016.08.23	120	2.09439510	2016.08.23.14.51
## 2807	80	26	2016.08.23	280	-1.39626340	2016.08.23.14.51
## 2808	80	27	2016.08.23	280	-1.39626340	2016.08.23.14.51
## 2809	60	28	2016.08.23	300	-1.04719755	2016.08.23.14.51
## 2810	240	29	2016.08.23	120	2.09439510	2016.08.23.14.51
## 2811	130	30	2016.08.23	230	-2.26892803	2016.08.23.14.51
## 2812	200	31	2016.08.23	160	2.79252680	2016.08.23.14.51
## 2813	160	32	2016.08.23	200	-2.79252680	2016.08.23.14.51
## 2814	200	33	2016.08.23	160	2.79252680	2016.08.23.14.51
## 2815	355	34	2016.08.23	5	0.08726646	2016.08.23.14.51
## 2816	325	35	2016.08.23	35	0.61086524	2016.08.23.14.51
## 2817	340	36	2016.08.23	20	0.34906585	2016.08.23.14.51
## 2818	0	37	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2819	320	38	2016.08.23	40	0.69813170	2016.08.23.14.51
## 2820	0	39	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2821	320	40	2016.08.23	40	0.69813170	2016.08.23.14.51
## 2822	355	41	2016.08.23	5	0.08726646	2016.08.23.14.51

## 2823	300	42	2016.08.23	60	1.04719755	2016.08.23.14.51
## 2824	325	43	2016.08.23	35	0.61086524	2016.08.23.14.51
## 2825	340	44	2016.08.23	20	0.34906585	2016.08.23.14.51
## 2826	0	45	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2827	325	46	2016.08.23	35	0.61086524	2016.08.23.14.51
## 2828	350	47	2016.08.23	10	0.17453293	2016.08.23.14.51
## 2829	0	48	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2830	320	49	2016.08.23	40	0.69813170	2016.08.23.14.51
## 2831	10	50	2016.08.23	350	-0.17453293	2016.08.23.14.51
## 2832	0	51	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2833	355	52	2016.08.23	5	0.08726646	2016.08.23.14.51
## 2834	0	53	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2835	330	54	2016.08.23	30	0.52359878	2016.08.23.14.51
## 2836	355	55	2016.08.23	5	0.08726646	2016.08.23.14.51
## 2837	330	56	2016.08.23	30	0.52359878	2016.08.23.14.51
## 2838	290	57	2016.08.23	70	1.22173048	2016.08.23.14.51
## 2839	0	58	2016.08.23	0	0.00000000	2016.08.23.14.51
## 2840	330	59	2016.08.23	30	0.52359878	2016.08.23.14.51
## 2841	355	60	2016.08.23	5	0.08726646	2016.08.23.14.51
## 2842	5	61	2016.08.23	355	-0.08726646	2016.08.23.14.51
## 2843	0	1	2016.08.23	0	0.00000000	2016.08.23.15.04
## 2844	5	2	2016.08.23	355	-0.08726646	2016.08.23.15.04
## 2845	30	3	2016.08.23	330	-0.52359878	2016.08.23.15.04
## 2846	15	4	2016.08.23	345	-0.26179939	2016.08.23.15.04
## 2847	345	5	2016.08.23	15	0.26179939	2016.08.23.15.04
## 2848	20	6	2016.08.23	340	-0.34906585	2016.08.23.15.04
## 2849	40	7	2016.08.23	320	-0.69813170	2016.08.23.15.04
## 2850	170	8	2016.08.23	190	-2.96705973	2016.08.23.15.04
## 2851	10	9	2016.08.23	350	-0.17453293	2016.08.23.15.04
## 2852	140	10	2016.08.23	220	-2.44346095	2016.08.23.15.04
## 2853	90	11	2016.08.23	270	-1.57079633	2016.08.23.15.04
## 2854	220	12	2016.08.23	140	2.44346095	2016.08.23.15.04
## 2855	30	13	2016.08.23	330	-0.52359878	2016.08.23.15.04
## 2856	65	14	2016.08.23	295	-1.13446401	2016.08.23.15.04
## 2857	200	15	2016.08.23	160	2.79252680	2016.08.23.15.04
## 2858	19	16	2016.08.23	341	-0.33161256	2016.08.23.15.04
## 2859	220	17	2016.08.23	140	2.44346095	2016.08.23.15.04
## 2860	155	18	2016.08.23	205	-2.70526034	2016.08.23.15.04
## 2861	180	19	2016.08.23	180	3.14159265	2016.08.23.15.04
## 2862	160	20	2016.08.23	200	-2.79252680	2016.08.23.15.04
## 2863	5	21	2016.08.23	355	-0.08726646	2016.08.23.15.04
## 2864	200	22	2016.08.23	160	2.79252680	2016.08.23.15.04
## 2865	190	23	2016.08.23	170	2.96705973	2016.08.23.15.04
## 2866	145	24	2016.08.23	215	-2.53072742	2016.08.23.15.04
## 2867	200	25	2016.08.23	160	2.79252680	2016.08.23.15.04
## 2868	200	26	2016.08.23	160	2.79252680	2016.08.23.15.04
## 2869	195	27	2016.08.23	165	2.87979327	2016.08.23.15.04
## 2870	140	28	2016.08.23	220	-2.44346095	2016.08.23.15.04
## 2871	145	29	2016.08.23	215	-2.53072742	2016.08.23.15.04
## 2872	165	30	2016.08.23	195	-2.87979327	2016.08.23.15.04
## 2873	125	31	2016.08.23	235	-2.18166156	2016.08.23.15.04
## 2874	30	32	2016.08.23	330	-0.52359878	2016.08.23.15.04
## 2875	170	33	2016.08.23	190	-2.96705973	2016.08.23.15.04
## 2876	155	34	2016.08.23	205	-2.70526034	2016.08.23.15.04

## 2877	0	35	2016.08.23	0	0.00000000	2016.08.23.15.04
## 2878	160	36	2016.08.23	200	-2.79252680	2016.08.23.15.04
## 2879	150	37	2016.08.23	210	-2.61799388	2016.08.23.15.04
## 2880	0	38	2016.08.23	0	0.00000000	2016.08.23.15.04
## 2881	190	39	2016.08.23	170	2.96705973	2016.08.23.15.04
## 2882	20	40	2016.08.23	340	-0.34906585	2016.08.23.15.04
## 2883	120	41	2016.08.23	240	-2.09439510	2016.08.23.15.04
## 2884	50	42	2016.08.23	310	-0.87266463	2016.08.23.15.04
## 2885	310	43	2016.08.23	50	0.87266463	2016.08.23.15.04
## 2886	325	44	2016.08.23	35	0.61086524	2016.08.23.15.04
## 2887	305	45	2016.08.23	55	0.95993109	2016.08.23.15.04
## 2888	340	46	2016.08.23	20	0.34906585	2016.08.23.15.04
## 2889	315	47	2016.08.23	45	0.78539816	2016.08.23.15.04
## 2890	340	48	2016.08.23	20	0.34906585	2016.08.23.15.04
## 2891	305	49	2016.08.23	55	0.95993109	2016.08.23.15.04
## 2892	320	50	2016.08.23	40	0.69813170	2016.08.23.15.04
## 2893	100	1	2016.08.23	260	-1.74532925	2016.08.23.15.13
## 2894	310	2	2016.08.23	50	0.87266463	2016.08.23.15.13
## 2895	260	3	2016.08.23	100	1.74532925	2016.08.23.15.13
## 2896	250	4	2016.08.23	110	1.91986218	2016.08.23.15.13
## 2897	240	5	2016.08.23	120	2.09439510	2016.08.23.15.13
## 2898	240	6	2016.08.23	120	2.09439510	2016.08.23.15.13
## 2899	160	7	2016.08.23	200	-2.79252680	2016.08.23.15.13
## 2900	220	8	2016.08.23	140	2.44346095	2016.08.23.15.13
## 2901	270	9	2016.08.23	90	1.57079633	2016.08.23.15.13
## 2902	115	10	2016.08.23	245	-2.00712864	2016.08.23.15.13
## 2903	110	11	2016.08.23	250	-1.91986218	2016.08.23.15.13
## 2904	305	12	2016.08.23	55	0.95993109	2016.08.23.15.13
## 2905	230	13	2016.08.23	130	2.26892803	2016.08.23.15.13
## 2906	275	14	2016.08.23	85	1.48352986	2016.08.23.15.13
## 2907	320	15	2016.08.23	40	0.69813170	2016.08.23.15.13
## 2908	240	16	2016.08.23	120	2.09439510	2016.08.23.15.13
## 2909	190	17	2016.08.23	170	2.96705973	2016.08.23.15.13
## 2910	250	18	2016.08.23	110	1.91986218	2016.08.23.15.13
## 2911	240	19	2016.08.23	120	2.09439510	2016.08.23.15.13
## 2912	250	20	2016.08.23	110	1.91986218	2016.08.23.15.13
## 2913	90	21	2016.08.23	270	-1.57079633	2016.08.23.15.13
## 2914	70	22	2016.08.23	290	-1.22173048	2016.08.23.15.13
## 2915	280	23	2016.08.23	80	1.39626340	2016.08.23.15.13
## 2916	300	24	2016.08.23	60	1.04719755	2016.08.23.15.13
## 2917	165	25	2016.08.23	195	-2.87979327	2016.08.23.15.13
## 2918	270	26	2016.08.23	90	1.57079633	2016.08.23.15.13
## 2919	260	27	2016.08.23	100	1.74532925	2016.08.23.15.13
## 2920	170	28	2016.08.23	190	-2.96705973	2016.08.23.15.13
## 2921	280	29	2016.08.23	80	1.39626340	2016.08.23.15.13
## 2922	10	30	2016.08.23	350	-0.17453293	2016.08.23.15.13
## 2923	345	31	2016.08.23	15	0.26179939	2016.08.23.15.13
## 2924	60	32	2016.08.23	300	-1.04719755	2016.08.23.15.13
## 2925	0	33	2016.08.23	0	0.00000000	2016.08.23.15.13
## 2926	15	34	2016.08.23	345	-0.26179939	2016.08.23.15.13
## 2927	10	35	2016.08.23	350	-0.17453293	2016.08.23.15.13
## 2928	40	36	2016.08.23	320	-0.69813170	2016.08.23.15.13
## 2929	35	37	2016.08.23	325	-0.61086524	2016.08.23.15.13
## 2930	50	38	2016.08.23	310	-0.87266463	2016.08.23.15.13

## 2931	30	39	2016.08.23	330	-0.52359878	2016.08.23.15.13
## 2932	15	40	2016.08.23	345	-0.26179939	2016.08.23.15.13
## 2933	20	41	2016.08.23	340	-0.34906585	2016.08.23.15.13
## 2934	20	42	2016.08.23	340	-0.34906585	2016.08.23.15.13
## 2935	20	43	2016.08.23	340	-0.34906585	2016.08.23.15.13
## 2936	210	1	2016.08.24	150	2.61799388	2016.08.24.14.30
## 2937	190	2	2016.08.24	170	2.96705973	2016.08.24.14.30
## 2938	190	3	2016.08.24	170	2.96705973	2016.08.24.14.30
## 2939	160	4	2016.08.24	200	-2.79252680	2016.08.24.14.30
## 2940	130	5	2016.08.24	230	-2.26892803	2016.08.24.14.30
## 2941	220	6	2016.08.24	140	2.44346095	2016.08.24.14.30
## 2942	90	7	2016.08.24	270	-1.57079633	2016.08.24.14.30
## 2943	140	8	2016.08.24	220	-2.44346095	2016.08.24.14.30
## 2944	130	9	2016.08.24	230	-2.26892803	2016.08.24.14.30
## 2945	110	10	2016.08.24	250	-1.91986218	2016.08.24.14.30
## 2946	40	11	2016.08.24	320	-0.69813170	2016.08.24.14.30
## 2947	110	12	2016.08.24	250	-1.91986218	2016.08.24.14.30
## 2948	50	13	2016.08.24	310	-0.87266463	2016.08.24.14.30
## 2949	0	14	2016.08.24	0	0.00000000	2016.08.24.14.30
## 2950	15	15	2016.08.24	345	-0.26179939	2016.08.24.14.30
## 2951	350	16	2016.08.24	10	0.17453293	2016.08.24.14.30
## 2952	40	17	2016.08.24	320	-0.69813170	2016.08.24.14.30
## 2953	5	18	2016.08.24	355	-0.08726646	2016.08.24.14.30
## 2954	20	19	2016.08.24	340	-0.34906585	2016.08.24.14.30
## 2955	355	20	2016.08.24	5	0.08726646	2016.08.24.14.30
## 2956	60	21	2016.08.24	300	-1.04719755	2016.08.24.14.30
## 2957	10	22	2016.08.24	350	-0.17453293	2016.08.24.14.30
## 2958	20	23	2016.08.24	340	-0.34906585	2016.08.24.14.30
## 2959	0	24	2016.08.24	0	0.00000000	2016.08.24.14.30
## 2960	30	25	2016.08.24	330	-0.52359878	2016.08.24.14.30
## 2961	355	26	2016.08.24	5	0.08726646	2016.08.24.14.30
## 2962	20	27	2016.08.24	340	-0.34906585	2016.08.24.14.30
## 2963	50	28	2016.08.24	310	-0.87266463	2016.08.24.14.30
## 2964	0	29	2016.08.24	0	0.00000000	2016.08.24.14.30
## 2965	40	30	2016.08.24	320	-0.69813170	2016.08.24.14.30
## 2966	30	31	2016.08.24	330	-0.52359878	2016.08.24.14.30
## 2967	215	1	2016.08.24	145	2.53072742	2016.08.24.14.40
## 2968	20	2	2016.08.24	340	-0.34906585	2016.08.24.14.40
## 2969	290	3	2016.08.24	70	1.22173048	2016.08.24.14.40
## 2970	10	4	2016.08.24	350	-0.17453293	2016.08.24.14.40
## 2971	230	5	2016.08.24	130	2.26892803	2016.08.24.14.40
## 2972	280	6	2016.08.24	80	1.39626340	2016.08.24.14.40
## 2973	10	7	2016.08.24	350	-0.17453293	2016.08.24.14.40
## 2974	340	8	2016.08.24	20	0.34906585	2016.08.24.14.40
## 2975	0	9	2016.08.24	0	0.00000000	2016.08.24.14.40
## 2976	0	10	2016.08.24	0	0.00000000	2016.08.24.14.40
## 2977	40	11	2016.08.24	320	-0.69813170	2016.08.24.14.40
## 2978	250	12	2016.08.24	110	1.91986218	2016.08.24.14.40
## 2979	240	13	2016.08.24	120	2.09439510	2016.08.24.14.40
## 2980	170	14	2016.08.24	190	-2.96705973	2016.08.24.14.40
## 2981	110	15	2016.08.24	250	-1.91986218	2016.08.24.14.40
## 2982	30	16	2016.08.24	330	-0.52359878	2016.08.24.14.40
## 2983	30	17	2016.08.24	330	-0.52359878	2016.08.24.14.40
## 2984	35	18	2016.08.24	325	-0.61086524	2016.08.24.14.40

## 2985	65	19	2016.08.24	295	-1.13446401	2016.08.24.14.40
## 2986	120	20	2016.08.24	240	-2.09439510	2016.08.24.14.40
## 2987	270	21	2016.08.24	90	1.57079633	2016.08.24.14.40
## 2988	320	22	2016.08.24	40	0.69813170	2016.08.24.14.40
## 2989	250	23	2016.08.24	110	1.91986218	2016.08.24.14.40
## 2990	5	24	2016.08.24	355	-0.08726646	2016.08.24.14.40
## 2991	55	25	2016.08.24	305	-0.95993109	2016.08.24.14.40
## 2992	180	26	2016.08.24	180	3.14159265	2016.08.24.14.40
## 2993	65	27	2016.08.24	295	-1.13446401	2016.08.24.14.40
## 2994	50	28	2016.08.24	310	-0.87266463	2016.08.24.14.40
## 2995	280	29	2016.08.24	80	1.39626340	2016.08.24.14.40
## 2996	0	30	2016.08.24	0	0.00000000	2016.08.24.14.40
## 2997	200	31	2016.08.24	160	2.79252680	2016.08.24.14.40
## 2998	45	32	2016.08.24	315	-0.78539816	2016.08.24.14.40
## 2999	190	33	2016.08.24	170	2.96705973	2016.08.24.14.40
## 3000	355	34	2016.08.24	5	0.08726646	2016.08.24.14.40
## 3001	330	35	2016.08.24	30	0.52359878	2016.08.24.14.40
## 3002	10	36	2016.08.24	350	-0.17453293	2016.08.24.14.40
## 3003	15	37	2016.08.24	345	-0.26179939	2016.08.24.14.40
## 3004	0	38	2016.08.24	0	0.00000000	2016.08.24.14.40
## 3005	350	39	2016.08.24	10	0.17453293	2016.08.24.14.40
## 3006	30	40	2016.08.24	330	-0.52359878	2016.08.24.14.40
## 3007	5	41	2016.08.24	355	-0.08726646	2016.08.24.14.40
## 3008	10	42	2016.08.24	350	-0.17453293	2016.08.24.14.40
## 3009	30	43	2016.08.24	330	-0.52359878	2016.08.24.14.40
## 3010	0	44	2016.08.24	0	0.00000000	2016.08.24.14.40
## 3011	130	1	2016.08.24	230	-2.26892803	2016.08.24.15.48
## 3012	330	2	2016.08.24	30	0.52359878	2016.08.24.15.48
## 3013	300	3	2016.08.24	60	1.04719755	2016.08.24.15.48
## 3014	70	4	2016.08.24	290	-1.22173048	2016.08.24.15.48
## 3015	15	5	2016.08.24	345	-0.26179939	2016.08.24.15.48
## 3016	355	6	2016.08.24	5	0.08726646	2016.08.24.15.48
## 3017	255	7	2016.08.24	105	1.83259571	2016.08.24.15.48
## 3018	140	8	2016.08.24	220	-2.44346095	2016.08.24.15.48
## 3019	80	9	2016.08.24	280	-1.39626340	2016.08.24.15.48
## 3020	355	10	2016.08.24	5	0.08726646	2016.08.24.15.48
## 3021	260	11	2016.08.24	100	1.74532925	2016.08.24.15.48
## 3022	60	12	2016.08.24	300	-1.04719755	2016.08.24.15.48
## 3023	200	13	2016.08.24	160	2.79252680	2016.08.24.15.48
## 3024	110	14	2016.08.24	250	-1.91986218	2016.08.24.15.48
## 3025	160	15	2016.08.24	200	-2.79252680	2016.08.24.15.48
## 3026	50	16	2016.08.24	310	-0.87266463	2016.08.24.15.48
## 3027	260	17	2016.08.24	100	1.74532925	2016.08.24.15.48
## 3028	280	18	2016.08.24	80	1.39626340	2016.08.24.15.48
## 3029	10	19	2016.08.24	350	-0.17453293	2016.08.24.15.48
## 3030	20	20	2016.08.24	340	-0.34906585	2016.08.24.15.48
## 3031	70	21	2016.08.24	290	-1.22173048	2016.08.24.15.48
## 3032	50	22	2016.08.24	310	-0.87266463	2016.08.24.15.48
## 3033	90	23	2016.08.24	270	-1.57079633	2016.08.24.15.48
## 3034	80	24	2016.08.24	280	-1.39626340	2016.08.24.15.48
## 3035	90	25	2016.08.24	270	-1.57079633	2016.08.24.15.48
## 3036	90	26	2016.08.24	270	-1.57079633	2016.08.24.15.48
## 3037	10	27	2016.08.24	350	-0.17453293	2016.08.24.15.48
## 3038	90	28	2016.08.24	270	-1.57079633	2016.08.24.15.48

## 3039	55	1	2016.08.24	305	-0.95993109	2016.08.24.16.09
## 3040	240	2	2016.08.24	120	2.09439510	2016.08.24.16.09
## 3041	220	3	2016.08.24	140	2.44346095	2016.08.24.16.09
## 3042	230	4	2016.08.24	130	2.26892803	2016.08.24.16.09
## 3043	270	5	2016.08.24	90	1.57079633	2016.08.24.16.09
## 3044	170	6	2016.08.24	190	-2.96705973	2016.08.24.16.09
## 3045	250	7	2016.08.24	110	1.91986218	2016.08.24.16.09
## 3046	240	8	2016.08.24	120	2.09439510	2016.08.24.16.09
## 3047	260	9	2016.08.24	100	1.74532925	2016.08.24.16.09
## 3048	240	10	2016.08.24	120	2.09439510	2016.08.24.16.09
## 3049	250	11	2016.08.24	110	1.91986218	2016.08.24.16.09
## 3050	190	12	2016.08.24	170	2.96705973	2016.08.24.16.09
## 3051	165	13	2016.08.24	195	-2.87979327	2016.08.24.16.09
## 3052	310	14	2016.08.24	50	0.87266463	2016.08.24.16.09
## 3053	170	15	2016.08.24	190	-2.96705973	2016.08.24.16.09
## 3054	150	16	2016.08.24	210	-2.61799388	2016.08.24.16.09
## 3055	320	17	2016.08.24	40	0.69813170	2016.08.24.16.09
## 3056	220	18	2016.08.24	140	2.44346095	2016.08.24.16.09
## 3057	250	19	2016.08.24	110	1.91986218	2016.08.24.16.09
## 3058	340	20	2016.08.24	20	0.34906585	2016.08.24.16.09
## 3059	340	21	2016.08.24	20	0.34906585	2016.08.24.16.09
## 3060	350	22	2016.08.24	10	0.17453293	2016.08.24.16.09
## 3061	175	23	2016.08.24	185	-3.05432619	2016.08.24.16.09
## 3062	270	24	2016.08.24	90	1.57079633	2016.08.24.16.09
## 3063	10	25	2016.08.24	350	-0.17453293	2016.08.24.16.09
## 3064	30	26	2016.08.24	330	-0.52359878	2016.08.24.16.09
## 3065	10	27	2016.08.24	350	-0.17453293	2016.08.24.16.09
## 3066	340	28	2016.08.24	20	0.34906585	2016.08.24.16.09
## 3067	50	29	2016.08.24	310	-0.87266463	2016.08.24.16.09
## 3068	25	30	2016.08.24	335	-0.43633231	2016.08.24.16.09
## 3069	60	31	2016.08.24	300	-1.04719755	2016.08.24.16.09
## 3070	100	1	2016.08.24	260	-1.74532925	2016.08.24.16.15
## 3071	10	2	2016.08.24	350	-0.17453293	2016.08.24.16.15
## 3072	90	3	2016.08.24	270	-1.57079633	2016.08.24.16.15
## 3073	10	4	2016.08.24	350	-0.17453293	2016.08.24.16.15
## 3074	90	5	2016.08.24	270	-1.57079633	2016.08.24.16.15
## 3075	95	6	2016.08.24	265	-1.65806279	2016.08.24.16.15
## 3076	25	7	2016.08.24	335	-0.43633231	2016.08.24.16.15
## 3077	10	8	2016.08.24	350	-0.17453293	2016.08.24.16.15
## 3078	35	9	2016.08.24	325	-0.61086524	2016.08.24.16.15
## 3079	200	10	2016.08.24	160	2.79252680	2016.08.24.16.15
## 3080	30	11	2016.08.24	330	-0.52359878	2016.08.24.16.15
## 3081	150	12	2016.08.24	210	-2.61799388	2016.08.24.16.15
## 3082	10	13	2016.08.24	350	-0.17453293	2016.08.24.16.15
## 3083	20	14	2016.08.24	340	-0.34906585	2016.08.24.16.15
## 3084	200	15	2016.08.24	160	2.79252680	2016.08.24.16.15
## 3085	190	16	2016.08.24	170	2.96705973	2016.08.24.16.15
## 3086	190	17	2016.08.24	170	2.96705973	2016.08.24.16.15
## 3087	340	18	2016.08.24	20	0.34906585	2016.08.24.16.15
## 3088	30	19	2016.08.24	330	-0.52359878	2016.08.24.16.15
## 3089	345	20	2016.08.24	15	0.26179939	2016.08.24.16.15
## 3090	50	21	2016.08.24	310	-0.87266463	2016.08.24.16.15
## 3091	90	22	2016.08.24	270	-1.57079633	2016.08.24.16.15
## 3092	185	23	2016.08.24	175	3.05432619	2016.08.24.16.15

## 3093	20	24	2016.08.24	340	-0.34906585	2016.08.24.16.15
## 3094	40	25	2016.08.24	320	-0.69813170	2016.08.24.16.15
## 3095	120	26	2016.08.24	240	-2.09439510	2016.08.24.16.15
## 3096	260	27	2016.08.24	100	1.74532925	2016.08.24.16.15
## 3097	260	28	2016.08.24	100	1.74532925	2016.08.24.16.15
## 3098	270	29	2016.08.24	90	1.57079633	2016.08.24.16.15
## 3099	265	30	2016.08.24	95	1.65806279	2016.08.24.16.15
## 3100	260	31	2016.08.24	100	1.74532925	2016.08.24.16.15
## 3101	270	32	2016.08.24	90	1.57079633	2016.08.24.16.15
## 3102	260	33	2016.08.24	100	1.74532925	2016.08.24.16.15
## 3103	265	34	2016.08.24	95	1.65806279	2016.08.24.16.15
## 3104	310	35	2016.08.24	50	0.87266463	2016.08.24.16.15
## 3105	270	36	2016.08.24	90	1.57079633	2016.08.24.16.15
## 3106	320	37	2016.08.24	40	0.69813170	2016.08.24.16.15
## 3107	290	38	2016.08.24	70	1.22173048	2016.08.24.16.15
## 3108	250	39	2016.08.24	110	1.91986218	2016.08.24.16.15
## 3109	250	40	2016.08.24	110	1.91986218	2016.08.24.16.15
## 3110	245	41	2016.08.24	115	2.00712864	2016.08.24.16.15
## 3111	275	42	2016.08.24	85	1.48352986	2016.08.24.16.15
## 3112	270	43	2016.08.24	90	1.57079633	2016.08.24.16.15
## 3113	320	44	2016.08.24	40	0.69813170	2016.08.24.16.15
## 3114	290	45	2016.08.24	70	1.22173048	2016.08.24.16.15
## 3115	270	46	2016.08.24	90	1.57079633	2016.08.24.16.15
## 3116	260	47	2016.08.24	100	1.74532925	2016.08.24.16.15
## 3117	50	1	2016.08.24	310	-0.87266463	2016.08.24.15.41
## 3118	60	2	2016.08.24	300	-1.04719755	2016.08.24.15.41
## 3119	60	3	2016.08.24	300	-1.04719755	2016.08.24.15.41
## 3120	270	4	2016.08.24	90	1.57079633	2016.08.24.15.41
## 3121	40	5	2016.08.24	320	-0.69813170	2016.08.24.15.41
## 3122	150	6	2016.08.24	210	-2.61799388	2016.08.24.15.41
## 3123	50	7	2016.08.24	310	-0.87266463	2016.08.24.15.41
## 3124	20	8	2016.08.24	340	-0.34906585	2016.08.24.15.41
## 3125	60	9	2016.08.24	300	-1.04719755	2016.08.24.15.41
## 3126	35	10	2016.08.24	325	-0.61086524	2016.08.24.15.41
## 3127	0	11	2016.08.24	0	0.00000000	2016.08.24.15.41
## 3128	20	12	2016.08.24	340	-0.34906585	2016.08.24.15.41
## 3129	10	13	2016.08.24	350	-0.17453293	2016.08.24.15.41
## 3130	10	14	2016.08.24	350	-0.17453293	2016.08.24.15.41
## 3131	10	15	2016.08.24	350	-0.17453293	2016.08.24.15.41
## 3132	250	16	2016.08.24	110	1.91986218	2016.08.24.15.41
## 3133	70	17	2016.08.24	290	-1.22173048	2016.08.24.15.41
## 3134	140	18	2016.08.24	220	-2.44346095	2016.08.24.15.41
## 3135	30	19	2016.08.24	330	-0.52359878	2016.08.24.15.41
## 3136	270	20	2016.08.24	90	1.57079633	2016.08.24.15.41
## 3137	160	21	2016.08.24	200	-2.79252680	2016.08.24.15.41
## 3138	330	22	2016.08.24	30	0.52359878	2016.08.24.15.41
## 3139	165	23	2016.08.24	195	-2.87979327	2016.08.24.15.41
## 3140	70	24	2016.08.24	290	-1.22173048	2016.08.24.15.41
## 3141	0	25	2016.08.24	0	0.00000000	2016.08.24.15.41
## 3142	70	26	2016.08.24	290	-1.22173048	2016.08.24.15.41
## 3143	175	27	2016.08.24	185	-3.05432619	2016.08.24.15.41
## 3144	340	28	2016.08.24	20	0.34906585	2016.08.24.15.41
## 3145	20	29	2016.08.24	340	-0.34906585	2016.08.24.15.41
## 3146	270	30	2016.08.24	90	1.57079633	2016.08.24.15.41

## 3147	110	31	2016.08.24	250	-1.91986218	2016.08.24.15.41
## 3148	170	32	2016.08.24	190	-2.96705973	2016.08.24.15.41
## 3149	260	1	2016.08.25	100	1.74532925	2016.08.25.13.01
## 3150	270	2	2016.08.25	90	1.57079633	2016.08.25.13.01
## 3151	260	3	2016.08.25	100	1.74532925	2016.08.25.13.01
## 3152	160	4	2016.08.25	200	-2.79252680	2016.08.25.13.01
## 3153	250	5	2016.08.25	110	1.91986218	2016.08.25.13.01
## 3154	130	6	2016.08.25	230	-2.26892803	2016.08.25.13.01
## 3155	150	7	2016.08.25	210	-2.61799388	2016.08.25.13.01
## 3156	5	8	2016.08.25	355	-0.08726646	2016.08.25.13.01
## 3157	150	9	2016.08.25	210	-2.61799388	2016.08.25.13.01
## 3158	150	10	2016.08.25	210	-2.61799388	2016.08.25.13.01
## 3159	90	11	2016.08.25	270	-1.57079633	2016.08.25.13.01
## 3160	210	12	2016.08.25	150	2.61799388	2016.08.25.13.01
## 3161	70	13	2016.08.25	290	-1.22173048	2016.08.25.13.01
## 3162	200	14	2016.08.25	160	2.79252680	2016.08.25.13.01
## 3163	190	15	2016.08.25	170	2.96705973	2016.08.25.13.01
## 3164	185	16	2016.08.25	175	3.05432619	2016.08.25.13.01
## 3165	310	17	2016.08.25	50	0.87266463	2016.08.25.13.01
## 3166	320	18	2016.08.25	40	0.69813170	2016.08.25.13.01
## 3167	270	19	2016.08.25	90	1.57079633	2016.08.25.13.01
## 3168	150	1	2016.08.25	210	-2.61799388	2016.08.25.13.14
## 3169	355	2	2016.08.25	5	0.08726646	2016.08.25.13.14
## 3170	150	3	2016.08.25	210	-2.61799388	2016.08.25.13.14
## 3171	110	4	2016.08.25	250	-1.91986218	2016.08.25.13.14
## 3172	90	5	2016.08.25	270	-1.57079633	2016.08.25.13.14
## 3173	130	6	2016.08.25	230	-2.26892803	2016.08.25.13.14
## 3174	165	7	2016.08.25	195	-2.87979327	2016.08.25.13.14
## 3175	90	8	2016.08.25	270	-1.57079633	2016.08.25.13.14
## 3176	100	9	2016.08.25	260	-1.74532925	2016.08.25.13.14
## 3177	160	10	2016.08.25	200	-2.79252680	2016.08.25.13.14
## 3178	240	11	2016.08.25	120	2.09439510	2016.08.25.13.14
## 3179	160	12	2016.08.25	200	-2.79252680	2016.08.25.13.14
## 3180	120	13	2016.08.25	240	-2.09439510	2016.08.25.13.14
## 3181	190	14	2016.08.25	170	2.96705973	2016.08.25.13.14
## 3182	140	15	2016.08.25	220	-2.44346095	2016.08.25.13.14
## 3183	160	16	2016.08.25	200	-2.79252680	2016.08.25.13.14
## 3184	90	17	2016.08.25	270	-1.57079633	2016.08.25.13.14
## 3185	270	18	2016.08.25	90	1.57079633	2016.08.25.13.14
## 3186	350	19	2016.08.25	10	0.17453293	2016.08.25.13.14
## 3187	160	20	2016.08.25	200	-2.79252680	2016.08.25.13.14
## 3188	275	21	2016.08.25	85	1.48352986	2016.08.25.13.14
## 3189	210	22	2016.08.25	150	2.61799388	2016.08.25.13.14
## 3190	210	23	2016.08.25	150	2.61799388	2016.08.25.13.14
## 3191	250	24	2016.08.25	110	1.91986218	2016.08.25.13.14
## 3192	85	25	2016.08.25	275	-1.48352986	2016.08.25.13.14
## 3193	140	26	2016.08.25	220	-2.44346095	2016.08.25.13.14
## 3194	340	27	2016.08.25	20	0.34906585	2016.08.25.13.14
## 3195	330	28	2016.08.25	30	0.52359878	2016.08.25.13.14
## 3196	100	29	2016.08.25	260	-1.74532925	2016.08.25.13.14
## 3197	320	30	2016.08.25	40	0.69813170	2016.08.25.13.14
## 3198	50	31	2016.08.25	310	-0.87266463	2016.08.25.13.14
## 3199	5	32	2016.08.25	355	-0.08726646	2016.08.25.13.14
## 3200	110	33	2016.08.25	250	-1.91986218	2016.08.25.13.14

## 3201	70	34	2016.08.25	290	-1.22173048	2016.08.25.13.14
## 3202	190	35	2016.08.25	170	2.96705973	2016.08.25.13.14
## 3203	275	36	2016.08.25	85	1.48352986	2016.08.25.13.14
## 3204	160	37	2016.08.25	200	-2.79252680	2016.08.25.13.14
## 3205	195	38	2016.08.25	165	2.87979327	2016.08.25.13.14
## 3206	355	39	2016.08.25	5	0.08726646	2016.08.25.13.14
## 3207	135	40	2016.08.25	225	-2.35619449	2016.08.25.13.14
## 3208	120	41	2016.08.25	240	-2.09439510	2016.08.25.13.14
## 3209	100	42	2016.08.25	260	-1.74532925	2016.08.25.13.14
## 3210	270	43	2016.08.25	90	1.57079633	2016.08.25.13.14
## 3211	250	44	2016.08.25	110	1.91986218	2016.08.25.13.14
## 3212	315	45	2016.08.25	45	0.78539816	2016.08.25.13.14
## 3213	75	46	2016.08.25	285	-1.30899694	2016.08.25.13.14
## 3214	255	47	2016.08.25	105	1.83259571	2016.08.25.13.14
## 3215	90	48	2016.08.25	270	-1.57079633	2016.08.25.13.14
## 3216	60	1	2016.08.25	300	-1.04719755	2016.08.25.13.23
## 3217	60	2	2016.08.25	300	-1.04719755	2016.08.25.13.23
## 3218	100	3	2016.08.25	260	-1.74532925	2016.08.25.13.23
## 3219	90	4	2016.08.25	270	-1.57079633	2016.08.25.13.23
## 3220	90	5	2016.08.25	270	-1.57079633	2016.08.25.13.23
## 3221	90	6	2016.08.25	270	-1.57079633	2016.08.25.13.23
## 3222	120	7	2016.08.25	240	-2.09439510	2016.08.25.13.23
## 3223	30	8	2016.08.25	330	-0.52359878	2016.08.25.13.23
## 3224	100	9	2016.08.25	260	-1.74532925	2016.08.25.13.23
## 3225	90	10	2016.08.25	270	-1.57079633	2016.08.25.13.23
## 3226	120	11	2016.08.25	240	-2.09439510	2016.08.25.13.23
## 3227	90	12	2016.08.25	270	-1.57079633	2016.08.25.13.23
## 3228	120	13	2016.08.25	240	-2.09439510	2016.08.25.13.23
## 3229	50	14	2016.08.25	310	-0.87266463	2016.08.25.13.23
## 3230	100	15	2016.08.25	260	-1.74532925	2016.08.25.13.23
## 3231	90	16	2016.08.25	270	-1.57079633	2016.08.25.13.23
## 3232	110	17	2016.08.25	250	-1.91986218	2016.08.25.13.23
## 3233	0	18	2016.08.25	0	0.00000000	2016.08.25.13.23
## 3234	85	19	2016.08.25	275	-1.48352986	2016.08.25.13.23
## 3235	40	20	2016.08.25	320	-0.69813170	2016.08.25.13.23
## 3236	80	21	2016.08.25	280	-1.39626340	2016.08.25.13.23
## 3237	170	22	2016.08.25	190	-2.96705973	2016.08.25.13.23
## 3238	130	23	2016.08.25	230	-2.26892803	2016.08.25.13.23
## 3239	85	24	2016.08.25	275	-1.48352986	2016.08.25.13.23
## 3240	85	25	2016.08.25	275	-1.48352986	2016.08.25.13.23
## 3241	260	26	2016.08.25	100	1.74532925	2016.08.25.13.23
## 3242	60	27	2016.08.25	300	-1.04719755	2016.08.25.13.23
## 3243	100	28	2016.08.25	260	-1.74532925	2016.08.25.13.23
## 3244	350	29	2016.08.25	10	0.17453293	2016.08.25.13.23
## 3245	130	30	2016.08.25	230	-2.26892803	2016.08.25.13.23
## 3246	140	31	2016.08.25	220	-2.44346095	2016.08.25.13.23
## 3247	220	32	2016.08.25	140	2.44346095	2016.08.25.13.23
## 3248	5	33	2016.08.25	355	-0.08726646	2016.08.25.13.23
## 3249	20	34	2016.08.25	340	-0.34906585	2016.08.25.13.23
## 3250	40	35	2016.08.25	320	-0.69813170	2016.08.25.13.23
## 3251	35	36	2016.08.25	325	-0.61086524	2016.08.25.13.23
## 3252	260	37	2016.08.25	100	1.74532925	2016.08.25.13.23
## 3253	330	38	2016.08.25	30	0.52359878	2016.08.25.13.23
## 3254	320	1	2016.08.26	40	0.69813170	2016.08.26.13.59

## 3255	20	2	2016.08.26	340	-0.34906585	2016.08.26.13.59
## 3256	5	3	2016.08.26	355	-0.08726646	2016.08.26.13.59
## 3257	330	4	2016.08.26	30	0.52359878	2016.08.26.13.59
## 3258	290	5	2016.08.26	70	1.22173048	2016.08.26.13.59
## 3259	5	6	2016.08.26	355	-0.08726646	2016.08.26.13.59
## 3260	310	7	2016.08.26	50	0.87266463	2016.08.26.13.59
## 3261	340	8	2016.08.26	20	0.34906585	2016.08.26.13.59
## 3262	60	9	2016.08.26	300	-1.04719755	2016.08.26.13.59
## 3263	300	10	2016.08.26	60	1.04719755	2016.08.26.13.59
## 3264	30	11	2016.08.26	330	-0.52359878	2016.08.26.13.59
## 3265	310	12	2016.08.26	50	0.87266463	2016.08.26.13.59
## 3266	20	13	2016.08.26	340	-0.34906585	2016.08.26.13.59
## 3267	260	14	2016.08.26	100	1.74532925	2016.08.26.13.59
## 3268	50	15	2016.08.26	310	-0.87266463	2016.08.26.13.59
## 3269	30	16	2016.08.26	330	-0.52359878	2016.08.26.13.59
## 3270	340	17	2016.08.26	20	0.34906585	2016.08.26.13.59
## 3271	60	18	2016.08.26	300	-1.04719755	2016.08.26.13.59
## 3272	70	19	2016.08.26	290	-1.22173048	2016.08.26.13.59
## 3273	125	20	2016.08.26	235	-2.18166156	2016.08.26.13.59
## 3274	335	21	2016.08.26	25	0.43633231	2016.08.26.13.59
## 3275	180	22	2016.08.26	180	3.14159265	2016.08.26.13.59
## 3276	325	23	2016.08.26	35	0.61086524	2016.08.26.13.59
## 3277	230	24	2016.08.26	130	2.26892803	2016.08.26.13.59
## 3278	240	25	2016.08.26	120	2.09439510	2016.08.26.13.59
## 3279	220	26	2016.08.26	140	2.44346095	2016.08.26.13.59
## 3280	215	27	2016.08.26	145	2.53072742	2016.08.26.13.59
## 3281	225	28	2016.08.26	135	2.35619449	2016.08.26.13.59
## 3282	355	1	2016.08.26	5	0.08726646	2016.08.26.14.39
## 3283	260	2	2016.08.26	100	1.74532925	2016.08.26.14.39
## 3284	105	3	2016.08.26	255	-1.83259571	2016.08.26.14.39
## 3285	130	4	2016.08.26	230	-2.26892803	2016.08.26.14.39
## 3286	205	5	2016.08.26	155	2.70526034	2016.08.26.14.39
## 3287	125	6	2016.08.26	235	-2.18166156	2016.08.26.14.39
## 3288	165	7	2016.08.26	195	-2.87979327	2016.08.26.14.39
## 3289	150	8	2016.08.26	210	-2.61799388	2016.08.26.14.39
## 3290	30	9	2016.08.26	330	-0.52359878	2016.08.26.14.39
## 3291	60	10	2016.08.26	300	-1.04719755	2016.08.26.14.39
## 3292	80	11	2016.08.26	280	-1.39626340	2016.08.26.14.39
## 3293	95	12	2016.08.26	265	-1.65806279	2016.08.26.14.39
## 3294	0	13	2016.08.26	0	0.00000000	2016.08.26.14.39
## 3295	50	14	2016.08.26	310	-0.87266463	2016.08.26.14.39
## 3296	350	15	2016.08.26	10	0.17453293	2016.08.26.14.39
## 3297	120	16	2016.08.26	240	-2.09439510	2016.08.26.14.39
## 3298	95	17	2016.08.26	265	-1.65806279	2016.08.26.14.39
## 3299	105	18	2016.08.26	255	-1.83259571	2016.08.26.14.39
## 3300	115	19	2016.08.26	245	-2.00712864	2016.08.26.14.39
## 3301	350	20	2016.08.26	10	0.17453293	2016.08.26.14.39
## 3302	80	21	2016.08.26	280	-1.39626340	2016.08.26.14.39
## 3303	60	22	2016.08.26	300	-1.04719755	2016.08.26.14.39
## 3304	185	1	2016.08.26	175	3.05432619	2016.08.26.15.03
## 3305	305	2	2016.08.26	55	0.95993109	2016.08.26.15.03
## 3306	125	3	2016.08.26	235	-2.18166156	2016.08.26.15.03
## 3307	130	4	2016.08.26	230	-2.26892803	2016.08.26.15.03
## 3308	290	5	2016.08.26	70	1.22173048	2016.08.26.15.03

## 3309	15	6	2016.08.26	345	-0.26179939	2016.08.26.15.03
## 3310	15	7	2016.08.26	345	-0.26179939	2016.08.26.15.03
## 3311	80	8	2016.08.26	280	-1.39626340	2016.08.26.15.03
## 3312	290	9	2016.08.26	70	1.22173048	2016.08.26.15.03
## 3313	25	10	2016.08.26	335	-0.43633231	2016.08.26.15.03
## 3314	240	11	2016.08.26	120	2.09439510	2016.08.26.15.03
## 3315	40	12	2016.08.26	320	-0.69813170	2016.08.26.15.03
## 3316	330	13	2016.08.26	30	0.52359878	2016.08.26.15.03
## 3317	170	14	2016.08.26	190	-2.96705973	2016.08.26.15.03
## 3318	310	15	2016.08.26	50	0.87266463	2016.08.26.15.03
## 3319	250	16	2016.08.26	110	1.91986218	2016.08.26.15.03
## 3320	250	17	2016.08.26	110	1.91986218	2016.08.26.15.03
## 3321	255	18	2016.08.26	105	1.83259571	2016.08.26.15.03
## 3322	50	19	2016.08.26	310	-0.87266463	2016.08.26.15.03
## 3323	335	20	2016.08.26	25	0.43633231	2016.08.26.15.03
## 3324	80	21	2016.08.26	280	-1.39626340	2016.08.26.15.03
## 3325	335	22	2016.08.26	25	0.43633231	2016.08.26.15.03
## 3326	335	23	2016.08.26	25	0.43633231	2016.08.26.15.03
## 3327	20	24	2016.08.26	340	-0.34906585	2016.08.26.15.03
## 3328	290	25	2016.08.26	70	1.22173048	2016.08.26.15.03
## 3329	15	26	2016.08.26	345	-0.26179939	2016.08.26.15.03
## 3330	260	27	2016.08.26	100	1.74532925	2016.08.26.15.03
## 3331	85	28	2016.08.26	275	-1.48352986	2016.08.26.15.03
## 3332	0	29	2016.08.26	0	0.00000000	2016.08.26.15.03
## 3333	10	30	2016.08.26	350	-0.17453293	2016.08.26.15.03
## 3334	345	31	2016.08.26	15	0.26179939	2016.08.26.15.03
## 3335	300	32	2016.08.26	60	1.04719755	2016.08.26.15.03
## 3336	30	33	2016.08.26	330	-0.52359878	2016.08.26.15.03
## 3337	320	34	2016.08.26	40	0.69813170	2016.08.26.15.03
## 3338	20	35	2016.08.26	340	-0.34906585	2016.08.26.15.03
## 3339	10	36	2016.08.26	350	-0.17453293	2016.08.26.15.03
## 3340	315	37	2016.08.26	45	0.78539816	2016.08.26.15.03
## 3341	20	38	2016.08.26	340	-0.34906585	2016.08.26.15.03
## 3342	290	39	2016.08.26	70	1.22173048	2016.08.26.15.03
## 3343	5	40	2016.08.26	355	-0.08726646	2016.08.26.15.03
## 3344	335	41	2016.08.26	25	0.43633231	2016.08.26.15.03
## 3345	345	42	2016.08.26	15	0.26179939	2016.08.26.15.03
## 3346	325	43	2016.08.26	35	0.61086524	2016.08.26.15.03
## 3347	10	44	2016.08.26	350	-0.17453293	2016.08.26.15.03
## 3348	340	45	2016.08.26	20	0.34906585	2016.08.26.15.03
## 3349	50	46	2016.08.26	310	-0.87266463	2016.08.26.15.03
## 3350	340	47	2016.08.26	20	0.34906585	2016.08.26.15.03
## 3351	150	48	2016.08.26	210	-2.61799388	2016.08.26.15.03
## 3352	345	49	2016.08.26	15	0.26179939	2016.08.26.15.03
## 3353	35	50	2016.08.26	325	-0.61086524	2016.08.26.15.03
## 3354	310	51	2016.08.26	50	0.87266463	2016.08.26.15.03
## 3355	0	52	2016.08.26	0	0.00000000	2016.08.26.15.03
## 3356	345	53	2016.08.26	15	0.26179939	2016.08.26.15.03
## 3357	285	54	2016.08.26	75	1.30899694	2016.08.26.15.03
## 3358	340	55	2016.08.26	20	0.34906585	2016.08.26.15.03
## 3359	330	56	2016.08.26	30	0.52359878	2016.08.26.15.03
## 3360	250	57	2016.08.26	110	1.91986218	2016.08.26.15.03
## 3361	335	58	2016.08.26	25	0.43633231	2016.08.26.15.03
## 3362	245	59	2016.08.26	115	2.00712864	2016.08.26.15.03

## 3363	345	60	2016.08.26	15	0.26179939	2016.08.26.15.03
## 3364	320	61	2016.08.26	40	0.69813170	2016.08.26.15.03
## 3365	340	62	2016.08.26	20	0.34906585	2016.08.26.15.03
## 3366	270	63	2016.08.26	90	1.57079633	2016.08.26.15.03
## 3367	330	64	2016.08.26	30	0.52359878	2016.08.26.15.03
## 3368	320	65	2016.08.26	40	0.69813170	2016.08.26.15.03
## 3369	355	66	2016.08.26	5	0.08726646	2016.08.26.15.03
## 3370	310	67	2016.08.26	50	0.87266463	2016.08.26.15.03
## 3371	320	68	2016.08.26	40	0.69813170	2016.08.26.15.03
## 3372	270	69	2016.08.26	90	1.57079633	2016.08.26.15.03
## 3373	340	70	2016.08.26	20	0.34906585	2016.08.26.15.03
## 3374	270	71	2016.08.26	90	1.57079633	2016.08.26.15.03
## 3375	350	72	2016.08.26	10	0.17453293	2016.08.26.15.03
## 3376	355	73	2016.08.26	5	0.08726646	2016.08.26.15.03
## 3377	340	74	2016.08.26	20	0.34906585	2016.08.26.15.03
## 3378	320	75	2016.08.26	40	0.69813170	2016.08.26.15.03
## 3379	185	1	2016.08.26	175	3.05432619	2016.08.26.15.05
## 3380	210	2	2016.08.26	150	2.61799388	2016.08.26.15.05
## 3381	170	3	2016.08.26	190	-2.96705973	2016.08.26.15.05
## 3382	220	4	2016.08.26	140	2.44346095	2016.08.26.15.05
## 3383	190	5	2016.08.26	170	2.96705973	2016.08.26.15.05
## 3384	260	6	2016.08.26	100	1.74532925	2016.08.26.15.05
## 3385	200	7	2016.08.26	160	2.79252680	2016.08.26.15.05
## 3386	270	8	2016.08.26	90	1.57079633	2016.08.26.15.05
## 3387	160	9	2016.08.26	200	-2.79252680	2016.08.26.15.05
## 3388	260	10	2016.08.26	100	1.74532925	2016.08.26.15.05
## 3389	140	11	2016.08.26	220	-2.44346095	2016.08.26.15.05
## 3390	205	12	2016.08.26	155	2.70526034	2016.08.26.15.05
## 3391	340	13	2016.08.26	20	0.34906585	2016.08.26.15.05
## 3392	340	14	2016.08.26	20	0.34906585	2016.08.26.15.05
## 3393	5	15	2016.08.26	355	-0.08726646	2016.08.26.15.05
## 3394	40	16	2016.08.26	320	-0.69813170	2016.08.26.15.05
## 3395	10	17	2016.08.26	350	-0.17453293	2016.08.26.15.05
## 3396	10	18	2016.08.26	350	-0.17453293	2016.08.26.15.05
## 3397	60	19	2016.08.26	300	-1.04719755	2016.08.26.15.05
## 3398	5	20	2016.08.26	355	-0.08726646	2016.08.26.15.05
## 3399	10	21	2016.08.26	350	-0.17453293	2016.08.26.15.05
## 3400	15	22	2016.08.26	345	-0.26179939	2016.08.26.15.05
## 3401	0	23	2016.08.26	0	0.00000000	2016.08.26.15.05
## 3402	50	24	2016.08.26	310	-0.87266463	2016.08.26.15.05
## 3403	340	25	2016.08.26	20	0.34906585	2016.08.26.15.05
## 3404	20	26	2016.08.26	340	-0.34906585	2016.08.26.15.05
## 3405	105	1	2016.08.26	255	-1.83259571	2016.08.26.14.52
## 3406	125	2	2016.08.26	235	-2.18166156	2016.08.26.14.52
## 3407	85	3	2016.08.26	275	-1.48352986	2016.08.26.14.52
## 3408	200	4	2016.08.26	160	2.79252680	2016.08.26.14.52
## 3409	50	5	2016.08.26	310	-0.87266463	2016.08.26.14.52
## 3410	145	6	2016.08.26	215	-2.53072742	2016.08.26.14.52
## 3411	135	7	2016.08.26	225	-2.35619449	2016.08.26.14.52
## 3412	140	8	2016.08.26	220	-2.44346095	2016.08.26.14.52
## 3413	160	9	2016.08.26	200	-2.79252680	2016.08.26.14.52
## 3414	155	10	2016.08.26	205	-2.70526034	2016.08.26.14.52
## 3415	130	11	2016.08.26	230	-2.26892803	2016.08.26.14.52
## 3416	200	12	2016.08.26	160	2.79252680	2016.08.26.14.52

## 3417	120	13	2016.08.26	240	-2.09439510	2016.08.26.14.52
## 3418	100	14	2016.08.26	260	-1.74532925	2016.08.26.14.52
## 3419	165	15	2016.08.26	195	-2.87979327	2016.08.26.14.52
## 3420	15	16	2016.08.26	345	-0.26179939	2016.08.26.14.52
## 3421	5	17	2016.08.26	355	-0.08726646	2016.08.26.14.52
## 3422	20	18	2016.08.26	340	-0.34906585	2016.08.26.14.52
## 3423	40	19	2016.08.26	320	-0.69813170	2016.08.26.14.52
## 3424	10	20	2016.08.26	350	-0.17453293	2016.08.26.14.52
## 3425	355	21	2016.08.26	5	0.08726646	2016.08.26.14.52
## 3426	20	22	2016.08.26	340	-0.34906585	2016.08.26.14.52
## 3427	30	23	2016.08.26	330	-0.52359878	2016.08.26.14.52
## 3428	350	24	2016.08.26	10	0.17453293	2016.08.26.14.52
## 3429	320	25	2016.08.26	40	0.69813170	2016.08.26.14.52
## 3430	0	26	2016.08.26	0	0.00000000	2016.08.26.14.52
## 3431	350	27	2016.08.26	10	0.17453293	2016.08.26.14.52
## 3432	355	28	2016.08.26	5	0.08726646	2016.08.26.14.52
## 3433	25	29	2016.08.26	335	-0.43633231	2016.08.26.14.52
## 3434	30	30	2016.08.26	330	-0.52359878	2016.08.26.14.52
## 3435	355	31	2016.08.26	5	0.08726646	2016.08.26.14.52
## 3436	330	32	2016.08.26	30	0.52359878	2016.08.26.14.52
## 3437	25	33	2016.08.26	335	-0.43633231	2016.08.26.14.52
## 3438	85	1	2016.08.31	275	-1.48352986	2016.08.31.15.16
## 3439	10	2	2016.08.31	350	-0.17453293	2016.08.31.15.16
## 3440	320	3	2016.08.31	40	0.69813170	2016.08.31.15.16
## 3441	95	4	2016.08.31	265	-1.65806279	2016.08.31.15.16
## 3442	155	5	2016.08.31	205	-2.70526034	2016.08.31.15.16
## 3443	180	6	2016.08.31	180	3.14159265	2016.08.31.15.16
## 3444	150	7	2016.08.31	210	-2.61799388	2016.08.31.15.16
## 3445	80	8	2016.08.31	280	-1.39626340	2016.08.31.15.16
## 3446	95	9	2016.08.31	265	-1.65806279	2016.08.31.15.16
## 3447	70	10	2016.08.31	290	-1.22173048	2016.08.31.15.16
## 3448	150	11	2016.08.31	210	-2.61799388	2016.08.31.15.16
## 3449	145	12	2016.08.31	215	-2.53072742	2016.08.31.15.16
## 3450	110	13	2016.08.31	250	-1.91986218	2016.08.31.15.16
## 3451	115	14	2016.08.31	245	-2.00712864	2016.08.31.15.16
## 3452	110	15	2016.08.31	250	-1.91986218	2016.08.31.15.16
## 3453	65	16	2016.08.31	295	-1.13446401	2016.08.31.15.16
## 3454	330	17	2016.08.31	30	0.52359878	2016.08.31.15.16
## 3455	340	18	2016.08.31	20	0.34906585	2016.08.31.15.16
## 3456	330	19	2016.08.31	30	0.52359878	2016.08.31.15.16
## 3457	320	20	2016.08.31	40	0.69813170	2016.08.31.15.16
## 3458	310	21	2016.08.31	50	0.87266463	2016.08.31.15.16
## 3459	290	22	2016.08.31	70	1.22173048	2016.08.31.15.16
## 3460	325	23	2016.08.31	35	0.61086524	2016.08.31.15.16
## 3461	310	24	2016.08.31	50	0.87266463	2016.08.31.15.16
## 3462	300	25	2016.08.31	60	1.04719755	2016.08.31.15.16
## 3463	340	26	2016.08.31	20	0.34906585	2016.08.31.15.16
## 3464	220	27	2016.08.31	140	2.44346095	2016.08.31.15.16
## 3465	80	28	2016.08.31	280	-1.39626340	2016.08.31.15.16
## 3466	70	29	2016.08.31	290	-1.22173048	2016.08.31.15.16
## 3467	90	30	2016.08.31	270	-1.57079633	2016.08.31.15.16
## 3468	150	31	2016.08.31	210	-2.61799388	2016.08.31.15.16
## 3469	145	32	2016.08.31	215	-2.53072742	2016.08.31.15.16
## 3470	140	33	2016.08.31	220	-2.44346095	2016.08.31.15.16

## 3471	140	34	2016.08.31	220	-2.44346095	2016.08.31.15.16
## 3472	90	35	2016.08.31	270	-1.57079633	2016.08.31.15.16
## 3473	95	36	2016.08.31	265	-1.65806279	2016.08.31.15.16
## 3474	120	37	2016.08.31	240	-2.09439510	2016.08.31.15.16
## 3475	355	38	2016.08.31	5	0.08726646	2016.08.31.15.16
## 3476	40	1	2016.08.31	320	-0.69813170	2016.08.31.15.35
## 3477	330	2	2016.08.31	30	0.52359878	2016.08.31.15.35
## 3478	90	3	2016.08.31	270	-1.57079633	2016.08.31.15.35
## 3479	340	4	2016.08.31	20	0.34906585	2016.08.31.15.35
## 3480	80	5	2016.08.31	280	-1.39626340	2016.08.31.15.35
## 3481	75	6	2016.08.31	285	-1.30899694	2016.08.31.15.35
## 3482	70	7	2016.08.31	290	-1.22173048	2016.08.31.15.35
## 3483	45	8	2016.08.31	315	-0.78539816	2016.08.31.15.35
## 3484	100	9	2016.08.31	260	-1.74532925	2016.08.31.15.35
## 3485	80	10	2016.08.31	280	-1.39626340	2016.08.31.15.35
## 3486	75	11	2016.08.31	285	-1.30899694	2016.08.31.15.35
## 3487	85	12	2016.08.31	275	-1.48352986	2016.08.31.15.35
## 3488	90	13	2016.08.31	270	-1.57079633	2016.08.31.15.35
## 3489	40	14	2016.08.31	320	-0.69813170	2016.08.31.15.35
## 3490	140	15	2016.08.31	220	-2.44346095	2016.08.31.15.35
## 3491	100	16	2016.08.31	260	-1.74532925	2016.08.31.15.35
## 3492	80	17	2016.08.31	280	-1.39626340	2016.08.31.15.35
## 3493	20	18	2016.08.31	340	-0.34906585	2016.08.31.15.35
## 3494	70	19	2016.08.31	290	-1.22173048	2016.08.31.15.35
## 3495	40	20	2016.08.31	320	-0.69813170	2016.08.31.15.35
## 3496	40	21	2016.08.31	320	-0.69813170	2016.08.31.15.35
## 3497	335	22	2016.08.31	25	0.43633231	2016.08.31.15.35
## 3498	70	23	2016.08.31	290	-1.22173048	2016.08.31.15.35
## 3499	20	24	2016.08.31	340	-0.34906585	2016.08.31.15.35
## 3500	240	25	2016.08.31	120	2.09439510	2016.08.31.15.35
## 3501	240	26	2016.08.31	120	2.09439510	2016.08.31.15.35
## 3502	5	27	2016.08.31	355	-0.08726646	2016.08.31.15.35
## 3503	220	28	2016.08.31	140	2.44346095	2016.08.31.15.35
## 3504	5	29	2016.08.31	355	-0.08726646	2016.08.31.15.35
## 3505	350	30	2016.08.31	10	0.17453293	2016.08.31.15.35
## 3506	210	31	2016.08.31	150	2.61799388	2016.08.31.15.35
## 3507	85	1	2016.09.01	275	-1.48352986	2016.09.01.13.19
## 3508	100	2	2016.09.01	260	-1.74532925	2016.09.01.13.19
## 3509	90	3	2016.09.01	270	-1.57079633	2016.09.01.13.19
## 3510	25	4	2016.09.01	335	-0.43633231	2016.09.01.13.19
## 3511	60	5	2016.09.01	300	-1.04719755	2016.09.01.13.19
## 3512	70	6	2016.09.01	290	-1.22173048	2016.09.01.13.19
## 3513	80	7	2016.09.01	280	-1.39626340	2016.09.01.13.19
## 3514	95	8	2016.09.01	265	-1.65806279	2016.09.01.13.19
## 3515	60	9	2016.09.01	300	-1.04719755	2016.09.01.13.19
## 3516	90	10	2016.09.01	270	-1.57079633	2016.09.01.13.19
## 3517	200	11	2016.09.01	160	2.79252680	2016.09.01.13.19
## 3518	30	12	2016.09.01	330	-0.52359878	2016.09.01.13.19
## 3519	210	13	2016.09.01	150	2.61799388	2016.09.01.13.19
## 3520	20	14	2016.09.01	340	-0.34906585	2016.09.01.13.19
## 3521	160	15	2016.09.01	200	-2.79252680	2016.09.01.13.19
## 3522	180	16	2016.09.01	180	3.14159265	2016.09.01.13.19
## 3523	55	17	2016.09.01	305	-0.95993109	2016.09.01.13.19
## 3524	280	18	2016.09.01	80	1.39626340	2016.09.01.13.19

## 3525	285	19	2016.09.01	75	1.30899694	2016.09.01.13.19
## 3526	280	20	2016.09.01	80	1.39626340	2016.09.01.13.19
## 3527	260	21	2016.09.01	100	1.74532925	2016.09.01.13.19
## 3528	290	22	2016.09.01	70	1.22173048	2016.09.01.13.19
## 3529	270	23	2016.09.01	90	1.57079633	2016.09.01.13.19
## 3530	330	24	2016.09.01	30	0.52359878	2016.09.01.13.19
## 3531	300	25	2016.09.01	60	1.04719755	2016.09.01.13.19
## 3532	330	26	2016.09.01	30	0.52359878	2016.09.01.13.19
## 3533	310	27	2016.09.01	50	0.87266463	2016.09.01.13.19
## 3534	320	28	2016.09.01	40	0.69813170	2016.09.01.13.19
## 3535	290	29	2016.09.01	70	1.22173048	2016.09.01.13.19
## 3536	300	30	2016.09.01	60	1.04719755	2016.09.01.13.19
## 3537	340	31	2016.09.01	20	0.34906585	2016.09.01.13.19
## 3538	260	32	2016.09.01	100	1.74532925	2016.09.01.13.19
## 3539	335	33	2016.09.01	25	0.43633231	2016.09.01.13.19
## 3540	320	34	2016.09.01	40	0.69813170	2016.09.01.13.19
## 3541	340	35	2016.09.01	20	0.34906585	2016.09.01.13.19
## 3542	300	36	2016.09.01	60	1.04719755	2016.09.01.13.19
## 3543	315	37	2016.09.01	45	0.78539816	2016.09.01.13.19
## 3544	285	38	2016.09.01	75	1.30899694	2016.09.01.13.19
## 3545	280	39	2016.09.01	80	1.39626340	2016.09.01.13.19
## 3546	350	40	2016.09.01	10	0.17453293	2016.09.01.13.19
## 3547	30	41	2016.09.01	330	-0.52359878	2016.09.01.13.19
## 3548	200	42	2016.09.01	160	2.79252680	2016.09.01.13.19
## 3549	105	43	2016.09.01	255	-1.83259571	2016.09.01.13.19
## 3550	0	44	2016.09.01	0	0.00000000	2016.09.01.13.19
## 3551	140	45	2016.09.01	220	-2.44346095	2016.09.01.13.19
## 3552	195	46	2016.09.01	165	2.87979327	2016.09.01.13.19
## 3553	120	47	2016.09.01	240	-2.09439510	2016.09.01.13.19
## 3554	70	48	2016.09.01	290	-1.22173048	2016.09.01.13.19
## 3555	330	49	2016.09.01	30	0.52359878	2016.09.01.13.19
## 3556	30	50	2016.09.01	330	-0.52359878	2016.09.01.13.19
## 3557	335	51	2016.09.01	25	0.43633231	2016.09.01.13.19
## 3558	25	52	2016.09.01	335	-0.43633231	2016.09.01.13.19
## 3559	265	53	2016.09.01	95	1.65806279	2016.09.01.13.19
## 3560	250	54	2016.09.01	110	1.91986218	2016.09.01.13.19
## 3561	150	55	2016.09.01	210	-2.61799388	2016.09.01.13.19
## 3562	170	56	2016.09.01	190	-2.96705973	2016.09.01.13.19
## 3563	310	57	2016.09.01	50	0.87266463	2016.09.01.13.19
## 3564	0	58	2016.09.01	0	0.00000000	2016.09.01.13.19
## 3565	120	59	2016.09.01	240	-2.09439510	2016.09.01.13.19
## 3566	100	60	2016.09.01	260	-1.74532925	2016.09.01.13.19
## 3567	95	61	2016.09.01	265	-1.65806279	2016.09.01.13.19
## 3568	355	62	2016.09.01	5	0.08726646	2016.09.01.13.19
## 3569	40	63	2016.09.01	320	-0.69813170	2016.09.01.13.19
## 3570	350	64	2016.09.01	10	0.17453293	2016.09.01.13.19
## 3571	330	65	2016.09.01	30	0.52359878	2016.09.01.13.19
## 3572	320	66	2016.09.01	40	0.69813170	2016.09.01.13.19
## 3573	105	1	2016.09.01	255	-1.83259571	2016.09.01.13.22
## 3574	30	2	2016.09.01	330	-0.52359878	2016.09.01.13.22
## 3575	335	3	2016.09.01	25	0.43633231	2016.09.01.13.22
## 3576	280	4	2016.09.01	80	1.39626340	2016.09.01.13.22
## 3577	75	5	2016.09.01	285	-1.30899694	2016.09.01.13.22
## 3578	220	6	2016.09.01	140	2.44346095	2016.09.01.13.22

## 3579	50	7	2016.09.01	310	-0.87266463	2016.09.01.13.22
## 3580	55	8	2016.09.01	305	-0.95993109	2016.09.01.13.22
## 3581	170	9	2016.09.01	190	-2.96705973	2016.09.01.13.22
## 3582	90	10	2016.09.01	270	-1.57079633	2016.09.01.13.22
## 3583	150	11	2016.09.01	210	-2.61799388	2016.09.01.13.22
## 3584	15	12	2016.09.01	345	-0.26179939	2016.09.01.13.22
## 3585	40	13	2016.09.01	320	-0.69813170	2016.09.01.13.22
## 3586	120	14	2016.09.01	240	-2.09439510	2016.09.01.13.22
## 3587	310	15	2016.09.01	50	0.87266463	2016.09.01.13.22
## 3588	230	16	2016.09.01	130	2.26892803	2016.09.01.13.22
## 3589	35	17	2016.09.01	325	-0.61086524	2016.09.01.13.22
## 3590	40	18	2016.09.01	320	-0.69813170	2016.09.01.13.22
## 3591	355	19	2016.09.01	5	0.08726646	2016.09.01.13.22
## 3592	70	20	2016.09.01	290	-1.22173048	2016.09.01.13.22
## 3593	110	21	2016.09.01	250	-1.91986218	2016.09.01.13.22
## 3594	80	22	2016.09.01	280	-1.39626340	2016.09.01.13.22
## 3595	290	23	2016.09.01	70	1.22173048	2016.09.01.13.22
## 3596	250	24	2016.09.01	110	1.91986218	2016.09.01.13.22
## 3597	240	25	2016.09.01	120	2.09439510	2016.09.01.13.22
## 3598	220	26	2016.09.01	140	2.44346095	2016.09.01.13.22
## 3599	265	27	2016.09.01	95	1.65806279	2016.09.01.13.22
## 3600	245	28	2016.09.01	115	2.00712864	2016.09.01.13.22
## 3601	250	29	2016.09.01	110	1.91986218	2016.09.01.13.22
## 3602	250	30	2016.09.01	110	1.91986218	2016.09.01.13.22
## 3603	260	31	2016.09.01	100	1.74532925	2016.09.01.13.22
## 3604	210	32	2016.09.01	150	2.61799388	2016.09.01.13.22
## 3605	310	33	2016.09.01	50	0.87266463	2016.09.01.13.22
## 3606	310	34	2016.09.01	50	0.87266463	2016.09.01.13.22
## 3607	300	35	2016.09.01	60	1.04719755	2016.09.01.13.22
## 3608	55	1	2016.09.01	305	-0.95993109	2016.09.01.13.29
## 3609	30	2	2016.09.01	330	-0.52359878	2016.09.01.13.29
## 3610	100	3	2016.09.01	260	-1.74532925	2016.09.01.13.29
## 3611	150	4	2016.09.01	210	-2.61799388	2016.09.01.13.29
## 3612	85	5	2016.09.01	275	-1.48352986	2016.09.01.13.29
## 3613	110	6	2016.09.01	250	-1.91986218	2016.09.01.13.29
## 3614	60	7	2016.09.01	300	-1.04719755	2016.09.01.13.29
## 3615	90	8	2016.09.01	270	-1.57079633	2016.09.01.13.29
## 3616	60	9	2016.09.01	300	-1.04719755	2016.09.01.13.29
## 3617	90	10	2016.09.01	270	-1.57079633	2016.09.01.13.29
## 3618	120	11	2016.09.01	240	-2.09439510	2016.09.01.13.29
## 3619	355	12	2016.09.01	5	0.08726646	2016.09.01.13.29
## 3620	345	13	2016.09.01	15	0.26179939	2016.09.01.13.29
## 3621	350	14	2016.09.01	10	0.17453293	2016.09.01.13.29
## 3622	0	15	2016.09.01	0	0.00000000	2016.09.01.13.29
## 3623	0	16	2016.09.01	0	0.00000000	2016.09.01.13.29
## 3624	345	17	2016.09.01	15	0.26179939	2016.09.01.13.29
## 3625	355	18	2016.09.01	5	0.08726646	2016.09.01.13.29
## 3626	0	19	2016.09.01	0	0.00000000	2016.09.01.13.29
## 3627	5	20	2016.09.01	355	-0.08726646	2016.09.01.13.29
## 3628	345	21	2016.09.01	15	0.26179939	2016.09.01.13.29
## 3629	0	22	2016.09.01	0	0.00000000	2016.09.01.13.29
## 3630	330	23	2016.09.01	30	0.52359878	2016.09.01.13.29
## 3631	5	24	2016.09.01	355	-0.08726646	2016.09.01.13.29
## 3632	340	25	2016.09.01	20	0.34906585	2016.09.01.13.29

## 3633	5	26	2016.09.01	355	-0.08726646	2016.09.01.13.29
## 3634	355	27	2016.09.01	5	0.08726646	2016.09.01.13.29
## 3635	0	28	2016.09.01	0	0.00000000	2016.09.01.13.29
## 3636	340	1	2016.09.01	20	0.34906585	2016.09.01.14.15
## 3637	310	2	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3638	350	3	2016.09.01	10	0.17453293	2016.09.01.14.15
## 3639	335	4	2016.09.01	25	0.43633231	2016.09.01.14.15
## 3640	345	5	2016.09.01	15	0.26179939	2016.09.01.14.15
## 3641	350	6	2016.09.01	10	0.17453293	2016.09.01.14.15
## 3642	350	7	2016.09.01	10	0.17453293	2016.09.01.14.15
## 3643	310	8	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3644	310	9	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3645	330	10	2016.09.01	30	0.52359878	2016.09.01.14.15
## 3646	325	11	2016.09.01	35	0.61086524	2016.09.01.14.15
## 3647	340	12	2016.09.01	20	0.34906585	2016.09.01.14.15
## 3648	290	13	2016.09.01	70	1.22173048	2016.09.01.14.15
## 3649	300	14	2016.09.01	60	1.04719755	2016.09.01.14.15
## 3650	350	15	2016.09.01	10	0.17453293	2016.09.01.14.15
## 3651	335	16	2016.09.01	25	0.43633231	2016.09.01.14.15
## 3652	110	17	2016.09.01	250	-1.91986218	2016.09.01.14.15
## 3653	225	18	2016.09.01	135	2.35619449	2016.09.01.14.15
## 3654	255	19	2016.09.01	105	1.83259571	2016.09.01.14.15
## 3655	355	20	2016.09.01	5	0.08726646	2016.09.01.14.15
## 3656	15	21	2016.09.01	345	-0.26179939	2016.09.01.14.15
## 3657	255	22	2016.09.01	105	1.83259571	2016.09.01.14.15
## 3658	90	23	2016.09.01	270	-1.57079633	2016.09.01.14.15
## 3659	30	24	2016.09.01	330	-0.52359878	2016.09.01.14.15
## 3660	330	25	2016.09.01	30	0.52359878	2016.09.01.14.15
## 3661	300	26	2016.09.01	60	1.04719755	2016.09.01.14.15
## 3662	40	27	2016.09.01	320	-0.69813170	2016.09.01.14.15
## 3663	45	28	2016.09.01	315	-0.78539816	2016.09.01.14.15
## 3664	80	29	2016.09.01	280	-1.39626340	2016.09.01.14.15
## 3665	220	30	2016.09.01	140	2.44346095	2016.09.01.14.15
## 3666	230	31	2016.09.01	130	2.26892803	2016.09.01.14.15
## 3667	220	32	2016.09.01	140	2.44346095	2016.09.01.14.15
## 3668	145	33	2016.09.01	215	-2.53072742	2016.09.01.14.15
## 3669	100	34	2016.09.01	260	-1.74532925	2016.09.01.14.15
## 3670	170	35	2016.09.01	190	-2.96705973	2016.09.01.14.15
## 3671	120	36	2016.09.01	240	-2.09439510	2016.09.01.14.15
## 3672	270	37	2016.09.01	90	1.57079633	2016.09.01.14.15
## 3673	340	38	2016.09.01	20	0.34906585	2016.09.01.14.15
## 3674	345	39	2016.09.01	15	0.26179939	2016.09.01.14.15
## 3675	345	40	2016.09.01	15	0.26179939	2016.09.01.14.15
## 3676	0	41	2016.09.01	0	0.00000000	2016.09.01.14.15
## 3677	350	42	2016.09.01	10	0.17453293	2016.09.01.14.15
## 3678	310	43	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3679	340	44	2016.09.01	20	0.34906585	2016.09.01.14.15
## 3680	300	45	2016.09.01	60	1.04719755	2016.09.01.14.15
## 3681	340	46	2016.09.01	20	0.34906585	2016.09.01.14.15
## 3682	310	47	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3683	110	48	2016.09.01	250	-1.91986218	2016.09.01.14.15
## 3684	280	49	2016.09.01	80	1.39626340	2016.09.01.14.15
## 3685	285	50	2016.09.01	75	1.30899694	2016.09.01.14.15
## 3686	0	51	2016.09.01	0	0.00000000	2016.09.01.14.15

## 3687	300	52	2016.09.01	60	1.04719755	2016.09.01.14.15
## 3688	265	53	2016.09.01	95	1.65806279	2016.09.01.14.15
## 3689	200	54	2016.09.01	160	2.79252680	2016.09.01.14.15
## 3690	80	55	2016.09.01	280	-1.39626340	2016.09.01.14.15
## 3691	310	56	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3692	255	57	2016.09.01	105	1.83259571	2016.09.01.14.15
## 3693	120	58	2016.09.01	240	-2.09439510	2016.09.01.14.15
## 3694	0	59	2016.09.01	0	0.00000000	2016.09.01.14.15
## 3695	270	60	2016.09.01	90	1.57079633	2016.09.01.14.15
## 3696	35	61	2016.09.01	325	-0.61086524	2016.09.01.14.15
## 3697	80	62	2016.09.01	280	-1.39626340	2016.09.01.14.15
## 3698	100	63	2016.09.01	260	-1.74532925	2016.09.01.14.15
## 3699	200	64	2016.09.01	160	2.79252680	2016.09.01.14.15
## 3700	220	65	2016.09.01	140	2.44346095	2016.09.01.14.15
## 3701	80	66	2016.09.01	280	-1.39626340	2016.09.01.14.15
## 3702	330	67	2016.09.01	30	0.52359878	2016.09.01.14.15
## 3703	330	68	2016.09.01	30	0.52359878	2016.09.01.14.15
## 3704	340	69	2016.09.01	20	0.34906585	2016.09.01.14.15
## 3705	10	70	2016.09.01	350	-0.17453293	2016.09.01.14.15
## 3706	350	71	2016.09.01	10	0.17453293	2016.09.01.14.15
## 3707	290	72	2016.09.01	70	1.22173048	2016.09.01.14.15
## 3708	30	73	2016.09.01	330	-0.52359878	2016.09.01.14.15
## 3709	10	74	2016.09.01	350	-0.17453293	2016.09.01.14.15
## 3710	310	75	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3711	25	76	2016.09.01	335	-0.43633231	2016.09.01.14.15
## 3712	330	77	2016.09.01	30	0.52359878	2016.09.01.14.15
## 3713	20	78	2016.09.01	340	-0.34906585	2016.09.01.14.15
## 3714	355	79	2016.09.01	5	0.08726646	2016.09.01.14.15
## 3715	310	80	2016.09.01	50	0.87266463	2016.09.01.14.15
## 3716	320	81	2016.09.01	40	0.69813170	2016.09.01.14.15
## 3717	105	1	2016.09.01	255	-1.83259571	2016.09.01.15.05
## 3718	160	2	2016.09.01	200	-2.79252680	2016.09.01.15.05
## 3719	170	3	2016.09.01	190	-2.96705973	2016.09.01.15.05
## 3720	250	4	2016.09.01	110	1.91986218	2016.09.01.15.05
## 3721	165	5	2016.09.01	195	-2.87979327	2016.09.01.15.05
## 3722	260	6	2016.09.01	100	1.74532925	2016.09.01.15.05
## 3723	95	7	2016.09.01	265	-1.65806279	2016.09.01.15.05
## 3724	160	8	2016.09.01	200	-2.79252680	2016.09.01.15.05
## 3725	100	9	2016.09.01	260	-1.74532925	2016.09.01.15.05
## 3726	120	10	2016.09.01	240	-2.09439510	2016.09.01.15.05
## 3727	120	11	2016.09.01	240	-2.09439510	2016.09.01.15.05
## 3728	100	12	2016.09.01	260	-1.74532925	2016.09.01.15.05
## 3729	320	13	2016.09.01	40	0.69813170	2016.09.01.15.05
## 3730	120	14	2016.09.01	240	-2.09439510	2016.09.01.15.05
## 3731	200	15	2016.09.01	160	2.79252680	2016.09.01.15.05
## 3732	10	16	2016.09.01	350	-0.17453293	2016.09.01.15.05
## 3733	40	17	2016.09.01	320	-0.69813170	2016.09.01.15.05
## 3734	5	18	2016.09.01	355	-0.08726646	2016.09.01.15.05
## 3735	45	19	2016.09.01	315	-0.78539816	2016.09.01.15.05
## 3736	350	20	2016.09.01	10	0.17453293	2016.09.01.15.05
## 3737	30	21	2016.09.01	330	-0.52359878	2016.09.01.15.05
## 3738	10	22	2016.09.01	350	-0.17453293	2016.09.01.15.05
## 3739	30	23	2016.09.01	330	-0.52359878	2016.09.01.15.05
## 3740	30	24	2016.09.01	330	-0.52359878	2016.09.01.15.05

## 3741	5	25	2016.09.01	355	-0.08726646	2016.09.01.15.05
## 3742	40	26	2016.09.01	320	-0.69813170	2016.09.01.15.05
## 3743	355	27	2016.09.01	5	0.08726646	2016.09.01.15.05
## 3744	130	1	2016.09.01	230	-2.26892803	2016.09.01.15.17
## 3745	155	2	2016.09.01	205	-2.70526034	2016.09.01.15.17
## 3746	105	3	2016.09.01	255	-1.83259571	2016.09.01.15.17
## 3747	70	4	2016.09.01	290	-1.22173048	2016.09.01.15.17
## 3748	160	5	2016.09.01	200	-2.79252680	2016.09.01.15.17
## 3749	125	6	2016.09.01	235	-2.18166156	2016.09.01.15.17
## 3750	265	7	2016.09.01	95	1.65806279	2016.09.01.15.17
## 3751	80	8	2016.09.01	280	-1.39626340	2016.09.01.15.17
## 3752	170	9	2016.09.01	190	-2.96705973	2016.09.01.15.17
## 3753	170	10	2016.09.01	190	-2.96705973	2016.09.01.15.17
## 3754	165	11	2016.09.01	195	-2.87979327	2016.09.01.15.17
## 3755	170	12	2016.09.01	190	-2.96705973	2016.09.01.15.17
## 3756	150	13	2016.09.01	210	-2.61799388	2016.09.01.15.17
## 3757	210	14	2016.09.01	150	2.61799388	2016.09.01.15.17
## 3758	210	15	2016.09.01	150	2.61799388	2016.09.01.15.17
## 3759	220	16	2016.09.01	140	2.44346095	2016.09.01.15.17
## 3760	300	17	2016.09.01	60	1.04719755	2016.09.01.15.17
## 3761	310	18	2016.09.01	50	0.87266463	2016.09.01.15.17
## 3762	220	19	2016.09.01	140	2.44346095	2016.09.01.15.17
## 3763	310	20	2016.09.01	50	0.87266463	2016.09.01.15.17
## 3764	50	21	2016.09.01	310	-0.87266463	2016.09.01.15.17
## 3765	260	22	2016.09.01	100	1.74532925	2016.09.01.15.17
## 3766	10	23	2016.09.01	350	-0.17453293	2016.09.01.15.17
## 3767	95	24	2016.09.01	265	-1.65806279	2016.09.01.15.17
## 3768	50	25	2016.09.01	310	-0.87266463	2016.09.01.15.17
## 3769	215	26	2016.09.01	145	2.53072742	2016.09.01.15.17
## 3770	300	27	2016.09.01	60	1.04719755	2016.09.01.15.17
## 3771	0	28	2016.09.01	0	0.00000000	2016.09.01.15.17
## 3772	350	29	2016.09.01	10	0.17453293	2016.09.01.15.17
## 3773	155	30	2016.09.01	205	-2.70526034	2016.09.01.15.17
## 3774	230	31	2016.09.01	130	2.26892803	2016.09.01.15.17
## 3775	140	32	2016.09.01	220	-2.44346095	2016.09.01.15.17
## 3776	70	33	2016.09.01	290	-1.22173048	2016.09.01.15.17
## 3777	300	34	2016.09.01	60	1.04719755	2016.09.01.15.17
## 3778	260	35	2016.09.01	100	1.74532925	2016.09.01.15.17
## 3779	25	36	2016.09.01	335	-0.43633231	2016.09.01.15.17
## 3780	110	37	2016.09.01	250	-1.91986218	2016.09.01.15.17
## 3781	25	38	2016.09.01	335	-0.43633231	2016.09.01.15.17
## 3782	0	39	2016.09.01	0	0.00000000	2016.09.01.15.17
## 3783	350	40	2016.09.01	10	0.17453293	2016.09.01.15.17
## 3784	355	41	2016.09.01	5	0.08726646	2016.09.01.15.17
## 3785	20	42	2016.09.01	340	-0.34906585	2016.09.01.15.17
## 3786	20	43	2016.09.01	340	-0.34906585	2016.09.01.15.17
## 3787	15	44	2016.09.01	345	-0.26179939	2016.09.01.15.17
## 3788	5	45	2016.09.01	355	-0.08726646	2016.09.01.15.17
## 3789	30	46	2016.09.01	330	-0.52359878	2016.09.01.15.17
## 3790	60	47	2016.09.01	300	-1.04719755	2016.09.01.15.17
## 3791	40	48	2016.09.01	320	-0.69813170	2016.09.01.15.17
## 3792	10	1	2016.09.02	350	-0.17453293	2016.09.02.13.39
## 3793	50	2	2016.09.02	310	-0.87266463	2016.09.02.13.39
## 3794	345	3	2016.09.02	15	0.26179939	2016.09.02.13.39

## 3795	320	4	2016.09.02	40	0.69813170	2016.09.02.13.39
## 3796	350	5	2016.09.02	10	0.17453293	2016.09.02.13.39
## 3797	115	6	2016.09.02	245	-2.00712864	2016.09.02.13.39
## 3798	290	7	2016.09.02	70	1.22173048	2016.09.02.13.39
## 3799	10	8	2016.09.02	350	-0.17453293	2016.09.02.13.39
## 3800	330	9	2016.09.02	30	0.52359878	2016.09.02.13.39
## 3801	335	10	2016.09.02	25	0.43633231	2016.09.02.13.39
## 3802	150	11	2016.09.02	210	-2.61799388	2016.09.02.13.39
## 3803	180	12	2016.09.02	180	3.14159265	2016.09.02.13.39
## 3804	165	13	2016.09.02	195	-2.87979327	2016.09.02.13.39
## 3805	210	14	2016.09.02	150	2.61799388	2016.09.02.13.39
## 3806	150	15	2016.09.02	210	-2.61799388	2016.09.02.13.39
## 3807	230	16	2016.09.02	130	2.26892803	2016.09.02.13.39
## 3808	160	17	2016.09.02	200	-2.79252680	2016.09.02.13.39
## 3809	165	18	2016.09.02	195	-2.87979327	2016.09.02.13.39
## 3810	180	19	2016.09.02	180	3.14159265	2016.09.02.13.39
## 3811	200	20	2016.09.02	160	2.79252680	2016.09.02.13.39
## 3812	160	21	2016.09.02	200	-2.79252680	2016.09.02.13.39
## 3813	215	22	2016.09.02	145	2.53072742	2016.09.02.13.39
## 3814	165	23	2016.09.02	195	-2.87979327	2016.09.02.13.39
## 3815	355	1	2016.09.02	5	0.08726646	2016.09.02.13.46
## 3816	350	2	2016.09.02	10	0.17453293	2016.09.02.13.46
## 3817	340	3	2016.09.02	20	0.34906585	2016.09.02.13.46
## 3818	0	4	2016.09.02	0	0.00000000	2016.09.02.13.46
## 3819	340	5	2016.09.02	20	0.34906585	2016.09.02.13.46
## 3820	300	6	2016.09.02	60	1.04719755	2016.09.02.13.46
## 3821	290	7	2016.09.02	70	1.22173048	2016.09.02.13.46
## 3822	310	8	2016.09.02	50	0.87266463	2016.09.02.13.46
## 3823	335	9	2016.09.02	25	0.43633231	2016.09.02.13.46
## 3824	330	10	2016.09.02	30	0.52359878	2016.09.02.13.46
## 3825	330	11	2016.09.02	30	0.52359878	2016.09.02.13.46
## 3826	305	12	2016.09.02	55	0.95993109	2016.09.02.13.46
## 3827	280	13	2016.09.02	80	1.39626340	2016.09.02.13.46
## 3828	345	14	2016.09.02	15	0.26179939	2016.09.02.13.46
## 3829	60	15	2016.09.02	300	-1.04719755	2016.09.02.13.46
## 3830	75	16	2016.09.02	285	-1.30899694	2016.09.02.13.46
## 3831	75	17	2016.09.02	285	-1.30899694	2016.09.02.13.46
## 3832	160	18	2016.09.02	200	-2.79252680	2016.09.02.13.46
## 3833	160	19	2016.09.02	200	-2.79252680	2016.09.02.13.46
## 3834	30	20	2016.09.02	330	-0.52359878	2016.09.02.13.46
## 3835	185	21	2016.09.02	175	3.05432619	2016.09.02.13.46
## 3836	110	22	2016.09.02	250	-1.91986218	2016.09.02.13.46
## 3837	155	23	2016.09.02	205	-2.70526034	2016.09.02.13.46
## 3838	210	24	2016.09.02	150	2.61799388	2016.09.02.13.46
## 3839	180	25	2016.09.02	180	3.14159265	2016.09.02.13.46
## 3840	140	26	2016.09.02	220	-2.44346095	2016.09.02.13.46
## 3841	150	27	2016.09.02	210	-2.61799388	2016.09.02.13.46
## 3842	180	28	2016.09.02	180	3.14159265	2016.09.02.13.46
## 3843	180	29	2016.09.02	180	3.14159265	2016.09.02.13.46
## 3844	215	30	2016.09.02	145	2.53072742	2016.09.02.13.46
## 3845	170	1	2016.09.06	190	-2.96705973	2016.09.06.14.19
## 3846	170	2	2016.09.06	190	-2.96705973	2016.09.06.14.19
## 3847	40	3	2016.09.06	320	-0.69813170	2016.09.06.14.19
## 3848	130	4	2016.09.06	230	-2.26892803	2016.09.06.14.19

## 3849	85	5	2016.09.06	275	-1.48352986	2016.09.06.14.19
## 3850	110	6	2016.09.06	250	-1.91986218	2016.09.06.14.19
## 3851	120	7	2016.09.06	240	-2.09439510	2016.09.06.14.19
## 3852	110	8	2016.09.06	250	-1.91986218	2016.09.06.14.19
## 3853	140	9	2016.09.06	220	-2.44346095	2016.09.06.14.19
## 3854	195	10	2016.09.06	165	2.87979327	2016.09.06.14.19
## 3855	120	11	2016.09.06	240	-2.09439510	2016.09.06.14.19
## 3856	150	12	2016.09.06	210	-2.61799388	2016.09.06.14.19
## 3857	95	13	2016.09.06	265	-1.65806279	2016.09.06.14.19
## 3858	100	14	2016.09.06	260	-1.74532925	2016.09.06.14.19
## 3859	200	15	2016.09.06	160	2.79252680	2016.09.06.14.19
## 3860	115	16	2016.09.06	245	-2.00712864	2016.09.06.14.19
## 3861	95	17	2016.09.06	265	-1.65806279	2016.09.06.14.19
## 3862	165	18	2016.09.06	195	-2.87979327	2016.09.06.14.19
## 3863	145	19	2016.09.06	215	-2.53072742	2016.09.06.14.19
## 3864	120	20	2016.09.06	240	-2.09439510	2016.09.06.14.19
## 3865	90	21	2016.09.06	270	-1.57079633	2016.09.06.14.19
## 3866	210	22	2016.09.06	150	2.61799388	2016.09.06.14.19
## 3867	170	23	2016.09.06	190	-2.96705973	2016.09.06.14.19
## 3868	100	24	2016.09.06	260	-1.74532925	2016.09.06.14.19
## 3869	5	25	2016.09.06	355	-0.08726646	2016.09.06.14.19
## 3870	30	26	2016.09.06	330	-0.52359878	2016.09.06.14.19
## 3871	110	27	2016.09.06	250	-1.91986218	2016.09.06.14.19
## 3872	175	28	2016.09.06	185	-3.05432619	2016.09.06.14.19
## 3873	170	29	2016.09.06	190	-2.96705973	2016.09.06.14.19
## 3874	160	30	2016.09.06	200	-2.79252680	2016.09.06.14.19
## 3875	135	31	2016.09.06	225	-2.35619449	2016.09.06.14.19
## 3876	170	32	2016.09.06	190	-2.96705973	2016.09.06.14.19
## 3877	90	33	2016.09.06	270	-1.57079633	2016.09.06.14.19
## 3878	175	34	2016.09.06	185	-3.05432619	2016.09.06.14.19
## 3879	120	35	2016.09.06	240	-2.09439510	2016.09.06.14.19
## 3880	130	36	2016.09.06	230	-2.26892803	2016.09.06.14.19
## 3881	140	37	2016.09.06	220	-2.44346095	2016.09.06.14.19
## 3882	175	38	2016.09.06	185	-3.05432619	2016.09.06.14.19
## 3883	115	39	2016.09.06	245	-2.00712864	2016.09.06.14.19
## 3884	125	40	2016.09.06	235	-2.18166156	2016.09.06.14.19
## 3885	115	41	2016.09.06	245	-2.00712864	2016.09.06.14.19
## 3886	145	42	2016.09.06	215	-2.53072742	2016.09.06.14.19
## 3887	155	43	2016.09.06	205	-2.70526034	2016.09.06.14.19
## 3888	165	44	2016.09.06	195	-2.87979327	2016.09.06.14.19
## 3889	220	45	2016.09.06	140	2.44346095	2016.09.06.14.19
## 3890	145	46	2016.09.06	215	-2.53072742	2016.09.06.14.19
## 3891	155	47	2016.09.06	205	-2.70526034	2016.09.06.14.19
## 3892	155	48	2016.09.06	205	-2.70526034	2016.09.06.14.19
## 3893	95	49	2016.09.06	265	-1.65806279	2016.09.06.14.19
## 3894	70	50	2016.09.06	290	-1.22173048	2016.09.06.14.19
## 3895	90	51	2016.09.06	270	-1.57079633	2016.09.06.14.19
## 3896	120	52	2016.09.06	240	-2.09439510	2016.09.06.14.19
## 3897	210	53	2016.09.06	150	2.61799388	2016.09.06.14.19
## 3898	105	54	2016.09.06	255	-1.83259571	2016.09.06.14.19
## 3899	95	55	2016.09.06	265	-1.65806279	2016.09.06.14.19
## 3900	140	56	2016.09.06	220	-2.44346095	2016.09.06.14.19
## 3901	140	57	2016.09.06	220	-2.44346095	2016.09.06.14.19
## 3902	160	58	2016.09.06	200	-2.79252680	2016.09.06.14.19

## 3903	150	59	2016.09.06	210	-2.61799388	2016.09.06.14.19
## 3904	80	60	2016.09.06	280	-1.39626340	2016.09.06.14.19
## 3905	15	61	2016.09.06	345	-0.26179939	2016.09.06.14.19
## 3906	60	62	2016.09.06	300	-1.04719755	2016.09.06.14.19
## 3907	50	63	2016.09.06	310	-0.87266463	2016.09.06.14.19
## 3908	140	1	2016.09.07	220	-2.44346095	2016.09.07.14.55
## 3909	50	2	2016.09.07	310	-0.87266463	2016.09.07.14.55
## 3910	45	3	2016.09.07	315	-0.78539816	2016.09.07.14.55
## 3911	70	4	2016.09.07	290	-1.22173048	2016.09.07.14.55
## 3912	125	5	2016.09.07	235	-2.18166156	2016.09.07.14.55
## 3913	65	6	2016.09.07	295	-1.13446401	2016.09.07.14.55
## 3914	60	7	2016.09.07	300	-1.04719755	2016.09.07.14.55
## 3915	55	8	2016.09.07	305	-0.95993109	2016.09.07.14.55
## 3916	70	9	2016.09.07	290	-1.22173048	2016.09.07.14.55
## 3917	65	10	2016.09.07	295	-1.13446401	2016.09.07.14.55
## 3918	340	11	2016.09.07	20	0.34906585	2016.09.07.14.55
## 3919	70	12	2016.09.07	290	-1.22173048	2016.09.07.14.55
## 3920	120	13	2016.09.07	240	-2.09439510	2016.09.07.14.55
## 3921	30	14	2016.09.07	330	-0.52359878	2016.09.07.14.55
## 3922	85	15	2016.09.07	275	-1.48352986	2016.09.07.14.55
## 3923	110	16	2016.09.07	250	-1.91986218	2016.09.07.14.55
## 3924	155	17	2016.09.07	205	-2.70526034	2016.09.07.14.55
## 3925	150	18	2016.09.07	210	-2.61799388	2016.09.07.14.55
## 3926	155	19	2016.09.07	205	-2.70526034	2016.09.07.14.55
## 3927	185	20	2016.09.07	175	3.05432619	2016.09.07.14.55
## 3928	160	21	2016.09.07	200	-2.79252680	2016.09.07.14.55
## 3929	155	22	2016.09.07	205	-2.70526034	2016.09.07.14.55
## 3930	110	23	2016.09.07	250	-1.91986218	2016.09.07.14.55
## 3931	320	24	2016.09.07	40	0.69813170	2016.09.07.14.55
## 3932	340	25	2016.09.07	20	0.34906585	2016.09.07.14.55
## 3933	345	26	2016.09.07	15	0.26179939	2016.09.07.14.55
## 3934	280	27	2016.09.07	80	1.39626340	2016.09.07.14.55
## 3935	0	28	2016.09.07	0	0.00000000	2016.09.07.14.55
## 3936	290	29	2016.09.07	70	1.22173048	2016.09.07.14.55
## 3937	5	30	2016.09.07	355	-0.08726646	2016.09.07.14.55
## 3938	295	31	2016.09.07	65	1.13446401	2016.09.07.14.55
## 3939	285	32	2016.09.07	75	1.30899694	2016.09.07.14.55
## 3940	335	33	2016.09.07	25	0.43633231	2016.09.07.14.55
## 3941	330	34	2016.09.07	30	0.52359878	2016.09.07.14.55
## 3942	350	35	2016.09.07	10	0.17453293	2016.09.07.14.55
## 3943	350	36	2016.09.07	10	0.17453293	2016.09.07.14.55
## 3944	320	37	2016.09.07	40	0.69813170	2016.09.07.14.55
## 3945	280	38	2016.09.07	80	1.39626340	2016.09.07.14.55
## 3946	345	39	2016.09.07	15	0.26179939	2016.09.07.14.55
## 3947	320	40	2016.09.07	40	0.69813170	2016.09.07.14.55
## 3948	320	41	2016.09.07	40	0.69813170	2016.09.07.14.55
## 3949	350	42	2016.09.07	10	0.17453293	2016.09.07.14.55
## 3950	10	43	2016.09.07	350	-0.17453293	2016.09.07.14.55
## 3951	300	44	2016.09.07	60	1.04719755	2016.09.07.14.55
## 3952	345	45	2016.09.07	15	0.26179939	2016.09.07.14.55
## 3953	330	46	2016.09.07	30	0.52359878	2016.09.07.14.55
## 3954	310	47	2016.09.07	50	0.87266463	2016.09.07.14.55
## 3955	5	1	2016.09.07	355	-0.08726646	2016.09.07.16.12
## 3956	15	2	2016.09.07	345	-0.26179939	2016.09.07.16.12

## 3957	65	3	2016.09.07	295	-1.13446401	2016.09.07.16.12
## 3958	70	4	2016.09.07	290	-1.22173048	2016.09.07.16.12
## 3959	95	5	2016.09.07	265	-1.65806279	2016.09.07.16.12
## 3960	105	6	2016.09.07	255	-1.83259571	2016.09.07.16.12
## 3961	40	7	2016.09.07	320	-0.69813170	2016.09.07.16.12
## 3962	90	8	2016.09.07	270	-1.57079633	2016.09.07.16.12
## 3963	100	9	2016.09.07	260	-1.74532925	2016.09.07.16.12
## 3964	50	10	2016.09.07	310	-0.87266463	2016.09.07.16.12
## 3965	40	11	2016.09.07	320	-0.69813170	2016.09.07.16.12
## 3966	15	12	2016.09.07	345	-0.26179939	2016.09.07.16.12
## 3967	90	13	2016.09.07	270	-1.57079633	2016.09.07.16.12
## 3968	300	14	2016.09.07	60	1.04719755	2016.09.07.16.12
## 3969	120	15	2016.09.07	240	-2.09439510	2016.09.07.16.12
## 3970	65	16	2016.09.07	295	-1.13446401	2016.09.07.16.12
## 3971	280	17	2016.09.07	80	1.39626340	2016.09.07.16.12
## 3972	330	18	2016.09.07	30	0.52359878	2016.09.07.16.12
## 3973	350	19	2016.09.07	10	0.17453293	2016.09.07.16.12
## 3974	350	20	2016.09.07	10	0.17453293	2016.09.07.16.12
## 3975	320	21	2016.09.07	40	0.69813170	2016.09.07.16.12
## 3976	320	22	2016.09.07	40	0.69813170	2016.09.07.16.12
## 3977	345	23	2016.09.07	15	0.26179939	2016.09.07.16.12
## 3978	335	24	2016.09.07	25	0.43633231	2016.09.07.16.12
## 3979	330	25	2016.09.07	30	0.52359878	2016.09.07.16.12
## 3980	295	1	2016.09.08	65	1.13446401	2016.09.08.14.07
## 3981	295	2	2016.09.08	65	1.13446401	2016.09.08.14.07
## 3982	240	3	2016.09.08	120	2.09439510	2016.09.08.14.07
## 3983	265	4	2016.09.08	95	1.65806279	2016.09.08.14.07
## 3984	250	5	2016.09.08	110	1.91986218	2016.09.08.14.07
## 3985	260	6	2016.09.08	100	1.74532925	2016.09.08.14.07
## 3986	320	7	2016.09.08	40	0.69813170	2016.09.08.14.07
## 3987	240	8	2016.09.08	120	2.09439510	2016.09.08.14.07
## 3988	280	9	2016.09.08	80	1.39626340	2016.09.08.14.07
## 3989	280	10	2016.09.08	80	1.39626340	2016.09.08.14.07
## 3990	220	11	2016.09.08	140	2.44346095	2016.09.08.14.07
## 3991	220	12	2016.09.08	140	2.44346095	2016.09.08.14.07
## 3992	300	13	2016.09.08	60	1.04719755	2016.09.08.14.07
## 3993	260	14	2016.09.08	100	1.74532925	2016.09.08.14.07
## 3994	290	15	2016.09.08	70	1.22173048	2016.09.08.14.07
## 3995	230	16	2016.09.08	130	2.26892803	2016.09.08.14.07
## 3996	240	17	2016.09.08	120	2.09439510	2016.09.08.14.07
## 3997	300	18	2016.09.08	60	1.04719755	2016.09.08.14.07
## 3998	230	19	2016.09.08	130	2.26892803	2016.09.08.14.07
## 3999	190	20	2016.09.08	170	2.96705973	2016.09.08.14.07
## 4000	110	1	2016.09.08	250	-1.91986218	2016.09.08.15.01
## 4001	190	2	2016.09.08	170	2.96705973	2016.09.08.15.01
## 4002	100	3	2016.09.08	260	-1.74532925	2016.09.08.15.01
## 4003	175	4	2016.09.08	185	-3.05432619	2016.09.08.15.01
## 4004	220	5	2016.09.08	140	2.44346095	2016.09.08.15.01
## 4005	205	6	2016.09.08	155	2.70526034	2016.09.08.15.01
## 4006	160	7	2016.09.08	200	-2.79252680	2016.09.08.15.01
## 4007	160	8	2016.09.08	200	-2.79252680	2016.09.08.15.01
## 4008	50	9	2016.09.08	310	-0.87266463	2016.09.08.15.01
## 4009	110	10	2016.09.08	250	-1.91986218	2016.09.08.15.01
## 4010	150	11	2016.09.08	210	-2.61799388	2016.09.08.15.01

## 4011	110	12	2016.09.08	250	-1.91986218	2016.09.08.15.01
## 4012	165	13	2016.09.08	195	-2.87979327	2016.09.08.15.01
## 4013	85	14	2016.09.08	275	-1.48352986	2016.09.08.15.01
## 4014	50	15	2016.09.08	310	-0.87266463	2016.09.08.15.01
## 4015	145	16	2016.09.08	215	-2.53072742	2016.09.08.15.01
## 4016	330	17	2016.09.08	30	0.52359878	2016.09.08.15.01
## 4017	355	18	2016.09.08	5	0.08726646	2016.09.08.15.01
## 4018	325	19	2016.09.08	35	0.61086524	2016.09.08.15.01
## 4019	0	20	2016.09.08	0	0.00000000	2016.09.08.15.01
## 4020	20	21	2016.09.08	340	-0.34906585	2016.09.08.15.01
## 4021	350	22	2016.09.08	10	0.17453293	2016.09.08.15.01
## 4022	20	23	2016.09.08	340	-0.34906585	2016.09.08.15.01
## 4023	20	24	2016.09.08	340	-0.34906585	2016.09.08.15.01
## 4024	20	25	2016.09.08	340	-0.34906585	2016.09.08.15.01
## 4025	330	26	2016.09.08	30	0.52359878	2016.09.08.15.01
## 4026	355	27	2016.09.08	5	0.08726646	2016.09.08.15.01
## 4027	0	28	2016.09.08	0	0.00000000	2016.09.08.15.01
## 4028	5	29	2016.09.08	355	-0.08726646	2016.09.08.15.01
## 4029	0	30	2016.09.08	0	0.00000000	2016.09.08.15.01
## 4030	30	31	2016.09.08	330	-0.52359878	2016.09.08.15.01
## 4031	350	32	2016.09.08	10	0.17453293	2016.09.08.15.01
## 4032	110	33	2016.09.08	250	-1.91986218	2016.09.08.15.01
## 4033	40	34	2016.09.08	320	-0.69813170	2016.09.08.15.01
## 4034	340	35	2016.09.08	20	0.34906585	2016.09.08.15.01
## 4035	40	36	2016.09.08	320	-0.69813170	2016.09.08.15.01
## 4036	355	37	2016.09.08	5	0.08726646	2016.09.08.15.01
## 4037	35	38	2016.09.08	325	-0.61086524	2016.09.08.15.01
## 4038	40	39	2016.09.08	320	-0.69813170	2016.09.08.15.01
## 4039	20	40	2016.09.08	340	-0.34906585	2016.09.08.15.01
## 4040	60	41	2016.09.08	300	-1.04719755	2016.09.08.15.01
## 4041	110	42	2016.09.08	250	-1.91986218	2016.09.08.15.01
## 4042	70	43	2016.09.08	290	-1.22173048	2016.09.08.15.01
## 4043	160	44	2016.09.08	200	-2.79252680	2016.09.08.15.01
## 4044	110	45	2016.09.08	250	-1.91986218	2016.09.08.15.01
## 4045	140	46	2016.09.08	220	-2.44346095	2016.09.08.15.01
## 4046	50	1	2016.09.08	310	-0.87266463	2016.09.08.15.15
## 4047	45	2	2016.09.08	315	-0.78539816	2016.09.08.15.15
## 4048	130	3	2016.09.08	230	-2.26892803	2016.09.08.15.15
## 4049	100	4	2016.09.08	260	-1.74532925	2016.09.08.15.15
## 4050	30	5	2016.09.08	330	-0.52359878	2016.09.08.15.15
## 4051	70	6	2016.09.08	290	-1.22173048	2016.09.08.15.15
## 4052	30	7	2016.09.08	330	-0.52359878	2016.09.08.15.15
## 4053	80	8	2016.09.08	280	-1.39626340	2016.09.08.15.15
## 4054	70	9	2016.09.08	290	-1.22173048	2016.09.08.15.15
## 4055	85	10	2016.09.08	275	-1.48352986	2016.09.08.15.15
## 4056	90	11	2016.09.08	270	-1.57079633	2016.09.08.15.15
## 4057	355	12	2016.09.08	5	0.08726646	2016.09.08.15.15
## 4058	350	13	2016.09.08	10	0.17453293	2016.09.08.15.15
## 4059	280	14	2016.09.08	80	1.39626340	2016.09.08.15.15
## 4060	320	15	2016.09.08	40	0.69813170	2016.09.08.15.15
## 4061	325	16	2016.09.08	35	0.61086524	2016.09.08.15.15
## 4062	330	17	2016.09.08	30	0.52359878	2016.09.08.15.15
## 4063	320	18	2016.09.08	40	0.69813170	2016.09.08.15.15
## 4064	310	19	2016.09.08	50	0.87266463	2016.09.08.15.15

## 4065	325	20	2016.09.08	35	0.61086524	2016.09.08.15.15
## 4066	330	21	2016.09.08	30	0.52359878	2016.09.08.15.15
## 4067	320	22	2016.09.08	40	0.69813170	2016.09.08.15.15
## 4068	335	23	2016.09.08	25	0.43633231	2016.09.08.15.15
## 4069	325	24	2016.09.08	35	0.61086524	2016.09.08.15.15
## 4070	345	25	2016.09.08	15	0.26179939	2016.09.08.15.15
## 4071	350	26	2016.09.08	10	0.17453293	2016.09.08.15.15
## 4072	310	27	2016.09.08	50	0.87266463	2016.09.08.15.15
## 4073	40	1	2016.09.08	320	-0.69813170	2016.09.08.15.56
## 4074	350	2	2016.09.08	10	0.17453293	2016.09.08.15.56
## 4075	10	3	2016.09.08	350	-0.17453293	2016.09.08.15.56
## 4076	10	4	2016.09.08	350	-0.17453293	2016.09.08.15.56
## 4077	40	5	2016.09.08	320	-0.69813170	2016.09.08.15.56
## 4078	25	6	2016.09.08	335	-0.43633231	2016.09.08.15.56
## 4079	60	7	2016.09.08	300	-1.04719755	2016.09.08.15.56
## 4080	10	8	2016.09.08	350	-0.17453293	2016.09.08.15.56
## 4081	80	9	2016.09.08	280	-1.39626340	2016.09.08.15.56
## 4082	75	10	2016.09.08	285	-1.30899694	2016.09.08.15.56
## 4083	70	11	2016.09.08	290	-1.22173048	2016.09.08.15.56
## 4084	70	12	2016.09.08	290	-1.22173048	2016.09.08.15.56
## 4085	90	13	2016.09.08	270	-1.57079633	2016.09.08.15.56
## 4086	100	14	2016.09.08	260	-1.74532925	2016.09.08.15.56
## 4087	70	15	2016.09.08	290	-1.22173048	2016.09.08.15.56
## 4088	30	16	2016.09.08	330	-0.52359878	2016.09.08.15.56
## 4089	85	17	2016.09.08	275	-1.48352986	2016.09.08.15.56
## 4090	45	18	2016.09.08	315	-0.78539816	2016.09.08.15.56
## 4091	90	19	2016.09.08	270	-1.57079633	2016.09.08.15.56
## 4092	60	20	2016.09.08	300	-1.04719755	2016.09.08.15.56
## 4093	10	21	2016.09.08	350	-0.17453293	2016.09.08.15.56
## 4094	120	22	2016.09.08	240	-2.09439510	2016.09.08.15.56
## 4095	55	23	2016.09.08	305	-0.95993109	2016.09.08.15.56
## 4096	10	24	2016.09.08	350	-0.17453293	2016.09.08.15.56
## 4097	120	25	2016.09.08	240	-2.09439510	2016.09.08.15.56
## 4098	330	26	2016.09.08	30	0.52359878	2016.09.08.15.56
## 4099	355	27	2016.09.08	5	0.08726646	2016.09.08.15.56
## 4100	70	28	2016.09.08	290	-1.22173048	2016.09.08.15.56
## 4101	65	29	2016.09.08	295	-1.13446401	2016.09.08.15.56
## 4102	160	30	2016.09.08	200	-2.79252680	2016.09.08.15.56
## 4103	180	1	2016.09.08	180	3.14159265	2016.09.08.16.58
## 4104	95	2	2016.09.08	265	-1.65806279	2016.09.08.16.58
## 4105	110	3	2016.09.08	250	-1.91986218	2016.09.08.16.58
## 4106	230	4	2016.09.08	130	2.26892803	2016.09.08.16.58
## 4107	230	5	2016.09.08	130	2.26892803	2016.09.08.16.58
## 4108	240	6	2016.09.08	120	2.09439510	2016.09.08.16.58
## 4109	210	7	2016.09.08	150	2.61799388	2016.09.08.16.58
## 4110	280	8	2016.09.08	80	1.39626340	2016.09.08.16.58
## 4111	220	9	2016.09.08	140	2.44346095	2016.09.08.16.58
## 4112	150	10	2016.09.08	210	-2.61799388	2016.09.08.16.58
## 4113	150	11	2016.09.08	210	-2.61799388	2016.09.08.16.58
## 4114	130	12	2016.09.08	230	-2.26892803	2016.09.08.16.58
## 4115	260	13	2016.09.08	100	1.74532925	2016.09.08.16.58
## 4116	70	14	2016.09.08	290	-1.22173048	2016.09.08.16.58
## 4117	40	15	2016.09.08	320	-0.69813170	2016.09.08.16.58
## 4118	75	16	2016.09.08	285	-1.30899694	2016.09.08.16.58

## 4119	70	17	2016.09.08	290	-1.22173048	2016.09.08.16.58
## 4120	90	18	2016.09.08	270	-1.57079633	2016.09.08.16.58
## 4121	75	19	2016.09.08	285	-1.30899694	2016.09.08.16.58
## 4122	60	20	2016.09.08	300	-1.04719755	2016.09.08.16.58
## 4123	90	21	2016.09.08	270	-1.57079633	2016.09.08.16.58
## 4124	60	22	2016.09.08	300	-1.04719755	2016.09.08.16.58
## 4125	70	23	2016.09.08	290	-1.22173048	2016.09.08.16.58
## 4126	40	24	2016.09.08	320	-0.69813170	2016.09.08.16.58
## 4127	170	1	2016.09.08	190	-2.96705973	2016.09.08.18.00
## 4128	205	2	2016.09.08	155	2.70526034	2016.09.08.18.00
## 4129	215	3	2016.09.08	145	2.53072742	2016.09.08.18.00
## 4130	170	4	2016.09.08	190	-2.96705973	2016.09.08.18.00
## 4131	190	5	2016.09.08	170	2.96705973	2016.09.08.18.00
## 4132	190	6	2016.09.08	170	2.96705973	2016.09.08.18.00
## 4133	210	7	2016.09.08	150	2.61799388	2016.09.08.18.00
## 4134	170	8	2016.09.08	190	-2.96705973	2016.09.08.18.00
## 4135	130	9	2016.09.08	230	-2.26892803	2016.09.08.18.00
## 4136	140	10	2016.09.08	220	-2.44346095	2016.09.08.18.00
## 4137	200	11	2016.09.08	160	2.79252680	2016.09.08.18.00
## 4138	70	12	2016.09.08	290	-1.22173048	2016.09.08.18.00
## 4139	70	13	2016.09.08	290	-1.22173048	2016.09.08.18.00
## 4140	80	14	2016.09.08	280	-1.39626340	2016.09.08.18.00
## 4141	30	15	2016.09.08	330	-0.52359878	2016.09.08.18.00
## 4142	70	16	2016.09.08	290	-1.22173048	2016.09.08.18.00
## 4143	30	17	2016.09.08	330	-0.52359878	2016.09.08.18.00
## 4144	35	18	2016.09.08	325	-0.61086524	2016.09.08.18.00
## 4145	80	19	2016.09.08	280	-1.39626340	2016.09.08.18.00
## 4146	90	20	2016.09.08	270	-1.57079633	2016.09.08.18.00
## 4147	70	21	2016.09.08	290	-1.22173048	2016.09.08.18.00
## 4148	20	22	2016.09.08	340	-0.34906585	2016.09.08.18.00
## 4149	120	23	2016.09.08	240	-2.09439510	2016.09.08.18.00
## 4150	145	24	2016.09.08	215	-2.53072742	2016.09.08.18.00
## 4151	155	25	2016.09.08	205	-2.70526034	2016.09.08.18.00
## 4152	150	26	2016.09.08	210	-2.61799388	2016.09.08.18.00
## 4153	300	1	2016.09.08	60	1.04719755	2016.09.08.18.21
## 4154	285	2	2016.09.08	75	1.30899694	2016.09.08.18.21
## 4155	280	3	2016.09.08	80	1.39626340	2016.09.08.18.21
## 4156	270	4	2016.09.08	90	1.57079633	2016.09.08.18.21
## 4157	290	5	2016.09.08	70	1.22173048	2016.09.08.18.21
## 4158	245	6	2016.09.08	115	2.00712864	2016.09.08.18.21
## 4159	300	7	2016.09.08	60	1.04719755	2016.09.08.18.21
## 4160	290	8	2016.09.08	70	1.22173048	2016.09.08.18.21
## 4161	270	9	2016.09.08	90	1.57079633	2016.09.08.18.21
## 4162	220	10	2016.09.08	140	2.44346095	2016.09.08.18.21
## 4163	235	11	2016.09.08	125	2.18166156	2016.09.08.18.21
## 4164	355	12	2016.09.08	5	0.08726646	2016.09.08.18.21
## 4165	70	13	2016.09.08	290	-1.22173048	2016.09.08.18.21
## 4166	20	14	2016.09.08	340	-0.34906585	2016.09.08.18.21
## 4167	60	15	2016.09.08	300	-1.04719755	2016.09.08.18.21
## 4168	350	16	2016.09.08	10	0.17453293	2016.09.08.18.21
## 4169	30	17	2016.09.08	330	-0.52359878	2016.09.08.18.21
## 4170	10	18	2016.09.08	350	-0.17453293	2016.09.08.18.21
## 4171	70	19	2016.09.08	290	-1.22173048	2016.09.08.18.21
## 4172	30	20	2016.09.08	330	-0.52359878	2016.09.08.18.21

## 4173	30	21	2016.09.08	330	-0.52359878	2016.09.08.18.21
## 4174	50	22	2016.09.08	310	-0.87266463	2016.09.08.18.21
## 4175	55	23	2016.09.08	305	-0.95993109	2016.09.08.18.21
## 4176	50	24	2016.09.08	310	-0.87266463	2016.09.08.18.21
## 4177	65	25	2016.09.08	295	-1.13446401	2016.09.08.18.21
## 4178	80	1	2016.09.12	280	-1.39626340	2016.09.12.13.50
## 4179	120	2	2016.09.12	240	-2.09439510	2016.09.12.13.50
## 4180	355	3	2016.09.12	5	0.08726646	2016.09.12.13.50
## 4181	130	4	2016.09.12	230	-2.26892803	2016.09.12.13.50
## 4182	45	5	2016.09.12	315	-0.78539816	2016.09.12.13.50
## 4183	355	6	2016.09.12	5	0.08726646	2016.09.12.13.50
## 4184	35	7	2016.09.12	325	-0.61086524	2016.09.12.13.50
## 4185	5	8	2016.09.12	355	-0.08726646	2016.09.12.13.50
## 4186	10	9	2016.09.12	350	-0.17453293	2016.09.12.13.50
## 4187	10	10	2016.09.12	350	-0.17453293	2016.09.12.13.50
## 4188	355	11	2016.09.12	5	0.08726646	2016.09.12.13.50
## 4189	340	12	2016.09.12	20	0.34906585	2016.09.12.13.50
## 4190	350	13	2016.09.12	10	0.17453293	2016.09.12.13.50
## 4191	345	14	2016.09.12	15	0.26179939	2016.09.12.13.50
## 4192	330	15	2016.09.12	30	0.52359878	2016.09.12.13.50
## 4193	350	16	2016.09.12	10	0.17453293	2016.09.12.13.50
## 4194	330	17	2016.09.12	30	0.52359878	2016.09.12.13.50
## 4195	5	18	2016.09.12	355	-0.08726646	2016.09.12.13.50
## 4196	230	1	2016.09.12	130	2.26892803	2016.09.12.14.26
## 4197	180	2	2016.09.12	180	3.14159265	2016.09.12.14.26
## 4198	50	3	2016.09.12	310	-0.87266463	2016.09.12.14.26
## 4199	30	4	2016.09.12	330	-0.52359878	2016.09.12.14.26
## 4200	100	5	2016.09.12	260	-1.74532925	2016.09.12.14.26
## 4201	50	6	2016.09.12	310	-0.87266463	2016.09.12.14.26
## 4202	70	7	2016.09.12	290	-1.22173048	2016.09.12.14.26
## 4203	60	8	2016.09.12	300	-1.04719755	2016.09.12.14.26
## 4204	75	9	2016.09.12	285	-1.30899694	2016.09.12.14.26
## 4205	80	10	2016.09.12	280	-1.39626340	2016.09.12.14.26
## 4206	110	11	2016.09.12	250	-1.91986218	2016.09.12.14.26
## 4207	330	12	2016.09.12	30	0.52359878	2016.09.12.14.26
## 4208	280	13	2016.09.12	80	1.39626340	2016.09.12.14.26
## 4209	355	14	2016.09.12	5	0.08726646	2016.09.12.14.26
## 4210	280	15	2016.09.12	80	1.39626340	2016.09.12.14.26
## 4211	345	16	2016.09.12	15	0.26179939	2016.09.12.14.26
## 4212	350	17	2016.09.12	10	0.17453293	2016.09.12.14.26
## 4213	355	18	2016.09.12	5	0.08726646	2016.09.12.14.26
## 4214	340	19	2016.09.12	20	0.34906585	2016.09.12.14.26
## 4215	340	20	2016.09.12	20	0.34906585	2016.09.12.14.26
## 4216	330	21	2016.09.12	30	0.52359878	2016.09.12.14.26
## 4217	340	22	2016.09.12	20	0.34906585	2016.09.12.14.26
## 4218	5	23	2016.09.12	355	-0.08726646	2016.09.12.14.26
## 4219	0	24	2016.09.12	0	0.00000000	2016.09.12.14.26
## 4220	355	25	2016.09.12	5	0.08726646	2016.09.12.14.26
## 4221	150	26	2016.09.12	210	-2.61799388	2016.09.12.14.26
## 4222	115	27	2016.09.12	245	-2.00712864	2016.09.12.14.26
## 4223	250	28	2016.09.12	110	1.91986218	2016.09.12.14.26
## 4224	300	1	2016.09.12	60	1.04719755	2016.09.12.14.41
## 4225	350	2	2016.09.12	10	0.17453293	2016.09.12.14.41
## 4226	305	3	2016.09.12	55	0.95993109	2016.09.12.14.41

## 4227	85	4	2016.09.12	275	-1.48352986	2016.09.12.14.41
## 4228	195	5	2016.09.12	165	2.87979327	2016.09.12.14.41
## 4229	330	6	2016.09.12	30	0.52359878	2016.09.12.14.41
## 4230	250	7	2016.09.12	110	1.91986218	2016.09.12.14.41
## 4231	60	8	2016.09.12	300	-1.04719755	2016.09.12.14.41
## 4232	140	9	2016.09.12	220	-2.44346095	2016.09.12.14.41
## 4233	155	10	2016.09.12	205	-2.70526034	2016.09.12.14.41
## 4234	240	11	2016.09.12	120	2.09439510	2016.09.12.14.41
## 4235	115	12	2016.09.12	245	-2.00712864	2016.09.12.14.41
## 4236	125	13	2016.09.12	235	-2.18166156	2016.09.12.14.41
## 4237	200	14	2016.09.12	160	2.79252680	2016.09.12.14.41
## 4238	220	15	2016.09.12	140	2.44346095	2016.09.12.14.41
## 4239	60	16	2016.09.12	300	-1.04719755	2016.09.12.14.41
## 4240	250	17	2016.09.12	110	1.91986218	2016.09.12.14.41
## 4241	300	18	2016.09.12	60	1.04719755	2016.09.12.14.41
## 4242	210	19	2016.09.12	150	2.61799388	2016.09.12.14.41
## 4243	285	20	2016.09.12	75	1.30899694	2016.09.12.14.41
## 4244	5	21	2016.09.12	355	-0.08726646	2016.09.12.14.41
## 4245	355	22	2016.09.12	5	0.08726646	2016.09.12.14.41
## 4246	5	23	2016.09.12	355	-0.08726646	2016.09.12.14.41
## 4247	10	24	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4248	0	25	2016.09.12	0	0.00000000	2016.09.12.14.41
## 4249	5	26	2016.09.12	355	-0.08726646	2016.09.12.14.41
## 4250	0	27	2016.09.12	0	0.00000000	2016.09.12.14.41
## 4251	350	28	2016.09.12	10	0.17453293	2016.09.12.14.41
## 4252	0	29	2016.09.12	0	0.00000000	2016.09.12.14.41
## 4253	10	30	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4254	355	31	2016.09.12	5	0.08726646	2016.09.12.14.41
## 4255	10	32	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4256	350	33	2016.09.12	10	0.17453293	2016.09.12.14.41
## 4257	10	34	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4258	355	35	2016.09.12	5	0.08726646	2016.09.12.14.41
## 4259	15	36	2016.09.12	345	-0.26179939	2016.09.12.14.41
## 4260	55	37	2016.09.12	305	-0.95993109	2016.09.12.14.41
## 4261	10	38	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4262	85	39	2016.09.12	275	-1.48352986	2016.09.12.14.41
## 4263	70	40	2016.09.12	290	-1.22173048	2016.09.12.14.41
## 4264	355	41	2016.09.12	5	0.08726646	2016.09.12.14.41
## 4265	40	42	2016.09.12	320	-0.69813170	2016.09.12.14.41
## 4266	90	43	2016.09.12	270	-1.57079633	2016.09.12.14.41
## 4267	10	44	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4268	355	45	2016.09.12	5	0.08726646	2016.09.12.14.41
## 4269	0	46	2016.09.12	0	0.00000000	2016.09.12.14.41
## 4270	10	47	2016.09.12	350	-0.17453293	2016.09.12.14.41
## 4271	40	48	2016.09.12	320	-0.69813170	2016.09.12.14.41
## 4272	0	49	2016.09.12	0	0.00000000	2016.09.12.14.41
## 4273	70	1	2016.09.12	290	-1.22173048	2016.09.12.14.54
## 4274	110	2	2016.09.12	250	-1.91986218	2016.09.12.14.54
## 4275	30	3	2016.09.12	330	-0.52359878	2016.09.12.14.54
## 4276	90	4	2016.09.12	270	-1.57079633	2016.09.12.14.54
## 4277	110	5	2016.09.12	250	-1.91986218	2016.09.12.14.54
## 4278	130	6	2016.09.12	230	-2.26892803	2016.09.12.14.54
## 4279	60	7	2016.09.12	300	-1.04719755	2016.09.12.14.54
## 4280	80	8	2016.09.12	280	-1.39626340	2016.09.12.14.54

## 4281	290	9	2016.09.12	70	1.22173048	2016.09.12.14.54
## 4282	105	10	2016.09.12	255	-1.83259571	2016.09.12.14.54
## 4283	100	11	2016.09.12	260	-1.74532925	2016.09.12.14.54
## 4284	240	12	2016.09.12	120	2.09439510	2016.09.12.14.54
## 4285	85	13	2016.09.12	275	-1.48352986	2016.09.12.14.54
## 4286	90	14	2016.09.12	270	-1.57079633	2016.09.12.14.54
## 4287	135	15	2016.09.12	225	-2.35619449	2016.09.12.14.54
## 4288	130	16	2016.09.12	230	-2.26892803	2016.09.12.14.54
## 4289	5	17	2016.09.12	355	-0.08726646	2016.09.12.14.54
## 4290	355	18	2016.09.12	5	0.08726646	2016.09.12.14.54
## 4291	130	19	2016.09.12	230	-2.26892803	2016.09.12.14.54
## 4292	320	20	2016.09.12	40	0.69813170	2016.09.12.14.54
## 4293	355	21	2016.09.12	5	0.08726646	2016.09.12.14.54
## 4294	5	22	2016.09.12	355	-0.08726646	2016.09.12.14.54
## 4295	130	23	2016.09.12	230	-2.26892803	2016.09.12.14.54
## 4296	10	24	2016.09.12	350	-0.17453293	2016.09.12.14.54
## 4297	355	25	2016.09.12	5	0.08726646	2016.09.12.14.54
## 4298	330	26	2016.09.12	30	0.52359878	2016.09.12.14.54
## 4299	35	27	2016.09.12	325	-0.61086524	2016.09.12.14.54
## 4300	40	28	2016.09.12	320	-0.69813170	2016.09.12.14.54
## 4301	355	29	2016.09.12	5	0.08726646	2016.09.12.14.54
## 4302	5	30	2016.09.12	355	-0.08726646	2016.09.12.14.54
## 4303	0	31	2016.09.12	0	0.00000000	2016.09.12.14.54
## 4304	355	32	2016.09.12	5	0.08726646	2016.09.12.14.54
## 4305	220	33	2016.09.12	140	2.44346095	2016.09.12.14.54
## 4306	220	34	2016.09.12	140	2.44346095	2016.09.12.14.54
## 4307	250	35	2016.09.12	110	1.91986218	2016.09.12.14.54
## 4308	200	36	2016.09.12	160	2.79252680	2016.09.12.14.54
## 4309	225	37	2016.09.12	135	2.35619449	2016.09.12.14.54
## 4310	280	38	2016.09.12	80	1.39626340	2016.09.12.14.54
## 4311	240	39	2016.09.12	120	2.09439510	2016.09.12.14.54
## 4312	230	40	2016.09.12	130	2.26892803	2016.09.12.14.54
## 4313	240	41	2016.09.12	120	2.09439510	2016.09.12.14.54
## 4314	190	1	2016.09.12	170	2.96705973	2016.09.12.15.38
## 4315	160	2	2016.09.12	200	-2.79252680	2016.09.12.15.38
## 4316	140	3	2016.09.12	220	-2.44346095	2016.09.12.15.38
## 4317	140	4	2016.09.12	220	-2.44346095	2016.09.12.15.38
## 4318	135	5	2016.09.12	225	-2.35619449	2016.09.12.15.38
## 4319	210	6	2016.09.12	150	2.61799388	2016.09.12.15.38
## 4320	120	7	2016.09.12	240	-2.09439510	2016.09.12.15.38
## 4321	160	8	2016.09.12	200	-2.79252680	2016.09.12.15.38
## 4322	190	9	2016.09.12	170	2.96705973	2016.09.12.15.38
## 4323	30	10	2016.09.12	330	-0.52359878	2016.09.12.15.38
## 4324	215	11	2016.09.12	145	2.53072742	2016.09.12.15.38
## 4325	100	12	2016.09.12	260	-1.74532925	2016.09.12.15.38
## 4326	130	13	2016.09.12	230	-2.26892803	2016.09.12.15.38
## 4327	70	14	2016.09.12	290	-1.22173048	2016.09.12.15.38
## 4328	60	15	2016.09.12	300	-1.04719755	2016.09.12.15.38
## 4329	260	16	2016.09.12	100	1.74532925	2016.09.12.15.38
## 4330	250	17	2016.09.12	110	1.91986218	2016.09.12.15.38
## 4331	250	18	2016.09.12	110	1.91986218	2016.09.12.15.38
## 4332	160	19	2016.09.12	200	-2.79252680	2016.09.12.15.38
## 4333	95	20	2016.09.12	265	-1.65806279	2016.09.12.15.38
## 4334	120	21	2016.09.12	240	-2.09439510	2016.09.12.15.38

## 4335	85	22	2016.09.12	275	-1.48352986	2016.09.12.15.38
## 4336	90	23	2016.09.12	270	-1.57079633	2016.09.12.15.38
## 4337	120	24	2016.09.12	240	-2.09439510	2016.09.12.15.38
## 4338	135	25	2016.09.12	225	-2.35619449	2016.09.12.15.38
## 4339	355	26	2016.09.12	5	0.08726646	2016.09.12.15.38
## 4340	40	27	2016.09.12	320	-0.69813170	2016.09.12.15.38
## 4341	50	28	2016.09.12	310	-0.87266463	2016.09.12.15.38
## 4342	30	29	2016.09.12	330	-0.52359878	2016.09.12.15.38
## 4343	60	30	2016.09.12	300	-1.04719755	2016.09.12.15.38
## 4344	30	31	2016.09.12	330	-0.52359878	2016.09.12.15.38
## 4345	60	32	2016.09.12	300	-1.04719755	2016.09.12.15.38
## 4346	20	33	2016.09.12	340	-0.34906585	2016.09.12.15.38
## 4347	30	34	2016.09.12	330	-0.52359878	2016.09.12.15.38
## 4348	30	35	2016.09.12	330	-0.52359878	2016.09.12.15.38
## 4349	40	36	2016.09.12	320	-0.69813170	2016.09.12.15.38
## 4350	40	37	2016.09.12	320	-0.69813170	2016.09.12.15.38
## 4351	20	38	2016.09.12	340	-0.34906585	2016.09.12.15.38
## 4352	30	39	2016.09.12	330	-0.52359878	2016.09.12.15.38
## 4353	10	40	2016.09.12	350	-0.17453293	2016.09.12.15.38
## 4354	85	1	2016.09.12	275	-1.48352986	2016.09.12.16.39
## 4355	60	2	2016.09.12	300	-1.04719755	2016.09.12.16.39
## 4356	60	3	2016.09.12	300	-1.04719755	2016.09.12.16.39
## 4357	90	4	2016.09.12	270	-1.57079633	2016.09.12.16.39
## 4358	5	5	2016.09.12	355	-0.08726646	2016.09.12.16.39
## 4359	60	6	2016.09.12	300	-1.04719755	2016.09.12.16.39
## 4360	95	7	2016.09.12	265	-1.65806279	2016.09.12.16.39
## 4361	60	8	2016.09.12	300	-1.04719755	2016.09.12.16.39
## 4362	30	9	2016.09.12	330	-0.52359878	2016.09.12.16.39
## 4363	90	10	2016.09.12	270	-1.57079633	2016.09.12.16.39
## 4364	50	11	2016.09.12	310	-0.87266463	2016.09.12.16.39
## 4365	330	12	2016.09.12	30	0.52359878	2016.09.12.16.39
## 4366	350	13	2016.09.12	10	0.17453293	2016.09.12.16.39
## 4367	355	14	2016.09.12	5	0.08726646	2016.09.12.16.39
## 4368	350	15	2016.09.12	10	0.17453293	2016.09.12.16.39
## 4369	355	1	2016.09.12	5	0.08726646	2016.09.12.16.41
## 4370	350	2	2016.09.12	10	0.17453293	2016.09.12.16.41
## 4371	355	3	2016.09.12	5	0.08726646	2016.09.12.16.41
## 4372	0	4	2016.09.12	0	0.00000000	2016.09.12.16.41
## 4373	10	5	2016.09.12	350	-0.17453293	2016.09.12.16.41
## 4374	5	6	2016.09.12	355	-0.08726646	2016.09.12.16.41
## 4375	350	7	2016.09.12	10	0.17453293	2016.09.12.16.41
## 4376	60	8	2016.09.12	300	-1.04719755	2016.09.12.16.41
## 4377	10	9	2016.09.12	350	-0.17453293	2016.09.12.16.41
## 4378	350	10	2016.09.12	10	0.17453293	2016.09.12.16.41
## 4379	60	11	2016.09.12	300	-1.04719755	2016.09.12.16.41
## 4380	280	12	2016.09.12	80	1.39626340	2016.09.12.16.41
## 4381	25	13	2016.09.12	335	-0.43633231	2016.09.12.16.41
## 4382	350	14	2016.09.12	10	0.17453293	2016.09.12.16.41
## 4383	340	15	2016.09.12	20	0.34906585	2016.09.12.16.41
## 4384	0	16	2016.09.12	0	0.00000000	2016.09.12.16.41
## 4385	340	17	2016.09.12	20	0.34906585	2016.09.12.16.41
## 4386	350	18	2016.09.12	10	0.17453293	2016.09.12.16.41
## 4387	130	1	2016.09.12	230	-2.26892803	2016.09.12.16.39
## 4388	120	2	2016.09.12	240	-2.09439510	2016.09.12.16.39

## 4389	110	3	2016.09.12	250	-1.91986218	2016.09.12.16.39
## 4390	120	4	2016.09.12	240	-2.09439510	2016.09.12.16.39
## 4391	105	5	2016.09.12	255	-1.83259571	2016.09.12.16.39
## 4392	120	6	2016.09.12	240	-2.09439510	2016.09.12.16.39
## 4393	140	7	2016.09.12	220	-2.44346095	2016.09.12.16.39
## 4394	130	8	2016.09.12	230	-2.26892803	2016.09.12.16.39
## 4395	120	9	2016.09.12	240	-2.09439510	2016.09.12.16.39
## 4396	120	10	2016.09.12	240	-2.09439510	2016.09.12.16.39
## 4397	110	11	2016.09.12	250	-1.91986218	2016.09.12.16.39
## 4398	115	12	2016.09.12	245	-2.00712864	2016.09.12.16.39
## 4399	130	13	2016.09.12	230	-2.26892803	2016.09.12.16.39
## 4400	150	14	2016.09.12	210	-2.61799388	2016.09.12.16.39
## 4401	40	15	2016.09.12	320	-0.69813170	2016.09.12.16.39
## 4402	50	16	2016.09.12	310	-0.87266463	2016.09.12.16.39
## 4403	40	17	2016.09.12	320	-0.69813170	2016.09.12.16.39
## 4404	50	18	2016.09.12	310	-0.87266463	2016.09.12.16.39
## 4405	50	19	2016.09.12	310	-0.87266463	2016.09.12.16.39
## 4406	40	20	2016.09.12	320	-0.69813170	2016.09.12.16.39
## 4407	60	21	2016.09.12	300	-1.04719755	2016.09.12.16.39
## 4408	40	22	2016.09.12	320	-0.69813170	2016.09.12.16.39
## 4409	70	23	2016.09.12	290	-1.22173048	2016.09.12.16.39
## 4410	350	1	2016.09.13	10	0.17453293	2016.09.13.13.55
## 4411	100	2	2016.09.13	260	-1.74532925	2016.09.13.13.55
## 4412	130	3	2016.09.13	230	-2.26892803	2016.09.13.13.55
## 4413	340	4	2016.09.13	20	0.34906585	2016.09.13.13.55
## 4414	95	5	2016.09.13	265	-1.65806279	2016.09.13.13.55
## 4415	85	6	2016.09.13	275	-1.48352986	2016.09.13.13.55
## 4416	10	7	2016.09.13	350	-0.17453293	2016.09.13.13.55
## 4417	310	8	2016.09.13	50	0.87266463	2016.09.13.13.55
## 4418	195	9	2016.09.13	165	2.87979327	2016.09.13.13.55
## 4419	230	10	2016.09.13	130	2.26892803	2016.09.13.13.55
## 4420	240	11	2016.09.13	120	2.09439510	2016.09.13.13.55
## 4421	185	12	2016.09.13	175	3.05432619	2016.09.13.13.55
## 4422	70	13	2016.09.13	290	-1.22173048	2016.09.13.13.55
## 4423	110	14	2016.09.13	250	-1.91986218	2016.09.13.13.55
## 4424	270	15	2016.09.13	90	1.57079633	2016.09.13.13.55
## 4425	240	16	2016.09.13	120	2.09439510	2016.09.13.13.55
## 4426	240	17	2016.09.13	120	2.09439510	2016.09.13.13.55
## 4427	265	18	2016.09.13	95	1.65806279	2016.09.13.13.55
## 4428	340	19	2016.09.13	20	0.34906585	2016.09.13.13.55
## 4429	330	20	2016.09.13	30	0.52359878	2016.09.13.13.55
## 4430	350	21	2016.09.13	10	0.17453293	2016.09.13.13.55
## 4431	350	22	2016.09.13	10	0.17453293	2016.09.13.13.55
## 4432	330	23	2016.09.13	30	0.52359878	2016.09.13.13.55
## 4433	350	24	2016.09.13	10	0.17453293	2016.09.13.13.55
## 4434	345	25	2016.09.13	15	0.26179939	2016.09.13.13.55
## 4435	355	26	2016.09.13	5	0.08726646	2016.09.13.13.55
## 4436	350	27	2016.09.13	10	0.17453293	2016.09.13.13.55
## 4437	340	28	2016.09.13	20	0.34906585	2016.09.13.13.55
## 4438	355	29	2016.09.13	5	0.08726646	2016.09.13.13.55
## 4439	335	30	2016.09.13	25	0.43633231	2016.09.13.13.55
## 4440	320	31	2016.09.13	40	0.69813170	2016.09.13.13.55
## 4441	190	1	2016.09.13	170	2.96705973	2016.09.13.14.11
## 4442	235	2	2016.09.13	125	2.18166156	2016.09.13.14.11

## 4443	105	3	2016.09.13	255	-1.83259571	2016.09.13.14.11
## 4444	220	4	2016.09.13	140	2.44346095	2016.09.13.14.11
## 4445	85	5	2016.09.13	275	-1.48352986	2016.09.13.14.11
## 4446	80	6	2016.09.13	280	-1.39626340	2016.09.13.14.11
## 4447	60	7	2016.09.13	300	-1.04719755	2016.09.13.14.11
## 4448	130	8	2016.09.13	230	-2.26892803	2016.09.13.14.11
## 4449	100	9	2016.09.13	260	-1.74532925	2016.09.13.14.11
## 4450	130	10	2016.09.13	230	-2.26892803	2016.09.13.14.11
## 4451	75	11	2016.09.13	285	-1.30899694	2016.09.13.14.11
## 4452	120	12	2016.09.13	240	-2.09439510	2016.09.13.14.11
## 4453	100	13	2016.09.13	260	-1.74532925	2016.09.13.14.11
## 4454	100	14	2016.09.13	260	-1.74532925	2016.09.13.14.11
## 4455	80	15	2016.09.13	280	-1.39626340	2016.09.13.14.11
## 4456	130	16	2016.09.13	230	-2.26892803	2016.09.13.14.11
## 4457	90	17	2016.09.13	270	-1.57079633	2016.09.13.14.11
## 4458	90	18	2016.09.13	270	-1.57079633	2016.09.13.14.11
## 4459	90	19	2016.09.13	270	-1.57079633	2016.09.13.14.11
## 4460	70	20	2016.09.13	290	-1.22173048	2016.09.13.14.11
## 4461	130	21	2016.09.13	230	-2.26892803	2016.09.13.14.11
## 4462	110	22	2016.09.13	250	-1.91986218	2016.09.13.14.11
## 4463	80	23	2016.09.13	280	-1.39626340	2016.09.13.14.11
## 4464	80	24	2016.09.13	280	-1.39626340	2016.09.13.14.11
## 4465	150	25	2016.09.13	210	-2.61799388	2016.09.13.14.11
## 4466	350	26	2016.09.13	10	0.17453293	2016.09.13.14.11
## 4467	20	27	2016.09.13	340	-0.34906585	2016.09.13.14.11
## 4468	10	28	2016.09.13	350	-0.17453293	2016.09.13.14.11
## 4469	355	29	2016.09.13	5	0.08726646	2016.09.13.14.11
## 4470	355	30	2016.09.13	5	0.08726646	2016.09.13.14.11
## 4471	355	31	2016.09.13	5	0.08726646	2016.09.13.14.11
## 4472	350	32	2016.09.13	10	0.17453293	2016.09.13.14.11
## 4473	10	33	2016.09.13	350	-0.17453293	2016.09.13.14.11
## 4474	350	34	2016.09.13	10	0.17453293	2016.09.13.14.11
## 4475	5	35	2016.09.13	355	-0.08726646	2016.09.13.14.11
## 4476	20	36	2016.09.13	340	-0.34906585	2016.09.13.14.11
## 4477	355	37	2016.09.13	5	0.08726646	2016.09.13.14.11
## 4478	60	38	2016.09.13	300	-1.04719755	2016.09.13.14.11
## 4479	5	39	2016.09.13	355	-0.08726646	2016.09.13.14.11
## 4480	15	40	2016.09.13	345	-0.26179939	2016.09.13.14.11
## 4481	10	41	2016.09.13	350	-0.17453293	2016.09.13.14.11
## 4482	40	42	2016.09.13	320	-0.69813170	2016.09.13.14.11
## 4483	350	43	2016.09.13	10	0.17453293	2016.09.13.14.11
## 4484	10	44	2016.09.13	350	-0.17453293	2016.09.13.14.11
## 4485	10	45	2016.09.13	350	-0.17453293	2016.09.13.14.11
## 4486	35	46	2016.09.13	325	-0.61086524	2016.09.13.14.11
## 4487	30	47	2016.09.13	330	-0.52359878	2016.09.13.14.11
## 4488	5	48	2016.09.13	355	-0.08726646	2016.09.13.14.11
## 4489	80	49	2016.09.13	280	-1.39626340	2016.09.13.14.11
## 4490	50	50	2016.09.13	310	-0.87266463	2016.09.13.14.11
## 4491	100	51	2016.09.13	260	-1.74532925	2016.09.13.14.11
## 4492	105	52	2016.09.13	255	-1.83259571	2016.09.13.14.11
## 4493	85	53	2016.09.13	275	-1.48352986	2016.09.13.14.11
## 4494	120	54	2016.09.13	240	-2.09439510	2016.09.13.14.11
## 4495	95	55	2016.09.13	265	-1.65806279	2016.09.13.14.11
## 4496	120	56	2016.09.13	240	-2.09439510	2016.09.13.14.11

## 4497	60	57	2016.09.13	300	-1.04719755	2016.09.13.14.11
## 4498	100	58	2016.09.13	260	-1.74532925	2016.09.13.14.11
## 4499	30	59	2016.09.13	330	-0.52359878	2016.09.13.14.11
## 4500	350	60	2016.09.13	10	0.17453293	2016.09.13.14.11
## 4501	50	61	2016.09.13	310	-0.87266463	2016.09.13.14.11
## 4502	90	62	2016.09.13	270	-1.57079633	2016.09.13.14.11
## 4503	90	1	2016.09.13	270	-1.57079633	2016.09.13.14.18
## 4504	185	2	2016.09.13	175	3.05432619	2016.09.13.14.18
## 4505	95	3	2016.09.13	265	-1.65806279	2016.09.13.14.18
## 4506	140	4	2016.09.13	220	-2.44346095	2016.09.13.14.18
## 4507	175	5	2016.09.13	185	-3.05432619	2016.09.13.14.18
## 4508	120	6	2016.09.13	240	-2.09439510	2016.09.13.14.18
## 4509	230	7	2016.09.13	130	2.26892803	2016.09.13.14.18
## 4510	110	8	2016.09.13	250	-1.91986218	2016.09.13.14.18
## 4511	160	9	2016.09.13	200	-2.79252680	2016.09.13.14.18
## 4512	90	10	2016.09.13	270	-1.57079633	2016.09.13.14.18
## 4513	200	11	2016.09.13	160	2.79252680	2016.09.13.14.18
## 4514	190	12	2016.09.13	170	2.96705973	2016.09.13.14.18
## 4515	120	13	2016.09.13	240	-2.09439510	2016.09.13.14.18
## 4516	90	14	2016.09.13	270	-1.57079633	2016.09.13.14.18
## 4517	70	15	2016.09.13	290	-1.22173048	2016.09.13.14.18
## 4518	305	16	2016.09.13	55	0.95993109	2016.09.13.14.18
## 4519	220	17	2016.09.13	140	2.44346095	2016.09.13.14.18
## 4520	355	18	2016.09.13	5	0.08726646	2016.09.13.14.18
## 4521	0	19	2016.09.13	0	0.00000000	2016.09.13.14.18
## 4522	350	20	2016.09.13	10	0.17453293	2016.09.13.14.18
## 4523	350	21	2016.09.13	10	0.17453293	2016.09.13.14.18
## 4524	330	22	2016.09.13	30	0.52359878	2016.09.13.14.18
## 4525	345	23	2016.09.13	15	0.26179939	2016.09.13.14.18
## 4526	350	24	2016.09.13	10	0.17453293	2016.09.13.14.18
## 4527	5	25	2016.09.13	355	-0.08726646	2016.09.13.14.18
## 4528	345	26	2016.09.13	15	0.26179939	2016.09.13.14.18
## 4529	0	27	2016.09.13	0	0.00000000	2016.09.13.14.18
## 4530	0	28	2016.09.13	0	0.00000000	2016.09.13.14.18
## 4531	355	29	2016.09.13	5	0.08726646	2016.09.13.14.18
## 4532	355	30	2016.09.13	5	0.08726646	2016.09.13.14.18
## 4533	340	31	2016.09.13	20	0.34906585	2016.09.13.14.18
## 4534	350	32	2016.09.13	10	0.17453293	2016.09.13.14.18
## 4535	350	33	2016.09.13	10	0.17453293	2016.09.13.14.18
## 4536	100	1	2016.09.13	260	-1.74532925	2016.09.13.14.35
## 4537	90	2	2016.09.13	270	-1.57079633	2016.09.13.14.35
## 4538	165	3	2016.09.13	195	-2.87979327	2016.09.13.14.35
## 4539	170	4	2016.09.13	190	-2.96705973	2016.09.13.14.35
## 4540	240	5	2016.09.13	120	2.09439510	2016.09.13.14.35
## 4541	70	6	2016.09.13	290	-1.22173048	2016.09.13.14.35
## 4542	95	7	2016.09.13	265	-1.65806279	2016.09.13.14.35
## 4543	100	8	2016.09.13	260	-1.74532925	2016.09.13.14.35
## 4544	130	9	2016.09.13	230	-2.26892803	2016.09.13.14.35
## 4545	135	10	2016.09.13	225	-2.35619449	2016.09.13.14.35
## 4546	150	11	2016.09.13	210	-2.61799388	2016.09.13.14.35
## 4547	200	12	2016.09.13	160	2.79252680	2016.09.13.14.35
## 4548	110	13	2016.09.13	250	-1.91986218	2016.09.13.14.35
## 4549	160	14	2016.09.13	200	-2.79252680	2016.09.13.14.35
## 4550	175	15	2016.09.13	185	-3.05432619	2016.09.13.14.35

## 4551	150	16	2016.09.13	210	-2.61799388	2016.09.13.14.35
## 4552	160	17	2016.09.13	200	-2.79252680	2016.09.13.14.35
## 4553	100	18	2016.09.13	260	-1.74532925	2016.09.13.14.35
## 4554	140	19	2016.09.13	220	-2.44346095	2016.09.13.14.35
## 4555	0	20	2016.09.13	0	0.00000000	2016.09.13.14.35
## 4556	0	21	2016.09.13	0	0.00000000	2016.09.13.14.35
## 4557	350	22	2016.09.13	10	0.17453293	2016.09.13.14.35
## 4558	350	23	2016.09.13	10	0.17453293	2016.09.13.14.35
## 4559	15	24	2016.09.13	345	-0.26179939	2016.09.13.14.35
## 4560	0	25	2016.09.13	0	0.00000000	2016.09.13.14.35
## 4561	10	26	2016.09.13	350	-0.17453293	2016.09.13.14.35
## 4562	20	27	2016.09.13	340	-0.34906585	2016.09.13.14.35
## 4563	30	28	2016.09.13	330	-0.52359878	2016.09.13.14.35
## 4564	340	29	2016.09.13	20	0.34906585	2016.09.13.14.35
## 4565	20	30	2016.09.13	340	-0.34906585	2016.09.13.14.35
## 4566	355	31	2016.09.13	5	0.08726646	2016.09.13.14.35
## 4567	185	32	2016.09.13	175	3.05432619	2016.09.13.14.35
## 4568	80	33	2016.09.13	280	-1.39626340	2016.09.13.14.35
## 4569	95	34	2016.09.13	265	-1.65806279	2016.09.13.14.35
## 4570	0	35	2016.09.13	0	0.00000000	2016.09.13.14.35
## 4571	130	36	2016.09.13	230	-2.26892803	2016.09.13.14.35
## 4572	95	37	2016.09.13	265	-1.65806279	2016.09.13.14.35
## 4573	80	38	2016.09.13	280	-1.39626340	2016.09.13.14.35
## 4574	105	39	2016.09.13	255	-1.83259571	2016.09.13.14.35
## 4575	140	40	2016.09.13	220	-2.44346095	2016.09.13.14.35
## 4576	110	41	2016.09.13	250	-1.91986218	2016.09.13.14.35
## 4577	100	42	2016.09.13	260	-1.74532925	2016.09.13.14.35
## 4578	90	43	2016.09.13	270	-1.57079633	2016.09.13.14.35
## 4579	100	44	2016.09.13	260	-1.74532925	2016.09.13.14.35
## 4580	120	45	2016.09.13	240	-2.09439510	2016.09.13.14.35
## 4581	85	46	2016.09.13	275	-1.48352986	2016.09.13.14.35
## 4582	90	47	2016.09.13	270	-1.57079633	2016.09.13.14.35
## 4583	110	48	2016.09.13	250	-1.91986218	2016.09.13.14.35
## 4584	110	49	2016.09.13	250	-1.91986218	2016.09.13.14.35
## 4585	10	50	2016.09.13	350	-0.17453293	2016.09.13.14.35
## 4586	85	51	2016.09.13	275	-1.48352986	2016.09.13.14.35
## 4587	75	52	2016.09.13	285	-1.30899694	2016.09.13.14.35
## 4588	80	53	2016.09.13	280	-1.39626340	2016.09.13.14.35
## 4589	85	54	2016.09.13	275	-1.48352986	2016.09.13.14.35
## 4590	130	55	2016.09.13	230	-2.26892803	2016.09.13.14.35
## 4591	100	56	2016.09.13	260	-1.74532925	2016.09.13.14.35
## 4592	120	57	2016.09.13	240	-2.09439510	2016.09.13.14.35
## 4593	90	58	2016.09.13	270	-1.57079633	2016.09.13.14.35
## 4594	90	59	2016.09.13	270	-1.57079633	2016.09.13.14.35
## 4595	120	60	2016.09.13	240	-2.09439510	2016.09.13.14.35
## 4596	105	61	2016.09.13	255	-1.83259571	2016.09.13.14.35
## 4597	10	62	2016.09.13	350	-0.17453293	2016.09.13.14.35
## 4598	10	63	2016.09.13	350	-0.17453293	2016.09.13.14.35
## 4599	50	64	2016.09.13	310	-0.87266463	2016.09.13.14.35
## 4600	80	65	2016.09.13	280	-1.39626340	2016.09.13.14.35
## 4601	15	66	2016.09.13	345	-0.26179939	2016.09.13.14.35
## 4602	350	67	2016.09.13	10	0.17453293	2016.09.13.14.35
## 4603	225	1	2016.09.13	135	2.35619449	2016.09.13.16.11
## 4604	215	2	2016.09.13	145	2.53072742	2016.09.13.16.11

## 4605	205	3	2016.09.13	155	2.70526034	2016.09.13.16.11
## 4606	240	4	2016.09.13	120	2.09439510	2016.09.13.16.11
## 4607	180	5	2016.09.13	180	3.14159265	2016.09.13.16.11
## 4608	140	6	2016.09.13	220	-2.44346095	2016.09.13.16.11
## 4609	190	7	2016.09.13	170	2.96705973	2016.09.13.16.11
## 4610	155	8	2016.09.13	205	-2.70526034	2016.09.13.16.11
## 4611	180	9	2016.09.13	180	3.14159265	2016.09.13.16.11
## 4612	180	10	2016.09.13	180	3.14159265	2016.09.13.16.11
## 4613	190	11	2016.09.13	170	2.96705973	2016.09.13.16.11
## 4614	150	12	2016.09.13	210	-2.61799388	2016.09.13.16.11
## 4615	195	13	2016.09.13	165	2.87979327	2016.09.13.16.11
## 4616	160	14	2016.09.13	200	-2.79252680	2016.09.13.16.11
## 4617	175	15	2016.09.13	185	-3.05432619	2016.09.13.16.11
## 4618	85	16	2016.09.13	275	-1.48352986	2016.09.13.16.11
## 4619	80	17	2016.09.13	280	-1.39626340	2016.09.13.16.11
## 4620	80	18	2016.09.13	280	-1.39626340	2016.09.13.16.11
## 4621	90	19	2016.09.13	270	-1.57079633	2016.09.13.16.11
## 4622	75	20	2016.09.13	285	-1.30899694	2016.09.13.16.11
## 4623	100	21	2016.09.13	260	-1.74532925	2016.09.13.16.11
## 4624	90	22	2016.09.13	270	-1.57079633	2016.09.13.16.11
## 4625	120	23	2016.09.13	240	-2.09439510	2016.09.13.16.11
## 4626	85	24	2016.09.13	275	-1.48352986	2016.09.13.16.11
## 4627	70	25	2016.09.13	290	-1.22173048	2016.09.13.16.11
## 4628	100	26	2016.09.13	260	-1.74532925	2016.09.13.16.11
## 4629	85	27	2016.09.13	275	-1.48352986	2016.09.13.16.11
## 4630	85	28	2016.09.13	275	-1.48352986	2016.09.13.16.11
## 4631	90	29	2016.09.13	270	-1.57079633	2016.09.13.16.11
## 4632	150	30	2016.09.13	210	-2.61799388	2016.09.13.16.11
## 4633	220	31	2016.09.13	140	2.44346095	2016.09.13.16.11
## 4634	185	32	2016.09.13	175	3.05432619	2016.09.13.16.11
## 4635	190	33	2016.09.13	170	2.96705973	2016.09.13.16.11
## 4636	20	34	2016.09.13	340	-0.34906585	2016.09.13.16.11
## 4637	15	35	2016.09.13	345	-0.26179939	2016.09.13.16.11
## 4638	80	36	2016.09.13	280	-1.39626340	2016.09.13.16.11
## 4639	70	37	2016.09.13	290	-1.22173048	2016.09.13.16.11
## 4640	240	38	2016.09.13	120	2.09439510	2016.09.13.16.11
## 4641	160	39	2016.09.13	200	-2.79252680	2016.09.13.16.11
## 4642	210	40	2016.09.13	150	2.61799388	2016.09.13.16.11
## 4643	220	41	2016.09.13	140	2.44346095	2016.09.13.16.11
## 4644	165	42	2016.09.13	195	-2.87979327	2016.09.13.16.11
## 4645	180	1	2016.09.13	180	3.14159265	2016.09.13.16.24
## 4646	250	2	2016.09.13	110	1.91986218	2016.09.13.16.24
## 4647	145	3	2016.09.13	215	-2.53072742	2016.09.13.16.24
## 4648	255	4	2016.09.13	105	1.83259571	2016.09.13.16.24
## 4649	130	5	2016.09.13	230	-2.26892803	2016.09.13.16.24
## 4650	130	6	2016.09.13	230	-2.26892803	2016.09.13.16.24
## 4651	150	7	2016.09.13	210	-2.61799388	2016.09.13.16.24
## 4652	135	8	2016.09.13	225	-2.35619449	2016.09.13.16.24
## 4653	300	9	2016.09.13	60	1.04719755	2016.09.13.16.24
## 4654	235	10	2016.09.13	125	2.18166156	2016.09.13.16.24
## 4655	290	11	2016.09.13	70	1.22173048	2016.09.13.16.24
## 4656	330	12	2016.09.13	30	0.52359878	2016.09.13.16.24
## 4657	180	13	2016.09.13	180	3.14159265	2016.09.13.16.24
## 4658	320	14	2016.09.13	40	0.69813170	2016.09.13.16.24

## 4659	280	15	2016.09.13	80	1.39626340	2016.09.13.16.24
## 4660	315	16	2016.09.13	45	0.78539816	2016.09.13.16.24
## 4661	220	17	2016.09.13	140	2.44346095	2016.09.13.16.24
## 4662	290	18	2016.09.13	70	1.22173048	2016.09.13.16.24
## 4663	250	19	2016.09.13	110	1.91986218	2016.09.13.16.24
## 4664	260	20	2016.09.13	100	1.74532925	2016.09.13.16.24
## 4665	175	21	2016.09.13	185	-3.05432619	2016.09.13.16.24
## 4666	170	22	2016.09.13	190	-2.96705973	2016.09.13.16.24
## 4667	190	23	2016.09.13	170	2.96705973	2016.09.13.16.24
## 4668	180	24	2016.09.13	180	3.14159265	2016.09.13.16.24
## 4669	165	25	2016.09.13	195	-2.87979327	2016.09.13.16.24
## 4670	170	26	2016.09.13	190	-2.96705973	2016.09.13.16.24
## 4671	160	27	2016.09.13	200	-2.79252680	2016.09.13.16.24
## 4672	180	28	2016.09.13	180	3.14159265	2016.09.13.16.24
## 4673	185	29	2016.09.13	175	3.05432619	2016.09.13.16.24
## 4674	155	30	2016.09.13	205	-2.70526034	2016.09.13.16.24
## 4675	20	1	2016.09.14	340	-0.34906585	2016.09.14.13.51
## 4676	100	2	2016.09.14	260	-1.74532925	2016.09.14.13.51
## 4677	50	3	2016.09.14	310	-0.87266463	2016.09.14.13.51
## 4678	100	4	2016.09.14	260	-1.74532925	2016.09.14.13.51
## 4679	85	5	2016.09.14	275	-1.48352986	2016.09.14.13.51
## 4680	190	6	2016.09.14	170	2.96705973	2016.09.14.13.51
## 4681	70	7	2016.09.14	290	-1.22173048	2016.09.14.13.51
## 4682	120	8	2016.09.14	240	-2.09439510	2016.09.14.13.51
## 4683	170	9	2016.09.14	190	-2.96705973	2016.09.14.13.51
## 4684	90	10	2016.09.14	270	-1.57079633	2016.09.14.13.51
## 4685	95	11	2016.09.14	265	-1.65806279	2016.09.14.13.51
## 4686	50	12	2016.09.14	310	-0.87266463	2016.09.14.13.51
## 4687	80	13	2016.09.14	280	-1.39626340	2016.09.14.13.51
## 4688	85	14	2016.09.14	275	-1.48352986	2016.09.14.13.51
## 4689	65	15	2016.09.14	295	-1.13446401	2016.09.14.13.51
## 4690	80	16	2016.09.14	280	-1.39626340	2016.09.14.13.51
## 4691	90	17	2016.09.14	270	-1.57079633	2016.09.14.13.51
## 4692	355	18	2016.09.14	5	0.08726646	2016.09.14.13.51
## 4693	350	19	2016.09.14	10	0.17453293	2016.09.14.13.51
## 4694	0	20	2016.09.14	0	0.00000000	2016.09.14.13.51
## 4695	10	21	2016.09.14	350	-0.17453293	2016.09.14.13.51
## 4696	45	22	2016.09.14	315	-0.78539816	2016.09.14.13.51
## 4697	355	23	2016.09.14	5	0.08726646	2016.09.14.13.51
## 4698	0	24	2016.09.14	0	0.00000000	2016.09.14.13.51
## 4699	355	25	2016.09.14	5	0.08726646	2016.09.14.13.51
## 4700	340	26	2016.09.14	20	0.34906585	2016.09.14.13.51
## 4701	335	27	2016.09.14	25	0.43633231	2016.09.14.13.51
## 4702	345	28	2016.09.14	15	0.26179939	2016.09.14.13.51
## 4703	340	29	2016.09.14	20	0.34906585	2016.09.14.13.51
## 4704	340	30	2016.09.14	20	0.34906585	2016.09.14.13.51
## 4705	350	31	2016.09.14	10	0.17453293	2016.09.14.13.51
## 4706	235	1	2016.09.14	125	2.18166156	2016.09.14.14.03
## 4707	290	2	2016.09.14	70	1.22173048	2016.09.14.14.03
## 4708	315	3	2016.09.14	45	0.78539816	2016.09.14.14.03
## 4709	5	4	2016.09.14	355	-0.08726646	2016.09.14.14.03
## 4710	340	5	2016.09.14	20	0.34906585	2016.09.14.14.03
## 4711	315	6	2016.09.14	45	0.78539816	2016.09.14.14.03
## 4712	60	7	2016.09.14	300	-1.04719755	2016.09.14.14.03

## 4713	275	8	2016.09.14	85	1.48352986	2016.09.14.14.03
## 4714	350	9	2016.09.14	10	0.17453293	2016.09.14.14.03
## 4715	320	10	2016.09.14	40	0.69813170	2016.09.14.14.03
## 4716	0	11	2016.09.14	0	0.00000000	2016.09.14.14.03
## 4717	320	12	2016.09.14	40	0.69813170	2016.09.14.14.03
## 4718	330	13	2016.09.14	30	0.52359878	2016.09.14.14.03
## 4719	280	14	2016.09.14	80	1.39626340	2016.09.14.14.03
## 4720	50	15	2016.09.14	310	-0.87266463	2016.09.14.14.03
## 4721	0	16	2016.09.14	0	0.00000000	2016.09.14.14.03
## 4722	330	17	2016.09.14	30	0.52359878	2016.09.14.14.03
## 4723	345	18	2016.09.14	15	0.26179939	2016.09.14.14.03
## 4724	260	19	2016.09.14	100	1.74532925	2016.09.14.14.03
## 4725	350	20	2016.09.14	10	0.17453293	2016.09.14.14.03
## 4726	260	21	2016.09.14	100	1.74532925	2016.09.14.14.03
## 4727	355	22	2016.09.14	5	0.08726646	2016.09.14.14.03
## 4728	275	23	2016.09.14	85	1.48352986	2016.09.14.14.03
## 4729	345	24	2016.09.14	15	0.26179939	2016.09.14.14.03
## 4730	260	25	2016.09.14	100	1.74532925	2016.09.14.14.03
## 4731	340	26	2016.09.14	20	0.34906585	2016.09.14.14.03
## 4732	280	27	2016.09.14	80	1.39626340	2016.09.14.14.03
## 4733	240	28	2016.09.14	120	2.09439510	2016.09.14.14.03
## 4734	175	29	2016.09.14	185	-3.05432619	2016.09.14.14.03
## 4735	175	30	2016.09.14	185	-3.05432619	2016.09.14.14.03
## 4736	160	31	2016.09.14	200	-2.79252680	2016.09.14.14.03
## 4737	160	32	2016.09.14	200	-2.79252680	2016.09.14.14.03
## 4738	190	33	2016.09.14	170	2.96705973	2016.09.14.14.03
## 4739	150	34	2016.09.14	210	-2.61799388	2016.09.14.14.03
## 4740	180	35	2016.09.14	180	3.14159265	2016.09.14.14.03
## 4741	170	36	2016.09.14	190	-2.96705973	2016.09.14.14.03
## 4742	180	37	2016.09.14	180	3.14159265	2016.09.14.14.03
## 4743	250	1	2016.09.14	110	1.91986218	2016.09.14.14.40
## 4744	265	2	2016.09.14	95	1.65806279	2016.09.14.14.40
## 4745	220	3	2016.09.14	140	2.44346095	2016.09.14.14.40
## 4746	220	4	2016.09.14	140	2.44346095	2016.09.14.14.40
## 4747	270	5	2016.09.14	90	1.57079633	2016.09.14.14.40
## 4748	275	6	2016.09.14	85	1.48352986	2016.09.14.14.40
## 4749	265	7	2016.09.14	95	1.65806279	2016.09.14.14.40
## 4750	210	8	2016.09.14	150	2.61799388	2016.09.14.14.40
## 4751	265	9	2016.09.14	95	1.65806279	2016.09.14.14.40
## 4752	275	10	2016.09.14	85	1.48352986	2016.09.14.14.40
## 4753	170	11	2016.09.14	190	-2.96705973	2016.09.14.14.40
## 4754	165	12	2016.09.14	195	-2.87979327	2016.09.14.14.40
## 4755	160	13	2016.09.14	200	-2.79252680	2016.09.14.14.40
## 4756	170	14	2016.09.14	190	-2.96705973	2016.09.14.14.40
## 4757	160	15	2016.09.14	200	-2.79252680	2016.09.14.14.40
## 4758	150	16	2016.09.14	210	-2.61799388	2016.09.14.14.40
## 4759	170	17	2016.09.14	190	-2.96705973	2016.09.14.14.40
## 4760	165	18	2016.09.14	195	-2.87979327	2016.09.14.14.40
## 4761	190	19	2016.09.14	170	2.96705973	2016.09.14.14.40
## 4762	175	20	2016.09.14	185	-3.05432619	2016.09.14.14.40
## 4763	165	21	2016.09.14	195	-2.87979327	2016.09.14.14.40
## 4764	160	22	2016.09.14	200	-2.79252680	2016.09.14.14.40
## 4765	230	23	2016.09.14	130	2.26892803	2016.09.14.14.40
## 4766	70	24	2016.09.14	290	-1.22173048	2016.09.14.14.40

## 4767	90	25	2016.09.14	270	-1.57079633	2016.09.14.14.40
## 4768	200	26	2016.09.14	160	2.79252680	2016.09.14.14.40
## 4769	340	27	2016.09.14	20	0.34906585	2016.09.14.14.40
## 4770	220	28	2016.09.14	140	2.44346095	2016.09.14.14.40
## 4771	145	29	2016.09.14	215	-2.53072742	2016.09.14.14.40
## 4772	180	30	2016.09.14	180	3.14159265	2016.09.14.14.40
## 4773	320	31	2016.09.14	40	0.69813170	2016.09.14.14.40
## 4774	190	32	2016.09.14	170	2.96705973	2016.09.14.14.40
## 4775	145	33	2016.09.14	215	-2.53072742	2016.09.14.14.40
## 4776	270	34	2016.09.14	90	1.57079633	2016.09.14.14.40
## 4777	210	35	2016.09.14	150	2.61799388	2016.09.14.14.40
## 4778	150	36	2016.09.14	210	-2.61799388	2016.09.14.14.40
## 4779	235	37	2016.09.14	125	2.18166156	2016.09.14.14.40
## 4780	250	38	2016.09.14	110	1.91986218	2016.09.14.14.40
## 4781	195	39	2016.09.14	165	2.87979327	2016.09.14.14.40
## 4782	330	40	2016.09.14	30	0.52359878	2016.09.14.14.40
## 4783	120	41	2016.09.14	240	-2.09439510	2016.09.14.14.40
## 4784	175	42	2016.09.14	185	-3.05432619	2016.09.14.14.40
## 4785	60	43	2016.09.14	300	-1.04719755	2016.09.14.14.40
## 4786	290	44	2016.09.14	70	1.22173048	2016.09.14.14.40
## 4787	30	45	2016.09.14	330	-0.52359878	2016.09.14.14.40
## 4788	60	1	2016.09.14	300	-1.04719755	2016.09.14.14.59
## 4789	200	2	2016.09.14	160	2.79252680	2016.09.14.14.59
## 4790	190	3	2016.09.14	170	2.96705973	2016.09.14.14.59
## 4791	140	4	2016.09.14	220	-2.44346095	2016.09.14.14.59
## 4792	200	5	2016.09.14	160	2.79252680	2016.09.14.14.59
## 4793	180	6	2016.09.14	180	3.14159265	2016.09.14.14.59
## 4794	120	7	2016.09.14	240	-2.09439510	2016.09.14.14.59
## 4795	185	8	2016.09.14	175	3.05432619	2016.09.14.14.59
## 4796	210	9	2016.09.14	150	2.61799388	2016.09.14.14.59
## 4797	260	10	2016.09.14	100	1.74532925	2016.09.14.14.59
## 4798	140	11	2016.09.14	220	-2.44346095	2016.09.14.14.59
## 4799	200	12	2016.09.14	160	2.79252680	2016.09.14.14.59
## 4800	215	13	2016.09.14	145	2.53072742	2016.09.14.14.59
## 4801	150	14	2016.09.14	210	-2.61799388	2016.09.14.14.59
## 4802	5	15	2016.09.14	355	-0.08726646	2016.09.14.14.59
## 4803	40	16	2016.09.14	320	-0.69813170	2016.09.14.14.59
## 4804	40	17	2016.09.14	320	-0.69813170	2016.09.14.14.59
## 4805	0	18	2016.09.14	0	0.00000000	2016.09.14.14.59
## 4806	30	19	2016.09.14	330	-0.52359878	2016.09.14.14.59
## 4807	30	20	2016.09.14	330	-0.52359878	2016.09.14.14.59
## 4808	10	21	2016.09.14	350	-0.17453293	2016.09.14.14.59
## 4809	0	22	2016.09.14	0	0.00000000	2016.09.14.14.59
## 4810	340	23	2016.09.14	20	0.34906585	2016.09.14.14.59
## 4811	10	24	2016.09.14	350	-0.17453293	2016.09.14.14.59
## 4812	5	25	2016.09.14	355	-0.08726646	2016.09.14.14.59
## 4813	10	26	2016.09.14	350	-0.17453293	2016.09.14.14.59
## 4814	350	27	2016.09.14	10	0.17453293	2016.09.14.14.59
## 4815	350	28	2016.09.14	10	0.17453293	2016.09.14.14.59
## 4816	340	29	2016.09.14	20	0.34906585	2016.09.14.14.59
## 4817	10	30	2016.09.14	350	-0.17453293	2016.09.14.14.59
## 4818	355	31	2016.09.14	5	0.08726646	2016.09.14.14.59
## 4819	0	32	2016.09.14	0	0.00000000	2016.09.14.14.59
## 4820	355	33	2016.09.14	5	0.08726646	2016.09.14.14.59

## 4821	270	34	2016.09.14	90	1.57079633	2016.09.14.14.59
## 4822	330	35	2016.09.14	30	0.52359878	2016.09.14.14.59
## 4823	270	36	2016.09.14	90	1.57079633	2016.09.14.14.59
## 4824	50	37	2016.09.14	310	-0.87266463	2016.09.14.14.59
## 4825	320	38	2016.09.14	40	0.69813170	2016.09.14.14.59
## 4826	290	39	2016.09.14	70	1.22173048	2016.09.14.14.59
## 4827	110	40	2016.09.14	250	-1.91986218	2016.09.14.14.59
## 4828	5	41	2016.09.14	355	-0.08726646	2016.09.14.14.59
## 4829	80	1	2016.09.14	280	-1.39626340	2016.09.14.15.47
## 4830	130	2	2016.09.14	230	-2.26892803	2016.09.14.15.47
## 4831	110	3	2016.09.14	250	-1.91986218	2016.09.14.15.47
## 4832	130	4	2016.09.14	230	-2.26892803	2016.09.14.15.47
## 4833	115	5	2016.09.14	245	-2.00712864	2016.09.14.15.47
## 4834	145	6	2016.09.14	215	-2.53072742	2016.09.14.15.47
## 4835	110	7	2016.09.14	250	-1.91986218	2016.09.14.15.47
## 4836	120	8	2016.09.14	240	-2.09439510	2016.09.14.15.47
## 4837	120	9	2016.09.14	240	-2.09439510	2016.09.14.15.47
## 4838	115	10	2016.09.14	245	-2.00712864	2016.09.14.15.47
## 4839	110	11	2016.09.14	250	-1.91986218	2016.09.14.15.47
## 4840	110	12	2016.09.14	250	-1.91986218	2016.09.14.15.47
## 4841	90	13	2016.09.14	270	-1.57079633	2016.09.14.15.47
## 4842	105	14	2016.09.14	255	-1.83259571	2016.09.14.15.47
## 4843	220	15	2016.09.14	140	2.44346095	2016.09.14.15.47
## 4844	340	16	2016.09.14	20	0.34906585	2016.09.14.15.47
## 4845	240	17	2016.09.14	120	2.09439510	2016.09.14.15.47
## 4846	240	18	2016.09.14	120	2.09439510	2016.09.14.15.47
## 4847	140	19	2016.09.14	220	-2.44346095	2016.09.14.15.47
## 4848	125	20	2016.09.14	235	-2.18166156	2016.09.14.15.47
## 4849	30	21	2016.09.14	330	-0.52359878	2016.09.14.15.47
## 4850	0	22	2016.09.14	0	0.00000000	2016.09.14.15.47
## 4851	15	23	2016.09.14	345	-0.26179939	2016.09.14.15.47
## 4852	40	24	2016.09.14	320	-0.69813170	2016.09.14.15.47
## 4853	20	25	2016.09.14	340	-0.34906585	2016.09.14.15.47
## 4854	355	26	2016.09.14	5	0.08726646	2016.09.14.15.47
## 4855	20	27	2016.09.14	340	-0.34906585	2016.09.14.15.47
## 4856	20	28	2016.09.14	340	-0.34906585	2016.09.14.15.47
## 4857	40	29	2016.09.14	320	-0.69813170	2016.09.14.15.47
## 4858	0	30	2016.09.14	0	0.00000000	2016.09.14.15.47
## 4859	350	31	2016.09.14	10	0.17453293	2016.09.14.15.47
## 4860	170	1	2016.09.14	190	-2.96705973	2016.09.14.16.07
## 4861	130	2	2016.09.14	230	-2.26892803	2016.09.14.16.07
## 4862	120	3	2016.09.14	240	-2.09439510	2016.09.14.16.07
## 4863	100	4	2016.09.14	260	-1.74532925	2016.09.14.16.07
## 4864	130	5	2016.09.14	230	-2.26892803	2016.09.14.16.07
## 4865	110	6	2016.09.14	250	-1.91986218	2016.09.14.16.07
## 4866	140	7	2016.09.14	220	-2.44346095	2016.09.14.16.07
## 4867	110	8	2016.09.14	250	-1.91986218	2016.09.14.16.07
## 4868	110	9	2016.09.14	250	-1.91986218	2016.09.14.16.07
## 4869	115	10	2016.09.14	245	-2.00712864	2016.09.14.16.07
## 4870	115	11	2016.09.14	245	-2.00712864	2016.09.14.16.07
## 4871	100	12	2016.09.14	260	-1.74532925	2016.09.14.16.07
## 4872	100	13	2016.09.14	260	-1.74532925	2016.09.14.16.07
## 4873	90	14	2016.09.14	270	-1.57079633	2016.09.14.16.07
## 4874	190	15	2016.09.14	170	2.96705973	2016.09.14.16.07

## 4875	90	16	2016.09.14	270	-1.57079633	2016.09.14.16.07
## 4876	130	17	2016.09.14	230	-2.26892803	2016.09.14.16.07
## 4877	85	18	2016.09.14	275	-1.48352986	2016.09.14.16.07
## 4878	150	19	2016.09.14	210	-2.61799388	2016.09.14.16.07
## 4879	60	20	2016.09.14	300	-1.04719755	2016.09.14.16.07
## 4880	80	21	2016.09.14	280	-1.39626340	2016.09.14.16.07
## 4881	10	22	2016.09.14	350	-0.17453293	2016.09.14.16.07
## 4882	65	23	2016.09.14	295	-1.13446401	2016.09.14.16.07
## 4883	40	24	2016.09.14	320	-0.69813170	2016.09.14.16.07
## 4884	290	1	2016.09.14	70	1.22173048	2016.09.14.16.19
## 4885	260	2	2016.09.14	100	1.74532925	2016.09.14.16.19
## 4886	300	3	2016.09.14	60	1.04719755	2016.09.14.16.19
## 4887	275	4	2016.09.14	85	1.48352986	2016.09.14.16.19
## 4888	320	5	2016.09.14	40	0.69813170	2016.09.14.16.19
## 4889	310	6	2016.09.14	50	0.87266463	2016.09.14.16.19
## 4890	310	7	2016.09.14	50	0.87266463	2016.09.14.16.19
## 4891	260	8	2016.09.14	100	1.74532925	2016.09.14.16.19
## 4892	320	9	2016.09.14	40	0.69813170	2016.09.14.16.19
## 4893	90	10	2016.09.14	270	-1.57079633	2016.09.14.16.19
## 4894	60	11	2016.09.14	300	-1.04719755	2016.09.14.16.19
## 4895	40	12	2016.09.14	320	-0.69813170	2016.09.14.16.19
## 4896	85	13	2016.09.14	275	-1.48352986	2016.09.14.16.19
## 4897	35	14	2016.09.14	325	-0.61086524	2016.09.14.16.19
## 4898	75	15	2016.09.14	285	-1.30899694	2016.09.14.16.19
## 4899	70	16	2016.09.14	290	-1.22173048	2016.09.14.16.19
## 4900	135	17	2016.09.14	225	-2.35619449	2016.09.14.16.19
## 4901	25	18	2016.09.14	335	-0.43633231	2016.09.14.16.19
## 4902	90	19	2016.09.14	270	-1.57079633	2016.09.14.16.19
## 4903	40	20	2016.09.14	320	-0.69813170	2016.09.14.16.19
## 4904	70	21	2016.09.14	290	-1.22173048	2016.09.14.16.19
## 4905	35	22	2016.09.14	325	-0.61086524	2016.09.14.16.19
## 4906	0	23	2016.09.14	0	0.00000000	2016.09.14.16.19
## 4907	0	24	2016.09.14	0	0.00000000	2016.09.14.16.19
## 4908	0	25	2016.09.14	0	0.00000000	2016.09.14.16.19
## 4909	5	26	2016.09.14	355	-0.08726646	2016.09.14.16.19
## 4910	10	27	2016.09.14	350	-0.17453293	2016.09.14.16.19
## 4911	50	28	2016.09.14	310	-0.87266463	2016.09.14.16.19
## 4912	10	29	2016.09.14	350	-0.17453293	2016.09.14.16.19
## 4913	10	30	2016.09.14	350	-0.17453293	2016.09.14.16.19
## 4914	10	31	2016.09.14	350	-0.17453293	2016.09.14.16.19
## 4915	20	32	2016.09.14	340	-0.34906585	2016.09.14.16.19
## 4916	0	33	2016.09.14	0	0.00000000	2016.09.14.16.19
## 4917	290	1	2016.09.14	70	1.22173048	2016.09.14.16.54
## 4918	300	2	2016.09.14	60	1.04719755	2016.09.14.16.54
## 4919	310	3	2016.09.14	50	0.87266463	2016.09.14.16.54
## 4920	175	4	2016.09.14	185	-3.05432619	2016.09.14.16.54
## 4921	310	5	2016.09.14	50	0.87266463	2016.09.14.16.54
## 4922	270	6	2016.09.14	90	1.57079633	2016.09.14.16.54
## 4923	330	7	2016.09.14	30	0.52359878	2016.09.14.16.54
## 4924	260	8	2016.09.14	100	1.74532925	2016.09.14.16.54
## 4925	320	9	2016.09.14	40	0.69813170	2016.09.14.16.54
## 4926	275	10	2016.09.14	85	1.48352986	2016.09.14.16.54
## 4927	320	11	2016.09.14	40	0.69813170	2016.09.14.16.54
## 4928	220	12	2016.09.14	140	2.44346095	2016.09.14.16.54

## 4929	325	13	2016.09.14	35	0.61086524	2016.09.14.16.54
## 4930	250	14	2016.09.14	110	1.91986218	2016.09.14.16.54
## 4931	310	15	2016.09.14	50	0.87266463	2016.09.14.16.54
## 4932	280	16	2016.09.14	80	1.39626340	2016.09.14.16.54
## 4933	320	17	2016.09.14	40	0.69813170	2016.09.14.16.54
## 4934	290	18	2016.09.14	70	1.22173048	2016.09.14.16.54
## 4935	350	19	2016.09.14	10	0.17453293	2016.09.14.16.54
## 4936	350	20	2016.09.14	10	0.17453293	2016.09.14.16.54
## 4937	200	21	2016.09.14	160	2.79252680	2016.09.14.16.54
## 4938	240	22	2016.09.14	120	2.09439510	2016.09.14.16.54
## 4939	200	23	2016.09.14	160	2.79252680	2016.09.14.16.54
## 4940	210	24	2016.09.14	150	2.61799388	2016.09.14.16.54
## 4941	220	25	2016.09.14	140	2.44346095	2016.09.14.16.54
## 4942	240	26	2016.09.14	120	2.09439510	2016.09.14.16.54
## 4943	230	27	2016.09.14	130	2.26892803	2016.09.14.16.54
## 4944	250	28	2016.09.14	110	1.91986218	2016.09.14.16.54
## 4945	190	29	2016.09.14	170	2.96705973	2016.09.14.16.54
## 4946	260	30	2016.09.14	100	1.74532925	2016.09.14.16.54
## 4947	225	31	2016.09.14	135	2.35619449	2016.09.14.16.54
## 4948	230	32	2016.09.14	130	2.26892803	2016.09.14.16.54
## 4949	215	33	2016.09.14	145	2.53072742	2016.09.14.16.54
## 4950	230	34	2016.09.14	130	2.26892803	2016.09.14.16.54
## 4951	210	35	2016.09.14	150	2.61799388	2016.09.14.16.54
## 4952	275	36	2016.09.14	85	1.48352986	2016.09.14.16.54
## 4953	155	37	2016.09.14	205	-2.70526034	2016.09.14.16.54
## 4954	220	38	2016.09.14	140	2.44346095	2016.09.14.16.54
## 4955	230	39	2016.09.14	130	2.26892803	2016.09.14.16.54
## 4956	220	40	2016.09.14	140	2.44346095	2016.09.14.16.54
## 4957	260	41	2016.09.14	100	1.74532925	2016.09.14.16.54
## 4958	240	42	2016.09.14	120	2.09439510	2016.09.14.16.54
## 4959	305	43	2016.09.14	55	0.95993109	2016.09.14.16.54
## 4960	340	44	2016.09.14	20	0.34906585	2016.09.14.16.54
## 4961	330	45	2016.09.14	30	0.52359878	2016.09.14.16.54
## 4962	210	46	2016.09.14	150	2.61799388	2016.09.14.16.54
## 4963	350	47	2016.09.14	10	0.17453293	2016.09.14.16.54
## 4964	280	48	2016.09.14	80	1.39626340	2016.09.14.16.54
## 4965	250	49	2016.09.14	110	1.91986218	2016.09.14.16.54
## 4966	200	50	2016.09.14	160	2.79252680	2016.09.14.16.54
## 4967	290	51	2016.09.14	70	1.22173048	2016.09.14.16.54
## 4968	205	52	2016.09.14	155	2.70526034	2016.09.14.16.54
## 4969	165	53	2016.09.14	195	-2.87979327	2016.09.14.16.54
## 4970	185	54	2016.09.14	175	3.05432619	2016.09.14.16.54
## 4971	165	1	2016.09.14	195	-2.87979327	2016.09.14.17.25
## 4972	275	2	2016.09.14	85	1.48352986	2016.09.14.17.25
## 4973	255	3	2016.09.14	105	1.83259571	2016.09.14.17.25
## 4974	230	4	2016.09.14	130	2.26892803	2016.09.14.17.25
## 4975	300	5	2016.09.14	60	1.04719755	2016.09.14.17.25
## 4976	235	6	2016.09.14	125	2.18166156	2016.09.14.17.25
## 4977	270	7	2016.09.14	90	1.57079633	2016.09.14.17.25
## 4978	230	8	2016.09.14	130	2.26892803	2016.09.14.17.25
## 4979	160	9	2016.09.14	200	-2.79252680	2016.09.14.17.25
## 4980	145	10	2016.09.14	215	-2.53072742	2016.09.14.17.25
## 4981	190	11	2016.09.14	170	2.96705973	2016.09.14.17.25
## 4982	100	12	2016.09.14	260	-1.74532925	2016.09.14.17.25

## 4983	280	13	2016.09.14	80	1.39626340	2016.09.14.17.25
## 4984	160	14	2016.09.14	200	-2.79252680	2016.09.14.17.25
## 4985	210	15	2016.09.14	150	2.61799388	2016.09.14.17.25
## 4986	270	16	2016.09.14	90	1.57079633	2016.09.14.17.25
## 4987	210	17	2016.09.14	150	2.61799388	2016.09.14.17.25
## 4988	180	18	2016.09.14	180	3.14159265	2016.09.14.17.25
## 4989	220	19	2016.09.14	140	2.44346095	2016.09.14.17.25
## 4990	230	20	2016.09.14	130	2.26892803	2016.09.14.17.25
## 4991	225	21	2016.09.14	135	2.35619449	2016.09.14.17.25
## 4992	240	22	2016.09.14	120	2.09439510	2016.09.14.17.25
## 4993	210	1	2016.09.27	150	2.61799388	2016.09.27.15.52
## 4994	250	2	2016.09.27	110	1.91986218	2016.09.27.15.52
## 4995	270	3	2016.09.27	90	1.57079633	2016.09.27.15.52
## 4996	330	4	2016.09.27	30	0.52359878	2016.09.27.15.52
## 4997	250	5	2016.09.27	110	1.91986218	2016.09.27.15.52
## 4998	340	6	2016.09.27	20	0.34906585	2016.09.27.15.52
## 4999	180	7	2016.09.27	180	3.14159265	2016.09.27.15.52
## 5000	290	8	2016.09.27	70	1.22173048	2016.09.27.15.52
## 5001	285	9	2016.09.27	75	1.30899694	2016.09.27.15.52
## 5002	160	10	2016.09.27	200	-2.79252680	2016.09.27.15.52
## 5003	240	11	2016.09.27	120	2.09439510	2016.09.27.15.52
## 5004	180	12	2016.09.27	180	3.14159265	2016.09.27.15.52
## 5005	190	13	2016.09.27	170	2.96705973	2016.09.27.15.52
## 5006	170	14	2016.09.27	190	-2.96705973	2016.09.27.15.52
## 5007	170	15	2016.09.27	190	-2.96705973	2016.09.27.15.52
## 5008	165	16	2016.09.27	195	-2.87979327	2016.09.27.15.52
## 5009	180	17	2016.09.27	180	3.14159265	2016.09.27.15.52
## 5010	170	18	2016.09.27	190	-2.96705973	2016.09.27.15.52
## 5011	120	19	2016.09.27	240	-2.09439510	2016.09.27.15.52
## 5012	165	20	2016.09.27	195	-2.87979327	2016.09.27.15.52
## 5013	170	21	2016.09.27	190	-2.96705973	2016.09.27.15.52
## 5014	35	1	2016.09.29	325	-0.61086524	2016.09.29.14.21
## 5015	190	2	2016.09.29	170	2.96705973	2016.09.29.14.21
## 5016	180	3	2016.09.29	180	3.14159265	2016.09.29.14.21
## 5017	45	4	2016.09.29	315	-0.78539816	2016.09.29.14.21
## 5018	340	5	2016.09.29	20	0.34906585	2016.09.29.14.21
## 5019	30	6	2016.09.29	330	-0.52359878	2016.09.29.14.21
## 5020	35	7	2016.09.29	325	-0.61086524	2016.09.29.14.21
## 5021	70	8	2016.09.29	290	-1.22173048	2016.09.29.14.21
## 5022	325	9	2016.09.29	35	0.61086524	2016.09.29.14.21
## 5023	110	10	2016.09.29	250	-1.91986218	2016.09.29.14.21
## 5024	120	11	2016.09.29	240	-2.09439510	2016.09.29.14.21
## 5025	140	12	2016.09.29	220	-2.44346095	2016.09.29.14.21
## 5026	40	13	2016.09.29	320	-0.69813170	2016.09.29.14.21
## 5027	20	14	2016.09.29	340	-0.34906585	2016.09.29.14.21
## 5028	355	15	2016.09.29	5	0.08726646	2016.09.29.14.21
## 5029	40	16	2016.09.29	320	-0.69813170	2016.09.29.14.21
## 5030	350	17	2016.09.29	10	0.17453293	2016.09.29.14.21
## 5031	340	18	2016.09.29	20	0.34906585	2016.09.29.14.21
## 5032	0	19	2016.09.29	0	0.00000000	2016.09.29.14.21
## 5033	100	1	2016.09.29	260	-1.74532925	2016.09.29.14.52
## 5034	190	2	2016.09.29	170	2.96705973	2016.09.29.14.52
## 5035	190	3	2016.09.29	170	2.96705973	2016.09.29.14.52
## 5036	115	4	2016.09.29	245	-2.00712864	2016.09.29.14.52

## 5037	70	5	2016.09.29	290	-1.22173048	2016.09.29.14.52
## 5038	230	6	2016.09.29	130	2.26892803	2016.09.29.14.52
## 5039	130	7	2016.09.29	230	-2.26892803	2016.09.29.14.52
## 5040	150	8	2016.09.29	210	-2.61799388	2016.09.29.14.52
## 5041	90	9	2016.09.29	270	-1.57079633	2016.09.29.14.52
## 5042	160	10	2016.09.29	200	-2.79252680	2016.09.29.14.52
## 5043	210	11	2016.09.29	150	2.61799388	2016.09.29.14.52
## 5044	100	12	2016.09.29	260	-1.74532925	2016.09.29.14.52
## 5045	70	13	2016.09.29	290	-1.22173048	2016.09.29.14.52
## 5046	70	14	2016.09.29	290	-1.22173048	2016.09.29.14.52
## 5047	110	15	2016.09.29	250	-1.91986218	2016.09.29.14.52
## 5048	120	16	2016.09.29	240	-2.09439510	2016.09.29.14.52
## 5049	105	17	2016.09.29	255	-1.83259571	2016.09.29.14.52
## 5050	60	18	2016.09.29	300	-1.04719755	2016.09.29.14.52
## 5051	30	19	2016.09.29	330	-0.52359878	2016.09.29.14.52
## 5052	90	20	2016.09.29	270	-1.57079633	2016.09.29.14.52
## 5053	350	21	2016.09.29	10	0.17453293	2016.09.29.14.52
## 5054	10	22	2016.09.29	350	-0.17453293	2016.09.29.14.52
## 5055	345	23	2016.09.29	15	0.26179939	2016.09.29.14.52
## 5056	30	24	2016.09.29	330	-0.52359878	2016.09.29.14.52
## 5057	20	25	2016.09.29	340	-0.34906585	2016.09.29.14.52
## 5058	355	26	2016.09.29	5	0.08726646	2016.09.29.14.52
## 5059	5	27	2016.09.29	355	-0.08726646	2016.09.29.14.52
## 5060	15	28	2016.09.29	345	-0.26179939	2016.09.29.14.52
## 5061	5	29	2016.09.29	355	-0.08726646	2016.09.29.14.52
## 5062	350	30	2016.09.29	10	0.17453293	2016.09.29.14.52
## 5063	350	31	2016.09.29	10	0.17453293	2016.09.29.14.52
## 5064	330	32	2016.09.29	30	0.52359878	2016.09.29.14.52
## 5065	60	33	2016.09.29	300	-1.04719755	2016.09.29.14.52
## 5066	90	1	2016.09.29	270	-1.57079633	2016.09.29.15.15
## 5067	70	2	2016.09.29	290	-1.22173048	2016.09.29.15.15
## 5068	115	3	2016.09.29	245	-2.00712864	2016.09.29.15.15
## 5069	65	4	2016.09.29	295	-1.13446401	2016.09.29.15.15
## 5070	0	5	2016.09.29	0	0.00000000	2016.09.29.15.15
## 5071	100	6	2016.09.29	260	-1.74532925	2016.09.29.15.15
## 5072	5	7	2016.09.29	355	-0.08726646	2016.09.29.15.15
## 5073	105	8	2016.09.29	255	-1.83259571	2016.09.29.15.15
## 5074	75	9	2016.09.29	285	-1.30899694	2016.09.29.15.15
## 5075	60	10	2016.09.29	300	-1.04719755	2016.09.29.15.15
## 5076	115	11	2016.09.29	245	-2.00712864	2016.09.29.15.15
## 5077	0	12	2016.09.29	0	0.00000000	2016.09.29.15.15
## 5078	100	13	2016.09.29	260	-1.74532925	2016.09.29.15.15
## 5079	10	14	2016.09.29	350	-0.17453293	2016.09.29.15.15
## 5080	350	15	2016.09.29	10	0.17453293	2016.09.29.15.15
## 5081	355	16	2016.09.29	5	0.08726646	2016.09.29.15.15
## 5082	0	17	2016.09.29	0	0.00000000	2016.09.29.15.15
## 5083	0	18	2016.09.29	0	0.00000000	2016.09.29.15.15
## 5084	40	19	2016.09.29	320	-0.69813170	2016.09.29.15.15
## 5085	0	20	2016.09.29	0	0.00000000	2016.09.29.15.15
## 5086	30	21	2016.09.29	330	-0.52359878	2016.09.29.15.15
## 5087	20	22	2016.09.29	340	-0.34906585	2016.09.29.15.15
## 5088	355	23	2016.09.29	5	0.08726646	2016.09.29.15.15
## 5089	30	24	2016.09.29	330	-0.52359878	2016.09.29.15.15
## 5090	335	25	2016.09.29	25	0.43633231	2016.09.29.15.15

## 5091	20	26	2016.09.29	340	-0.34906585	2016.09.29.15.15
## 5092	330	27	2016.09.29	30	0.52359878	2016.09.29.15.15
## 5093	355	28	2016.09.29	5	0.08726646	2016.09.29.15.15
## 5094	35	29	2016.09.29	325	-0.61086524	2016.09.29.15.15
## 5095	70	1	2016.09.29	290	-1.22173048	2016.09.29.15.43
## 5096	90	2	2016.09.29	270	-1.57079633	2016.09.29.15.43
## 5097	90	3	2016.09.29	270	-1.57079633	2016.09.29.15.43
## 5098	110	4	2016.09.29	250	-1.91986218	2016.09.29.15.43
## 5099	60	5	2016.09.29	300	-1.04719755	2016.09.29.15.43
## 5100	135	6	2016.09.29	225	-2.35619449	2016.09.29.15.43
## 5101	160	7	2016.09.29	200	-2.79252680	2016.09.29.15.43
## 5102	135	8	2016.09.29	225	-2.35619449	2016.09.29.15.43
## 5103	110	9	2016.09.29	250	-1.91986218	2016.09.29.15.43
## 5104	75	10	2016.09.29	285	-1.30899694	2016.09.29.15.43
## 5105	100	11	2016.09.29	260	-1.74532925	2016.09.29.15.43
## 5106	105	12	2016.09.29	255	-1.83259571	2016.09.29.15.43
## 5107	90	13	2016.09.29	270	-1.57079633	2016.09.29.15.43
## 5108	85	14	2016.09.29	275	-1.48352986	2016.09.29.15.43
## 5109	120	15	2016.09.29	240	-2.09439510	2016.09.29.15.43
## 5110	45	16	2016.09.29	315	-0.78539816	2016.09.29.15.43
## 5111	110	17	2016.09.29	250	-1.91986218	2016.09.29.15.43
## 5112	320	18	2016.09.29	40	0.69813170	2016.09.29.15.43
## 5113	355	19	2016.09.29	5	0.08726646	2016.09.29.15.43
## 5114	30	20	2016.09.29	330	-0.52359878	2016.09.29.15.43
## 5115	340	21	2016.09.29	20	0.34906585	2016.09.29.15.43
## 5116	20	22	2016.09.29	340	-0.34906585	2016.09.29.15.43
## 5117	350	23	2016.09.29	10	0.17453293	2016.09.29.15.43
## 5118	30	24	2016.09.29	330	-0.52359878	2016.09.29.15.43
## 5119	350	25	2016.09.29	10	0.17453293	2016.09.29.15.43
## 5120	5	26	2016.09.29	355	-0.08726646	2016.09.29.15.43
## 5121	40	27	2016.09.29	320	-0.69813170	2016.09.29.15.43
## 5122	350	28	2016.09.29	10	0.17453293	2016.09.29.15.43
## 5123	10	29	2016.09.29	350	-0.17453293	2016.09.29.15.43
## 5124	355	30	2016.09.29	5	0.08726646	2016.09.29.15.43
## 5125	5	31	2016.09.29	355	-0.08726646	2016.09.29.15.43
## 5126	15	32	2016.09.29	345	-0.26179939	2016.09.29.15.43
## 5127	335	33	2016.09.29	25	0.43633231	2016.09.29.15.43
## 5128	30	34	2016.09.29	330	-0.52359878	2016.09.29.15.43
## 5129	25	35	2016.09.29	335	-0.43633231	2016.09.29.15.43
## 5130	30	36	2016.09.29	330	-0.52359878	2016.09.29.15.43
## 5131	50	37	2016.09.29	310	-0.87266463	2016.09.29.15.43
## 5132	355	38	2016.09.29	5	0.08726646	2016.09.29.15.43
## 5133	50	39	2016.09.29	310	-0.87266463	2016.09.29.15.43
## 5134	110	1	2016.06.02	250	-1.91986218	2016.06.02.12.51
## 5135	90	2	2016.06.02	270	-1.57079633	2016.06.02.12.51
## 5136	90	3	2016.06.02	270	-1.57079633	2016.06.02.12.51
## 5137	85	4	2016.06.02	275	-1.48352986	2016.06.02.12.51
## 5138	40	5	2016.06.02	320	-0.69813170	2016.06.02.12.51
## 5139	70	6	2016.06.02	290	-1.22173048	2016.06.02.12.51
## 5140	275	7	2016.06.02	85	1.48352986	2016.06.02.12.51
## 5141	275	8	2016.06.02	85	1.48352986	2016.06.02.12.51
## 5142	60	9	2016.06.02	300	-1.04719755	2016.06.02.12.51
## 5143	295	10	2016.06.02	65	1.13446401	2016.06.02.12.51
## 5144	320	11	2016.06.02	40	0.69813170	2016.06.02.12.51

## 5145	345	12	2016.06.02	15	0.26179939	2016.06.02.12.51
## 5146	40	13	2016.06.02	320	-0.69813170	2016.06.02.12.51
## 5147	0	14	2016.06.02	0	0.00000000	2016.06.02.12.51
## 5148	10	15	2016.06.02	350	-0.17453293	2016.06.02.12.51
## 5149	5	16	2016.06.02	355	-0.08726646	2016.06.02.12.51
## 5150	0	17	2016.06.02	0	0.00000000	2016.06.02.12.51
## 5151	10	18	2016.06.02	350	-0.17453293	2016.06.02.12.51
## 5152	0	19	2016.06.02	0	0.00000000	2016.06.02.12.51
## 5153	0	20	2016.06.02	0	0.00000000	2016.06.02.12.51
## 5154	10	21	2016.06.02	350	-0.17453293	2016.06.02.12.51
## 5155	5	22	2016.06.02	355	-0.08726646	2016.06.02.12.51
## 5156	0	23	2016.06.02	0	0.00000000	2016.06.02.12.51
## 5157	100	24	2016.06.02	260	-1.74532925	2016.06.02.12.51
## 5158	90	25	2016.06.02	270	-1.57079633	2016.06.02.12.51
## 5159	15	26	2016.06.02	345	-0.26179939	2016.06.02.12.51
## 5160	65	1	2016.06.02	295	-1.13446401	2016.06.02.13.07
## 5161	80	2	2016.06.02	280	-1.39626340	2016.06.02.13.07
## 5162	205	3	2016.06.02	155	2.70526034	2016.06.02.13.07
## 5163	20	4	2016.06.02	340	-0.34906585	2016.06.02.13.07
## 5164	70	5	2016.06.02	290	-1.22173048	2016.06.02.13.07
## 5165	300	6	2016.06.02	60	1.04719755	2016.06.02.13.07
## 5166	95	7	2016.06.02	265	-1.65806279	2016.06.02.13.07
## 5167	115	8	2016.06.02	245	-2.00712864	2016.06.02.13.07
## 5168	320	9	2016.06.02	40	0.69813170	2016.06.02.13.07
## 5169	100	10	2016.06.02	260	-1.74532925	2016.06.02.13.07
## 5170	95	11	2016.06.02	265	-1.65806279	2016.06.02.13.07
## 5171	355	12	2016.06.02	5	0.08726646	2016.06.02.13.07
## 5172	5	13	2016.06.02	355	-0.08726646	2016.06.02.13.07
## 5173	0	14	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5174	0	15	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5175	5	16	2016.06.02	355	-0.08726646	2016.06.02.13.07
## 5176	20	17	2016.06.02	340	-0.34906585	2016.06.02.13.07
## 5177	350	18	2016.06.02	10	0.17453293	2016.06.02.13.07
## 5178	5	19	2016.06.02	355	-0.08726646	2016.06.02.13.07
## 5179	0	20	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5180	20	21	2016.06.02	340	-0.34906585	2016.06.02.13.07
## 5181	15	22	2016.06.02	345	-0.26179939	2016.06.02.13.07
## 5182	5	23	2016.06.02	355	-0.08726646	2016.06.02.13.07
## 5183	15	24	2016.06.02	345	-0.26179939	2016.06.02.13.07
## 5184	0	25	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5185	355	26	2016.06.02	5	0.08726646	2016.06.02.13.07
## 5186	5	27	2016.06.02	355	-0.08726646	2016.06.02.13.07
## 5187	355	28	2016.06.02	5	0.08726646	2016.06.02.13.07
## 5188	0	29	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5189	350	30	2016.06.02	10	0.17453293	2016.06.02.13.07
## 5190	350	31	2016.06.02	10	0.17453293	2016.06.02.13.07
## 5191	10	32	2016.06.02	350	-0.17453293	2016.06.02.13.07
## 5192	355	33	2016.06.02	5	0.08726646	2016.06.02.13.07
## 5193	350	34	2016.06.02	10	0.17453293	2016.06.02.13.07
## 5194	25	35	2016.06.02	335	-0.43633231	2016.06.02.13.07
## 5195	30	36	2016.06.02	330	-0.52359878	2016.06.02.13.07
## 5196	20	37	2016.06.02	340	-0.34906585	2016.06.02.13.07
## 5197	0	38	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5198	345	39	2016.06.02	15	0.26179939	2016.06.02.13.07

## 5199	10	40	2016.06.02	350	-0.17453293	2016.06.02.13.07
## 5200	0	41	2016.06.02	0	0.00000000	2016.06.02.13.07
## 5201	20	42	2016.06.02	340	-0.34906585	2016.06.02.13.07
## 5202	0	1	2016.06.02	0	0.00000000	2016.06.02.13.15
## 5203	355	2	2016.06.02	5	0.08726646	2016.06.02.13.15
## 5204	0	3	2016.06.02	0	0.00000000	2016.06.02.13.15
## 5205	5	4	2016.06.02	355	-0.08726646	2016.06.02.13.15
## 5206	60	5	2016.06.02	300	-1.04719755	2016.06.02.13.15
## 5207	5	6	2016.06.02	355	-0.08726646	2016.06.02.13.15
## 5208	25	7	2016.06.02	335	-0.43633231	2016.06.02.13.15
## 5209	350	8	2016.06.02	10	0.17453293	2016.06.02.13.15
## 5210	15	9	2016.06.02	345	-0.26179939	2016.06.02.13.15
## 5211	15	10	2016.06.02	345	-0.26179939	2016.06.02.13.15
## 5212	320	11	2016.06.02	40	0.69813170	2016.06.02.13.15
## 5213	50	12	2016.06.02	310	-0.87266463	2016.06.02.13.15
## 5214	65	13	2016.06.02	295	-1.13446401	2016.06.02.13.15
## 5215	340	14	2016.06.02	20	0.34906585	2016.06.02.13.15
## 5216	10	15	2016.06.02	350	-0.17453293	2016.06.02.13.15
## 5217	355	16	2016.06.02	5	0.08726646	2016.06.02.13.15
## 5218	25	17	2016.06.02	335	-0.43633231	2016.06.02.13.15
## 5219	15	18	2016.06.02	345	-0.26179939	2016.06.02.13.15
## 5220	15	19	2016.06.02	345	-0.26179939	2016.06.02.13.15
## 5221	5	20	2016.06.02	355	-0.08726646	2016.06.02.13.15
## 5222	350	21	2016.06.02	10	0.17453293	2016.06.02.13.15
## 5223	30	22	2016.06.02	330	-0.52359878	2016.06.02.13.15
## 5224	350	23	2016.06.02	10	0.17453293	2016.06.02.13.15
## 5225	0	24	2016.06.02	0	0.00000000	2016.06.02.13.15
## 5226	10	25	2016.06.02	350	-0.17453293	2016.06.02.13.15
## 5227	0	1	2016.06.02	0	0.00000000	2016.06.02.13.27
## 5228	0	2	2016.06.02	0	0.00000000	2016.06.02.13.27
## 5229	355	3	2016.06.02	5	0.08726646	2016.06.02.13.27
## 5230	60	4	2016.06.02	300	-1.04719755	2016.06.02.13.27
## 5231	5	5	2016.06.02	355	-0.08726646	2016.06.02.13.27
## 5232	65	6	2016.06.02	295	-1.13446401	2016.06.02.13.27
## 5233	345	7	2016.06.02	15	0.26179939	2016.06.02.13.27
## 5234	70	8	2016.06.02	290	-1.22173048	2016.06.02.13.27
## 5235	20	9	2016.06.02	340	-0.34906585	2016.06.02.13.27
## 5236	30	10	2016.06.02	330	-0.52359878	2016.06.02.13.27
## 5237	320	11	2016.06.02	40	0.69813170	2016.06.02.13.27
## 5238	50	12	2016.06.02	310	-0.87266463	2016.06.02.13.27
## 5239	60	13	2016.06.02	300	-1.04719755	2016.06.02.13.27
## 5240	335	14	2016.06.02	25	0.43633231	2016.06.02.13.27
## 5241	355	15	2016.06.02	5	0.08726646	2016.06.02.13.27
## 5242	350	16	2016.06.02	10	0.17453293	2016.06.02.13.27
## 5243	15	17	2016.06.02	345	-0.26179939	2016.06.02.13.27
## 5244	355	18	2016.06.02	5	0.08726646	2016.06.02.13.27
## 5245	5	19	2016.06.02	355	-0.08726646	2016.06.02.13.27
## 5246	0	20	2016.06.02	0	0.00000000	2016.06.02.13.27
## 5247	355	21	2016.06.02	5	0.08726646	2016.06.02.13.27
## 5248	25	22	2016.06.02	335	-0.43633231	2016.06.02.13.27
## 5249	345	23	2016.06.02	15	0.26179939	2016.06.02.13.27
## 5250	5	24	2016.06.02	355	-0.08726646	2016.06.02.13.27
## 5251	20	25	2016.06.02	340	-0.34906585	2016.06.02.13.27
## 5252	345	26	2016.06.02	15	0.26179939	2016.06.02.13.27

## 5253	295	1	2016.06.14	65	1.13446401	2016.06.14.15.05
## 5254	350	2	2016.06.14	10	0.17453293	2016.06.14.15.05
## 5255	350	3	2016.06.14	10	0.17453293	2016.06.14.15.05
## 5256	5	4	2016.06.14	355	-0.08726646	2016.06.14.15.05
## 5257	180	5	2016.06.14	180	3.14159265	2016.06.14.15.05
## 5258	30	6	2016.06.14	330	-0.52359878	2016.06.14.15.05
## 5259	170	7	2016.06.14	190	-2.96705973	2016.06.14.15.05
## 5260	60	8	2016.06.14	300	-1.04719755	2016.06.14.15.05
## 5261	45	9	2016.06.14	315	-0.78539816	2016.06.14.15.05
## 5262	30	10	2016.06.14	330	-0.52359878	2016.06.14.15.05
## 5263	85	11	2016.06.14	275	-1.48352986	2016.06.14.15.05
## 5264	45	12	2016.06.14	315	-0.78539816	2016.06.14.15.05
## 5265	300	13	2016.06.14	60	1.04719755	2016.06.14.15.05
## 5266	300	14	2016.06.14	60	1.04719755	2016.06.14.15.05
## 5267	310	15	2016.06.14	50	0.87266463	2016.06.14.15.05
## 5268	290	16	2016.06.14	70	1.22173048	2016.06.14.15.05
## 5269	280	17	2016.06.14	80	1.39626340	2016.06.14.15.05
## 5270	275	18	2016.06.14	85	1.48352986	2016.06.14.15.05
## 5271	320	19	2016.06.14	40	0.69813170	2016.06.14.15.05
## 5272	325	20	2016.06.14	35	0.61086524	2016.06.14.15.05
## 5273	300	21	2016.06.14	60	1.04719755	2016.06.14.15.05
## 5274	325	22	2016.06.14	35	0.61086524	2016.06.14.15.05
## 5275	295	23	2016.06.14	65	1.13446401	2016.06.14.15.05
## 5276	290	24	2016.06.14	70	1.22173048	2016.06.14.15.05
## 5277	295	25	2016.06.14	65	1.13446401	2016.06.14.15.05
## 5278	280	26	2016.06.14	80	1.39626340	2016.06.14.15.05
## 5279	355	1	2016.06.20	5	0.08726646	2016.06.20.13.17
## 5280	140	2	2016.06.20	220	-2.44346095	2016.06.20.13.17
## 5281	345	3	2016.06.20	15	0.26179939	2016.06.20.13.17
## 5282	340	4	2016.06.20	20	0.34906585	2016.06.20.13.17
## 5283	0	5	2016.06.20	0	0.00000000	2016.06.20.13.17
## 5284	335	6	2016.06.20	25	0.43633231	2016.06.20.13.17
## 5285	340	7	2016.06.20	20	0.34906585	2016.06.20.13.17
## 5286	280	8	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5287	330	9	2016.06.20	30	0.52359878	2016.06.20.13.17
## 5288	190	10	2016.06.20	170	2.96705973	2016.06.20.13.17
## 5289	340	11	2016.06.20	20	0.34906585	2016.06.20.13.17
## 5290	320	12	2016.06.20	40	0.69813170	2016.06.20.13.17
## 5291	330	13	2016.06.20	30	0.52359878	2016.06.20.13.17
## 5292	280	14	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5293	225	15	2016.06.20	135	2.35619449	2016.06.20.13.17
## 5294	280	16	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5295	270	17	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5296	285	18	2016.06.20	75	1.30899694	2016.06.20.13.17
## 5297	280	19	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5298	290	20	2016.06.20	70	1.22173048	2016.06.20.13.17
## 5299	280	21	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5300	280	22	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5301	260	23	2016.06.20	100	1.74532925	2016.06.20.13.17
## 5302	270	24	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5303	250	25	2016.06.20	110	1.91986218	2016.06.20.13.17
## 5304	270	26	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5305	275	27	2016.06.20	85	1.48352986	2016.06.20.13.17
## 5306	275	28	2016.06.20	85	1.48352986	2016.06.20.13.17

## 5307	270	29	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5308	270	30	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5309	270	31	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5310	275	32	2016.06.20	85	1.48352986	2016.06.20.13.17
## 5311	270	33	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5312	260	34	2016.06.20	100	1.74532925	2016.06.20.13.17
## 5313	280	35	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5314	285	36	2016.06.20	75	1.30899694	2016.06.20.13.17
## 5315	275	37	2016.06.20	85	1.48352986	2016.06.20.13.17
## 5316	300	38	2016.06.20	60	1.04719755	2016.06.20.13.17
## 5317	345	39	2016.06.20	15	0.26179939	2016.06.20.13.17
## 5318	290	40	2016.06.20	70	1.22173048	2016.06.20.13.17
## 5319	325	41	2016.06.20	35	0.61086524	2016.06.20.13.17
## 5320	270	42	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5321	275	43	2016.06.20	85	1.48352986	2016.06.20.13.17
## 5322	225	44	2016.06.20	135	2.35619449	2016.06.20.13.17
## 5323	240	45	2016.06.20	120	2.09439510	2016.06.20.13.17
## 5324	260	46	2016.06.20	100	1.74532925	2016.06.20.13.17
## 5325	290	47	2016.06.20	70	1.22173048	2016.06.20.13.17
## 5326	280	48	2016.06.20	80	1.39626340	2016.06.20.13.17
## 5327	270	49	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5328	300	50	2016.06.20	60	1.04719755	2016.06.20.13.17
## 5329	250	51	2016.06.20	110	1.91986218	2016.06.20.13.17
## 5330	270	52	2016.06.20	90	1.57079633	2016.06.20.13.17
## 5331	290	53	2016.06.20	70	1.22173048	2016.06.20.13.17
## 5332	0	1	2016.06.20	0	0.00000000	2016.06.20.13.22
## 5333	320	2	2016.06.20	40	0.69813170	2016.06.20.13.22
## 5334	15	3	2016.06.20	345	-0.26179939	2016.06.20.13.22
## 5335	315	4	2016.06.20	45	0.78539816	2016.06.20.13.22
## 5336	15	5	2016.06.20	345	-0.26179939	2016.06.20.13.22
## 5337	345	6	2016.06.20	15	0.26179939	2016.06.20.13.22
## 5338	355	7	2016.06.20	5	0.08726646	2016.06.20.13.22
## 5339	200	8	2016.06.20	160	2.79252680	2016.06.20.13.22
## 5340	350	9	2016.06.20	10	0.17453293	2016.06.20.13.22
## 5341	0	10	2016.06.20	0	0.00000000	2016.06.20.13.22
## 5342	350	11	2016.06.20	10	0.17453293	2016.06.20.13.22
## 5343	0	12	2016.06.20	0	0.00000000	2016.06.20.13.22
## 5344	15	13	2016.06.20	345	-0.26179939	2016.06.20.13.22
## 5345	10	14	2016.06.20	350	-0.17453293	2016.06.20.13.22
## 5346	270	15	2016.06.20	90	1.57079633	2016.06.20.13.22
## 5347	275	16	2016.06.20	85	1.48352986	2016.06.20.13.22
## 5348	300	17	2016.06.20	60	1.04719755	2016.06.20.13.22
## 5349	280	18	2016.06.20	80	1.39626340	2016.06.20.13.22
## 5350	270	19	2016.06.20	90	1.57079633	2016.06.20.13.22
## 5351	290	20	2016.06.20	70	1.22173048	2016.06.20.13.22
## 5352	300	21	2016.06.20	60	1.04719755	2016.06.20.13.22
## 5353	300	22	2016.06.20	60	1.04719755	2016.06.20.13.22
## 5354	275	23	2016.06.20	85	1.48352986	2016.06.20.13.22
## 5355	270	24	2016.06.20	90	1.57079633	2016.06.20.13.22
## 5356	270	25	2016.06.20	90	1.57079633	2016.06.20.13.22
## 5357	270	26	2016.06.20	90	1.57079633	2016.06.20.13.22
## 5358	270	27	2016.06.20	90	1.57079633	2016.06.20.13.22
## 5359	265	28	2016.06.20	95	1.65806279	2016.06.20.13.22
## 5360	300	1	2016.06.20	60	1.04719755	2016.06.20.13.33

## 5361	330	2	2016.06.20	30	0.52359878	2016.06.20.13.33
## 5362	340	3	2016.06.20	20	0.34906585	2016.06.20.13.33
## 5363	25	4	2016.06.20	335	-0.43633231	2016.06.20.13.33
## 5364	20	5	2016.06.20	340	-0.34906585	2016.06.20.13.33
## 5365	70	6	2016.06.20	290	-1.22173048	2016.06.20.13.33
## 5366	10	7	2016.06.20	350	-0.17453293	2016.06.20.13.33
## 5367	330	8	2016.06.20	30	0.52359878	2016.06.20.13.33
## 5368	10	9	2016.06.20	350	-0.17453293	2016.06.20.13.33
## 5369	260	10	2016.06.20	100	1.74532925	2016.06.20.13.33
## 5370	280	11	2016.06.20	80	1.39626340	2016.06.20.13.33
## 5371	300	12	2016.06.20	60	1.04719755	2016.06.20.13.33
## 5372	290	13	2016.06.20	70	1.22173048	2016.06.20.13.33
## 5373	290	14	2016.06.20	70	1.22173048	2016.06.20.13.33
## 5374	290	15	2016.06.20	70	1.22173048	2016.06.20.13.33
## 5375	295	16	2016.06.20	65	1.13446401	2016.06.20.13.33
## 5376	270	17	2016.06.20	90	1.57079633	2016.06.20.13.33
## 5377	280	18	2016.06.20	80	1.39626340	2016.06.20.13.33
## 5378	270	19	2016.06.20	90	1.57079633	2016.06.20.13.33
## 5379	285	20	2016.06.20	75	1.30899694	2016.06.20.13.33
## 5380	260	21	2016.06.20	100	1.74532925	2016.06.20.13.33
## 5381	250	22	2016.06.20	110	1.91986218	2016.06.20.13.33
## 5382	320	23	2016.06.20	40	0.69813170	2016.06.20.13.33
## 5383	280	24	2016.06.20	80	1.39626340	2016.06.20.13.33
## 5384	270	25	2016.06.20	90	1.57079633	2016.06.20.13.33
## 5385	275	26	2016.06.20	85	1.48352986	2016.06.20.13.33
## 5386	295	27	2016.06.20	65	1.13446401	2016.06.20.13.33
## 5387	265	28	2016.06.20	95	1.65806279	2016.06.20.13.33
## 5388	285	29	2016.06.20	75	1.30899694	2016.06.20.13.33
## 5389	280	30	2016.06.20	80	1.39626340	2016.06.20.13.33
## 5390	295	31	2016.06.20	65	1.13446401	2016.06.20.13.33
## 5391	260	32	2016.06.20	100	1.74532925	2016.06.20.13.33
## 5392	370	33	2016.06.20	350	-0.17453293	2016.06.20.13.33
## 5393	330	34	2016.06.20	30	0.52359878	2016.06.20.13.33
## 5394	265	35	2016.06.20	95	1.65806279	2016.06.20.13.33
## 5395	50	1	2016.06.20	310	-0.87266463	2016.06.20.13.42
## 5396	20	2	2016.06.20	340	-0.34906585	2016.06.20.13.42
## 5397	10	3	2016.06.20	350	-0.17453293	2016.06.20.13.42
## 5398	240	4	2016.06.20	120	2.09439510	2016.06.20.13.42
## 5399	110	5	2016.06.20	250	-1.91986218	2016.06.20.13.42
## 5400	160	6	2016.06.20	200	-2.79252680	2016.06.20.13.42
## 5401	35	7	2016.06.20	325	-0.61086524	2016.06.20.13.42
## 5402	40	8	2016.06.20	320	-0.69813170	2016.06.20.13.42
## 5403	50	9	2016.06.20	310	-0.87266463	2016.06.20.13.42
## 5404	35	10	2016.06.20	325	-0.61086524	2016.06.20.13.42
## 5405	40	11	2016.06.20	320	-0.69813170	2016.06.20.13.42
## 5406	330	12	2016.06.20	30	0.52359878	2016.06.20.13.42
## 5407	270	13	2016.06.20	90	1.57079633	2016.06.20.13.42
## 5408	310	14	2016.06.20	50	0.87266463	2016.06.20.13.42
## 5409	320	15	2016.06.20	40	0.69813170	2016.06.20.13.42
## 5410	295	16	2016.06.20	65	1.13446401	2016.06.20.13.42
## 5411	290	17	2016.06.20	70	1.22173048	2016.06.20.13.42
## 5412	275	18	2016.06.20	85	1.48352986	2016.06.20.13.42
## 5413	310	19	2016.06.20	50	0.87266463	2016.06.20.13.42
## 5414	310	20	2016.06.20	50	0.87266463	2016.06.20.13.42

## 5415	245	21	2016.06.20	115	2.00712864	2016.06.20.13.42
## 5416	270	22	2016.06.20	90	1.57079633	2016.06.20.13.42
## 5417	250	23	2016.06.20	110	1.91986218	2016.06.20.13.42
## 5418	265	24	2016.06.20	95	1.65806279	2016.06.20.13.42
## 5419	305	25	2016.06.20	55	0.95993109	2016.06.20.13.42
## 5420	290	26	2016.06.20	70	1.22173048	2016.06.20.13.42
## 5421	275	27	2016.06.20	85	1.48352986	2016.06.20.13.42
## 5422	300	28	2016.06.20	60	1.04719755	2016.06.20.13.42
## 5423	5	1	2016.06.20	355	-0.08726646	2016.06.20.13.48
## 5424	340	2	2016.06.20	20	0.34906585	2016.06.20.13.48
## 5425	175	3	2016.06.20	185	-3.05432619	2016.06.20.13.48
## 5426	255	4	2016.06.20	105	1.83259571	2016.06.20.13.48
## 5427	270	5	2016.06.20	90	1.57079633	2016.06.20.13.48
## 5428	150	6	2016.06.20	210	-2.61799388	2016.06.20.13.48
## 5429	230	7	2016.06.20	130	2.26892803	2016.06.20.13.48
## 5430	90	8	2016.06.20	270	-1.57079633	2016.06.20.13.48
## 5431	350	9	2016.06.20	10	0.17453293	2016.06.20.13.48
## 5432	110	10	2016.06.20	250	-1.91986218	2016.06.20.13.48
## 5433	260	11	2016.06.20	100	1.74532925	2016.06.20.13.48
## 5434	280	12	2016.06.20	80	1.39626340	2016.06.20.13.48
## 5435	240	13	2016.06.20	120	2.09439510	2016.06.20.13.48
## 5436	270	14	2016.06.20	90	1.57079633	2016.06.20.13.48
## 5437	245	15	2016.06.20	115	2.00712864	2016.06.20.13.48
## 5438	270	16	2016.06.20	90	1.57079633	2016.06.20.13.48
## 5439	265	17	2016.06.20	95	1.65806279	2016.06.20.13.48
## 5440	270	18	2016.06.20	90	1.57079633	2016.06.20.13.48
## 5441	270	19	2016.06.20	90	1.57079633	2016.06.20.13.48
## 5442	20	1	2016.06.20	340	-0.34906585	2016.06.20.14.01
## 5443	355	2	2016.06.20	5	0.08726646	2016.06.20.14.01
## 5444	15	3	2016.06.20	345	-0.26179939	2016.06.20.14.01
## 5445	50	4	2016.06.20	310	-0.87266463	2016.06.20.14.01
## 5446	55	5	2016.06.20	305	-0.95993109	2016.06.20.14.01
## 5447	120	6	2016.06.20	240	-2.09439510	2016.06.20.14.01
## 5448	20	7	2016.06.20	340	-0.34906585	2016.06.20.14.01
## 5449	180	8	2016.06.20	180	3.14159265	2016.06.20.14.01
## 5450	40	9	2016.06.20	320	-0.69813170	2016.06.20.14.01
## 5451	60	10	2016.06.20	300	-1.04719755	2016.06.20.14.01
## 5452	220	11	2016.06.20	140	2.44346095	2016.06.20.14.01
## 5453	40	12	2016.06.20	320	-0.69813170	2016.06.20.14.01
## 5454	280	13	2016.06.20	80	1.39626340	2016.06.20.14.01
## 5455	90	14	2016.06.20	270	-1.57079633	2016.06.20.14.01
## 5456	280	15	2016.06.20	80	1.39626340	2016.06.20.14.01
## 5457	310	16	2016.06.20	50	0.87266463	2016.06.20.14.01
## 5458	290	17	2016.06.20	70	1.22173048	2016.06.20.14.01
## 5459	330	18	2016.06.20	30	0.52359878	2016.06.20.14.01
## 5460	290	19	2016.06.20	70	1.22173048	2016.06.20.14.01
## 5461	270	20	2016.06.20	90	1.57079633	2016.06.20.14.01
## 5462	310	21	2016.06.20	50	0.87266463	2016.06.20.14.01
## 5463	290	22	2016.06.20	70	1.22173048	2016.06.20.14.01
## 5464	310	23	2016.06.20	50	0.87266463	2016.06.20.14.01
## 5465	325	24	2016.06.20	35	0.61086524	2016.06.20.14.01
## 5466	300	25	2016.06.20	60	1.04719755	2016.06.20.14.01
## 5467	320	26	2016.06.20	40	0.69813170	2016.06.20.14.01
## 5468	280	27	2016.06.20	80	1.39626340	2016.06.20.14.01

## 5469	275	28	2016.06.20	85	1.48352986	2016.06.20.14.01
## 5470	310	29	2016.06.20	50	0.87266463	2016.06.20.14.01
## 5471	290	30	2016.06.20	70	1.22173048	2016.06.20.14.01
## 5472	330	31	2016.06.20	30	0.52359878	2016.06.20.14.01
## 5473	320	32	2016.06.20	40	0.69813170	2016.06.20.14.01
## 5474	310	33	2016.06.20	50	0.87266463	2016.06.20.14.01
## 5475	330	34	2016.06.20	30	0.52359878	2016.06.20.14.01
## 5476	295	35	2016.06.20	65	1.13446401	2016.06.20.14.01
## 5477	295	36	2016.06.20	65	1.13446401	2016.06.20.14.01
## 5478	300	37	2016.06.20	60	1.04719755	2016.06.20.14.01
## 5479	220	1	2016.06.20	140	2.44346095	2016.06.20.14.24
## 5480	125	2	2016.06.20	235	-2.18166156	2016.06.20.14.24
## 5481	30	3	2016.06.20	330	-0.52359878	2016.06.20.14.24
## 5482	5	4	2016.06.20	355	-0.08726646	2016.06.20.14.24
## 5483	330	5	2016.06.20	30	0.52359878	2016.06.20.14.24
## 5484	335	6	2016.06.20	25	0.43633231	2016.06.20.14.24
## 5485	340	7	2016.06.20	20	0.34906585	2016.06.20.14.24
## 5486	45	8	2016.06.20	315	-0.78539816	2016.06.20.14.24
## 5487	50	9	2016.06.20	310	-0.87266463	2016.06.20.14.24
## 5488	40	10	2016.06.20	320	-0.69813170	2016.06.20.14.24
## 5489	230	11	2016.06.20	130	2.26892803	2016.06.20.14.24
## 5490	10	12	2016.06.20	350	-0.17453293	2016.06.20.14.24
## 5491	210	13	2016.06.20	150	2.61799388	2016.06.20.14.24
## 5492	30	14	2016.06.20	330	-0.52359878	2016.06.20.14.24
## 5493	340	15	2016.06.20	20	0.34906585	2016.06.20.14.24
## 5494	40	16	2016.06.20	320	-0.69813170	2016.06.20.14.24
## 5495	350	17	2016.06.20	10	0.17453293	2016.06.20.14.24
## 5496	250	18	2016.06.20	110	1.91986218	2016.06.20.14.24
## 5497	290	19	2016.06.20	70	1.22173048	2016.06.20.14.24
## 5498	300	20	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5499	300	21	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5500	295	22	2016.06.20	65	1.13446401	2016.06.20.14.24
## 5501	285	23	2016.06.20	75	1.30899694	2016.06.20.14.24
## 5502	270	24	2016.06.20	90	1.57079633	2016.06.20.14.24
## 5503	280	25	2016.06.20	80	1.39626340	2016.06.20.14.24
## 5504	320	26	2016.06.20	40	0.69813170	2016.06.20.14.24
## 5505	280	27	2016.06.20	80	1.39626340	2016.06.20.14.24
## 5506	270	28	2016.06.20	90	1.57079633	2016.06.20.14.24
## 5507	300	29	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5508	290	30	2016.06.20	70	1.22173048	2016.06.20.14.24
## 5509	305	31	2016.06.20	55	0.95993109	2016.06.20.14.24
## 5510	310	32	2016.06.20	50	0.87266463	2016.06.20.14.24
## 5511	280	33	2016.06.20	80	1.39626340	2016.06.20.14.24
## 5512	305	34	2016.06.20	55	0.95993109	2016.06.20.14.24
## 5513	300	35	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5514	290	36	2016.06.20	70	1.22173048	2016.06.20.14.24
## 5515	300	37	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5516	290	38	2016.06.20	70	1.22173048	2016.06.20.14.24
## 5517	280	39	2016.06.20	80	1.39626340	2016.06.20.14.24
## 5518	280	40	2016.06.20	80	1.39626340	2016.06.20.14.24
## 5519	290	41	2016.06.20	70	1.22173048	2016.06.20.14.24
## 5520	300	42	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5521	290	43	2016.06.20	70	1.22173048	2016.06.20.14.24
## 5522	310	44	2016.06.20	50	0.87266463	2016.06.20.14.24

## 5523	270	45	2016.06.20	90	1.57079633	2016.06.20.14.24
## 5524	300	46	2016.06.20	60	1.04719755	2016.06.20.14.24
## 5525	315	47	2016.06.20	45	0.78539816	2016.06.20.14.24
## 5526	15	48	2016.06.20	345	-0.26179939	2016.06.20.14.24
## 5527	340	49	2016.06.20	20	0.34906585	2016.06.20.14.24
## 5528	195	1	2016.06.20	165	2.87979327	2016.06.20.14.52
## 5529	220	2	2016.06.20	140	2.44346095	2016.06.20.14.52
## 5530	345	3	2016.06.20	15	0.26179939	2016.06.20.14.52
## 5531	220	4	2016.06.20	140	2.44346095	2016.06.20.14.52
## 5532	210	5	2016.06.20	150	2.61799388	2016.06.20.14.52
## 5533	5	6	2016.06.20	355	-0.08726646	2016.06.20.14.52
## 5534	160	7	2016.06.20	200	-2.79252680	2016.06.20.14.52
## 5535	220	8	2016.06.20	140	2.44346095	2016.06.20.14.52
## 5536	170	9	2016.06.20	190	-2.96705973	2016.06.20.14.52
## 5537	120	10	2016.06.20	240	-2.09439510	2016.06.20.14.52
## 5538	190	11	2016.06.20	170	2.96705973	2016.06.20.14.52
## 5539	330	12	2016.06.20	30	0.52359878	2016.06.20.14.52
## 5540	230	13	2016.06.20	130	2.26892803	2016.06.20.14.52
## 5541	290	14	2016.06.20	70	1.22173048	2016.06.20.14.52
## 5542	330	15	2016.06.20	30	0.52359878	2016.06.20.14.52
## 5543	200	16	2016.06.20	160	2.79252680	2016.06.20.14.52
## 5544	195	17	2016.06.20	165	2.87979327	2016.06.20.14.52
## 5545	215	18	2016.06.20	145	2.53072742	2016.06.20.14.52
## 5546	260	19	2016.06.20	100	1.74532925	2016.06.20.14.52
## 5547	340	20	2016.06.20	20	0.34906585	2016.06.20.14.52
## 5548	265	21	2016.06.20	95	1.65806279	2016.06.20.14.52
## 5549	330	22	2016.06.20	30	0.52359878	2016.06.20.14.52
## 5550	260	23	2016.06.20	100	1.74532925	2016.06.20.14.52
## 5551	255	24	2016.06.20	105	1.83259571	2016.06.20.14.52
## 5552	275	25	2016.06.20	85	1.48352986	2016.06.20.14.52
## 5553	280	26	2016.06.20	80	1.39626340	2016.06.20.14.52
## 5554	270	27	2016.06.20	90	1.57079633	2016.06.20.14.52
## 5555	260	28	2016.06.20	100	1.74532925	2016.06.20.14.52
## 5556	320	29	2016.06.20	40	0.69813170	2016.06.20.14.52
## 5557	315	30	2016.06.20	45	0.78539816	2016.06.20.14.52
## 5558	310	31	2016.06.20	50	0.87266463	2016.06.20.14.52
## 5559	330	32	2016.06.20	30	0.52359878	2016.06.20.14.52
## 5560	330	33	2016.06.20	30	0.52359878	2016.06.20.14.52
## 5561	325	34	2016.06.20	35	0.61086524	2016.06.20.14.52
## 5562	330	35	2016.06.20	30	0.52359878	2016.06.20.14.52
## 5563	280	36	2016.06.20	80	1.39626340	2016.06.20.14.52
## 5564	320	37	2016.06.20	40	0.69813170	2016.06.20.14.52
## 5565	275	38	2016.06.20	85	1.48352986	2016.06.20.14.52
## 5566	280	39	2016.06.20	80	1.39626340	2016.06.20.14.52
## 5567	290	40	2016.06.20	70	1.22173048	2016.06.20.14.52
## 5568	280	41	2016.06.20	80	1.39626340	2016.06.20.14.52
## 5569	275	42	2016.06.20	85	1.48352986	2016.06.20.14.52
## 5570	285	43	2016.06.20	75	1.30899694	2016.06.20.14.52
## 5571	275	44	2016.06.20	85	1.48352986	2016.06.20.14.52
## 5572	305	45	2016.06.20	55	0.95993109	2016.06.20.14.52
## 5573	275	46	2016.06.20	85	1.48352986	2016.06.20.14.52
## 5574	290	47	2016.06.20	70	1.22173048	2016.06.20.14.52
## 5575	290	48	2016.06.20	70	1.22173048	2016.06.20.14.52
## 5576	190	1	2016.06.21	170	2.96705973	2016.06.21.14.36

## 5577	250	2	2016.06.21	110	1.91986218	2016.06.21.14.36
## 5578	180	3	2016.06.21	180	3.14159265	2016.06.21.14.36
## 5579	320	4	2016.06.21	40	0.69813170	2016.06.21.14.36
## 5580	175	5	2016.06.21	185	-3.05432619	2016.06.21.14.36
## 5581	20	6	2016.06.21	340	-0.34906585	2016.06.21.14.36
## 5582	235	7	2016.06.21	125	2.18166156	2016.06.21.14.36
## 5583	350	8	2016.06.21	10	0.17453293	2016.06.21.14.36
## 5584	180	9	2016.06.21	180	3.14159265	2016.06.21.14.36
## 5585	160	10	2016.06.21	200	-2.79252680	2016.06.21.14.36
## 5586	190	11	2016.06.21	170	2.96705973	2016.06.21.14.36
## 5587	320	12	2016.06.21	40	0.69813170	2016.06.21.14.36
## 5588	240	13	2016.06.21	120	2.09439510	2016.06.21.14.36
## 5589	160	14	2016.06.21	200	-2.79252680	2016.06.21.14.36
## 5590	265	15	2016.06.21	95	1.65806279	2016.06.21.14.36
## 5591	285	16	2016.06.21	75	1.30899694	2016.06.21.14.36
## 5592	275	17	2016.06.21	85	1.48352986	2016.06.21.14.36
## 5593	300	18	2016.06.21	60	1.04719755	2016.06.21.14.36
## 5594	260	19	2016.06.21	100	1.74532925	2016.06.21.14.36
## 5595	305	20	2016.06.21	55	0.95993109	2016.06.21.14.36
## 5596	310	21	2016.06.21	50	0.87266463	2016.06.21.14.36
## 5597	320	22	2016.06.21	40	0.69813170	2016.06.21.14.36
## 5598	290	23	2016.06.21	70	1.22173048	2016.06.21.14.36
## 5599	310	24	2016.06.21	50	0.87266463	2016.06.21.14.36
## 5600	270	25	2016.06.21	90	1.57079633	2016.06.21.14.36
## 5601	270	26	2016.06.21	90	1.57079633	2016.06.21.14.36
## 5602	350	1	2016.06.21	10	0.17453293	2016.06.21.14.40
## 5603	200	2	2016.06.21	160	2.79252680	2016.06.21.14.40
## 5604	210	3	2016.06.21	150	2.61799388	2016.06.21.14.40
## 5605	190	4	2016.06.21	170	2.96705973	2016.06.21.14.40
## 5606	225	5	2016.06.21	135	2.35619449	2016.06.21.14.40
## 5607	310	6	2016.06.21	50	0.87266463	2016.06.21.14.40
## 5608	210	7	2016.06.21	150	2.61799388	2016.06.21.14.40
## 5609	0	8	2016.06.21	0	0.00000000	2016.06.21.14.40
## 5610	200	9	2016.06.21	160	2.79252680	2016.06.21.14.40
## 5611	175	10	2016.06.21	185	-3.05432619	2016.06.21.14.40
## 5612	300	11	2016.06.21	60	1.04719755	2016.06.21.14.40
## 5613	300	12	2016.06.21	60	1.04719755	2016.06.21.14.40
## 5614	290	13	2016.06.21	70	1.22173048	2016.06.21.14.40
## 5615	305	14	2016.06.21	55	0.95993109	2016.06.21.14.40
## 5616	280	15	2016.06.21	80	1.39626340	2016.06.21.14.40
## 5617	275	16	2016.06.21	85	1.48352986	2016.06.21.14.40
## 5618	275	17	2016.06.21	85	1.48352986	2016.06.21.14.40
## 5619	305	18	2016.06.21	55	0.95993109	2016.06.21.14.40
## 5620	300	19	2016.06.21	60	1.04719755	2016.06.21.14.40
## 5621	320	20	2016.06.21	40	0.69813170	2016.06.21.14.40
## 5622	280	21	2016.06.21	80	1.39626340	2016.06.21.14.40
## 5623	275	22	2016.06.21	85	1.48352986	2016.06.21.14.40
## 5624	305	23	2016.06.21	55	0.95993109	2016.06.21.14.40
## 5625	295	24	2016.06.21	65	1.13446401	2016.06.21.14.40
## 5626	270	25	2016.06.21	90	1.57079633	2016.06.21.14.40
## 5627	245	26	2016.06.21	115	2.00712864	2016.06.21.14.40
## 5628	260	27	2016.06.21	100	1.74532925	2016.06.21.14.40
## 5629	255	28	2016.06.21	105	1.83259571	2016.06.21.14.40
## 5630	250	29	2016.06.21	110	1.91986218	2016.06.21.14.40

## 5631	265	30	2016.06.21	95	1.65806279	2016.06.21.14.40
## 5632	240	31	2016.06.21	120	2.09439510	2016.06.21.14.40
## 5633	265	32	2016.06.21	95	1.65806279	2016.06.21.14.40
## 5634	270	33	2016.06.21	90	1.57079633	2016.06.21.14.40
## 5635	290	34	2016.06.21	70	1.22173048	2016.06.21.14.40
## 5636	300	35	2016.06.21	60	1.04719755	2016.06.21.14.40
## 5637	270	36	2016.06.21	90	1.57079633	2016.06.21.14.40
## 5638	295	37	2016.06.21	65	1.13446401	2016.06.21.14.40
## 5639	330	1	2016.06.21	30	0.52359878	2016.06.21.14.52
## 5640	20	2	2016.06.21	340	-0.34906585	2016.06.21.14.52
## 5641	350	3	2016.06.21	10	0.17453293	2016.06.21.14.52
## 5642	190	4	2016.06.21	170	2.96705973	2016.06.21.14.52
## 5643	265	5	2016.06.21	95	1.65806279	2016.06.21.14.52
## 5644	260	6	2016.06.21	100	1.74532925	2016.06.21.14.52
## 5645	265	7	2016.06.21	95	1.65806279	2016.06.21.14.52
## 5646	260	8	2016.06.21	100	1.74532925	2016.06.21.14.52
## 5647	285	9	2016.06.21	75	1.30899694	2016.06.21.14.52
## 5648	30	10	2016.06.21	330	-0.52359878	2016.06.21.14.52
## 5649	15	11	2016.06.21	345	-0.26179939	2016.06.21.14.52
## 5650	345	12	2016.06.21	15	0.26179939	2016.06.21.14.52
## 5651	330	13	2016.06.21	30	0.52359878	2016.06.21.14.52
## 5652	280	14	2016.06.21	80	1.39626340	2016.06.21.14.52
## 5653	300	15	2016.06.21	60	1.04719755	2016.06.21.14.52
## 5654	290	16	2016.06.21	70	1.22173048	2016.06.21.14.52
## 5655	275	17	2016.06.21	85	1.48352986	2016.06.21.14.52
## 5656	280	18	2016.06.21	80	1.39626340	2016.06.21.14.52
## 5657	330	19	2016.06.21	30	0.52359878	2016.06.21.14.52
## 5658	280	20	2016.06.21	80	1.39626340	2016.06.21.14.52
## 5659	325	21	2016.06.21	35	0.61086524	2016.06.21.14.52
## 5660	310	22	2016.06.21	50	0.87266463	2016.06.21.14.52
## 5661	310	23	2016.06.21	50	0.87266463	2016.06.21.14.52
## 5662	325	24	2016.06.21	35	0.61086524	2016.06.21.14.52
## 5663	310	25	2016.06.21	50	0.87266463	2016.06.21.14.52
## 5664	330	26	2016.06.21	30	0.52359878	2016.06.21.14.52
## 5665	325	27	2016.06.21	35	0.61086524	2016.06.21.14.52
## 5666	305	28	2016.06.21	55	0.95993109	2016.06.21.14.52
## 5667	245	1	2016.06.21	115	2.00712864	2016.06.21.15.24
## 5668	0	2	2016.06.21	0	0.00000000	2016.06.21.15.24
## 5669	205	3	2016.06.21	155	2.70526034	2016.06.21.15.24
## 5670	5	4	2016.06.21	355	-0.08726646	2016.06.21.15.24
## 5671	25	5	2016.06.21	335	-0.43633231	2016.06.21.15.24
## 5672	220	6	2016.06.21	140	2.44346095	2016.06.21.15.24
## 5673	40	7	2016.06.21	320	-0.69813170	2016.06.21.15.24
## 5674	320	8	2016.06.21	40	0.69813170	2016.06.21.15.24
## 5675	310	9	2016.06.21	50	0.87266463	2016.06.21.15.24
## 5676	330	10	2016.06.21	30	0.52359878	2016.06.21.15.24
## 5677	340	11	2016.06.21	20	0.34906585	2016.06.21.15.24
## 5678	330	12	2016.06.21	30	0.52359878	2016.06.21.15.24
## 5679	330	13	2016.06.21	30	0.52359878	2016.06.21.15.24
## 5680	325	14	2016.06.21	35	0.61086524	2016.06.21.15.24
## 5681	300	15	2016.06.21	60	1.04719755	2016.06.21.15.24
## 5682	310	16	2016.06.21	50	0.87266463	2016.06.21.15.24
## 5683	320	17	2016.06.21	40	0.69813170	2016.06.21.15.24
## 5684	320	18	2016.06.21	40	0.69813170	2016.06.21.15.24

## 5685	305	19	2016.06.21	55	0.95993109	2016.06.21.15.24
## 5686	345	20	2016.06.21	15	0.26179939	2016.06.21.15.24
## 5687	0	21	2016.06.21	0	0.00000000	2016.06.21.15.24
## 5688	330	22	2016.06.21	30	0.52359878	2016.06.21.15.24
## 5689	30	1	2016.06.21	330	-0.52359878	2016.06.21.15.29
## 5690	230	2	2016.06.21	130	2.26892803	2016.06.21.15.29
## 5691	70	3	2016.06.21	290	-1.22173048	2016.06.21.15.29
## 5692	0	4	2016.06.21	0	0.00000000	2016.06.21.15.29
## 5693	190	5	2016.06.21	170	2.96705973	2016.06.21.15.29
## 5694	120	6	2016.06.21	240	-2.09439510	2016.06.21.15.29
## 5695	190	7	2016.06.21	170	2.96705973	2016.06.21.15.29
## 5696	60	8	2016.06.21	300	-1.04719755	2016.06.21.15.29
## 5697	55	9	2016.06.21	305	-0.95993109	2016.06.21.15.29
## 5698	80	10	2016.06.21	280	-1.39626340	2016.06.21.15.29
## 5699	330	11	2016.06.21	30	0.52359878	2016.06.21.15.29
## 5700	5	12	2016.06.21	355	-0.08726646	2016.06.21.15.29
## 5701	105	13	2016.06.21	255	-1.83259571	2016.06.21.15.29
## 5702	240	14	2016.06.21	120	2.09439510	2016.06.21.15.29
## 5703	260	15	2016.06.21	100	1.74532925	2016.06.21.15.29
## 5704	325	16	2016.06.21	35	0.61086524	2016.06.21.15.29
## 5705	300	17	2016.06.21	60	1.04719755	2016.06.21.15.29
## 5706	320	18	2016.06.21	40	0.69813170	2016.06.21.15.29
## 5707	310	19	2016.06.21	50	0.87266463	2016.06.21.15.29
## 5708	305	20	2016.06.21	55	0.95993109	2016.06.21.15.29
## 5709	310	21	2016.06.21	50	0.87266463	2016.06.21.15.29
## 5710	310	22	2016.06.21	50	0.87266463	2016.06.21.15.29
## 5711	320	23	2016.06.21	40	0.69813170	2016.06.21.15.29
## 5712	320	24	2016.06.21	40	0.69813170	2016.06.21.15.29
## 5713	315	25	2016.06.21	45	0.78539816	2016.06.21.15.29
## 5714	335	26	2016.06.21	25	0.43633231	2016.06.21.15.29
## 5715	330	27	2016.06.21	30	0.52359878	2016.06.21.15.29
## 5716	325	28	2016.06.21	35	0.61086524	2016.06.21.15.29
## 5717	340	29	2016.06.21	20	0.34906585	2016.06.21.15.29
## 5718	335	30	2016.06.21	25	0.43633231	2016.06.21.15.29
## 5719	315	31	2016.06.21	45	0.78539816	2016.06.21.15.29
## 5720	295	32	2016.06.21	65	1.13446401	2016.06.21.15.29
## 5721	340	33	2016.06.21	20	0.34906585	2016.06.21.15.29
## 5722	330	34	2016.06.21	30	0.52359878	2016.06.21.15.29
## 5723	310	35	2016.06.21	50	0.87266463	2016.06.21.15.29
## 5724	340	36	2016.06.21	20	0.34906585	2016.06.21.15.29
## 5725	350	37	2016.06.21	10	0.17453293	2016.06.21.15.29
## 5726	350	38	2016.06.21	10	0.17453293	2016.06.21.15.29
## 5727	335	39	2016.06.21	25	0.43633231	2016.06.21.15.29
## 5728	300	40	2016.06.21	60	1.04719755	2016.06.21.15.29
## 5729	290	41	2016.06.21	70	1.22173048	2016.06.21.15.29
## 5730	310	42	2016.06.21	50	0.87266463	2016.06.21.15.29
## 5731	60	1	2016.06.21	300	-1.04719755	2016.06.21.15.50
## 5732	60	2	2016.06.21	300	-1.04719755	2016.06.21.15.50
## 5733	105	3	2016.06.21	255	-1.83259571	2016.06.21.15.50
## 5734	45	4	2016.06.21	315	-0.78539816	2016.06.21.15.50
## 5735	260	5	2016.06.21	100	1.74532925	2016.06.21.15.50
## 5736	70	6	2016.06.21	290	-1.22173048	2016.06.21.15.50
## 5737	30	7	2016.06.21	330	-0.52359878	2016.06.21.15.50
## 5738	60	8	2016.06.21	300	-1.04719755	2016.06.21.15.50

## 5739	305	9	2016.06.21	55	0.95993109	2016.06.21.15.50
## 5740	320	10	2016.06.21	40	0.69813170	2016.06.21.15.50
## 5741	320	11	2016.06.21	40	0.69813170	2016.06.21.15.50
## 5742	270	12	2016.06.21	90	1.57079633	2016.06.21.15.50
## 5743	40	13	2016.06.21	320	-0.69813170	2016.06.21.15.50
## 5744	5	14	2016.06.21	355	-0.08726646	2016.06.21.15.50
## 5745	5	15	2016.06.21	355	-0.08726646	2016.06.21.15.50
## 5746	0	16	2016.06.21	0	0.00000000	2016.06.21.15.50
## 5747	10	17	2016.06.21	350	-0.17453293	2016.06.21.15.50
## 5748	5	18	2016.06.21	355	-0.08726646	2016.06.21.15.50
## 5749	5	19	2016.06.21	355	-0.08726646	2016.06.21.15.50
## 5750	175	1	2016.06.07	185	-3.05432619	2016.06.07.13.59
## 5751	155	2	2016.06.07	205	-2.70526034	2016.06.07.13.59
## 5752	225	3	2016.06.07	135	2.35619449	2016.06.07.13.59
## 5753	210	4	2016.06.07	150	2.61799388	2016.06.07.13.59
## 5754	215	5	2016.06.07	145	2.53072742	2016.06.07.13.59
## 5755	185	6	2016.06.07	175	3.05432619	2016.06.07.13.59
## 5756	160	7	2016.06.07	200	-2.79252680	2016.06.07.13.59
## 5757	175	8	2016.06.07	185	-3.05432619	2016.06.07.13.59
## 5758	130	9	2016.06.07	230	-2.26892803	2016.06.07.13.59
## 5759	350	10	2016.06.07	10	0.17453293	2016.06.07.13.59
## 5760	350	11	2016.06.07	10	0.17453293	2016.06.07.13.59
## 5761	0	12	2016.06.07	0	0.00000000	2016.06.07.13.59
## 5762	5	13	2016.06.07	355	-0.08726646	2016.06.07.13.59
## 5763	350	14	2016.06.07	10	0.17453293	2016.06.07.13.59
## 5764	345	15	2016.06.07	15	0.26179939	2016.06.07.13.59
## 5765	350	16	2016.06.07	10	0.17453293	2016.06.07.13.59
## 5766	335	17	2016.06.07	25	0.43633231	2016.06.07.13.59
## 5767	5	18	2016.06.07	355	-0.08726646	2016.06.07.13.59
## 5768	340	19	2016.06.07	20	0.34906585	2016.06.07.13.59
## 5769	310	20	2016.06.07	50	0.87266463	2016.06.07.13.59
## 5770	30	21	2016.06.07	330	-0.52359878	2016.06.07.13.59
## 5771	0	22	2016.06.07	0	0.00000000	2016.06.07.13.59
## 5772	15	23	2016.06.07	345	-0.26179939	2016.06.07.13.59
## 5773	10	24	2016.06.07	350	-0.17453293	2016.06.07.13.59
## 5774	30	25	2016.06.07	330	-0.52359878	2016.06.07.13.59
## 5775	340	26	2016.06.07	20	0.34906585	2016.06.07.13.59
## 5776	355	27	2016.06.07	5	0.08726646	2016.06.07.13.59
## 5777	270	1	2016.06.07	90	1.57079633	2016.06.07.14.05
## 5778	220	2	2016.06.07	140	2.44346095	2016.06.07.14.05
## 5779	260	3	2016.06.07	100	1.74532925	2016.06.07.14.05
## 5780	250	4	2016.06.07	110	1.91986218	2016.06.07.14.05
## 5781	190	5	2016.06.07	170	2.96705973	2016.06.07.14.05
## 5782	185	6	2016.06.07	175	3.05432619	2016.06.07.14.05
## 5783	175	7	2016.06.07	185	-3.05432619	2016.06.07.14.05
## 5784	200	8	2016.06.07	160	2.79252680	2016.06.07.14.05
## 5785	140	9	2016.06.07	220	-2.44346095	2016.06.07.14.05
## 5786	160	10	2016.06.07	200	-2.79252680	2016.06.07.14.05
## 5787	10	11	2016.06.07	350	-0.17453293	2016.06.07.14.05
## 5788	10	12	2016.06.07	350	-0.17453293	2016.06.07.14.05
## 5789	5	13	2016.06.07	355	-0.08726646	2016.06.07.14.05
## 5790	0	14	2016.06.07	0	0.00000000	2016.06.07.14.05
## 5791	15	15	2016.06.07	345	-0.26179939	2016.06.07.14.05
## 5792	0	16	2016.06.07	0	0.00000000	2016.06.07.14.05

## 5793	5	17	2016.06.07	355	-0.08726646	2016.06.07.14.05
## 5794	15	18	2016.06.07	345	-0.26179939	2016.06.07.14.05
## 5795	40	19	2016.06.07	320	-0.69813170	2016.06.07.14.05
## 5796	5	20	2016.06.07	355	-0.08726646	2016.06.07.14.05
## 5797	10	21	2016.06.07	350	-0.17453293	2016.06.07.14.05
## 5798	355	22	2016.06.07	5	0.08726646	2016.06.07.14.05
## 5799	0	23	2016.06.07	0	0.00000000	2016.06.07.14.05
## 5800	5	24	2016.06.07	355	-0.08726646	2016.06.07.14.05
## 5801	350	25	2016.06.07	10	0.17453293	2016.06.07.14.05
## 5802	10	26	2016.06.07	350	-0.17453293	2016.06.07.14.05
## 5803	0	27	2016.06.07	0	0.00000000	2016.06.07.14.05
## 5804	15	28	2016.06.07	345	-0.26179939	2016.06.07.14.05
## 5805	5	29	2016.06.07	355	-0.08726646	2016.06.07.14.05
## 5806	160	1	2016.06.07	200	-2.79252680	2016.06.07.14.10
## 5807	190	2	2016.06.07	170	2.96705973	2016.06.07.14.10
## 5808	150	3	2016.06.07	210	-2.61799388	2016.06.07.14.10
## 5809	180	4	2016.06.07	180	3.14159265	2016.06.07.14.10
## 5810	75	5	2016.06.07	285	-1.30899694	2016.06.07.14.10
## 5811	240	6	2016.06.07	120	2.09439510	2016.06.07.14.10
## 5812	90	7	2016.06.07	270	-1.57079633	2016.06.07.14.10
## 5813	200	8	2016.06.07	160	2.79252680	2016.06.07.14.10
## 5814	145	9	2016.06.07	215	-2.53072742	2016.06.07.14.10
## 5815	170	10	2016.06.07	190	-2.96705973	2016.06.07.14.10
## 5816	160	11	2016.06.07	200	-2.79252680	2016.06.07.14.10
## 5817	145	12	2016.06.07	215	-2.53072742	2016.06.07.14.10
## 5818	165	13	2016.06.07	195	-2.87979327	2016.06.07.14.10
## 5819	125	14	2016.06.07	235	-2.18166156	2016.06.07.14.10
## 5820	120	15	2016.06.07	240	-2.09439510	2016.06.07.14.10
## 5821	30	16	2016.06.07	330	-0.52359878	2016.06.07.14.10
## 5822	355	17	2016.06.07	5	0.08726646	2016.06.07.14.10
## 5823	5	18	2016.06.07	355	-0.08726646	2016.06.07.14.10
## 5824	10	19	2016.06.07	350	-0.17453293	2016.06.07.14.10
## 5825	5	20	2016.06.07	355	-0.08726646	2016.06.07.14.10
## 5826	240	1	2016.06.07	120	2.09439510	2016.06.07.14.20
## 5827	250	2	2016.06.07	110	1.91986218	2016.06.07.14.20
## 5828	265	3	2016.06.07	95	1.65806279	2016.06.07.14.20
## 5829	270	4	2016.06.07	90	1.57079633	2016.06.07.14.20
## 5830	265	5	2016.06.07	95	1.65806279	2016.06.07.14.20
## 5831	270	6	2016.06.07	90	1.57079633	2016.06.07.14.20
## 5832	270	7	2016.06.07	90	1.57079633	2016.06.07.14.20
## 5833	255	8	2016.06.07	105	1.83259571	2016.06.07.14.20
## 5834	260	9	2016.06.07	100	1.74532925	2016.06.07.14.20
## 5835	240	10	2016.06.07	120	2.09439510	2016.06.07.14.20
## 5836	250	11	2016.06.07	110	1.91986218	2016.06.07.14.20
## 5837	280	12	2016.06.07	80	1.39626340	2016.06.07.14.20
## 5838	300	13	2016.06.07	60	1.04719755	2016.06.07.14.20
## 5839	165	14	2016.06.07	195	-2.87979327	2016.06.07.14.20
## 5840	200	15	2016.06.07	160	2.79252680	2016.06.07.14.20
## 5841	245	16	2016.06.07	115	2.00712864	2016.06.07.14.20
## 5842	225	17	2016.06.07	135	2.35619449	2016.06.07.14.20
## 5843	275	18	2016.06.07	85	1.48352986	2016.06.07.14.20
## 5844	250	19	2016.06.07	110	1.91986218	2016.06.07.14.20
## 5845	280	20	2016.06.07	80	1.39626340	2016.06.07.14.20
## 5846	180	21	2016.06.07	180	3.14159265	2016.06.07.14.20

## 5847	355	22	2016.06.07	5	0.08726646	2016.06.07.14.20
## 5848	340	23	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5849	300	24	2016.06.07	60	1.04719755	2016.06.07.14.20
## 5850	340	25	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5851	355	26	2016.06.07	5	0.08726646	2016.06.07.14.20
## 5852	330	27	2016.06.07	30	0.52359878	2016.06.07.14.20
## 5853	355	28	2016.06.07	5	0.08726646	2016.06.07.14.20
## 5854	350	29	2016.06.07	10	0.17453293	2016.06.07.14.20
## 5855	355	30	2016.06.07	5	0.08726646	2016.06.07.14.20
## 5856	350	31	2016.06.07	10	0.17453293	2016.06.07.14.20
## 5857	0	32	2016.06.07	0	0.00000000	2016.06.07.14.20
## 5858	330	33	2016.06.07	30	0.52359878	2016.06.07.14.20
## 5859	340	34	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5860	325	35	2016.06.07	35	0.61086524	2016.06.07.14.20
## 5861	350	36	2016.06.07	10	0.17453293	2016.06.07.14.20
## 5862	320	37	2016.06.07	40	0.69813170	2016.06.07.14.20
## 5863	5	38	2016.06.07	355	-0.08726646	2016.06.07.14.20
## 5864	340	39	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5865	350	40	2016.06.07	10	0.17453293	2016.06.07.14.20
## 5866	0	41	2016.06.07	0	0.00000000	2016.06.07.14.20
## 5867	0	42	2016.06.07	0	0.00000000	2016.06.07.14.20
## 5868	330	43	2016.06.07	30	0.52359878	2016.06.07.14.20
## 5869	335	44	2016.06.07	25	0.43633231	2016.06.07.14.20
## 5870	340	45	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5871	340	46	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5872	340	47	2016.06.07	20	0.34906585	2016.06.07.14.20
## 5873	45	48	2016.06.07	315	-0.78539816	2016.06.07.14.20
## 5874	270	49	2016.06.07	90	1.57079633	2016.06.07.14.20
## 5875	15	50	2016.06.07	345	-0.26179939	2016.06.07.14.20
## 5876	310	51	2016.06.07	50	0.87266463	2016.06.07.14.20
## 5877	5	52	2016.06.07	355	-0.08726646	2016.06.07.14.20
## 5878	0	53	2016.06.07	0	0.00000000	2016.06.07.14.20
## 5879	10	54	2016.06.07	350	-0.17453293	2016.06.07.14.20
## 5880	0	55	2016.06.07	0	0.00000000	2016.06.07.14.20
## 5881	315	56	2016.06.07	45	0.78539816	2016.06.07.14.20
## 5882	355	57	2016.06.07	5	0.08726646	2016.06.07.14.20
## 5883	250	58	2016.06.07	110	1.91986218	2016.06.07.14.20
## 5884	230	59	2016.06.07	130	2.26892803	2016.06.07.14.20
## 5885	230	60	2016.06.07	130	2.26892803	2016.06.07.14.20
## 5886	250	61	2016.06.07	110	1.91986218	2016.06.07.14.20
## 5887	330	62	2016.06.07	30	0.52359878	2016.06.07.14.20
## 5888	90	63	2016.06.07	270	-1.57079633	2016.06.07.14.20
## 5889	225	64	2016.06.07	135	2.35619449	2016.06.07.14.20
## 5890	215	65	2016.06.07	145	2.53072742	2016.06.07.14.20
## 5891	290	66	2016.06.07	70	1.22173048	2016.06.07.14.20
## 5892	90	67	2016.06.07	270	-1.57079633	2016.06.07.14.20
## 5893	240	68	2016.06.07	120	2.09439510	2016.06.07.14.20
## 5894	270	69	2016.06.07	90	1.57079633	2016.06.07.14.20
## 5895	300	70	2016.06.07	60	1.04719755	2016.06.07.14.20
## 5896	85	71	2016.06.07	275	-1.48352986	2016.06.07.14.20
## 5897	5	72	2016.06.07	355	-0.08726646	2016.06.07.14.20
## 5898	120	1	2016.06.07	240	-2.09439510	2016.06.07.14.32
## 5899	295	2	2016.06.07	65	1.13446401	2016.06.07.14.32
## 5900	120	3	2016.06.07	240	-2.09439510	2016.06.07.14.32

## 5901	180	4	2016.06.07	180	3.14159265	2016.06.07.14.32
## 5902	90	5	2016.06.07	270	-1.57079633	2016.06.07.14.32
## 5903	185	6	2016.06.07	175	3.05432619	2016.06.07.14.32
## 5904	270	7	2016.06.07	90	1.57079633	2016.06.07.14.32
## 5905	140	8	2016.06.07	220	-2.44346095	2016.06.07.14.32
## 5906	165	9	2016.06.07	195	-2.87979327	2016.06.07.14.32
## 5907	210	10	2016.06.07	150	2.61799388	2016.06.07.14.32
## 5908	150	11	2016.06.07	210	-2.61799388	2016.06.07.14.32
## 5909	145	12	2016.06.07	215	-2.53072742	2016.06.07.14.32
## 5910	140	13	2016.06.07	220	-2.44346095	2016.06.07.14.32
## 5911	190	14	2016.06.07	170	2.96705973	2016.06.07.14.32
## 5912	165	15	2016.06.07	195	-2.87979327	2016.06.07.14.32
## 5913	110	16	2016.06.07	250	-1.91986218	2016.06.07.14.32
## 5914	105	17	2016.06.07	255	-1.83259571	2016.06.07.14.32
## 5915	180	18	2016.06.07	180	3.14159265	2016.06.07.14.32
## 5916	230	19	2016.06.07	130	2.26892803	2016.06.07.14.32
## 5917	145	20	2016.06.07	215	-2.53072742	2016.06.07.14.32
## 5918	155	21	2016.06.07	205	-2.70526034	2016.06.07.14.32
## 5919	95	22	2016.06.07	265	-1.65806279	2016.06.07.14.32
## 5920	85	23	2016.06.07	275	-1.48352986	2016.06.07.14.32
## 5921	155	24	2016.06.07	205	-2.70526034	2016.06.07.14.32
## 5922	90	25	2016.06.07	270	-1.57079633	2016.06.07.14.32
## 5923	15	26	2016.06.07	345	-0.26179939	2016.06.07.14.32
## 5924	10	27	2016.06.07	350	-0.17453293	2016.06.07.14.32
## 5925	20	28	2016.06.07	340	-0.34906585	2016.06.07.14.32
## 5926	5	29	2016.06.07	355	-0.08726646	2016.06.07.14.32
## 5927	340	30	2016.06.07	20	0.34906585	2016.06.07.14.32
## 5928	30	31	2016.06.07	330	-0.52359878	2016.06.07.14.32
## 5929	0	32	2016.06.07	0	0.00000000	2016.06.07.14.32
## 5930	20	33	2016.06.07	340	-0.34906585	2016.06.07.14.32
## 5931	5	34	2016.06.07	355	-0.08726646	2016.06.07.14.32
## 5932	355	35	2016.06.07	5	0.08726646	2016.06.07.14.32
## 5933	5	36	2016.06.07	355	-0.08726646	2016.06.07.14.32
## 5934	0	37	2016.06.07	0	0.00000000	2016.06.07.14.32
## 5935	0	38	2016.06.07	0	0.00000000	2016.06.07.14.32
## 5936	0	39	2016.06.07	0	0.00000000	2016.06.07.14.32
## 5937	340	40	2016.06.07	20	0.34906585	2016.06.07.14.32
## 5938	0	41	2016.06.07	0	0.00000000	2016.06.07.14.32
## 5939	355	42	2016.06.07	5	0.08726646	2016.06.07.14.32
## 5940	10	43	2016.06.07	350	-0.17453293	2016.06.07.14.32
## 5941	350	44	2016.06.07	10	0.17453293	2016.06.07.14.32
## 5942	25	45	2016.06.07	335	-0.43633231	2016.06.07.14.32
## 5943	355	46	2016.06.07	5	0.08726646	2016.06.07.14.32
## 5944	350	47	2016.06.07	10	0.17453293	2016.06.07.14.32
## 5945	220	48	2016.06.07	140	2.44346095	2016.06.07.14.32
## 5946	50	49	2016.06.07	310	-0.87266463	2016.06.07.14.32
## 5947	320	50	2016.06.07	40	0.69813170	2016.06.07.14.32
## 5948	340	51	2016.06.07	20	0.34906585	2016.06.07.14.32
## 5949	320	52	2016.06.07	40	0.69813170	2016.06.07.14.32
## 5950	10	53	2016.06.07	350	-0.17453293	2016.06.07.14.32
## 5951	340	54	2016.06.07	20	0.34906585	2016.06.07.14.32
## 5952	30	55	2016.06.07	330	-0.52359878	2016.06.07.14.32
## 5953	45	56	2016.06.07	315	-0.78539816	2016.06.07.14.32
## 5954	355	57	2016.06.07	5	0.08726646	2016.06.07.14.32

## 5955	55	58	2016.06.07	305	-0.95993109	2016.06.07.14.32
## 5956	260	59	2016.06.07	100	1.74532925	2016.06.07.14.32
## 5957	355	60	2016.06.07	5	0.08726646	2016.06.07.14.32
## 5958	270	61	2016.06.07	90	1.57079633	2016.06.07.14.32
## 5959	355	62	2016.06.07	5	0.08726646	2016.06.07.14.32
## 5960	350	63	2016.06.07	10	0.17453293	2016.06.07.14.32
## 5961	50	64	2016.06.07	310	-0.87266463	2016.06.07.14.32
## 5962	160	1	2016.06.07	200	-2.79252680	2016.06.07.14.52
## 5963	130	2	2016.06.07	230	-2.26892803	2016.06.07.14.52
## 5964	155	3	2016.06.07	205	-2.70526034	2016.06.07.14.52
## 5965	105	4	2016.06.07	255	-1.83259571	2016.06.07.14.52
## 5966	130	5	2016.06.07	230	-2.26892803	2016.06.07.14.52
## 5967	225	6	2016.06.07	135	2.35619449	2016.06.07.14.52
## 5968	200	7	2016.06.07	160	2.79252680	2016.06.07.14.52
## 5969	175	8	2016.06.07	185	-3.05432619	2016.06.07.14.52
## 5970	160	9	2016.06.07	200	-2.79252680	2016.06.07.14.52
## 5971	135	10	2016.06.07	225	-2.35619449	2016.06.07.14.52
## 5972	185	11	2016.06.07	175	3.05432619	2016.06.07.14.52
## 5973	160	12	2016.06.07	200	-2.79252680	2016.06.07.14.52
## 5974	295	13	2016.06.07	65	1.13446401	2016.06.07.14.52
## 5975	240	14	2016.06.07	120	2.09439510	2016.06.07.14.52
## 5976	0	15	2016.06.07	0	0.00000000	2016.06.07.14.52
## 5977	5	16	2016.06.07	355	-0.08726646	2016.06.07.14.52
## 5978	5	17	2016.06.07	355	-0.08726646	2016.06.07.14.52
## 5979	30	18	2016.06.07	330	-0.52359878	2016.06.07.14.52
## 5980	0	19	2016.06.07	0	0.00000000	2016.06.07.14.52
## 5981	355	20	2016.06.07	5	0.08726646	2016.06.07.14.52
## 5982	5	21	2016.06.07	355	-0.08726646	2016.06.07.14.52
## 5983	20	22	2016.06.07	340	-0.34906585	2016.06.07.14.52
## 5984	20	23	2016.06.07	340	-0.34906585	2016.06.07.14.52
## 5985	355	24	2016.06.07	5	0.08726646	2016.06.07.14.52
## 5986	10	25	2016.06.07	350	-0.17453293	2016.06.07.14.52
## 5987	20	26	2016.06.07	340	-0.34906585	2016.06.07.14.52
## 5988	25	27	2016.06.07	335	-0.43633231	2016.06.07.14.52
## 5989	5	28	2016.06.07	355	-0.08726646	2016.06.07.14.52
## 5990	50	29	2016.06.07	310	-0.87266463	2016.06.07.14.52
## 5991	35	30	2016.06.07	325	-0.61086524	2016.06.07.14.52
## 5992	30	31	2016.06.07	330	-0.52359878	2016.06.07.14.52
## 5993	40	1	2016.06.07	320	-0.69813170	2016.06.07.14.59
## 5994	70	2	2016.06.07	290	-1.22173048	2016.06.07.14.59
## 5995	85	3	2016.06.07	275	-1.48352986	2016.06.07.14.59
## 5996	30	4	2016.06.07	330	-0.52359878	2016.06.07.14.59
## 5997	45	5	2016.06.07	315	-0.78539816	2016.06.07.14.59
## 5998	95	6	2016.06.07	265	-1.65806279	2016.06.07.14.59
## 5999	340	7	2016.06.07	20	0.34906585	2016.06.07.14.59
## 6000	65	8	2016.06.07	295	-1.13446401	2016.06.07.14.59
## 6001	85	9	2016.06.07	275	-1.48352986	2016.06.07.14.59
## 6002	25	10	2016.06.07	335	-0.43633231	2016.06.07.14.59
## 6003	75	11	2016.06.07	285	-1.30899694	2016.06.07.14.59
## 6004	130	12	2016.06.07	230	-2.26892803	2016.06.07.14.59
## 6005	95	13	2016.06.07	265	-1.65806279	2016.06.07.14.59
## 6006	80	14	2016.06.07	280	-1.39626340	2016.06.07.14.59
## 6007	60	15	2016.06.07	300	-1.04719755	2016.06.07.14.59
## 6008	30	16	2016.06.07	330	-0.52359878	2016.06.07.14.59

## 6009	35	17	2016.06.07	325	-0.61086524	2016.06.07.14.59
## 6010	70	18	2016.06.07	290	-1.22173048	2016.06.07.14.59
## 6011	65	19	2016.06.07	295	-1.13446401	2016.06.07.14.59
## 6012	30	20	2016.06.07	330	-0.52359878	2016.06.07.14.59
## 6013	40	21	2016.06.07	320	-0.69813170	2016.06.07.14.59
## 6014	15	22	2016.06.07	345	-0.26179939	2016.06.07.14.59
## 6015	50	23	2016.06.07	310	-0.87266463	2016.06.07.14.59
## 6016	30	24	2016.06.07	330	-0.52359878	2016.06.07.14.59
## 6017	45	25	2016.06.07	315	-0.78539816	2016.06.07.14.59
## 6018	45	26	2016.06.07	315	-0.78539816	2016.06.07.14.59
## 6019	30	27	2016.06.07	330	-0.52359878	2016.06.07.14.59
## 6020	25	28	2016.06.07	335	-0.43633231	2016.06.07.14.59
## 6021	55	29	2016.06.07	305	-0.95993109	2016.06.07.14.59
## 6022	30	30	2016.06.07	330	-0.52359878	2016.06.07.14.59
## 6023	30	31	2016.06.07	330	-0.52359878	2016.06.07.14.59
## 6024	40	32	2016.06.07	320	-0.69813170	2016.06.07.14.59
## 6025	25	33	2016.06.07	335	-0.43633231	2016.06.07.14.59
## 6026	10	34	2016.06.07	350	-0.17453293	2016.06.07.14.59
## 6027	55	35	2016.06.07	305	-0.95993109	2016.06.07.14.59
## 6028	70	36	2016.06.07	290	-1.22173048	2016.06.07.14.59
## 6029	45	37	2016.06.07	315	-0.78539816	2016.06.07.14.59
## 6030	0	38	2016.06.07	0	0.00000000	2016.06.07.14.59
## 6031	55	39	2016.06.07	305	-0.95993109	2016.06.07.14.59
## 6032	50	40	2016.06.07	310	-0.87266463	2016.06.07.14.59
## 6033	180	1	2016.06.07	180	3.14159265	2016.06.07.15.24
## 6034	185	2	2016.06.07	175	3.05432619	2016.06.07.15.24
## 6035	205	3	2016.06.07	155	2.70526034	2016.06.07.15.24
## 6036	120	4	2016.06.07	240	-2.09439510	2016.06.07.15.24
## 6037	130	5	2016.06.07	230	-2.26892803	2016.06.07.15.24
## 6038	125	6	2016.06.07	235	-2.18166156	2016.06.07.15.24
## 6039	155	7	2016.06.07	205	-2.70526034	2016.06.07.15.24
## 6040	150	8	2016.06.07	210	-2.61799388	2016.06.07.15.24
## 6041	195	9	2016.06.07	165	2.87979327	2016.06.07.15.24
## 6042	145	10	2016.06.07	215	-2.53072742	2016.06.07.15.24
## 6043	160	11	2016.06.07	200	-2.79252680	2016.06.07.15.24
## 6044	175	12	2016.06.07	185	-3.05432619	2016.06.07.15.24
## 6045	175	13	2016.06.07	185	-3.05432619	2016.06.07.15.24
## 6046	140	14	2016.06.07	220	-2.44346095	2016.06.07.15.24
## 6047	180	15	2016.06.07	180	3.14159265	2016.06.07.15.24
## 6048	95	16	2016.06.07	265	-1.65806279	2016.06.07.15.24
## 6049	160	17	2016.06.07	200	-2.79252680	2016.06.07.15.24
## 6050	90	18	2016.06.07	270	-1.57079633	2016.06.07.15.24
## 6051	100	19	2016.06.07	260	-1.74532925	2016.06.07.15.24
## 6052	90	20	2016.06.07	270	-1.57079633	2016.06.07.15.24
## 6053	70	21	2016.06.07	290	-1.22173048	2016.06.07.15.24
## 6054	80	22	2016.06.07	280	-1.39626340	2016.06.07.15.24
## 6055	60	23	2016.06.07	300	-1.04719755	2016.06.07.15.24
## 6056	40	24	2016.06.07	320	-0.69813170	2016.06.07.15.24
## 6057	230	1	2016.06.07	130	2.26892803	2016.06.07.15.28
## 6058	270	2	2016.06.07	90	1.57079633	2016.06.07.15.28
## 6059	90	3	2016.06.07	270	-1.57079633	2016.06.07.15.28
## 6060	190	4	2016.06.07	170	2.96705973	2016.06.07.15.28
## 6061	160	5	2016.06.07	200	-2.79252680	2016.06.07.15.28
## 6062	110	6	2016.06.07	250	-1.91986218	2016.06.07.15.28

## 6063	95	7	2016.06.07	265	-1.65806279	2016.06.07.15.28
## 6064	190	8	2016.06.07	170	2.96705973	2016.06.07.15.28
## 6065	190	9	2016.06.07	170	2.96705973	2016.06.07.15.28
## 6066	355	10	2016.06.07	5	0.08726646	2016.06.07.15.28
## 6067	345	11	2016.06.07	15	0.26179939	2016.06.07.15.28
## 6068	15	12	2016.06.07	345	-0.26179939	2016.06.07.15.28
## 6069	50	13	2016.06.07	310	-0.87266463	2016.06.07.15.28
## 6070	25	14	2016.06.07	335	-0.43633231	2016.06.07.15.28
## 6071	170	15	2016.06.07	190	-2.96705973	2016.06.07.15.28
## 6072	100	16	2016.06.07	260	-1.74532925	2016.06.07.15.28
## 6073	310	17	2016.06.07	50	0.87266463	2016.06.07.15.28
## 6074	25	18	2016.06.07	335	-0.43633231	2016.06.07.15.28
## 6075	355	19	2016.06.07	5	0.08726646	2016.06.07.15.28
## 6076	340	20	2016.06.07	20	0.34906585	2016.06.07.15.28
## 6077	335	21	2016.06.07	25	0.43633231	2016.06.07.15.28
## 6078	355	22	2016.06.07	5	0.08726646	2016.06.07.15.28
## 6079	35	23	2016.06.07	325	-0.61086524	2016.06.07.15.28
## 6080	5	24	2016.06.07	355	-0.08726646	2016.06.07.15.28
## 6081	10	25	2016.06.07	350	-0.17453293	2016.06.07.15.28
## 6082	40	26	2016.06.07	320	-0.69813170	2016.06.07.15.28
## 6083	20	27	2016.06.07	340	-0.34906585	2016.06.07.15.28
## 6084	20	28	2016.06.07	340	-0.34906585	2016.06.07.15.28
## 6085	25	29	2016.06.07	335	-0.43633231	2016.06.07.15.28
## 6086	30	30	2016.06.07	330	-0.52359878	2016.06.07.15.28
## 6087	40	31	2016.06.07	320	-0.69813170	2016.06.07.15.28
## 6088	80	32	2016.06.07	280	-1.39626340	2016.06.07.15.28
## 6089	45	33	2016.06.07	315	-0.78539816	2016.06.07.15.28
## 6090	50	34	2016.06.07	310	-0.87266463	2016.06.07.15.28
## 6091	25	35	2016.06.07	335	-0.43633231	2016.06.07.15.28
## 6092	50	36	2016.06.07	310	-0.87266463	2016.06.07.15.28
## 6093	35	37	2016.06.07	325	-0.61086524	2016.06.07.15.28
## 6094	80	38	2016.06.07	280	-1.39626340	2016.06.07.15.28
## 6095	35	39	2016.06.07	325	-0.61086524	2016.06.07.15.28
## 6096	65	40	2016.06.07	295	-1.13446401	2016.06.07.15.28
## 6097	50	41	2016.06.07	310	-0.87266463	2016.06.07.15.28
## 6098	55	42	2016.06.07	305	-0.95993109	2016.06.07.15.28
## 6099	40	43	2016.06.07	320	-0.69813170	2016.06.07.15.28
## 6100	70	44	2016.06.07	290	-1.22173048	2016.06.07.15.28
## 6101	355	45	2016.06.07	5	0.08726646	2016.06.07.15.28
## 6102	350	46	2016.06.07	10	0.17453293	2016.06.07.15.28
## 6103	15	47	2016.06.07	345	-0.26179939	2016.06.07.15.28
## 6104	5	48	2016.06.07	355	-0.08726646	2016.06.07.15.28
## 6105	185	49	2016.06.07	175	3.05432619	2016.06.07.15.28
## 6106	335	1	2016.06.07	25	0.43633231	2016.06.07.15.52
## 6107	15	2	2016.06.07	345	-0.26179939	2016.06.07.15.52
## 6108	0	3	2016.06.07	0	0.00000000	2016.06.07.15.52
## 6109	330	4	2016.06.07	30	0.52359878	2016.06.07.15.52
## 6110	10	5	2016.06.07	350	-0.17453293	2016.06.07.15.52
## 6111	350	6	2016.06.07	10	0.17453293	2016.06.07.15.52
## 6112	300	7	2016.06.07	60	1.04719755	2016.06.07.15.52
## 6113	355	8	2016.06.07	5	0.08726646	2016.06.07.15.52
## 6114	0	9	2016.06.07	0	0.00000000	2016.06.07.15.52
## 6115	310	10	2016.06.07	50	0.87266463	2016.06.07.15.52
## 6116	330	11	2016.06.07	30	0.52359878	2016.06.07.15.52

## 6117	345	12	2016.06.07	15	0.26179939	2016.06.07.15.52
## 6118	340	13	2016.06.07	20	0.34906585	2016.06.07.15.52
## 6119	30	14	2016.06.07	330	-0.52359878	2016.06.07.15.52
## 6120	340	15	2016.06.07	20	0.34906585	2016.06.07.15.52
## 6121	40	16	2016.06.07	320	-0.69813170	2016.06.07.15.52
## 6122	45	17	2016.06.07	315	-0.78539816	2016.06.07.15.52
## 6123	40	18	2016.06.07	320	-0.69813170	2016.06.07.15.52
## 6124	50	19	2016.06.07	310	-0.87266463	2016.06.07.15.52
## 6125	60	20	2016.06.07	300	-1.04719755	2016.06.07.15.52
## 6126	50	21	2016.06.07	310	-0.87266463	2016.06.07.15.52
## 6127	40	22	2016.06.07	320	-0.69813170	2016.06.07.15.52
## 6128	50	23	2016.06.07	310	-0.87266463	2016.06.07.15.52
## 6129	30	24	2016.06.07	330	-0.52359878	2016.06.07.15.52
## 6130	60	25	2016.06.07	300	-1.04719755	2016.06.07.15.52
## 6131	40	26	2016.06.07	320	-0.69813170	2016.06.07.15.52
## 6132	45	27	2016.06.07	315	-0.78539816	2016.06.07.15.52
## 6133	45	28	2016.06.07	315	-0.78539816	2016.06.07.15.52
## 6134	35	29	2016.06.07	325	-0.61086524	2016.06.07.15.52
## 6135	15	30	2016.06.07	345	-0.26179939	2016.06.07.15.52
## 6136	10	31	2016.06.07	350	-0.17453293	2016.06.07.15.52
## 6137	220	1	2016.06.07	140	2.44346095	2016.06.07.16.04
## 6138	250	2	2016.06.07	110	1.91986218	2016.06.07.16.04
## 6139	230	3	2016.06.07	130	2.26892803	2016.06.07.16.04
## 6140	250	4	2016.06.07	110	1.91986218	2016.06.07.16.04
## 6141	140	5	2016.06.07	220	-2.44346095	2016.06.07.16.04
## 6142	120	6	2016.06.07	240	-2.09439510	2016.06.07.16.04
## 6143	120	7	2016.06.07	240	-2.09439510	2016.06.07.16.04
## 6144	150	8	2016.06.07	210	-2.61799388	2016.06.07.16.04
## 6145	125	9	2016.06.07	235	-2.18166156	2016.06.07.16.04
## 6146	195	10	2016.06.07	165	2.87979327	2016.06.07.16.04
## 6147	255	11	2016.06.07	105	1.83259571	2016.06.07.16.04
## 6148	185	12	2016.06.07	175	3.05432619	2016.06.07.16.04
## 6149	280	13	2016.06.07	80	1.39626340	2016.06.07.16.04
## 6150	145	14	2016.06.07	215	-2.53072742	2016.06.07.16.04
## 6151	185	15	2016.06.07	175	3.05432619	2016.06.07.16.04
## 6152	240	16	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6153	230	17	2016.06.07	130	2.26892803	2016.06.07.16.04
## 6154	225	18	2016.06.07	135	2.35619449	2016.06.07.16.04
## 6155	230	19	2016.06.07	130	2.26892803	2016.06.07.16.04
## 6156	225	20	2016.06.07	135	2.35619449	2016.06.07.16.04
## 6157	255	21	2016.06.07	105	1.83259571	2016.06.07.16.04
## 6158	245	22	2016.06.07	115	2.00712864	2016.06.07.16.04
## 6159	240	23	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6160	240	24	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6161	245	25	2016.06.07	115	2.00712864	2016.06.07.16.04
## 6162	240	26	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6163	230	27	2016.06.07	130	2.26892803	2016.06.07.16.04
## 6164	220	28	2016.06.07	140	2.44346095	2016.06.07.16.04
## 6165	240	29	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6166	225	30	2016.06.07	135	2.35619449	2016.06.07.16.04
## 6167	240	31	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6168	220	32	2016.06.07	140	2.44346095	2016.06.07.16.04
## 6169	245	33	2016.06.07	115	2.00712864	2016.06.07.16.04
## 6170	280	34	2016.06.07	80	1.39626340	2016.06.07.16.04

## 6171	250	35	2016.06.07	110	1.91986218	2016.06.07.16.04
## 6172	245	36	2016.06.07	115	2.00712864	2016.06.07.16.04
## 6173	290	37	2016.06.07	70	1.22173048	2016.06.07.16.04
## 6174	240	38	2016.06.07	120	2.09439510	2016.06.07.16.04
## 6175	210	39	2016.06.07	150	2.61799388	2016.06.07.16.04
## 6176	185	40	2016.06.07	175	3.05432619	2016.06.07.16.04
## 6177	235	41	2016.06.07	125	2.18166156	2016.06.07.16.04
## 6178	160	42	2016.06.07	200	-2.79252680	2016.06.07.16.04
## 6179	245	43	2016.06.07	115	2.00712864	2016.06.07.16.04
## 6180	230	44	2016.06.07	130	2.26892803	2016.06.07.16.04
## 6181	160	45	2016.06.07	200	-2.79252680	2016.06.07.16.04
## 6182	235	46	2016.06.07	125	2.18166156	2016.06.07.16.04
## 6183	200	47	2016.06.07	160	2.79252680	2016.06.07.16.04
## 6184	265	48	2016.06.07	95	1.65806279	2016.06.07.16.04
## 6185	210	49	2016.06.07	150	2.61799388	2016.06.07.16.04
## 6186	265	50	2016.06.07	95	1.65806279	2016.06.07.16.04
## 6187	260	51	2016.06.07	100	1.74532925	2016.06.07.16.04
## 6188	280	52	2016.06.07	80	1.39626340	2016.06.07.16.04
## 6189	235	53	2016.06.07	125	2.18166156	2016.06.07.16.04
## 6190	220	54	2016.06.07	140	2.44346095	2016.06.07.16.04
## 6191	210	55	2016.06.07	150	2.61799388	2016.06.07.16.04
## 6192	135	1	2016.06.07	225	-2.35619449	2016.06.07.____
## 6193	30	2	2016.06.07	330	-0.52359878	2016.06.07.____
## 6194	5	3	2016.06.07	355	-0.08726646	2016.06.07.____
## 6195	350	4	2016.06.07	10	0.17453293	2016.06.07.____
## 6196	15	5	2016.06.07	345	-0.26179939	2016.06.07.____
## 6197	325	6	2016.06.07	35	0.61086524	2016.06.07.____
## 6198	330	7	2016.06.07	30	0.52359878	2016.06.07.____
## 6199	340	8	2016.06.07	20	0.34906585	2016.06.07.____
## 6200	295	9	2016.06.07	65	1.13446401	2016.06.07.____
## 6201	305	10	2016.06.07	55	0.95993109	2016.06.07.____
## 6202	350	11	2016.06.07	10	0.17453293	2016.06.07.____
## 6203	175	12	2016.06.07	185	-3.05432619	2016.06.07.____
## 6204	185	13	2016.06.07	175	3.05432619	2016.06.07.____
## 6205	190	14	2016.06.07	170	2.96705973	2016.06.07.____
## 6206	170	15	2016.06.07	190	-2.96705973	2016.06.07.____
## 6207	180	16	2016.06.07	180	3.14159265	2016.06.07.____
## 6208	175	17	2016.06.07	185	-3.05432619	2016.06.07.____
## 6209	180	18	2016.06.07	180	3.14159265	2016.06.07.____
## 6210	185	19	2016.06.07	175	3.05432619	2016.06.07.____
## 6211	160	20	2016.06.07	200	-2.79252680	2016.06.07.____
## 6212	185	21	2016.06.07	175	3.05432619	2016.06.07.____
## 6213	180	22	2016.06.07	180	3.14159265	2016.06.07.____
## 6214	190	23	2016.06.07	170	2.96705973	2016.06.07.____
## 6215	170	24	2016.06.07	190	-2.96705973	2016.06.07.____
## 6216	180	25	2016.06.07	180	3.14159265	2016.06.07.____
## 6217	265	26	2016.06.07	95	1.65806279	2016.06.07.____
## 6218	180	27	2016.06.07	180	3.14159265	2016.06.07.____
## 6219	190	28	2016.06.07	170	2.96705973	2016.06.07.____
## 6220	60	29	2016.06.07	300	-1.04719755	2016.06.07.____
## 6221	250	30	2016.06.07	110	1.91986218	2016.06.07.____
## 6222	245	31	2016.06.07	115	2.00712864	2016.06.07.____
## 6223	150	32	2016.06.07	210	-2.61799388	2016.06.07.____
## 6224	220	33	2016.06.07	140	2.44346095	2016.06.07.____

## 6225	170	34	2016.06.07	190	-2.96705973	2016.06.07.---	---
## 6226	235	35	2016.06.07	125	2.18166156	2016.06.07.---	---
## 6227	200	36	2016.06.07	160	2.79252680	2016.06.07.---	---
## 6228	210	37	2016.06.07	150	2.61799388	2016.06.07.---	---
## 6229	185	38	2016.06.07	175	3.05432619	2016.06.07.---	---
## 6230	165	39	2016.06.07	195	-2.87979327	2016.06.07.---	---
## 6231	170	40	2016.06.07	190	-2.96705973	2016.06.07.---	---
##	brightn	colour	cet_time	sun_az	sun_el	sun_el_rad	sun_az_rad
## 1	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 2	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 3	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 4	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 5	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 6	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 7	l	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 8	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 9	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 10	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 11	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 12	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 13	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 14	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 15	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 16	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 17	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 18	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 19	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 20	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 21	h	u	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 22	h	g	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 23	h	g	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 24	h	g	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 25	h	g	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 26	h	g	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 27	h	g	2016-05-31 14:24:00	208.3263	58.81733	1.0265560	-2.647205
## 28	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 29	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 30	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 31	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 32	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 33	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 34	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 35	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 36	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 37	l	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 38	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 39	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 40	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 41	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 42	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 43	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 44	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 45	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008
## 46	h	u	2016-05-31 14:53:00	219.9114	56.24790	0.9817111	-2.445008

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

##	533		h	u	2016-06-28	14:12:00	201.	2400	61.	15642	1.0673808	-2.770885
##	534		h	u	2016-06-28	14:12:00	201.	2400	61.	15642	1.0673808	-2.770885
##	535		h	u	2016-06-28	14:12:00	201.	2400	61.	15642	1.0673808	-2.770885
##	536		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	537		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	538		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	539		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	540		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	541		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	542		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	543		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	544		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	545		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	546		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	547		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	548		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	549		l	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	550		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	551		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	552		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	553		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	554		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	555		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	556		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	557		h	u	2016-06-28	14:41:00	213.	9226	59.	03113	1.0302875	-2.549532
##	558		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	559		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	560		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	561		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	562		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	563		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	564		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	565		l	u	2016-06-28	15:27:00	230.	9742	54.	11640	0.9445094	-2.251926
##	566		l	u	2							

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## 911	l	u	<NA>	NA	NA	NA	NA
## 912	l	u	<NA>	NA	NA	NA	NA
## 913	l	u	<NA>	NA	NA	NA	NA
## 914	l	u	<NA>	NA	NA	NA	NA
## 915	l	u	<NA>	NA	NA	NA	NA
## 916	l	u	<NA>	NA	NA	NA	NA
## 917	l	u	<NA>	NA	NA	NA	NA
## 918	l	u	<NA>	NA	NA	NA	NA
## 919	h	u	<NA>	NA	NA	NA	NA
## 920	h	u	<NA>	NA	NA	NA	NA
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## 922	h	u	<NA>	NA	NA	NA	NA
## 923	h	u	<NA>	NA	NA	NA	NA
## 924	h	u	<NA>	NA	NA	NA	NA
## 925	h	u	<NA>	NA	NA	NA	NA
## 926	h	u	<NA>	NA	NA	NA	NA
## 927	h	u	<NA>	NA	NA	NA	NA
## 928	h	u	<NA>	NA	NA	NA	NA
## 929	h	u	<NA>	NA	NA	NA	NA
## 930	h	u	<NA>	NA	NA	NA	NA
## 931	h	u	<NA>	NA	NA	NA	NA
## 932	h	u	<NA>	NA	NA	NA	NA
## 933	h	u	<NA>	NA	NA	NA	NA
## 934	h	u	<NA>	NA	NA	NA	NA
## 935	h	u	<NA>	NA	NA	NA	NA
## 936	h	u	<NA>	NA	NA	NA	NA
## 937	h	g	<NA>	NA	NA	NA	NA
## 938	h	g	<NA>	NA	NA	NA	NA
## 939	h	g	<NA>	NA	NA	NA	NA
## 940	h	g	<NA>	NA	NA	NA	NA
## 941	h	g	<NA>	NA	NA	NA	NA
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## 946	h	g	<NA>	NA	NA	NA	NA
## 947	h	g	<NA>	NA	NA	NA	NA
## 948	h	g	<NA>	NA	NA	NA	NA
## 949	h	g	<NA>	NA	NA	NA	NA
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## 956	h	g	<NA>	NA	NA	NA	NA
## 957	l	g	<NA>	NA	NA	NA	NA
## 958	l	g	<NA>	NA	NA	NA	NA
## 959	l	g	<NA>	NA	NA	NA	NA
## 960	l	g	<NA>	NA	NA	NA	NA
## 961	l	g	<NA>	NA	NA	NA	NA
## 962	l	g	<NA>	NA	NA	NA	NA
## 963	l	g	<NA>	NA	NA	NA	NA
## 964	l	g	<NA>	NA	NA	NA	NA

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##	1181	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1182	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1183	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1184	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1185	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1186	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1187	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1188	h	u	2016-07-07	13:47:00	188.3641	61.50626	1.0734868	-2.995611
##	1189	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1190	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1191	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1192	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1193	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1194	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1195	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1196	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1197	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1198	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1199	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1200	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1201	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1202	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1203	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1204	l	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1205	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1206	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1207	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1208	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1209	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1210	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1211	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1212	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1213	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1214	h	u	2016-07-07	14:14:00	201.0314	60.42006	1.0545289	-2.774525
##	1215	l	u	2016-07-08	13:32:00	181.0122	61.58841	1.0749206	-3.123927
##	1216	l	u	2016-07-08	13:32:00	181.0122	61.58841	1.0749206	-3.123927
##	1217	l	u	2016-07-08	13:32:00	181			

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##	1505	h	g	2016-07-12	13:41:00	185.0460	60.99399	1.0645460	-3.053523
##	1506	h	g	2016-07-12	13:41:00	185.0460	60.99399	1.0645460	-3.053523
##	1507	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1508	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1509	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1510	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1511	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1512	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1513	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1514	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1515	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1516	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1517	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1518	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1519	l	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1520	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1521	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1522	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1523	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1524	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1525	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1526	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1527	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1528	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1529	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1530	h	u	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1531	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1532	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1533	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1534	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1535	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1536	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1537	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1538	h	g	2016-07-12	14:19:00	202.6326	59.54888	1.0393241	-2.746579
##	1539	l	u	2016-07-12	14:43:00	212.7959	57.78201	1.0084863	-2.569197
##	1540	l	u	2016-07-12	14:43:00	212.7959	57.78201	1.0084863	-2.569197
##	1541	l	u	2016-07-12	14:43:00	212			

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##	4098	l	g	2016-09-08	15:56:00	228.6351	34.46512	0.6015297	-2.292750
##	4099	l	g	2016-09-08	15:56:00	228.6351	34.46512	0.6015297	-2.292750
##	4100	l	g	2016-09-08	15:56:00	228.6351	34.46512	0.6015297	-2.292750
##	4101	l	g	2016-09-08	15:56:00	228.6351	34.46512	0.6015297	-2.292750
##	4102	l	g	2016-09-08	15:56:00	228.6351	34.46512	0.6015297	-2.292750
##	4103	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4104	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4105	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4106	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4107	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4108	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4109	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4110	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4111	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4112	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4113	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4114	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4115	l	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4116	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4117	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4118	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4119	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4120	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4121	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4122	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4123	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4124	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4125	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4126	h	u	2016-09-08	16:58:00	243.8616	26.31986	0.4593681	-2.026998
##	4127	l	u	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4128	l	u	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4129	l	u	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4130	l	u	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4131	l	u	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4132	l	u	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4133	l	u	2016-09-08	18:00:00	257			

##	4151	l	g	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4152	l	g	2016-09-08	18:00:00	257.1945	17.10279	0.2985000	-1.794294
##	4153	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4154	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4155	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4156	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4157	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4158	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4159	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4160	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4161	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4162	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4163	l	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4164	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4165	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4166	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4167	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4168	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4169	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4170	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4171	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4172	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4173	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4174	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4175	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4176	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4177	h	u	2016-09-08	18:21:00	261.4316	13.85026	0.2417326	-1.720342
##	4178	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4179	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4180	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4181	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4182	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4183	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4184	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4185	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4186	l	g	2016-09-12	13:50:00	189.9102	42.71603	0.7455353	-2.968627
##	4187	h	g	2016-09-12	13:50:00	189			

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##	4961	l	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4962	l	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4963	l	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4964	l	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4965	h	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4966	h	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4967	h	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4968	h	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4969	h	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4970	h	g	2016-09-14	16:54:00	242.0032	24.70077	0.4311098	-2.059433
##	4971	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4972	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4973	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4974	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4975	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4976	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4977	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4978	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4979	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4980	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4981	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4982	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4983	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4984	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4985	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4986	l	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4987	h	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4988	h	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4989	h	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4990	h	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4991	h	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4992	h	u	2016-09-14	17:25:00	248.8508	20.24892	0.3534104	-1.939919
##	4993	l	u	2016-09-27	15:52:00	225.3685	27.66289	0.4828086	-2.349762
##	4994	l	u	2016-09-27	15:52:00	225.3685	27.66289	0.4828086	-2.349762
##	4995	l	u	2016-09-27	15:52:00	225.3685	27.66289	0.4828086	-2.349762
##	4996	l	u	2016-09-27	15:52:00	225.3685	27.66289	0.4828086	-2.349762
##	4997	l	u	2016-09-27	15:52:00	225			

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## 5123	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5124	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5125	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5126	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5127	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5128	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5129	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5130	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5131	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5132	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5133	h	u	2016-09-29	15:43:00	222.8706	27.88340	0.4866572	-2.393360
## 5134	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5135	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5136	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5137	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5138	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5139	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5140	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5141	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5142	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5143	l	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5144	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5145	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5146	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5147	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5148	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5149	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5150	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5151	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5152	h	u	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5153	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5154	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5155	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5156	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5157	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5158	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5159	l	g	2016-06-02	12:51:00	164.7835	60.80519	1.0612507	2.876015
## 5160	l	u	2016-06-02	13:07:00	172.3627	61.30752	1.0700181	3.0

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##	5663	h	u	2016-06-21	14:52:00	219.0833	58.03822	1.0129580	-2.459460
##	5664	h	g	2016-06-21	14:52:00	219.0833	58.03822	1.0129580	-2.459460
##	5665	h	g	2016-06-21	14:52:00	219.0833	58.03822	1.0129580	-2.459460
##	5666	h	g	2016-06-21	14:52:00	219.0833	58.03822	1.0129580	-2.459460
##	5667	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5668	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5669	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5670	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5671	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5672	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5673	l	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5674	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5675	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5676	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5677	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5678	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5679	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5680	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5681	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5682	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5683	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5684	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5685	h	u	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5686	h	g	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5687	h	g	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5688	h	g	2016-06-21	15:24:00	230.6497	54.47031	0.9506862	-2.257589
##	5689	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5690	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5691	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5692	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5693	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5694	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5695	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5696	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5697	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5698	l	u	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5699	l	u	2016-06-21	15:29:00	232			

##	5717	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5718	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5719	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5720	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5721	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5722	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5723	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5724	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5725	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5726	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5727	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5728	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5729	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5730	h	g	2016-06-21	15:29:00	232.2992	53.85265	0.9399061	-2.228800
##	5731	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5732	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5733	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5734	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5735	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5736	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5737	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5738	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5739	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5740	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5741	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5742	l	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5743	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5744	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5745	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5746	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5747	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5748	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5749	h	u	2016-06-21	15:50:00	238.8116	51.11791	0.8921758	-2.115136
##	5750	l	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5751	l	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5752	l	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5753	l	u	2016-06-07	13:59:00	197			

##	5771	h	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5772	h	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5773	h	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5774	h	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5775	h	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5776	h	u	2016-06-07	13:59:00	197.0537	61.18983	1.0679640	-2.843949
##	5777	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5778	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5779	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5780	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5781	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5782	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5783	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
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##	5785	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5786	l	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5787	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
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##	5790	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
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##	5795	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5796	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5797	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
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##	5799	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5800	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5801	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5802	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5803	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5804	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5805	h	u	2016-06-07	14:05:00	199.8404	60.89037	1.0627375	-2.795312
##	5806	l	u	2016-06-07	14:10:00	202.1256	60.60794	1.0578081	-2.755429
##	5807	l	u	2016-06-07	14:10:00	202			

350

[illegible]

352

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## 5319	35
## 5320	90
## 5321	85
## 5322	135
## 5323	120
## 5324	100
## 5325	70
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## 5327	90
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## 5334	-15
## 5335	45
## 5336	-15
## 5337	15
## 5338	5
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## 5340	10
## 5341	0
## 5342	10
## 5343	0
## 5344	-15
## 5345	-10
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## 5348	60
## 5349	80
## 5350	90
## 5351	70
## 5352	60
## 5353	60
## 5354	85
## 5355	90
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## 5362	20
## 5363	-25
## 5364	-20
## 5365	-70
## 5366	-10
## 5367	30
## 5368	-10
## 5369	100
## 5370	80

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## 5374	70
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## 5395	-50
## 5396	-20
## 5397	-10
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## 5401	-35
## 5402	-40
## 5403	-50
## 5404	-35
## 5405	-40
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## 5409	40
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## 5417	110
## 5418	95
## 5419	55
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## 5421	85
## 5422	60
## 5423	-5
## 5424	20

## 5425	-175
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## 5427	90
## 5428	-150
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## 5430	-90
## 5431	10
## 5432	-110
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## 5436	90
## 5437	115
## 5438	90
## 5439	95
## 5440	90
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## 5444	-15
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## 5448	-20
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## 5451	-60
## 5452	140
## 5453	-40
## 5454	80
## 5455	-90
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## 5488	-40
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## 5490	-10
## 5491	150
## 5492	-30
## 5493	20
## 5494	-40
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## 5523	90
## 5524	60
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## 5530	15
## 5531	140
## 5532	150

## 5533	-5
## 5534	-160
## 5535	140
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## 5537	-120
## 5538	170
## 5539	30
## 5540	130
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## 5542	30
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## 5544	165
## 5545	145
## 5546	100
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## 5568	80
## 5569	85
## 5570	75
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## 5572	55
## 5573	85
## 5574	70
## 5575	70
## 5576	170
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## 5578	180
## 5579	40
## 5580	-175
## 5581	-20
## 5582	125
## 5583	10
## 5584	180
## 5585	-160
## 5586	170

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## 5601	90
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## 5605	170
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## 5608	150
## 5609	0
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## 5629	105
## 5630	110
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## 5694	-120

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## 5696	-60
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## 5726	10
## 5727	25
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## 5730	50
## 5731	-60
## 5732	-60
## 5733	-105
## 5734	-45
## 5735	100
## 5736	-70
## 5737	-30
## 5738	-60
## 5739	55
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## 5741	40
## 5742	90
## 5743	-40
## 5744	-5
## 5745	-5
## 5746	0
## 5747	-10
## 5748	-5

## 5749	-5
## 5750	-175
## 5751	-155
## 5752	135
## 5753	150
## 5754	145
## 5755	175
## 5756	-160
## 5757	-175
## 5758	-130
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## 5769	50
## 5770	-30
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## 5773	-10
## 5774	-30
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## 5779	100
## 5780	110
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## 5782	175
## 5783	-175
## 5784	160
## 5785	-140
## 5786	-160
## 5787	-10
## 5788	-10
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## 5801	10
## 5802	-10

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## 5805	-5
## 5806	-160
## 5807	170
## 5808	-150
## 5809	180
## 5810	-75
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## 5814	-145
## 5815	-170
## 5816	-160
## 5817	-145
## 5818	-165
## 5819	-125
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## 5821	-30
## 5822	5
## 5823	-5
## 5824	-10
## 5825	-5
## 5826	120
## 5827	110
## 5828	95
## 5829	90
## 5830	95
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## 5834	100
## 5835	120
## 5836	110
## 5837	80
## 5838	60
## 5839	-165
## 5840	160
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## 5842	135
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## 5844	110
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## 5879	-10
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## 5886	110
## 5887	30
## 5888	-90
## 5889	135
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## 5892	-90
## 5893	120
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## 5895	60
## 5896	-85
## 5897	-5
## 5898	-120
## 5899	65
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## 5901	180
## 5902	-90
## 5903	175
## 5904	90
## 5905	-140
## 5906	-165
## 5907	150
## 5908	-150
## 5909	-145
## 5910	-140

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## 5913	-110
## 5914	-105
## 5915	180
## 5916	130
## 5917	-145
## 5918	-155
## 5919	-95
## 5920	-85
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## 5922	-90
## 5923	-15
## 5924	-10
## 5925	-20
## 5926	-5
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## 5928	-30
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## 5954	5
## 5955	-55
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## 5957	5
## 5958	90
## 5959	5
## 5960	10
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## 5962	-160
## 5963	-130
## 5964	-155

## 5965	-105
## 5966	-130
## 5967	135
## 5968	160
## 5969	-175
## 5970	-160
## 5971	-135
## 5972	175
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## 5978	-5
## 5979	-30
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## 5982	-5
## 5983	-20
## 5984	-20
## 5985	5
## 5986	-10
## 5987	-20
## 5988	-25
## 5989	-5
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## 5992	-30
## 5993	-40
## 5994	-70
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## 5996	-30
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## 5998	-95
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## 6026	-10
## 6027	-55
## 6028	-70
## 6029	-45
## 6030	0
## 6031	-55
## 6032	-50
## 6033	180
## 6034	175
## 6035	155
## 6036	-120
## 6037	-130
## 6038	-125
## 6039	-155
## 6040	-150
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## 6043	-160
## 6044	-175
## 6045	-175
## 6046	-140
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## 6049	-160
## 6050	-90
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## 6053	-70
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## 6055	-60
## 6056	-40
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## 6062	-110
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## 6086	-30
## 6087	-40
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## 6091	-25
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## 6099	-40
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## 6103	-15
## 6104	-5
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## 6187	100
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## 6193	-30
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## 6204	175
## 6205	170
## 6206	-170
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## 6219	170
## 6220	-60
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## 6222	115
## 6223	-150
## 6224	140
## 6225	-170
## 6226	125
## 6227	160
## 6228	150
## 6229	175
## 6230	-165
## 6231	-170

```
#Edrich et al. 1979
print(ed)
```



```

##          xx intensity      nn accuracy wavelength colour log10_intensity
## 1  -31.208  2.15e+09 13.010      0.14        354      UV          9.332438
## 2  -20.263  3.69e+09 14.249      0.15        535 Ogreen          9.567026
## 3   -8.217  6.67e+09 19.135      0.20        354      UV          9.824126
## 4   -1.342  9.36e+09 15.350      0.16        535 Ogreen          9.971276
## 5   15.118  2.10e+10 30.769      0.33        354      UV         10.322219
## 6   24.548  3.35e+10 14.731      0.16        535 Ogreen         10.525045
## 7   32.671  4.99e+10 70.349      0.74        354      UV         10.698101
## 8   45.199  9.25e+10 37.928      0.40        535 Ogreen         10.966142
## 9   42.996  8.30e+10 10.463      0.11        429     blue         10.919078
## 10  61.994  2.12e+11 75.030      0.79        354      UV         11.326336
## 11  69.360  3.04e+11 70.349      0.74        535 Ogreen         11.482874
## 12  71.254  3.34e+11 48.460      0.51        429     blue         11.523746
## 13  96.412  1.15e+12 85.355      0.90        535 Ogreen         12.060698
## 14  95.448  1.10e+12 72.208      0.76        429     blue         12.041393
## 15 104.741  1.74e+12 78.059      0.83        354      UV         12.240549
## 16 118.026  3.34e+12 80.468      0.85        429     blue         12.523746
## 17 120.986  3.86e+12 89.657      0.95        535 Ogreen         12.586587
## 18 141.233  1.05e+13 90.189      0.95        535 Ogreen         13.021189
## 19 141.352  1.05e+13 83.978      0.89        429     blue         13.021189
## 20 165.252  3.41e+13 91.722      0.97        535 Ogreen         13.532754
##      logit_accuracy
## 1      -1.81528997
## 2      -1.73460106
## 3      -1.38629436
## 4      -1.65822808
## 5      -0.70818506
## 6      -1.65822808
## 7       1.04596856
## 8      -0.40546511
## 9      -2.09074110
## 10     1.32492541
## 11     1.04596856
## 12     0.04000533
## 13     2.19722458
## 14     1.15267951
## 15     1.58562726
## 16     1.73460106
## 17     2.94443898
## 18     2.94443898
## 19     2.09074110
## 20     3.47609869

```

Session Information

Click to expand session metadata

```

## R version 4.5.1 (2025-06-13)
## Platform: x86_64-apple-darwin20
## Running under: macOS Sequoia 15.7.9
##
## Matrix products: default
## BLAS:      /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRblas.0.dylib

```

```

## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRlapack.dylib; LAPACK
##
## locale:
## [1] en_GB.UTF-8/en_GB.UTF-8/en_GB.UTF-8/C/en_GB.UTF-8/en_GB.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] geosphere_1.5-20 emmeans_1.11.2-8 ARTool_0.11.2   brms_2.23.0
## [5] Rcpp_1.1.0      CircMLE_0.3.0   circular_0.5-2
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1      dplyr_1.1.4          farver_2.1.2
## [4] loo_2.8.0             S7_0.2.0             fastmap_1.2.0
## [7] tensorA_0.36.2.1     digest_0.6.37        estimability_1.5.1
## [10] lifecycle_1.0.4      StanHeaders_2.32.10  processx_3.8.6
## [13] magrittr_2.0.4       posterior_1.6.1      compiler_4.5.1
## [16] rlang_1.1.6          tools_4.5.1          yaml_2.3.10
## [19] data.table_1.17.8    knitr_1.50           labeling_0.4.3
## [22] bridgesampling_1.1-2 sp_2.2-0             pkgbuild_1.4.8
## [25] plyr_1.8.9           RColorBrewer_1.1-3   cmdstanr_0.9.0.9000
## [28] abind_1.4-8          withr_3.0.2          purrr_1.1.0
## [31] grid_4.5.1           stats4_4.5.1         inline_0.3.21
## [34] ggplot2_4.0.0        scales_1.4.0         MASS_7.3-65
## [37] cli_3.6.5           mvtnorm_1.3-3        rmarkdown_2.30
## [40] reformulas_0.4.3.1   generics_0.1.4       RcppParallel_5.1.11-1
## [43] reshape2_1.4.4       minqa_1.2.8          energy_1.7-12
## [46] rstan_2.32.7         stringr_1.5.2        splines_4.5.1
## [49] bayesplot_1.14.0     parallel_4.5.1       matrixStats_1.5.0
## [52] vctr_0.6.5           boot_1.3-31          Matrix_1.7-3
## [55] jsonlite_2.0.0       carData_3.0-5        callr_3.7.6
## [58] car_3.1-3            pbkrtest_0.5.5       Formula_1.2-5
## [61] tidyr_1.3.1          glue_1.8.0           nloptr_2.2.1
## [64] codetools_0.2-20     ps_1.9.1             distributional_0.5.0
## [67] stringi_1.8.7        gtable_0.3.6         QuickJSR_1.8.1
## [70] lme4_1.1-37          gsl_2.1-8            tibble_3.3.0
## [73] pillar_1.11.1        htmltools_0.5.8.1    Brodingtonag_1.2-9
## [76] R6_2.6.1            Rdpack_2.6.4         evaluate_1.0.5
## [79] lattice_0.22-7       rbibutils_2.3        backports_1.5.0
## [82] broom_1.0.10         rstantools_2.5.0     gridExtra_2.3
## [85] coda_0.19-4.1        nlme_3.1-168         checkmate_2.3.3
## [88] xfun_0.57            pkgconfig_2.0.3

```